Technical Appendices

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This revision updates my mathematical appendix from *How To Build Conscious Machines: Deriving Intelligence, Life and Mind from Time* [Bennett, 2027]. Current book definitions, statements and proofs govern shared material; independent earlier results are retained. The elements of Φ are now called distinctions and its subsets possible facts. A finite set of possible facts is a thing, a possible thing when its intersection is nonempty and an impossible thing otherwise. My earlier papers call a distinction a state, a possible fact a program, a possible thing a statement or an aspect, and a thing a finite bundle. A freer possible thing was called a weaker one. Claims and notation precede formal development. Experimental results follow. Earlier foundations appear in my doctoral thesis [Bennett, 2025b].

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Notation guide

Notation	Meaning and formal source
Φ, $\mathcal{P}$, $\mathfrak{v}$	Environment, possible facts, vocabulary; Definition 1. Earlier papers call distinctions states and possible facts programs.
$L_{\mathfrak{v}}$, l, $T(l)$	Embodied language, thing or possible thing, truth set; Definition 4. Earlier papers call a thing a finite bundle and a possible thing a statement or an aspect.
$\mathrm{Ext}(l)$, $\mathrm{Ext}(X)$, $\mathrm{ff}(l)$	Extension of a possible thing, extension of a set, and freedom; Definition 5. Earlier notation: E_l, E_X, $w(l)$. Earlier papers call freedom weakness and call a freer possible thing weaker.
$<_{\mathrm{ff}}$, $<_s$	Freedom and simplicity selectors; Definition 19.
$\mathrm{Enc}_{\mathfrak{v}}$, $S_{\mathfrak{v}}$, $\kappa_{\mathfrak{v}}$, $\kappa^{\tau}_{\mathfrak{v}}$	Encoding, layer-space count, layer-time count and induced clock; Definition 8.
τ, h, $\tau_{\leq t}$	Timeline, finite fragment and prefix; Definition 2.
objective time, tick, time	Objective time is the ordering of environmental distinctions, each of which is a change. Each index u of a timeline τ is a time, a position in the timeline, which can be thought of as a tick of the objective clock. The timeline assigns one distinction $\tau(u)$ to each tick u. A thing is true at tick u when it is true given $\tau(u)$. A tick of subjective time in an abstraction layer is a tick at which the encoding in that layer's vocabulary differs from the encoding at the tick before; Definitions 2 and 8.
$\rho_{\tau,[u_1,u_2]}$, $\geq_{\tau,[u_1,u_2]}$	Tenure, the period of objective time for which a thing persists, and the persistence ordering; Definitions 14 and 15.
α, I_α, O_α, $\Gamma_{\mathfrak{v}}$	Task, inputs, correct outputs and task space; Definition 16.
π, Π_α	Policy and correct-policy set; Definition 17.
$\sqsubset$, $\ell_{\mathfrak{v}}$, λ_ρ	Strict child relation, task height and disembodied task; Definitions 16 and 18.
U, B_π, $G_P(\pi)$	Unseen region, policy buffer and demand-survival probability; Definitions 20 and 22.
μ, ff_μ	Reference measure and measured freedom; Definition 22.
β_γ, $\epsilon(\gamma)$	Unseen-buffer value and utility of intelligence; Definition 25.
$\mathfrak{f}$, χ_i, g_i^{χ}	Abstractor, grounding choice and one-step grounding; Definitions 6 and 7.
$\mathfrak{v}^i$, π^i, λ^i	Vocabulary, policy and disembodied task at stack layer i; Definition 26.
$K_\rho(M)$, R_ρ, $\Delta_{\mathcal{A}}(M)$	Irreducible regret, normalised regret and the loss from restricting the available policies to $\mathcal{A}$; Theorem 11 and Corollary 18. Here ρ denotes weights.
$\mathcal{V}$, $F(\pi)$, $I(\pi)$	Compared viable policies, survival probability and functional information; Theorem 14.
$\Delta(c)$, P_τ, P_N	Continuation demands, induced novelty law and exogenous novelty law; Definitions 29 and 31.
δ, $\mathbb{D}_H$, Q_H	Damage history, history space and damage law; Definition 30.
$R_\pi(H)$, $q_\pi(H)$, m_π, $k(\pi)$	Mend, probability of mending, finite capacity to mend damage and failure-point count; Definition 30.
$P_H(\pi)$, $\Lambda_H(\pi)$	Joint persistence probability and log persistence; Definition 31.
μ, μ_j, Ω_π, $F_2(\pi)$	Viability, the viability of part j of a body, viable completions and viability divergence; Corollary 14 and Theorem 18. A part of a body can be anything from a part of a cell up to the whole body. This μ is distinct from the reference measure above.
INT, OBS, $\mathcal{C}(INT, OBS)$	Interventions, matched observations and causal identities; Definitions 32 and 33.
$\mathfrak{o}$, $\mathfrak{p}_{\mathfrak{o}}$, $\mu_{\mathfrak{o}}$, $<_{\mathfrak{o}}$	Organism, retained policies, viability task and preferences; Definition 35.
$\mathfrak{o}^1$, $\mathfrak{o}^n$, $c^{\mathfrak{r}_1\cdots\mathfrak{r}_k}_{\mathfrak{a}}$	First/higher-order selves and nested causal identities; Definitions 37–39.

Notation	Meaning and formal source
$\mathfrak{s}_{\mathfrak{o}}$, $\mathfrak{S}_{\mathfrak{o}}(i)$, $\alpha^{\star}$	Protosymbol system, signified protosymbols and selected interpretation; Definitions 40 and 41.
Π'_{ω}, Π'_{ω_j}	Restricted collective and component policy sets; Definition 45.
ℓ, Sites_{ℓ}, Graph_{ℓ}, c_{conc}, θ	Moment, physical sites, interaction graph, concurrency capacity and physical time window; Definitions 47 and 48.
$D(\ell)$, $L_{\max}$, v, hop	Diameter, longest required edge, maximum signal speed and hop diameter; Definition 49.
$\mathsf{I}(\mathfrak{v})$, $\mathsf{I}_{\max}(\mathcal{R})$	Encoding capacity and physical patch ceiling; Definitions 12 and 13.
$\beta_{\lambda_\rho,V}$	Semantic Unagi score; Definition 27.
O_A, $\mathfrak{v}^{\mathrm{conf}}_{M,S}$	Oracle fact and bounded-machine configuration vocabulary; Definitions 52 and 10.
φ, (W,b), $\mathcal{U}$, PP	Feature map, shared readout, evaluation units and pair proxy; Definitions 57, 59 and 62.

Claims map

Here is the map of the claims made in my book [Bennett, 2027], along with the evidence type that supports each claim[1]. Each claim is worded as in the book and followed by further explanation. In that explanation, "this book" and the names of chapters and appendices refer to the book, while theorem, proposition, lemma and corollary numbers are those of this document.

Status: M, mathematical result; C, conditional deduction from premises about what its subject is; E, empirical report; I, interpretation; H, hypothesis or design proposal; Q, conjecture. Formal statements specify the premises. Entries may combine statuses. An interpretation may be the conclusion of a cumulative argument supported by several results. The constitutive explanation of consciousness in Section 20 draws on the mathematical results to explain qualitative character, the subject and reflective awareness together.

Claim	Status	Statements, proofs and evidence
The Cosmic Ought. Every timeline orders possible things by persistence. This persistence ordering is the discriminator from which all other discriminations, such as norms, are derived. Put another way, it is a selection process that precedes natural selection.	M, I	Lemma 6; Corollary 4; Propositions 8 and 9.
Change induces timelines. Timelines preserve some things and destroy others. Persistence orderings are then a minimal source of ought before organisms appear. Timelines that have different distinctions at the same time rank possible things differently at that time (in every vocabulary in which those distinctions have different encodings). One possible thing entails another when the second is true given every distinction of which the first is true. In that case the entailed possible thing persists at least as long as the entailing possible thing in every timeline. These are the only rankings shared by all timelines. If neither of two possible things entails the other, each persists longer than the other in some timeline. No possible thing persists longer than another in every timeline.		
The freest correct policy has the highest probability of satisfying unseen demands under every distribution thereof.	M	Theorem 2; Lemma 7; Corollary 6.
Freest correct generalisation. On every task with an input and a correct policy, the freest correct policy is unique and its extension contains that of every other correct policy. Hence the freest correct policy maximises the probability of satisfying unseen demands under every distribution of those demands. Under exchangeable demands, freedom also ranks the other correct policies. In my earlier work where these results were first presented freedom was called weakness of constraint, and freest correct was weakest correct.		
Bennett's Razor and Unagi. Explanations should be no more specific than necessary. Unagi is an upper bound upon intelligence, attained by pairing the ideal embodied vocabulary with the freest correct policy in that vocabulary.	M, H	Proposition 23; Corollaries 6 and 18; Section 14.
Simplicity is encoding dependent. Within a fixed embodied language, freedom of function is invariant under function-preserving reparameterisation. Unagi is a proposed upper bound on intelligence based on the razor. It is defined in the notation guide, with its proofs in the appendix to Chapter VII and the appendix to Chapter VIII.		
Compression is not necessary for generalisation. Nor is it sufficient.	M, E	Proposition 11; Section 28.5.
In the embodied setting, I prove that compression is neither necessary nor sufficient for generalisation, and I report the binary arithmetic experiments.		

[1] Most of these claims I've already presented elsewhere, in papers or my PhD thesis, but it helps (me) to be clear about what exactly this book says.

The reason generalisability is often correlated with simplicity is because in a bounded system function must be encoded in a finite number of forms, and a selection process that rewards versatility within the local environment will cause the most used things to have the simplest forms. This usage bias confounds simplicity with generalisation.	M, I	Corollaries 9, 10, 11 and 12; Proposition 15.

A usage distribution assigns a probability to each completion of a policy. This is the probability that the body embodying the policy uses that completion. Say we have an abstraction layer above that level of the body. This higher abstraction layer has one possible fact for each of *some* of those completions in the layer below, and the rest have either more complex forms in the layer above, or cannot be expressed at all (put another way, the things in the layer below are assigned forms in the layer above). Suppose freer completions are used at least as often as more constrained ones. Then among layers of a given size, one that gives one-fact forms to the most used completions gives them to the freest completions. In that layer, if a completion has a one-fact form, so does every freer completion (Corollary 10). In simpler terms, weaker constraints below are embodied in simpler forms above. Under the assumptions of Theorem 2, a simple form above then correlates with generalisation because freedom influences both (Corollary 11). A finite vocabulary is needed for this, since in the absence of abstraction (when $\mathfrak{v} = \mathcal{P}$) every truth set has a one-fact form (Proposition 15 and Corollary 9). Usage is local. A selection process imposes demands and shortens the forms of the possible things reused across those demands. Hence simplicity tracks freedom within the part of the environment (local environment) that did the selecting and need not elsewhere. This is why Ockham's Razor works where it works. Simplicity is an effect of selection for freedom rather than a cause of generalisation, so it predicts generalisation only under the demands that selected the forms. The interpretive aspect of this claim is that this is why humans are adaptable within the terrestrial environment and fragile outside it, why mathematicians perceive reusable structure as elegant, and why an artificial intelligence can attain superhuman competence in many areas, while being less adaptable than humans in general. A computer is built with memory to spare, to embody a narrow slice of human behaviour.

Freedom of function is invariant under any reparameterisation that preserves a network's function, while measures of form like simplicity or flatness are not invariant. Networks selected for freedom attained higher mean test accuracy.	M, E	Propositions 13, 53 and 54; Sections 28.1, 28.2 and 28.3.

Function-level freedom in neural networks. I report provisional experiments in this book merely to illustrate the theory, and note that these experiments are still under review and subject to change. I show that freedom is invariant under function-preserving reparameterisation. Selecting networks for freedom during evolution gives higher mean test accuracy than selecting them by leading methods such as approximate marginal likelihood or by GdScore on all five image and language benchmarks tested [Bennett, 2026h]. Finally, I show that a generative pretraining method increases freedom, which improves its performance. By having it select for freedom directly, I further improve its performance [Bennett, 2026h,g].

Regret (in reinforcement learning terms) is weighted freedom lost in a degenerate vocabulary. In any other vocabulary, regret is a lower bound on weighted freedom lost.	M	Theorem 11; Corollary 18; Proposition 24; Section 13.

Regret is freedom lost. For the fixed-distribution normalised regret setup defined in Chapter VIII, the least regret of any policy with a given memory equals that memory's irreducible regret. In the degenerate vocabulary, it also equals minimum weighted freedom lost. Every policy's regret splits into this minimum and avoidable prediction error. For restricted policies in other vocabularies, the irreducible regret is a lower bound on weighted freedom lost.

Energy, matter and information amount to freedom of function.	M, I	Propositions 59, 17, 18 and 24; Theorems 11 and 4; Corollary 18; Section 27.

Adaptation is an exchange of freedom of function in the present for freedom of function in the future. It

is done by gathering resources, and a resource is energy, material or information. I prove that each of the three is exchanged for freedom of function. Committing to a policy requires free energy, meaning energy and material available for work, and the least amount is $-k_B\Theta_0$ times the logarithm of the fraction of viable weighted freedom that remains after the commitment (Proposition 17). Irreducible regret is weighted freedom lost, and a memory that stores more information has no greater irreducible regret (Theorem 11, Proposition 24). What is gained is a policy that meets unseen demands with a probability that rises with its freedom and falls with the irreducible regret of its memory (Theorem 4, Corollary 18). Any resource is stored freedom in the vocabulary of what it can be converted into, and a conversion that cannot be undone lowers that freedom (Proposition 59). The first sentence is the interpretive claim, made in Chapter VIII.

The Law of the Stack. Adaptability at a higher abstraction layer is bounded by freedom at the lower layers.	M, I	Proposition 16; Definition 25.

The application to biological and organisational systems is interpretive.

A timeline induces a clock per abstraction layer, and time moves slower at higher levels of abstraction.	M, I	Theorem 10; Propositions 1 and 3.

Time in a stack. Every tick of a higher layer's clock is also a tick of the clock of the layer below, while changes in lower layers may cause no change in higher layers (the clock can progress at lower levels of abstraction, while the clocks above remain stilled, meaning there is more time in lower layers).

Biological systems delegate adaptation further than conventional AI. Given the Law of the Stack, this delegation means they adapt more efficiently overall.	M, I	Proposition 16; Corollary 14; physical identification.

Delegation and biological intelligence. Under the finite viability construction of Chapter VIII, delegation is required to lower the viability divergence F_2 across layers. The scientific interpretation is that biology currently delegates further than mainstream AI.

The orthogonality thesis has a number of interpretations. I refute the strongest of these (that intelligence and final goals are entirely independent). Intelligence and final goals cannot be combined freely, because success is a joint property of software, body and environment.	M, I	Proposition 37.

Strong orthogonality fails in embodied systems. Realised success (your actual success or failure, not perceived) is a property of software, body, and goal environment together. So if a measure of intelligence depends on software alone, it cannot be paired freely with independent final goals. Alignment is therefore framed as the design of the weakest sufficient boundary conditions.

No selves, no alignment. Alignment with humans depends on three orders of self. Causal responsibility requires a 1ST-order-self, understanding of intent requires 2ND-order-selves and integrity requires a 3RD-order-self.	M, I	Theorems 20, 21 and 22; Section 19.

Alignment with humans is alignment on causal responsibility, communication and commitment, since those are prerequisites to co-operating with humans [Bennett, 2026i]. Theorems 20, 21 and 22 show in turn that causal responsibility requires a 1ST-order-self, communication a 2ND-order-self and commitment a 3RD-order-self. Hence an aligned autonomous system needs at least the first three orders of self.

The Psychophysical Principle of Causality. A robustly adaptive system divides the world into objects and properties according to the causes of valence. Its seemingly neutral representations are constructed from attractive and repulsive impulses.	M, I	Theorems 18 and 19; Corollary 2; Section 20.

The top-down proof rests upon viability and the viability divergence F_2, and the bottom-up proof upon

persistence orderings. Both show that value neutral representational commitments beyond viability are costly. Hence a generalising system never acquires them, and stays blind to whatever would have fallen under those neutral classifications. This is probably the most important result in the book for consciousness. Chapter XII and Chapter XIII develop these results into a constitutive explanation of experience. Patterns of embodied evaluation explain qualitative character, the first-order self supplies the subject, and higher orders of self explain reflective awareness and narrative. Identifying this organisation with experience is the conclusion of the cumulative argument.

Evaluation precedes representation. A system that delegates control to the lowest possible level of abstraction constructs its representations out of evaluations, which are compositions of attraction and repulsion.	M, I	Corollaries 24 and 25.

This is effectively a restatement of the Psychophysical Principle of Causality in other terms (Corollary 24). Delegate processing of information to the lowest possible level of abstraction to maximise adaptability, and you end up constructing your representations out of evaluations (compositions of attractive and repulsive states). All evolved concepts are *thick*, to use the parlance of Williams [1985].

A collective's identity is a policy shared by all of its members. When a collective is constrained so tightly that its members share no policy, it breaks into splinters. This formalises the bioelectric theory of cancer, and reframes AI alignment as a matter of cancer suppression.	M, I, E	Proposition 35; Corollary 28; Section 28.

Collective identity and splintering. A collective's identity is a policy shared by its members, and it exists only while it is weak (free) enough for every member to abide by it. When over-constraint from above or within removes every shared policy, the collective splinters (Proposition 35 and Corollary 28). The interpretive aspect is that a splinter is cancer, and alignment is cancer suppression. In Chapter XI I illustrate splintering and its suppression with simulations of synthetic collectives [Bennett, 2026j].

Recursive self impairment. Members of an unhealthy collective bind themselves to a facade of unity amid the decay of their shared meanings / policies. This self-binding corrupts the sub-groups, and then individual members of the collective.	Q	Conjecture on recursive impairment of collective coordination; Section 19.

Self-binding in unhealthy collectives may corrupt their splinters and the individuals within them. Maintaining a facade recruits the 2ND-order-self and 3RD-order self-binding to preserve appearances while shared meaning and mutual understanding decay. In Chapter XI I develop this proposed mechanism from the self-binding and splintering results, preference falsification, and informational parasitism.

No human-like access consciousness without 1ST and 2ND-order-self.	M	Theorems 20 and 21; Corollary 27; Section 20.

Access consciousness is availability for reasoning and report [Block, 1995]. Human-like access consciousness is availability for report of what one means to listeners who may take one's words differently and who also respond to other speakers. Corollary 27 shows that such report requires a 1ST-order-self and a 2ND-order-self. Gricean report, causal reasoning, deception, trust, and audience-calibrated communication require the relevant self orders under the stated behavioural conditions.

Global broadcast is neither necessary nor sufficient for human-like access consciousness.	M	Corollaries 27 and 32; Section 20.

A global workspace is a workspace of limited capacity for which contents compete, with the winning content broadcast to processes for reasoning, memory, action and speech. Corollary 32 shows that a system with a global workspace can lack human-like access consciousness, and that a system can have human-like access consciousness with no workspace at all, if each content reaches those processes by a dedicated channel. If broadcast means no more than availability to those processes, then the claim that broadcast is necessary is true by definition and explains nothing about how content becomes available

[Bennett, 2026l].

Efficient biological access consciousness implies feeling and awareness of feeling. A perfect philosophical zombie is therefore impossible.	M, I	Theorems 18 and 21; Corollary 24; Section 20.

An evolved system that supports human-like report will first build 1ST and 2ND-order-selves from valence rather than neutral symbols. The mathematical parts are the Psychophysical Principle and the self-order necessity theorems. Chapter XII explains how this evaluative organisation constitutes feeling and awareness of feeling. Qualitative character, the subject and reflective awareness are accounted for together. The identification of this organisation with experience is the conclusion of that explanation.

Freedom of function determines persistence, and freedom is ordinarily correlated with simplicity. Complex life arises and persists because the two can become anti-correlated under self-repair. Self-repair requires a stable abstraction layer below, to which a system can specialise in order to extract resources.	M, I, E	Proposition 25; Definition 31; Section 28.4.

A simple explanation of life. Novelty consists of demands that have not arisen before, and damage consists of disruptions in which possible facts become false. Under exchangeable novelty, persistence favours unseen-buffer size and therefore freedom. In every timeline, the freest correct policy persists at least as long as every other correct policy. When novelty and damage occur independently, the chance that an existing thing persists is the chance that its buffer contains every new demand, multiplied by the probability that the thing can mend the damage. Under recurring damage, complex existing things that do not mend themselves, or have only finite resources with which to mend themselves, must eventually be destroyed. Those that mend both themselves and their means of mending can have a positive chance of persisting forever. The loop simulation illustrates this mechanism in a small case.

The Law of Increasing Functional Information. I propose and prove a formal analogue of the informal law, proposed by Wong et al, purported to explain the origins of life.	M, I	Lemma 13; Proposition 25; Theorem 14.

I formalise the law proposed by Wong et al. [Wong et al., 2023], taking the degree of function to be survival, so that functional information is how rare a policy's survival probability is among the viable policies. Conditioning on full persistence raises the expected functional information of a policy, strictly unless every viable policy has the same survival probability (Theorem 14). The translation of the informal law into these terms is the interpretive aspect.

Over long horizons, a complex system with only a finite capacity for repair cannot persist indefinitely. Nothing persists for long in the void of the unviable, which lies between inert simplicity and self-maintaining life.	M, I	Lemmas 14 and 15; Theorems 15 and 16.

Closure necessity and the void of the unviable. If unseen demands exceed the buffer bound b_s of static policies, complex policies with finite capacity to mend damage will be eliminated over long horizons, while closed policies with an abiding chance of repair can persist. The void is the resulting gap in persistence between static passive systems with few points of failure (low k) and closed complex systems.

Finite Recursive Self Improvement. A system can recursively improve its own learning procedure with the resources left over after repair. Suppose that in each cycle of acquisition and repair, the system gains more resources on average than the cycle costs. Then it survives indefinitely with positive probability. A finite body can embody only finitely many learning procedures. If the system spends part of each gain upon trials of those procedures, and survives, then it almost certainly installs the best-performing of them.	M	Proposition 27; Corollary 20; Proposition 28.

Suppose an acquisition and repair process has bounded independent cycle outcomes, positive mean surplus and sufficient initial reserve. Then it has positive probability of remaining funded indefinitely. Suppose comparisons between learners are correct, and each funded trial has at least a fixed positive chance of discovering and installing a best learner. Then maximal learning performance within a fixed finite family will almost certainly be reached in the event of perpetual operation. In mathematics a family is a set, called a family when its members are themselves sets or functions. Here it is a set of learning procedures.

The Autopoietic Theorem. Under recurring damage and unbounded novelty, a system that stays viable indefinitely almost certainly repairs itself endlessly, renews the conditions for repair endlessly, and adopts policies beyond every fixed finite class.	M, I	Theorem 17; Proposition 29; Corollary 21.

Given finite embodied vocabularies, recurring damage, finite capacity to mend damage after a final response or renewal, demands beyond every fixed finite class of policies, and positive probability of indefinite viability, almost every indefinitely viable timeline contains endless repair, endless renewal of the conditions for repair, and policies beyond every fixed finite class. Static persistence is the first of the three orders of persistence, and these conclusions describe the other two. The connection to biological life is the interpretive aspect of this claim.

Time is space. Within an abstraction layer, time can be traded for space and space for time. The division between them depends upon the vocabulary.	M, I	Proposition 4; Corollary 3.

If a current abstraction layer has earlier distinctions separately available given one distinction, it needs at least one true possible fact for each distinction. Equality holds when those possible facts are the whole relevant encoding. If fewer current possible facts are true, some earlier distinctions have been merged. Tone is the finite number of distinctions a layer can draw. The interpretive aspect is that a vocabulary represents tone in two ways, which are time and space.

P versus NP depends on the vocabulary.	M, I	Propositions 43, 44 and 45; Corollary 33; Section 21.

Every task is a satisfiability problem. A body solves such a problem through a sequence of changes, measured in its own layer time, as explained in chapters IV and V. This gives each body its own P and NP, which coincide with their conventional definitions for the Turing machine formalised in Chapter V. A machine uses a possible fact as an oracle when its next step depends on whether that possible fact is true. In the absence of abstraction (when $\mathfrak{v} = \mathcal{P}$, the set of every possible fact), any language can be decided with a single oracle query (Proposition 44), so P = NP. In contrast, a Turing machine operates on its usual states, tape cells and head position. It can be given one additional possible fact that it queries in a single step. With the possible fact constructed in Proposition 45, P $\neq$ NP. More generally, for every vocabulary made by adding countably many such possible facts to the configuration vocabulary, one more can be chosen so that P $\neq$ NP, and another so that P = NP (Corollary 33). I interpret computational complexity as relative to vocabulary. It remains an open question whether P equals NP in the bog standard configuration vocabulary. I'm not claiming I solved P versus NP, just pointing out it is a subjective question whose answer depends on vocabulary choice.

Multilayer functionalism. Two systems may appear the same at one level of abstraction, while differing in function in lower layers. Two systems that give the same reports can therefore differ in the functions that constitute experience. Those functions must match at lower levels to ensure equivalence. We cannot verify the lower layers from our place at the top of the stack, so top-down verification is impossible. I instead derive from the bottom up what must be true at every layer of any evolved system, such as a human, namely that evaluation precedes representation.	M, I, H	Corollaries 1 and 31; Proposition 7; Section 20.

Corollary 1 shows that two systems can agree at one layer and differ in function beneath it, and Proposition 7 shows that one such pair defeats a claimed equivalence. My proposal is that an implementation preserves an experience only if it preserves the evaluations that constitute the experience at the layer that grounds them.

Suppose a moment of experience has several contents, such as a sight and a sound. A body can have those contents at the same time, as a chord, or one after another, as an arpeggio. A body that experiences an arpeggio must store a record of each earlier content until the last content occurs. Every stored record constrains the body. Unless viability requires that record, the body has fewer ways to stay viable. Hence a body that experiences in chords is more adaptable than a body that experiences in arpeggios.	M	Propositions 38, 39 and 40; Section 20.

If the contents of a conscious moment could instead exist at different times (Arpeggio), then every timeline fragment that visits two distinctions would contain a consciousness, and that consciousness could include arbitrary subsystems.	M, I	Theorems 23 and 24.

Arpeggio collapse and inclusion drift. Any fragment visiting two distinctions contains a moment (of subjective experience) that satisfies Arpeggio while failing Chord. Such a moment appears to be a chord subjectively, but is objectively not a chord, meaning its contents are smeared across time. If Arpeggio suffices, that moment qualifies as an Arpeggiated consciousness. Adding a 1ST-order-self requirement removes only part of that local collapse. With the vocabulary and the time window enlarged, arbitrary subsystems can still be included in an Arpeggiated consciousness. Chord stops this drift by requiring co-instantiation and interaction within the time window.

The contents of a conscious moment must physically interact. This is a consequence of special relativity. If interaction is not required, then a conscious moment could unite contents originating at opposite sides of the universe. Such unity would mean information travels faster than light, in violation of special relativity. Hence sites must physically interact.	M, I	Section 20.

The contents of a conscious moment therefore exist at the same time and interact within a time window (Chord). Hence signal speed and the duration of that window bound the physical diameter of a conscious machine.	M, H	Definition 47; Theorem 25; Corollary 30.

Chord and the spacetime diameter bound. A conscious moment therefore requires co-instantiated contents whose sites interact within a time window, so Theorem 25 bounds the diameter of a conscious machine's sites when only the topology of interaction is known. The bound relates required edge length,

diameter, signal speed, interaction graph, and window duration.

A conscious machine needs a freedom maximising, delegated, collective, self-organising, solid-brain architecture in which evaluation precedes representation. It needs the hierarchy of 1ST, 2ND and 3RD-order-selves. It needs enough concurrency for the contents of a moment to exist at the same time, and the sites of those contents must physically interact.	M, I, H	Proposition 38; Theorem 25; Corollaries 29 and 30; Section 20.

Criteria for conscious contraptions. By my interpretation of the results of this book, these four criteria are necessary for a conscious machine. Co-instantiation is derived (Proposition 38). While no premises are required for the foundational claims of this book, the Bekenstein bound and special relativity play a role in proving the above to be necessary. Sufficiency is a somewhat more open question. For a system to be sufficient for human-like consciousness, I hypothesise that it also needs the capabilities of an artificial scientist, enumerated in the final chapter.

Further results and applications

Result or application	Status	Statements, proofs and evidence
Grounding preserves truth and persistence	M	Lemma 1; Theorem 1; Corollary 2.
Product-vocabulary alignment criterion	M	Lemma 8; Theorem 3; Corollary 7.
Encoding capacity and physical ceilings	M, C	Definitions 11–13; Lemma 5; Section 4.
PAC-Bayes demand and completion certificates	M	Theorems 8 and 9; Corollary 8; Proposition 14.
Neural compatibility and measurement	M	Propositions 53 and 54; Section 24.
Permitted-output freedom formula	M	Section 28.2.
Assembly-index relativity and endpoint non-identification	M	Proposition 46; Theorem 26.
Engineered life niches	C	Corollaries 22 and 23; Section 15.
Agreement of first-order selves	M	Proposition 33.
All-orders self acquisition	M	Definitions 36 and 39; Proposition 32.
Deception and credible self-binding	M	Corollary 26; Theorem 22.
Cooperation with advance detection	M	Proposition 34.
Mirror Symbol Hypothesis	H, I	Preference reuse and empathy; Section 18.
IIT sufficiency and unit-choice counterexamples	I	Proposition 7; Section 20.
Organisational-invariance application	C, I	Proposition 39; Section 20.
Inert-record cost and present evaluation	M	Proposition 38; Corollary 29; trace conditions.
Occurrence does not determine temporal order	M	Proposition 41.
Repeated functional-information selection	C	Theorem 14; Section 15.
Empirical self hierarchy	E	Section 28.
Qualia, attention and narrative	I	Definition 46; Section 20.

Canonical version table

The canonical version table and the proof dependency map come from my book [Bennett, 2027]. In both, “this book” and the names of chapters, appendices and results refer to the book, while definition and result numbers are those of this document.

Table 4 records the canonical formulation used in this book and the earlier formulations it replaces.

Table 4: Canonical Stack Theory formulations used in this book and their relation to earlier papers. Definition, lemma and proposition numbers are those of this document.

Topic	Earlier paper formulation	Canonical formulation in this book
Primitives	Earlier papers call the elements of Φ states and the subsets of Φ programs or declarative programs.	Definition 1. The elements of Φ are distinctions and the subsets of Φ are possible facts. The symbols Φ, $\mathcal{P} = 2^{\Phi}$ and $p \subseteq \Phi$ are unchanged, and the oracle program of a set of strings is the oracle fact O_A of Definition 52.
Things	Earlier papers call a satisfiable set of possible facts an aspect, a finite one inside a vocabulary a statement, and a finite set of possible facts that may be unsatisfiable a finite bundle.	Definitions 1 and 4. A thing is a finite set of possible facts. It is a possible thing if its intersection is nonempty and an impossible thing otherwise, and a possible thing is an existing thing given the distinctions in its intersection. The symbols $L_{\mathfrak{v}}$, $T(l)$ and $\mathrm{Ext}(l)$ are unchanged.
Core language	Several papers split environment, language, extension, and freedom across separate notation blocks.	Definitions 1, 4, and 5. Possible things are finite sets of possible facts from the vocabulary that do not contradict. Freedom is the number of compatible completions of a possible thing in its embodied language.
Freedom	Earlier papers call freedom weakness and write $w(l)$, E_l and E_X. A freer possible thing was called a weaker one.	Definition 5. Freedom is written $\mathrm{ff}(l)$, and extensions are written $\mathrm{Ext}(l)$ and $\mathrm{Ext}(X)$. Freer and more constrained replace weaker and stronger.
Extension equivalence	Early texts defined equivalence through completion structure.	In Definition 5, equivalence is equal freedom, $x \equiv_{\mathfrak{v}} y$ if and only if $\mathrm{ff}(x) = \mathrm{ff}(y)$ inside $L_{\mathfrak{v}}$, so it remains vocabulary-relative. Equality of truth sets is stated separately when objective truth sets are compared.
Tasks and learning	Early work used task triples, a uniform distribution over tasks, and intensional or ostensive solution language.	Definitions 16, 17, 18, 19, and 20. Generalisation is satisfaction of further demands that lie in the policy's unseen buffer.
Organism	Earlier papers used longer tuples containing history tasks or reflex-policy data.	Definition 35. The organism is the four-tuple $\langle \mathfrak{v}_{\mathfrak{o}}, \mu_{\mathfrak{o}}, \mathfrak{p}_{\mathfrak{o}}, <_{\mathfrak{o}} \rangle$. Other data are auxiliary.
Meaning	Earlier work used an undefined rough-equivalence symbol between different protosymbol systems.	Definition 43. Communication succeeds when the listener's response, as registered by the speaker, satisfies the speaker's intended protosymbol because the listener recognised the intention.
Bagi	Earlier fixed-embodiment formulations used the same freest correct principle with changing notation.	Bagi chooses a freest correct policy in one embodied language.
Unagi	*The Wrong Razor* [Bennett, 2026f] gives an operational outer loop that maximises utility of intelligence, minimises K_ρ among utility ties, then applies Bagi.	Definition 27 gives semantic Unagi's score, which relies on objective truth sets that can never be observed. The number of future demands, distinct in truth set, that a policy's completions express is at most that score. In the degenerate vocabulary of Chapter VIII, minimising K_ρ maximises the weighted freedom available to a policy. When each permitted policy assigns at most one prediction per group, K_ρ is a lower bound on weighted freedom lost.
Temporal semantics	Earlier papers introduced separate occurrence, co-instantiation, lift, and persistence operators.	Definition 47 and Proposition 40 are stated with direct quantified conditions. The earlier operators are derived notation.
Layer-time and layer-space counts	Identifying a tone with one true possible fact conflates true possible facts with visible transitions.	Tone is the finite number of distinctions a layer can draw, and layer space and layer time share this number. $S_{\mathfrak{v}}(\phi)$ is the number of true possible facts at one distinct time. $\kappa_{\mathfrak{v}}(h)$ is the number of adjacent transitions whose encodings differ. Proposition 4 relates the layer-space count to the layer-time count.
Timelines and bounded machines	Earlier papers used *trajectory*, *history*, repeated halting configurations, and product coordinates for temporal positions.	In Definition 2, timeline is the canonical term, and I may be $\mathbb{N}$ or a finite initial segment $\{0, \ldots, N\}$. Consecutive environmental distinctions differ, while a distinction may recur after an intervening change. The bounded Turing machine is an embodied possible thing, and a complete run ends at halting. Proposition 4 states the result on recording a fragment in one present distinction. Separately available temporal distinctions require separately active present possible facts.
Oracle facts and runs on inputs	None. These are new in this book.	Definition 52 admits a query step on any possible fact, with O_A the oracle fact of a set of strings, and by Lemma 17 relativised classes are classes measured in layer time. Definition 50 gives a task a run and a run time along a timeline, Definition 51 gives a vocabulary, under permitted timelines, a P and an NP, and Proposition 43 identifies the Turing machine's classes with P and NP and its run with the run of the task.
Encoding limits	The encoding limits first appeared in application papers and the extended appendix.	Definitions 11–13 and Lemma 5 are the canonical general statements.

Proof dependency map

Each entry takes its name from my claims map where one exists, gives each result's printed title and states what that result depends on. Entries follow chapter order.

Time is space. Proposition 4 (Embodied recoding of temporal distinctions) is stated for a fixed finite vocabulary, a fragment $h = (\phi_0, \ldots, \phi_n)$, a current distinction ψ and a current vocabulary $\mathfrak{w}$. Injection $e_A : A_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$ implies

$$S_{\mathfrak{w}}(\psi) \geq \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t).$$

Injection $e_C : C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$ instead implies

$$S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h).$$

Surjectivity of each injection entails equality in its bound. Both calculations appear in the appendix to Chapter V, Temporal Recoding. Each rests upon one distinct true possible fact for each recorded item. Tone is the finite number of distinctions a layer can draw, and layer space and layer time share this number. Bounded Turing-machine reconstruction represents controller, head, tape and usage facts as possible facts over Φ, with transition constraints on environmental changes. Theorem 3 (when simplicity and freedom agree) states when simplicity and freedom coincide on product vocabularies.

Multilayer functionalism. Proposition 7 (Functional sufficiency) rests upon a description map and a functional condition. Equal descriptions determine that condition if and only if its value agrees across all implementations that share a description. Temporal Gap exhibits a counterexample when descriptions record separate occurrence and a target property requires co-instantiation.

Freest-correct generalisation. Theorem 2 (Freest-correct is Bayes optimal under exchangeable task growth) depends on finite embodied language, correctness on observed inputs, and exchangeable unseen demands. Theorem 4 (continuous freest correct optimality) further depends on a measured embodied language, finite $\mu(O_\alpha)$ for ranking by buffer measure, and μ-exchangeability. Theorem 8 (PAC-Bayes future-demand certificate) instead depends on independently sampled demand sets and a prior fixed before those demand sets are sampled. Lemma 7 (The freest correct policy permits every completion of every correct policy) depends on a task with an input and a correct policy, on every input having a correct output, and on every subset of a possible thing being a possible thing. Corollary 6 (Every demand set that a correct policy survives is also survived by the freest correct policy) further depends on Definition 20 for its parent tasks and on a finite vocabulary for its tally of parent tasks.

Unagi. Proposition 23 (Semantic Unagi optimisation) depends on a nonempty set of admissible pairs of a vocabulary and a correct policy, finite vocabularies in the domain of the chosen λ-task, and an attained maximum of semantic Unagi's score across admissible pairs. Its lower bound from future demands additionally depends on the distinct truth sets of those demands. Corollary 18 (Semantic and operational Unagi) further depends on a task in a degenerate vocabulary, on which minimising K_ρ maximises W^*. Its agreement with semantic Unagi further depends on a pairing of each $M_{\mathfrak{v}}$-based policy with a correct policy whose distinct unseen truth sets correspond one to one with the unseen outputs in the buffer of the $M_{\mathfrak{v}}$-based policy, all with equal weight. Its decomposition with $\Delta_{\mathcal{A}}$ additionally depends on a fixed representation and on a nonempty finite family of policies, each permitting outputs from at most one prediction per group. When two such families are compared, the groups and weights are the same for both.

P versus NP depends on the vocabulary. Proposition 42 (Every task is a satisfiability problem) rests upon Definitions 4 and 5 alone. Definition 50 (run on an input) rests upon timelines and layer time from Chapter V. Definition 51 (complexity classes relative to a vocabulary) rests upon Definition 50 and the number of possible facts in an input. Proposition 43 (P and NP are the classes of the Turing machine) depends on a complete bounded machine that halts on every input, a separator symbol in the input alphabet, Proposition 1 and the standard definitions of P and NP [Sipser, 2012]. Lemma 17 (oracle steps are layer time) depends on Definition 52, Proposition 3 and, for its converse, on a complete bounded machine. Proposition 44 (Satisfiability on the tape) depends on Proposition 43 and, for satisfiability, a standard encoding of formulas over the input alphabet without the separator, injective, computable and decodable in polynomial time, and of polynomial length. Proposition 45 (A vocabulary in which P $\neq$ NP) is the construction of Baker, Gill and Solovay in the oracle vocabulary and depends on Lemma 17 and, for the classes relative to that vocabulary, on Proposition 43 relative to M^R with its hypotheses, a complete machine that halts on every input and a separator [Baker et al., 1975]. Corollary 33 (Either answer is one oracle fact away) depends on the construction in the proof of Proposition 45, on the countability of the oracle facts added, and on a set complete for polynomial space relative to their join [Baker et al., 1975]. The barrier against a proof in the configuration vocabulary rests upon their collapse relative to a PSPACE-complete set, and on Arora, Impagliazzo and Vazirani for the identification of locality [Arora et al., 1992]. The cost of a decision depends on the vocabulary, and this dependence rests upon Propositions 15 and 4.

Time in a stack. Theorem 10 (Time in a stack) depends on the layer-time definition and the abstractor relation $\mathfrak{v}^{i+1} \subseteq \{T(o) \mid o \in \mathrm{Ext}(\pi^i)\}$. A possible fact in a higher layer can change truth value only when at least one possible fact of its grounding in the lower layer changes truth value. The same step shows that equal lower encodings of two distinctions give equal higher encodings, so every uninterrupted block of constant lower encoding lies within one block of constant higher encoding.

Regret is freedom lost. Theorem 11 (Regret is freedom lost) depends on a finite evaluation distribution over interaction-record and test pairs, normalised loss weights derived from conditional prediction probabilities, and policies with one answer per group defined by memory state and prediction task. Its freedom identity rests upon independent predictions for each group in the degenerate vocabulary. For other nonempty sets of permitted policies with at most one prediction per group, K_ρ lower-bounds weighted freedom lost. Probability $\exp(-K_\rho(M))$ additionally depends on independent demand probabilities $1 - \exp(-\rho_z m_z)$. Proposition 24 concerns a zero-regret partition of the interaction records that is coarser than every other zero-regret partition. Such a partition exists if and only if compatibility is transitive on the records with positive regret weight.

The Law of Increasing Functional Information. Theorem 14 (The Law of Increasing Functional Information) depends on a finite vocabulary, a child task, a finite viable policy set with a probability distribution over it that is independent of the continuation, a chosen timeline prefix, admissible continuations, and the novelty distribution induced by those continuations.

Closure necessity and the void of the unviable. Proposition 26 (Closure provideth an abiding chance) depends on organisational closure relative to one period and one damage distribution for that period, with each period's damage mended immediately after that period. It also depends on a horizon divided into periods whose damage is drawn independently. The distribution for each period gives positive probability only to damage histories that have positive probability under the damage distribution of the closure premise. Finally,

its abiding chance depends on a summable sequence of chances of damage outside the mend. In Corollary 19 (A widening mend provideth an abiding chance) one prefix set and one mend are replaced by a sequence of prefix sets and period mends. Under the independent damage model, the summable sequence there follows from a per-period capacity to mend damage that reaches the number of opportunities in a period. Theorem 15 (Closure Necessity Theorem) depends on finite embodied language, unseen buffer, independent persistent damage, mends, a static bound b_s on unseen buffers that the demand set exceeds with positive probability, a static bound k_s on points of failure that complex policies exceed, finite capacity to mend damage for the comparison class, and a policy with an abiding chance whose buffer contains the demand set with positive probability. The appendix to Chapter IX, Persistence, contains the calculations of the proofs of Proposition 26, Lemma 15 and the Autopoietic Theorem. Theorem 16 (Void of the Unviable) further depends on uniform novelty, a policy with buffer b_c and an abiding chance $q_c > 0$, a finite comparison class of complex policies with finite capacity to mend damage, and a static policy with buffer b_s of one of two kinds. Either the static policy's probability of mending is a constant q_s and $b_c + \log_2 q_c > b_s + \log_2 q_s$, or the static policy cannot mend damage and has at least one and at most k_s points of failure.

Finite Recursive Self Improvement. Proposition 27 (Surplus sustains repair) depends on a process that completes every funded cycle, returns to the prefix set, has a common bounded cycle distribution with fresh independent outcomes and positive mean net gain, and can store every finite reserve. The initial reserve exceeds an operating requirement that covers every cost within a cycle. Corollary 20 (Surplus accumulates) follows from the strong law on the resulting positive-probability event of perpetual operation. Proposition 28 (Surplus funds learner improvement) further depends on a separate allocation for finite-cost trials, a preserved cycle distribution, a fixed finite family of learning procedures, correct performance comparisons, and a uniform positive conditional chance of discovering and installing a maximiser. The installed procedure performs subsequent learning and trials. The claim that the installed maximiser also maximises freedom additionally depends on uniform novelty on a common task.

The Autopoietic Theorem. Theorem 17 (Autopoietic Theorem) depends on finite embodied vocabularies that may change with time, a fixed countable set $\mathcal{P}_*$ of possible higher-layer policies, and four assumptions. Damage recurs with conditional probability at least p_0. Each finite viable prefix h determines finite capacities $m_0(h)$ and $m_1(h)$. Along every timeline confined to a fixed finite policy class, a viability-relevant demand beyond that class is imposed with conditional probability at least $\varepsilon_{\mathcal{K}}$. Indefinite viability has positive probability. No independence is assumed. Full-support exploration is a separate sufficient condition connecting a buffer to the novelty expressible by the policy itself. Corollary 21 replaces the fixed bounds p_0 and $\varepsilon_{\mathcal{K}}$ with the condition that the conditional probabilities sum to infinity on the relevant timelines that remain viable, and leaves the other assumptions unchanged.

Assembly limits. Proposition 46 (Layer-relative assembly index) depends on stated fine and coarse basic types, concatenation, string equality, and a map that forgets fine types and preserves joins. Theorem 26 (Endpoint non-identification) depends on a unary join relation, an inventory of occurrences that joins consume, a stopping rule, and observation of an endpoint. It is true for every choice of production probabilities on the space. Proposition 46 is stated across a change of layer. Theorem 26 is stated for a fixed assembly space with changed production probabilities.

The Cosmic Ought. Proposition 8 (Timelines rank differently) depends on two timelines on one environment, a time shared by both, and the encodings of their distinctions at that time.

Its last claim further depends on the singleton possible fact of a distinction. Proposition 9 (Timelines agree only on entailment) depends on Lemma 6 and on one timeline through each distinction, such as the single-distinction timelines, which satisfy Definition 2. Its extension to a family of timelines depends only on the set of distinctions visited by the family.

The Psychophysical Principle of Causality. Theorem 18 (The Psychophysical Principle of Causality) depends on viability and F_2. Theorem 19 (Bottom-up Psychophysical Principle from persistence) depends on persistence orderings and monotonicity under added commitments. Corollary 2 (Grounding preserves persistence) rests upon grounding, which preserves truth sets, at fixed ticks. Corollary 25 (Value-first construction order) further depends on the requirement that each primitive commitment be zero-cost for some part across viable timelines and, for its claim about later layers, on a primitive layer whose possible facts are all evaluative, with each later layer built as in Definition 26.

Orders of self. Theorem 20 (Intervention control requires a represented distinction) depends on a finite decision problem whose cases pair a possible thing with whether the organism brought it about, and on a matched pair of cases with no optimal action in common. Theorem 21 (A speaker needs a model of the listener) depends on a set of decoders the speaker admits as possible and on a meaning that no single utterance conveys to all of them, and its binary case on a hidden decoder drawn uniformly and on one probe. Theorem 22 (Separating audience modelling from self-binding) depends on the game of Definition 44, on subgame-perfect equilibrium and on a binding cost $0 < c < 1$. Corollary 27 (Human-like report requires 1ST and 2ND-order-selves) depends on Theorems 20 and 21. Its second item further depends on a listener who responds to another speaker's utterance with the same decoder, on occasions that have positive probability and are drawn independently of the meaning and the decoder.

Same possible fact above, different control below. Corollary 31 (Same possible fact above, different control below) depends on a policy at layer i of a stack, a possible fact R at layer $i + 1$, at least two and finitely many distinctions of $T(\pi^i)$ outside R, and a layer-i vocabulary containing R and the possible fact $\Phi \setminus \{s\}$ for each of those distinctions.

Succession. Proposition 41 (Occurrence does not determine order) is stated for a fixed vocabulary and a permutation whose original and permuted fragments are admissible. Content-wise occurrence is permutation invariant, while actual induced clocks preserve objective order.

Present moment. Corollary 29 (Subject and moment) rests upon compositional grounding and truth-set intersection. Under its phenomenal interpretation the present moment is constitutive evaluation. This identification is developed through the cumulative argument in Chapter XII and Chapter XIII. Proposition 38 (Arpeggio shortens tenure, and thus threatens viability)[2] depends on Corollary 4, Theorem 18, Corollary 2, Proposition 40 and the premise that a chordal realisation adds no possible fact to the policy when the contents of the moment become true at the same time. On the persistence route of Proposition 40, the case for Chord rests upon the test of whether a present representation implies its moment. Proposition 39 (Temporal Gap separation) rests upon Definition 47, and its separation upon a fragment and a vocabulary in which the contents of a moment are true at different times and never at the same time. Given sufficient concurrency, co-instantiation follows from freedom maximisation. Interaction follows from special relativity.

Chord and the spacetime diameter bound. Theorem 25 (Edge length and spacetime diameter bounds) depends on a nonempty set of sites, every required edge completed within

[2]Tenure is the period of objective time for which a thing persists.

the time window, a physical model that registers the emission and receipt of each signal, a maximum signal speed, and a physical duration θ assigned to the time window of a finite timeline fragment. Its edge length bound is $\theta \geq L_{\max}(\mathrm{Graph}_\ell, d)/v$, and its diameter bound depends further on a connected interaction graph and the triangle inequality for d, and otherwise only on the shape of the interaction graph. Corollary 30 (Feasible window) further depends on a guaranteed persistence duration for every content, on Chord being required for every permitted timing, and on a permitted timing in which one content lapses at that duration while another first becomes true at window end.

Arpeggio collapse and inclusion drift. Theorem 23 (Arpeggio collapse) rests upon a fragment visiting at least two distinctions and a distinction it never visits, and a vocabulary containing the constructed possible facts. Theorem 24 (Arpeggiated inclusion drift) rests upon a finite vocabulary containing A's possible facts and a possible fact q_X of X, a nonempty set of causal identities from Definition 33, and visits to $T(int_A)$ for some $int_A \in INT_A$ and to q_X. The attribution of an Arpeggiated consciousness in each theorem rests upon the assumption that Arpeggio suffices. 1ST-order-self attribution rests upon Definition 37's differentiation and alignment preconditions. Chord additionally requires co-instantiation, capacity and interaction within the time window. It excludes an enlargement whenever that enlargement violates one of those conditions.

Global broadcast is neither necessary nor sufficient for human-like access. Corollary 32 (Global broadcast is neither necessary nor sufficient for human-like access consciousness) depends on Corollary 27, on the binary case of Theorem 21, and on a global workspace as described by GWT, with contents competing for a workspace of limited capacity and the winning content broadcast to processes for reasoning, memory, action and speech.

1 Environment and embodied language

Definition 1 (environment, possible facts, things, and vocabularies)**.**
An *environment* is a nonempty set Φ of mutually exclusive distinctions[3]. It is assumed infinite by default. A *possible fact* is any set of distinctions $p \subseteq \Phi$[4]. For any set X, I write 2^X for the set of all subsets of X. In particular, let $\mathcal{P} := 2^\Phi$ denote the set of all possible facts. A possible fact is a fact when one of its distinctions is the present time, and is false otherwise.

I say a possible fact is a *fact* if it is true. A *thing* is a set x of possible facts. It is a *possible thing* if its possible facts do not contradict, meaning their intersection is nonempty, $x \in \{\, l \subseteq \mathcal{P} \mid \bigcap_{p\in l} p \neq \emptyset \,\}$, and an *impossible thing* otherwise. Given distinction ϕ, a possible thing x is an *existing thing* if all of its possible facts are facts, meaning $\phi \in \bigcap_{p\in x} p$.

A *vocabulary* is any set of possible facts $\mathfrak{v} \subseteq \mathcal{P}$. Unlike a possible thing, the possible facts in a vocabulary can contradict, so its intersection can be empty, and usually is. I call a vocabulary with a non-empty intersection degenerate. Unless specified otherwise, vocabularies and things are assumed finite.

Definition 2 (Timeline and finite timeline fragment)**.**
A **timeline** is a function

$$\tau : I \to \Phi,$$

where I is either $\mathbb{N}$ or a finite initial segment $\{0, \ldots, N\}$, such that

$$\tau(u+1) \neq \tau(u) \qquad \text{whenever } u, u+1 \in I.$$

Each index $u \in I$ is a **time**, a position in the timeline, which can be thought of as a **tick** of the objective clock. I write tick u for the time with index u. The timeline assigns one distinction $\tau(u)$ to each tick u. A possible fact or thing is **true at tick** u when it is true given $\tau(u)$, and it is true at a time when it is true given the distinction at that time. Each abstraction layer, defined in Section 3, can be understood as a subjective clock. The distinction at each tick has an encoding in the layer, and subjective time passes in that layer when the encoding changes[5]. Every distinction at a tick is a change. The distinction at every tick after the first also differs from the immediately preceding distinction. A distinction may recur after one or more intervening ticks of objective time. Distinctions at different ticks may nevertheless have the same encoding in a vocabulary, a notion that I define in Section 3. For a starting tick $u \in I$ and an integer $n \geq 0$ with $u + n \in I$, the **finite timeline fragment**

$$h = \tau_{[u,u+n]} := (\tau(u), \tau(u+1), \ldots, \tau(u+n)) \in \Phi^{n+1}$$

is the restriction of that timeline to $n+1$ consecutive ticks of objective time. Its consecutive entries satisfy $\tau(u+j+1) \neq \tau(u+j)$ for every $0 \leq j < n$, while nonconsecutive entries may coincide. A **timeline prefix** is the special case, for $t \in I$,

$$\tau_{\leq t} := \tau_{[0,t]} = (\tau(0), \ldots, \tau(t)).$$

[3]For readers coming from elsewhere. In Spencer-Brown's calculus the distinction is the sole primitive and the void is the absence of any distinction [Spencer-Brown, 1969]. In possible-worlds semantics a proposition is a set of worlds, true at a world when that world is a member [Stalnaker, 1984, Lewis, 1986]. A possible world is a timeline here, so a proposition in that sense is a set of timelines. Existence, the set of all timelines, is the plurality of worlds. Wittgenstein's world, everything that is the case, is the timeline that occurs [Wittgenstein, 1922]. In reinforcement learning the environment is what lies outside the agent [Sutton and Barto, 2018]. Here it has no outside. Every agent and observer is in it. It is embedded, enacted. In this way, I avoid the nasty dualism problems and can focus on my precious.

[4]My earlier papers call a distinction a state, a possible fact a program, and a possible thing a statement or an aspect.

[5]Subjective ticks of time in a layer are encoding changes in said layer.

Definition 3 (Objective time)**.**
For a timeline $\tau : I \to \Phi$, **objective time** is the order of its changes. Each $\tau(u)$ is a distinction and therefore a change at position u in that order. The index u records order rather than adding another object to the environment. Requirement $\tau(u+1) \neq \tau(u)$ whenever $u, u+1 \in I$ means that consecutive changes differ. It does not forbid a later return to a distinction seen before.

Definition 4 (abstraction layer, things, and truth sets)**.**
An **abstraction layer** is an environment Φ considered through a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$. A **thing** is any finite $l \subseteq \mathfrak{v}$. It is a **possible thing** if its intersection is nonempty, and an **impossible thing** if that intersection is empty. I define the truth set of a thing by

$$T(l) := \bigcap_{p \in l} p, \qquad T(\emptyset) := \Phi.$$

The layer induces the embodied language

$$L_{\mathfrak{v}} = \{\, l \subseteq \mathfrak{v} \mid l \text{ is finite and } T(l) \neq \emptyset \,\}.$$

Elements $l \in L_{\mathfrak{v}}$ are the possible things of the layer. A possible thing **exists** given $\phi \in \Phi$, and I also say it is **true** given ϕ, if and only if $\phi \in T(l)$, meaning every possible fact in it is a fact given ϕ. An **existing thing** is a possible thing that exists given the present distinction. The possible things of the layer are the possible things contained in the vocabulary:

$$L_{\mathfrak{v}} = \{x \subseteq \mathfrak{v} \mid x \text{ is a possible thing}\}.$$

Definition 5 (completions, extensions, and freedom)**.**
A **completion** of $x \in L_{\mathfrak{v}}$ is any $y \in L_{\mathfrak{v}}$ with $x \subseteq y$. The **extension** of x is

$$\mathrm{Ext}(x) := \{y \in L_{\mathfrak{v}} \mid x \subseteq y\}.$$

For $X \subseteq L_{\mathfrak{v}}$, I write

$$\mathrm{Ext}(X) := \bigcup_{x \in X} \mathrm{Ext}(x).$$

The **freedom** of x is the cardinality of the extension

$$\mathrm{ff}(x) := |\,\mathrm{Ext}(x)|.$$

Freedom is also called the **weakness** of constraints. Here x is a constraint, and a weaker constraint allows more possibilities. These possibilities are the completions of x, which together form the extension of x. Freedom is used to order possible things by

$$x <_{\mathrm{ff}} y \quad \Longleftrightarrow \quad \mathrm{ff}(x) < \mathrm{ff}(y).$$

This ordering ranks y ahead of x when y has more completions. A possible thing is freer, or less constrained, when it has greater freedom, and it is ranked higher by $<_{\mathrm{ff}}$. Just to be extra clear, freedom does not require one extension to contain another. The two possible things just need to be in the same embodied language. I say possible things x and y are **equivalent in** $L_{\mathfrak{v}}$ iff

$$\mathrm{ff}(x) = \mathrm{ff}(y).$$

This equivalence means they have the same rank in terms of weakness. It is relative to the embodied language because both extension sets are computed inside $L_{\mathfrak{v}}$.

2 Abstraction and grounding

Definition 6 (abstractor function)**.**
An **abstractor function** $\mathfrak{f}$ is defined on pairs $(\mathfrak{v}, l)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary and $l \in L_{\mathfrak{v}}$ is a possible thing. It returns the sub-vocabulary

$$\mathfrak{f}(\mathfrak{v}, l) \;:=\; \{T(o) \mid o \in \mathrm{Ext}(l)\} \subseteq \mathcal{P}.$$

In words, $\mathfrak{f}(\mathfrak{v}, l)$ is the set of truth sets of all completions of l.

Definition 7 (one-step grounding)**.**
Let $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, l_i)$. A **grounding choice** is a map from possible facts in the higher layer to completions in the lower layer

$$\chi_i : \mathfrak{v}^{i+1} \to \mathrm{Ext}(l_i)$$

such that $T(\chi_i(q)) = q$ for every $q \in \mathfrak{v}^{i+1}$. The completion $\chi_i(q)$ is the **grounding** of the possible fact q. For a thing $a \subseteq \mathfrak{v}^{i+1}$ in layer $i{+}1$, its **one-step grounding** under χ_i is

$$g_i^{\chi}(a) := \bigcup_{q \in a} \chi_i(q) \subseteq \mathfrak{v}^i.$$

Lemma 1 (one-step truth preservation)**.**
For every thing $a \subseteq \mathfrak{v}^{i+1}$, $g_i^{\chi}(a) \subseteq \mathfrak{v}^i$ is a thing with

$$T(g_i^{\chi}(a)) = \bigcap_{q \in a} q = T(a),$$

and $g_i^{\chi}(a) \in L_{\mathfrak{v}^i}$ if and only if $a \in L_{\mathfrak{v}^{i+1}}$. Thus one-step grounding preserves the truth set of the thing in the higher layer.

Proof. Each $\chi_i(q)$ lies in $L_{\mathfrak{v}^i}$, so is a finite subset of $\mathfrak{v}^i$, and the union $g_i^{\chi}(a)$ of finitely many of them is a thing in $\mathfrak{v}^i$. By definition,

$$T(g_i^{\chi}(a)) = \bigcap_{q \in a} T(\chi_i(q)).$$

The grounding choice satisfies $T(\chi_i(q)) = q$ for every $q \in a$. By substitution, $T(g_i^{\chi}(a)) = \bigcap_{q \in a} q = T(a)$. Both $g_i^{\chi}(a)$ and a are things with this truth set, so $g_i^{\chi}(a) \in L_{\mathfrak{v}^i}$ if and only if $a \in L_{\mathfrak{v}^{i+1}}$. □

Theorem 1 (compositional grounding)**.**
Let a finite chain of abstractor steps

$$\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, l_i), \qquad i = 0, \ldots, n-1,$$

have grounding choices χ_i, and let $a_n \subseteq \mathfrak{v}^n$ be a top-layer thing with the recursively defined groundings

$$a_i := g_i^{\chi}(a_{i+1}), \qquad i = n-1, \ldots, 0.$$

Then each grounding step preserves the same truth set, so the top-layer thing is grounded at layer 0 by a_0:

$$T(a_0) = T(a_1) = \cdots = T(a_n),$$

and $a_0 \in L_{\mathfrak{v}^0}$ if and only if $a_n \in L_{\mathfrak{v}^n}$.

Proof. I apply Lemma 1 at each adjacent pair of layers. By induction over $i = n-1, \ldots, 0$, each a_i is a thing in $\mathfrak{v}^i$ and $T(a_0) = T(a_1) = \cdots = T(a_n)$. Since a_0 and a_n are things with the same truth set, $a_0 \in L_{\mathfrak{v}^0}$ if and only if $a_n \in L_{\mathfrak{v}^n}$. □

Corollary 1 (Functional agreement across layers)**.**
Two systems can compute the same input–output relation at layer $i+1$ while computing different input–output relations at layer i. Two groundings at layer i of one possible fact at layer $i+1$ can have the same truth set and strictly nested extensions, hence unequal freedom.

Proof. For the first claim, I choose distinctions $\phi_1 \neq \phi_2$ and possible facts $h = \{\phi_1, \phi_2\}$, $c_1 = \{\phi_1\}$ and $c_2 = \{\phi_2\}$. The first system has layer-i vocabulary $\{h, c_1\}$, the second has $\{h, c_2\}$, and each realises the policy $\{h\}$. The completions of $\{h\}$ are $\{h\}$ and $\{h, c_k\}$, with truth sets h and $h \cap c_k = c_k$. Hence the abstractor returns $\{h, c_1\}$ for the first system and $\{h, c_2\}$ for the second. Each system can therefore embody the layer-(i+1) vocabulary $\{h\}$ (Definition 26), and at layer $i+1$ both compute one relation, the task with input $\{h\}$ and correct output $\{h\}$. At layer i the completions of $\{h\}$ differ, with $\{h, c_1\}$ true given ϕ_1 alone in the first system and $\{h, c_2\}$ true given ϕ_2 alone in the second. Hence the systems compute different relations at layer i. For the second claim, I choose distinct possible facts $p, q \in \mathfrak{v}^i$ with $q = \Phi$ and $\varnothing \neq p \subsetneq \Phi$. The possible things $x = \{p\}$ and $y = \{p, q\}$ have the same truth set, since $T(x) = p = T(y)$. Either can ground the same possible fact at layer $i+1$. Yet $x \subsetneq y$, so $\mathrm{Ext}(y) \subsetneq \mathrm{Ext}(x)$ and $\mathrm{ff}(y) < \mathrm{ff}(x)$. The groundings agree functionally at layer $i+1$ and differ functionally at layer i. □

Corollary 2 (Grounding preserves persistence)**.**
Fix a finite chain as in Theorem 1, and let l, c be things at its highest layer. I use g_l, g_c for their groundings at its lowest layer, and ρ for tenure[6]. For any timeline $\tau : I \to \Phi$ and $u_1, u_2 \in I$ with $u_1 \leq u_2$,

$$\begin{aligned}\rho_{\tau,[u_1,u_2]}(g_l \cup g_c) &= \rho_{\tau,[u_1,u_2]}(l \cup c) \\ &\leq \rho_{\tau,[u_1,u_2]}(l) = \rho_{\tau,[u_1,u_2]}(g_l).\end{aligned}$$

The inequality is strict if and only if τ visits $T(l) \setminus T(c)$ during that interval.

Proof. By Theorem 1, $T(g_l) = T(l)$ and $T(g_c) = T(c)$. Hence $T(g_l \cup g_c) = T(l \cup c)$, so both equalities follow from Definition 14. By Corollary 4, the weak inequality is true. The two sets of ticks in the weak inequality differ by

$$\{u \in \{u_1, \ldots, u_2\} : \tau(u) \in T(l) \setminus T(c)\},$$

so the inequality is strict if and only if this set is nonempty. □

3 Encoding, space and time

Definition 8 (encoding, layer space, and layer time)**.**
Fix a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ of possible facts. For a distinction $\phi \in \Phi$, the **encoding** of ϕ in $\mathfrak{v}$ is

$$\mathrm{Enc}_{\mathfrak{v}}(\phi) := \{p \in \mathfrak{v} : \phi \in p\}.$$

The **layer-space count** is

$$S_{\mathfrak{v}}(\phi) := |\,\mathrm{Enc}_{\mathfrak{v}}(\phi)|.$$

For a finite timeline fragment $h = (\phi_0, \ldots, \phi_n)$, the **layer-time count** is

$$\kappa_{\mathfrak{v}}(h) := |\{j \in \{1, \ldots, n\} : \mathrm{Enc}_{\mathfrak{v}}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{j-1})\}|\,.$$

For a timeline $\tau : I \to \Phi$ and $t \in I$, the clock induced at layer $\mathfrak{v}$ is

$$\kappa^{\tau}_{\mathfrak{v}}(t) := \kappa_{\mathfrak{v}}(\tau_{\leq t}).$$

[6]Tenure is the period of objective time for which a thing persists.

A tick $u \geq 1$ of τ is a **tick of subjective time** in layer $\mathfrak{v}$ when $\mathrm{Enc}_{\mathfrak{v}}(\tau(u)) \neq \mathrm{Enc}_{\mathfrak{v}}(\tau(u-1))$, so $\kappa^{\tau}_{\mathfrak{v}}(t)$ is the number of ticks of subjective time in $\mathfrak{v}$ up to tick t. Objective time advances at every tick of τ. Layer time advances only at ticks of subjective time, the ticks at which the layer's encoding changes.

Definition 9 (bounded and embodied Turing machine)**.**
A deterministic one-tape Turing machine is

$$M = (Q, \Sigma, \Gamma, b, \delta, q_0, q_{\mathrm{acc}}, q_{\mathrm{rej}}).$$

Here Q is a finite set of **states**, meaning labels of the machine's controller rather than environmental distinctions in Φ. Alphabets are sets of **symbols**. The finite **input alphabet** Σ is contained in the finite **tape alphabet** Γ. The symbol $b \in \Gamma \setminus \Sigma$ is blank. The pairwise distinct labels q_0, q_{acc}, and q_{rej} are the start, accept, and reject states. The transition function

$$\delta : (Q \setminus \{q_{\mathrm{acc}}, q_{\mathrm{rej}}\}) \times \Gamma \longrightarrow Q \times \Gamma \times \{L, R\}$$

specifies one machine step. If $\delta(q, a) = (q', a', D)$, the machine reads a under its head, writes a', changes state from q to q', and moves the head one cell left or right according to D. A computation starts in q_0 with its input beginning at tape cell 1, blank symbols elsewhere, and the head on cell 1. The computation ends when it reaches q_{acc} or q_{rej}.

Fix a tape bound $S \in \mathbb{N}$ with $S \geq 2$. An **embodied S-bounded implementation** of M in Φ consists of a nonempty possible fact $\varnothing \neq \mathcal{M}_{M,S} \subseteq \Phi$, the distinctions that realise the machine, and possible facts

$$Q_q, \quad H_i, \quad C_{j,a}, \quad U_j \subseteq \Phi$$

indexed by $q \in Q$, $1 \leq i, j \leq S$, and $a \in \Gamma$. On $\mathcal{M}_{M,S}$, the state facts partition its environmental distinctions, the head facts partition its environmental distinctions, and for each tape cell j the symbol facts partition its environmental distinctions

$$\mathcal{M}_{M,S} = \bigsqcup_{q \in Q} (Q_q \cap \mathcal{M}_{M,S}) = \bigsqcup_{i=1}^{S} (H_i \cap \mathcal{M}_{M,S}),$$

$$\mathcal{M}_{M,S} = \bigsqcup_{a \in \Gamma} (C_{j,a} \cap \mathcal{M}_{M,S}) \qquad (1 \leq j \leq S).$$

Thus every $\phi \in \mathcal{M}_{M,S}$ determines unique values

$$q(\phi) \in Q, \qquad i(\phi) \in \{1, \ldots, S\}, \qquad T_\phi(j) \in \Gamma,$$

and usage values $u_\phi(j) \in \{0, 1\}$ defined by $u_\phi(j) = 1$ when $\phi \in U_j$.

A pair $\phi, \psi \in \mathcal{M}_{M,S}$ is a **machine step** when $q(\phi)$ is nonhalting and, writing

$$\delta(q(\phi), T_\phi(i(\phi))) = (q', a', D),$$

the next distinction satisfies

$$q(\psi) = q', \qquad i(\psi) = \begin{cases} i(\phi) - 1, & D = L, \\ i(\phi) + 1, & D = R, \end{cases}$$

with $1 \leq i(\psi) \leq S$, and

$$T_\psi(j) = \begin{cases} a', & j = i(\phi), \\ T_\phi(j), & j \neq i(\phi). \end{cases}$$

Cells already marked as used remain marked, and the newly visited head cell is also marked as used

$$u_\psi(j) = 1 \quad \Longleftrightarrow \quad u_\phi(j) = 1 \text{ or } j = i(\psi).$$

A **run of** M is a finite timeline fragment $r = (\phi_0, \dots, \phi_n)$ in $\mathcal{M}_{M,S}$ with the following properties. Its first distinction ϕ_0 has state q_0, head on cell 1 and the input written from cell 1 with blanks elsewhere. Every consecutive pair is a machine step. Its initial usage facts identify the initial head cell. The run ends when it reaches a halting state or a chosen final tick. The machine is therefore an embodied possible thing. Its step number does not define another clock, but clicks along inside the environment. You know, like a real computer does.

Definition 10 (bounded Turing vocabularies)**.**
For an embodied S-bounded implementation, the **configuration vocabulary** and **usage vocabulary** are

$$\mathfrak{v}^{\mathrm{conf}}_{M,S} := \{Q_q : q \in Q\} \cup \{H_i : 1 \le i \le S\} \cup \{C_{j,a} : 1 \le j \le S,\ a \in \Gamma\},$$

$$\mathfrak{v}^{\mathrm{use}}_{M,S} := \{U_j : 1 \le j \le S\}.$$

Both are vocabularies over the original environment because every member is a subset of Φ. For $\phi \in \mathcal{M}_{M,S}$,

$$\mathrm{Enc}_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(\phi) = \{Q_{q(\phi)}, H_{i(\phi)}\} \cup \{C_{j,T_\phi(j)} : 1 \le j \le S\},$$

$$\mathrm{Enc}_{\mathfrak{v}^{\mathrm{use}}_{M,S}}(\phi) = \{U_j : u_\phi(j) = 1\}.$$

Proposition 1 (machine time is layer time)**.**
Let

$$r = (\phi_0, \dots, \phi_n)$$

be a finite timeline fragment in $\mathcal{M}_{M,S}$ whose consecutive pairs are machine steps. I define its machine time by $\mathrm{Time}_M(r) := n$, the number of machine steps. Then for every finite vocabulary $\mathfrak{v}$ containing $H_1, \dots, H_S$,

$$\kappa_{\mathfrak{v}}(r) = \mathrm{Time}_M(r) = n.$$

In particular this holds for every run r of M and for $\mathfrak{v} = \mathfrak{v}^{\mathrm{conf}}_{M,S}$.

Proof. In every machine step the head moves from some cell i to a different cell i'. Hence the true head-position fact changes from H_i to $H_{i'}$, so

$$\mathrm{Enc}_{\{H_1,\dots,H_S\}}(\phi_j) \neq \mathrm{Enc}_{\{H_1,\dots,H_S\}}(\phi_{j-1})$$

for every $j = 1, \dots, n$. The fragment contains n consecutive pairs, so $\kappa_{\{H_1,\dots,H_S\}}(r) = n$. Since $\{H_1, \dots, H_S\} \subseteq \mathfrak{v}$, the encoding in $\mathfrak{v}$ changes at every pair with a change in the head encoding. Hence $\kappa_{\mathfrak{v}}(r) \ge n$, and $\kappa_{\mathfrak{v}}(r) \le n$ because r has n consecutive pairs. □

Proposition 2 (machine space is layer space)**.**
Let $r = (\phi_0, \dots, \phi_n)$ be a run of M. For $0 \le t \le n$, define used space by

$$\mathrm{Space}_M(r, t) := |\{j : u_{\phi_t}(j) = 1\}|.$$

Then

$$\mathrm{Space}_M(r, t) = S_{\mathfrak{v}^{\mathrm{use}}_{M,S}}(\phi_t).$$

Proof. The usage fact U_j is true given ϕ_t when $u_{\phi_t}(j) = 1$. The number of true usage facts is the number of cells used at that time in the run. It is the amount of layer-space used by the usage vocabulary. □

Proposition 3 (monotonicity of layer time)**.**
Let $\mathfrak{v}' \subseteq \mathfrak{v}$ be finite vocabularies over Φ. For every finite timeline fragment h, every tick of h at which the encoding in $\mathfrak{v}'$ changes is a tick at which the encoding in $\mathfrak{v}$ changes, so

$$\kappa_{\mathfrak{v}'}(h) \le \kappa_{\mathfrak{v}}(h),$$

with equality if and only if every tick of h at which the encoding in $\mathfrak{v}$ changes is a tick at which the encoding in $\mathfrak{v}'$ changes. Consequently, for every timeline $\tau : I \to \Phi$ and $t \in I$,

$$\kappa^{\tau}_{\mathfrak{v}'}(t) \leq \kappa^{\tau}_{\mathfrak{v}}(t).$$

Proof. Let $h = (\phi_0, \ldots, \phi_n)$. If tick j of h is a tick of subjective time in the smaller vocabulary, then

$$\mathrm{Enc}_{\mathfrak{v}'}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}'}(\phi_{j-1}).$$

Hence some possible fact $p \in \mathfrak{v}'$ is true at one of the two distinctions and false at the other. Since $\mathfrak{v}' \subseteq \mathfrak{v}$, that same possible fact belongs to $\mathfrak{v}$, so

$$\mathrm{Enc}_{\mathfrak{v}}(\phi_j) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{j-1}).$$

Every tick counted by $\kappa_{\mathfrak{v}'}(h)$ is therefore counted by $\kappa_{\mathfrak{v}}(h)$. A subset of a finite set has as many elements as that set if and only if the two are equal. Hence $\kappa_{\mathfrak{v}'}(h) = \kappa_{\mathfrak{v}}(h)$ if and only if every tick counted by $\kappa_{\mathfrak{v}}(h)$ is counted by $\kappa_{\mathfrak{v}'}(h)$. Taking $h = \tau_{\leq t}$, I get $\kappa^{\tau}_{\mathfrak{v}'}(t) \leq \kappa^{\tau}_{\mathfrak{v}}(t)$. □

Corollary 3 (layer-space bound)**.**
For every finite vocabulary $\mathfrak{v}$ and distinction $\phi \in \Phi$,

$$S_{\mathfrak{v}}(\phi) \leq |\mathfrak{v}|.$$

Proof. $\mathrm{Enc}_{\mathfrak{v}}(\phi) \subseteq \mathfrak{v}$. □

Proposition 4 (Embodied recoding of temporal distinctions)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a finite timeline fragment

$$h = (\phi_0, \ldots, \phi_n).$$

I write

$$A_{\mathfrak{v}}(h) := \{(p, t) \mid 0 \leq t \leq n,\ p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\}$$

for the pairs of a possible fact and a tick at which it is true in the fragment, and

$$C_{\mathfrak{v}}(h) := \{t \in \{1, \ldots, n\} \mid \mathrm{Enc}_{\mathfrak{v}}(\phi_t) \neq \mathrm{Enc}_{\mathfrak{v}}(\phi_{t-1})\}$$

for the visible ticks, the ticks at which the encoding changes. Let $\psi \in \Phi$ be a current distinction and let $\mathfrak{w}$ be a finite vocabulary over Φ. The current encoding is the encoding of ψ in $\mathfrak{w}$.

If the current encoding contains a different true possible fact for every member of $A_{\mathfrak{v}}(h)$, meaning there is an injection

$$e_A : A_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi),$$

then

$$S_{\mathfrak{w}}(\psi) \geq \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t).$$

If the current encoding contains a different true possible fact for every member of $C_{\mathfrak{v}}(h)$, meaning there is an injection

$$e_C : C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi),$$

then

$$S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h).$$

Bound $S_{\mathfrak{w}}(\psi) \geq \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$ is an equality if and only if $e_A(A_{\mathfrak{v}}(h)) = \mathrm{Enc}_{\mathfrak{w}}(\psi)$. Bound $S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h)$ is an equality if and only if $e_C(C_{\mathfrak{v}}(h)) = \mathrm{Enc}_{\mathfrak{w}}(\psi)$. If $S_{\mathfrak{w}}(\psi) < \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$, no injection e_A exists. If $S_{\mathfrak{w}}(\psi) < \kappa_{\mathfrak{v}}(h)$, no injection e_C exists.

Proof. Because each pair contains its time index, the sets

$$\{(p,t) \mid p \in \mathrm{Enc}_{\mathfrak{v}}(\phi_t)\}$$

are disjoint for different t. Hence

$$|A_{\mathfrak{v}}(h)| = \sum_{t=0}^{n} |\,\mathrm{Enc}_{\mathfrak{v}}(\phi_t)| = \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t).$$

An injection of $A_{\mathfrak{v}}(h)$ into $\mathrm{Enc}_{\mathfrak{w}}(\psi)$ implies

$$S_{\mathfrak{w}}(\psi) = |\,\mathrm{Enc}_{\mathfrak{w}}(\psi)| \geq |A_{\mathfrak{v}}(h)|.$$

Also, by Definition 8,

$$|C_{\mathfrak{v}}(h)| = \kappa_{\mathfrak{v}}(h).$$

Injection $e_C : C_{\mathfrak{v}}(h) \hookrightarrow \mathrm{Enc}_{\mathfrak{w}}(\psi)$ requires $S_{\mathfrak{w}}(\psi) \geq \kappa_{\mathfrak{v}}(h)$. Surjectivity of e_A implies $S_{\mathfrak{w}}(\psi) = \sum_{t=0}^{n} S_{\mathfrak{v}}(\phi_t)$, while surjectivity of e_C entails $S_{\mathfrak{w}}(\psi) = \kappa_{\mathfrak{v}}(h)$. Conversely, an injection between finite sets of equal size is a surjection, so equality in either bound forces the image of its injection to be all of $\mathrm{Enc}_{\mathfrak{w}}(\psi)$. If $|\,\mathrm{Enc}_{\mathfrak{w}}(\psi)|$ is smaller than the size of a source set, then by the pigeonhole principle no injection from that set exists. □

Lemma 2 (Encoding is the maximal true possible thing)**.**
For every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(\phi)$ is true given ϕ. If $l \in L_{\mathfrak{v}}$ is also true given ϕ, then $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$.

Proof. Let $\phi \in \Phi$. By definition, if $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$ then $\phi \in p$. Therefore $\phi \in \bigcap_{p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)} p = T(\mathrm{Enc}_{\mathfrak{v}}(\phi))$.

Now let $l \in L_{\mathfrak{v}}$ and assume $\phi \in T(l) = \bigcap_{p \in l} p$. Then for every $p \in l$ I have $\phi \in p$, hence $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$. So $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Proposition 5 (Layer time can be arbitrarily sparse)**.**
For any $M \in \mathbb{N}_{>0}$ there exist an environment Φ, a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, and a timeline $\tau : \mathbb{N} \to \Phi$ such that its encoding $\mathrm{Enc}_{\mathfrak{v}}(\tau(u))$ changes at most once during the first M ticks of objective time.

Proof. I take $\Phi = \{0,1\}$, $\mathfrak{v} = \{\Phi\}$ and $\tau(u) = u \bmod 2$. Consecutive environmental distinctions differ, while $\mathrm{Enc}_{\mathfrak{v}}(\tau(u)) = \{\Phi\}$ at every tick. Layer time remains zero for every finite fragment. □

Lemma 3 (The language is the union of all local powersets)**.**
Fix an environment Φ and a finite vocabulary $\mathfrak{v} \subseteq 2^{\Phi}$. Then

$$L_{\mathfrak{v}} = \bigcup_{\phi \in \Phi} 2^{\mathrm{Enc}_{\mathfrak{v}}(\phi)} = \{l \subseteq \mathfrak{v} \mid \exists\, \phi \in \Phi :\ l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)\}.$$

Proof. If $l \in L_{\mathfrak{v}}$, choose $\phi \in T(l)$. Then Definition 8 gives $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$, so $l \in 2^{\mathrm{Enc}_{\mathfrak{v}}(\phi)}$. Conversely, if $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$ for some ϕ, then Definition 8 gives $\phi \in T(l)$. So $T(l) \neq \emptyset$ and $l \in L_{\mathfrak{v}}$. □

Proposition 6 (Extensions are unions of encoding intervals)**.**
Assume $\mathfrak{v}$ is finite. For a possible thing $l \in L_{\mathfrak{v}}$, write $\mathcal{M}(l) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in T(l)\}$ for the set of realised maximal descriptions compatible with l. Then

$$\mathrm{Ext}(l) = \bigcup_{m \in \mathcal{M}(l)} \{y \subseteq \mathfrak{v} \mid l \subseteq y \subseteq m\}.$$

Consequently, $\max_{m \in \mathcal{M}(l)} 2^{|m|-|l|} \leq \mathrm{ff}(l) \leq \sum_{m \in \mathcal{M}(l)} 2^{|m|-|l|}$.

Proof. Let $y \in \mathrm{Ext}(l)$. Then $l \subseteq y$ and $y \in L_{\mathfrak{v}}$. Choose $\phi \in T(y)$. Then $\phi \in T(l)$ since $T(y) \subseteq T(l)$. By Definition 8, $y \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. So y lies between l and $m := \mathrm{Enc}_{\mathfrak{v}}(\phi)$, and $m \in \mathcal{M}(l)$.

Conversely, suppose $m \in \mathcal{M}(l)$ and $l \subseteq y \subseteq m$. By definition of $\mathcal{M}(l)$, there exists $\phi \in T(l)$ with $m = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. Then $y \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. By Definition 8, $\phi \in T(y)$. So $y \in L_{\mathfrak{v}}$ and $y \in \mathrm{Ext}(l)$.

For the bounds, each set $\{y \mid l \subseteq y \subseteq m\}$ has exactly $2^{|m|-|l|}$ elements. The union contains each such set, so its size is at least the maximum. The size of a union is at most the sum. □

Lemma 4 (Ordered recursion for freedom)**.**
Fix a finite vocabulary $\mathfrak{v} = \{p_0, \ldots, p_{n-1}\}$ under any fixed total order. For $l \in L_{\mathfrak{v}}$ and $j \in \{0, \ldots, n\}$, define

$$\mathrm{ff}_j(l) := \left|\{y \in \mathrm{Ext}(l) \mid y \setminus l \subseteq \{p_j, p_{j+1}, \ldots, p_{n-1}\}\}\right|.$$

This counts completions that only add possible facts with index $\geq j$. Then

$$\mathrm{ff}_j(l) = 1 + \sum_{\substack{k \in \{j,\ldots,n-1\} \\ p_k \notin l \\ l \cup \{p_k\} \in L_{\mathfrak{v}}}} \mathrm{ff}_{k+1}(l \cup \{p_k\}).$$

In particular, $\mathrm{ff}(l) = \mathrm{ff}_0(l)$.

Proof. Every y counted by $\mathrm{ff}_j(l)$ satisfies $l \subseteq y$ and $y \setminus l \subseteq \{p_j, \ldots, p_{n-1}\}$. If $y = l$, it contributes the initial 1. If $y \supsetneq l$, let p_k be the smallest-index possible fact in $y \setminus l$. Then $k \geq j$ (by the index restriction on ff_j), $p_k \notin l$, $l \cup \{p_k\} \subseteq y$, and $l \cup \{p_k\} \in L_{\mathfrak{v}}$ (since y is a possible thing and $l \cup \{p_k\} \subseteq y$). The remaining possible facts in $y \setminus (l \cup \{p_k\})$ all have index $> k$ (because p_k was the smallest). So y is counted exactly once in $\mathrm{ff}_{k+1}(l \cup \{p_k\})$. The sets $\{y \in \mathrm{Ext}_j(l) : y \supsetneq l, \text{ min-index of } y \setminus l = k\}$ for different k are disjoint, and together with $\{l\}$ they partition the set counted by $\mathrm{ff}_j(l)$. Summing over all valid k gives the recursion. The base case $j = n$ gives $\mathrm{ff}_n(l) = 1$ (only $y = l$ survives). Setting $j = 0$ recovers $\mathrm{ff}(l) = \mathrm{ff}_0(l)$. □

4 Encoding capacity and functional sufficiency

Definition 11 (Realised encoding set)**.**
For a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, the **realised encoding set** is the image of the encoding map

$$\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}}) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in \Phi\} \subseteq L_{\mathfrak{v}}.$$

Definition 12 (Encoding capacity)**.**
The **encoding capacity** of a finite vocabulary $\mathfrak{v}$ is

$$\mathsf{I}(\mathfrak{v}) := \log_2 \left|\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}})\right|.$$

Definition 13 (Patch capacity ceiling)**.**
If a bounded physical region $\mathcal{R}$ can sustain at most $N_{\max}(\mathcal{R})$ reliably distinguishable internal states, its bit ceiling is

$$\mathsf{I}_{\max}(\mathcal{R}) := \log_2 N_{\max}(\mathcal{R}).$$

A vocabulary is physically realisable inside $\mathcal{R}$ only if

$$\mathsf{I}(\mathfrak{v}) \leq \mathsf{I}_{\max}(\mathcal{R}).$$

For a complete, isolated, bounded, weakly self-gravitating system with circumscribing radius $r_{\mathcal{R}}$ and total rest-frame energy E_{tot} including rest energy, the Bekenstein bound[7] gives

$$S_{\max} \leq \frac{2\pi k_B r_{\mathcal{R}} E_{\text{tot}}}{\hbar c_{\text{light}}}.$$

If $2^{\mathsf{I}_{\max}(\mathcal{R})} \leq \exp(S_{\max}/k_B)$ and $\mathfrak{v}$ is physically realisable in $\mathcal{R}$, then

$$\mathsf{I}(\mathfrak{v}) \leq \mathsf{I}_{\max}(\mathcal{R}) \leq \frac{2\pi r_{\mathcal{R}} E_{\text{tot}}}{\hbar c_{\text{light}} \ln 2}.$$

This is the stated conditional capacity bound [Bekenstein, 1981]. For a memory reset, let n_{erase} be the decrease of its Shannon entropy in bits. If the memory is uncorrelated with retained available side information, the implementation is cyclic with no compensating stored-energy change, and a thermal environment at Θ_0 is the entropy sink, then

$$Q_{\text{diss}} \geq n_{\text{erase}} k_B \Theta_0 \ln 2.$$

This is Landauer's bound under the stated reset conditions [Landauer, 1961]. The quantity n_{erase} is not generally freedom, functional information, or a change in either.

Lemma 5 (Union capacity is subadditive)**.**
For finite vocabularies $\mathfrak{v}_1$ and $\mathfrak{v}_2$ over one environment,

$$\mathsf{I}(\mathfrak{v}_1 \cup \mathfrak{v}_2) = \log_2 \left|\{(\text{Enc}_{\mathfrak{v}_1}(\phi), \text{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}\right|,$$

so

$$\max\{\mathsf{I}(\mathfrak{v}_1), \mathsf{I}(\mathfrak{v}_2)\} \leq \mathsf{I}(\mathfrak{v}_1 \cup \mathfrak{v}_2) \leq \mathsf{I}(\mathfrak{v}_1) + \mathsf{I}(\mathfrak{v}_2),$$

with equality on the right if and only if

$$\{(\text{Enc}_{\mathfrak{v}_1}(\phi), \text{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\} = \text{Im}(\text{Enc}_{\mathfrak{v}_1}) \times \text{Im}(\text{Enc}_{\mathfrak{v}_2}).$$

Proof. Each realised encoding $c = \text{Enc}_{\mathfrak{v}_1 \cup \mathfrak{v}_2}(\phi)$ of their union determines the pair

$$(c \cap \mathfrak{v}_1, c \cap \mathfrak{v}_2) = (\text{Enc}_{\mathfrak{v}_1}(\phi), \text{Enc}_{\mathfrak{v}_2}(\phi)).$$

The map $c \mapsto (c \cap \mathfrak{v}_1, c \cap \mathfrak{v}_2)$ into $\text{Im}(\text{Enc}_{\mathfrak{v}_1}) \times \text{Im}(\text{Enc}_{\mathfrak{v}_2})$ is injective, because $c = (c \cap \mathfrak{v}_1) \cup (c \cap \mathfrak{v}_2)$, and its image is $\{(\text{Enc}_{\mathfrak{v}_1}(\phi), \text{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}$. Hence

$$|\,\text{Im}(\text{Enc}_{\mathfrak{v}_1 \cup \mathfrak{v}_2})| = \left|\{(\text{Enc}_{\mathfrak{v}_1}(\phi), \text{Enc}_{\mathfrak{v}_2}(\phi)) \mid \phi \in \Phi\}\right| \leq |\,\text{Im}(\text{Enc}_{\mathfrak{v}_1})|\,|\,\text{Im}(\text{Enc}_{\mathfrak{v}_2})|,$$

with equality if and only if the image is the whole product, both being finite. The image covers all of $\text{Im}(\text{Enc}_{\mathfrak{v}_1})$ in its first coordinate and all of $\text{Im}(\text{Enc}_{\mathfrak{v}_2})$ in its second coordinate. So the image has at least $\max\{|\,\text{Im}(\text{Enc}_{\mathfrak{v}_1})|, |\,\text{Im}(\text{Enc}_{\mathfrak{v}_2})|\}$ elements. Applying $\log_2$, I obtain the identity and both bounds. □

[7]The Boltzmann constant k_B is a physical constant, about 1.38×10^{-23} joules per kelvin. Heat is the random motion of molecules, and k_B is roughly the energy of that motion in one molecule of a gas for each kelvin of temperature. A temperature multiplied by k_B is therefore an energy. Physicists prize the constant because it relates the properties of a whole system to the microscopic states of that system. Thermodynamic entropy is one such property. When the microscopic states of a system are equally probable, its entropy is k_B multiplied by the natural logarithm of how many such states there are. Entropy divided by k_B is thus the natural logarithm of a number of states, and the logarithm to base two of that number is a number of bits. The Bekenstein bound upon entropy thereby becomes a bound upon bits. Landauer's bound below relates bits to energy instead. Erasing one bit at a given temperature dissipates at least $k_B \ln 2$ multiplied by that temperature as heat.

Proposition 7 (Functional sufficiency)**.**
Let $\mathcal{S}$ be a set of implementations, let $A : \mathcal{S} \to \mathcal{D}$ assign a description to each implementation, and let $\Psi : \mathcal{S} \to Y$ be a functional condition with values in a set Y. There exists $F : A(\mathcal{S}) \to Y$ satisfying $\Psi = F \circ A$ if and only if

$$A(s) = A(t) \quad \Longrightarrow \quad \Psi(s) = \Psi(t) \qquad (s, t \in \mathcal{S}).$$

The descriptions are sufficient for Ψ when such an F exists. Thus one pair with equal descriptions and unequal Ψ values disproves sufficiency of those descriptions for Ψ.

Proof. If $\Psi = F \circ A$ and $A(s) = A(t)$, then by substitution $\Psi(s) = F(A(s)) = F(A(t)) = \Psi(t)$. Conversely, assume that equal descriptions imply equal Ψ values. For each $d \in A(\mathcal{S})$, I define $F(d)$ as that common value on $\{s : A(s) = d\}$. This set is nonempty and all its members have the same Ψ value, so F is well defined and $F(A(s)) = \Psi(s)$ for every s. □

5 Persistence order

Definition 14 (Tenure)**.**
Fix a vocabulary $\mathfrak{v}$, a timeline $\tau : I \to \Phi$, and $u_1, u_2 \in I$ with $u_1 \leq u_2$. The tenure of a thing $x \subseteq \mathfrak{v}$ is

$$\rho_{\tau,[u_1,u_2]}(x) := |\{u \in \{u_1, \ldots, u_2\} \mid \tau(u) \in T(x)\}|.$$

If x is inconsistent, then $T(x) = \emptyset$ and its tenure is zero.

Definition 15 (Persistence ordering)**.**
Fix a vocabulary $\mathfrak{v}$, a timeline $\tau : I \to \Phi$, and $u_1, u_2 \in I$ with $u_1 \leq u_2$. For things $l, l' \subseteq \mathfrak{v}$ I write

$$l \geq_{\tau,[u_1,u_2]} l' \quad \text{iff} \quad \rho_{\tau,[u_1,u_2]}(l) \geq \rho_{\tau,[u_1,u_2]}(l').$$

The strict part is written $l >_{\tau,[u_1,u_2]} l'$ and means

$$\rho_{\tau,[u_1,u_2]}(l) > \rho_{\tau,[u_1,u_2]}(l').$$

The relation $\geq_{\tau,[u_1,u_2]}$ is a total preorder on things over $\mathfrak{v}$. It is total because every pair of tenures can be compared. It is a preorder because equal tenures can tie distinct things.

I say that l' **entails** l when $T(l') \subseteq T(l)$, meaning l is true given every distinction of which l' is true.

Lemma 6 (A thing cannot exist for longer than the things it entails)**.**
Fix $\mathfrak{v}$, τ, and $[u_1, u_2]$. If $l, l' \subseteq \mathfrak{v}$ are things and $T(l') \subseteq T(l)$, then

$$\rho_{\tau,[u_1,u_2]}(l') \leq \rho_{\tau,[u_1,u_2]}(l).$$

Strict inequality $\rho_{\tau,[u_1,u_2]}(l') < \rho_{\tau,[u_1,u_2]}(l)$ holds if and only if $\tau(u) \in T(l) \setminus T(l')$ for some $u \in \{u_1, \ldots, u_2\}$.

Proof. For every tick u, $\tau(u) \in T(l')$ implies $\tau(u) \in T(l)$ because $T(l') \subseteq T(l)$. Therefore the set of ticks counted by l' is a subset of the set of ticks counted by l. Taking cardinalities, I obtain the weak inequality. The difference of the two tick sets is $\{u \in \{u_1, \ldots, u_2\} : \tau(u) \in T(l) \setminus T(l')\}$. Hence the subset is proper if and only if the timeline visits a distinction in $T(l) \setminus T(l')$ during the interval. □

Corollary 4 (One cannot extend tenure by committing to unnecessary things)**.**
Fix a vocabulary $\mathfrak{v}$, a timeline $\tau : I \to \Phi$, and $u_1, u_2 \in I$ with $u_1 \leq u_2$. Let $l, c \subseteq \mathfrak{v}$ be things, and read $T(l \cup c) = \emptyset$ when $l \cup c$ is inconsistent. Then

$$\rho_{\tau,[u_1,u_2]}(l \cup c) \leq \rho_{\tau,[u_1,u_2]}(l),$$

with strict inequality if and only if the timeline visits a distinction in $T(l) \setminus T(l \cup c)$ during the interval.

Proof. Since $T(l \cup c) = T(l) \cap T(c)$, we have $T(l \cup c) \subseteq T(l)$. I apply Lemma 6. □

Proposition 8 (Timelines rank differently)**.**
Let τ and τ' be timelines on one environment Φ, fix a vocabulary $\mathfrak{v}$, and let u^* be a tick in both index sets. The persistence orderings $\geq_{\tau,[u^*,u^*]}$ and $\geq_{\tau',[u^*,u^*]}$ on things over $\mathfrak{v}$ are equal if and only if $\tau(u^*)$ and $\tau'(u^*)$ have the same encoding in $\mathfrak{v}$. If a possible fact $p \in \mathfrak{v}$ contains $\tau(u^*)$ and not $\tau'(u^*)$, then

$$\{p\} \geq_{\tau,[u^*,u^*]} \emptyset \qquad \text{while} \qquad \{p\} \not\geq_{\tau',[u^*,u^*]} \emptyset.$$

Whenever $\tau(u^*) \neq \tau'(u^*)$, the two distinctions have different encodings in the vocabulary whose only possible fact is $\{\tau(u^*)\}$. Hence timelines that have different distinctions at the same time rank possible things differently at that time in some vocabulary.

Proof. For each thing $x \subseteq \mathfrak{v}$, the distinction $\tau(u^*)$ is in $T(x)$ if and only if each possible fact of x contains $\tau(u^*)$, which means $x \subseteq \mathrm{Enc}_{\mathfrak{v}}(\tau(u^*))$. Hence the tenure $\rho_{\tau,[u^*,u^*]}(x)$ is 1 when $x \subseteq \mathrm{Enc}_{\mathfrak{v}}(\tau(u^*))$ and 0 otherwise, and the same is true for τ'. If the two encodings are equal, every thing has the same tenure in both timelines, so the two orderings are equal. Now suppose that a possible fact $p \in \mathfrak{v}$ contains $\tau(u^*)$ and not $\tau'(u^*)$. The thing $\{p\}$ has tenure 1 in τ and 0 in τ'. The empty thing has tenure 1 in both timelines, because $T(\emptyset) = \Phi$. Hence $\{p\} \geq_{\tau,[u^*,u^*]} \emptyset$ while $\{p\} \not\geq_{\tau',[u^*,u^*]} \emptyset$, which is the displayed claim. If the two encodings differ, some possible fact $p \in \mathfrak{v}$ contains just one of the two distinctions. I name the timelines so that p contains $\tau(u^*)$. The displayed claim then shows that the orderings differ. For the last claim, the possible fact $\{\tau(u^*)\}$ contains $\tau(u^*)$ and not $\tau'(u^*)$. □

Lemma 6 is one half of the next result.

Proposition 9 (Timelines agree only on entailment)**.**
Fix a vocabulary $\mathfrak{v}$. For possible things $l, l' \in L_{\mathfrak{v}}$,

1. $l \geq_{\tau,[u_1,u_2]} l'$ for every timeline τ and every interval $[u_1, u_2]$ if and only if l' entails l;
2. if neither possible thing entails the other, then $l >_{\tau,[u,u]} l'$ for some timeline τ and tick u, and $l' >_{\tau',[u',u']} l$ for some timeline τ' and tick u';
3. no strict ranking $l >_{\tau,[u_1,u_2]} l'$ is true for every timeline and every interval.

Proof. For each distinction ϕ, the map $\tau_\phi : \{0\} \to \Phi$ with $\tau_\phi(0) = \phi$ is a timeline by Definition 2. On the interval $[0, 0]$, a possible thing has tenure 1 under τ_ϕ when ϕ is in its truth set, and 0 otherwise.

For the first item, if l' entails l, then Lemma 6 shows that $\rho_{\tau,[u_1,u_2]}(l') \leq \rho_{\tau,[u_1,u_2]}(l)$ for every timeline and interval. If l' does not entail l, I choose a distinction ϕ in $T(l') \setminus T(l)$. Under τ_ϕ the tenure of l' is 1 and that of l is 0. Hence the strict ranking $l' >_{\tau_\phi,[0,0]} l$ is true, and the weak ranking $l \geq_{\tau_\phi,[0,0]} l'$ is false.

For the second item, suppose that neither possible thing entails the other. Since l' does not entail l, the set $T(l') \setminus T(l)$ contains a distinction ϕ. As in the first item, the strict ranking

$l' >_{\tau_\phi,[0,0]} l$ is true. Since l does not entail l', the set $T(l) \setminus T(l')$ contains a distinction ψ. For the same reason, the strict ranking $l >_{\tau_\psi,[0,0]} l'$ is true.

For the third item, suppose that the strict ranking $l >_{\tau,[u_1,u_2]} l'$ is true for every timeline and interval. The weak ranking is then true for every timeline and interval, so l' entails l by the first item. Since l' is a possible thing, its truth set contains a distinction ϕ. Under τ_ϕ both possible things have tenure 1, because l' entails l. Hence the strict ranking is false under τ_ϕ, contrary to the supposition. □

The Cosmic Ought. Each timeline induces the persistence ordering above, and different timelines may induce different orderings (Propositions 8 and 9). Added commitments cannot extend tenure in any fixed fragment. The Cosmic Ought interprets this ordering as the environment's normativity preceding organisms [Bennett, 2026e].

6 Tasks, inference and learning

Definition 16 ($\mathfrak{v}$-task)**.**
Let $\mathfrak{v} \subseteq \mathcal{P}$ be a vocabulary. A **$\mathfrak{v}$-task** is a pair

$$\alpha = \langle I_\alpha, O_\alpha \rangle$$

where

$$I_\alpha \subseteq L_\mathfrak{v} \qquad \text{and} \qquad O_\alpha \subseteq \mathrm{Ext}(I_\alpha).$$

The members of I_α are **inputs**. The members of O_α are **correct outputs**. I require every input to have at least one correct output, meaning every $i \in I_\alpha$ has some $o \in O_\alpha$ with $i \subseteq o$. Set $\Gamma_\mathfrak{v}$ consists of all $\mathfrak{v}$-tasks.

For tasks α and ω, I call α a **child** of ω, written $\alpha \sqsubset \omega$, iff

$$I_\alpha \subsetneq I_\omega \qquad \text{and} \qquad O_\alpha \subseteq O_\omega.$$

I use $\alpha \sqsubseteq \omega$ when $\alpha = \omega$ or $\alpha \sqsubset \omega$. If $\mathfrak{v}$ is finite, define the **level** or **height** of a task ω by

$$\ell_\mathfrak{v}(\omega) := \max \left\{ k \in \mathbb{N} \;\middle|\; \begin{array}{l} \exists \langle \alpha_0, \ldots, \alpha_k \rangle, \quad \alpha_0 = \omega, \\ \alpha_{i+1} \sqsubset \alpha_i \text{ for } 0 \le i < k \end{array} \right\}.$$

The maximum exists because a finite vocabulary induces a finite task space. The level counts how far the task can be specialised through strict child steps. A child contains fewer input cases and no correct outputs outside those of its parent. Learning from a child to a parent therefore means generalising from fewer cases to a broader task.

Definition 17 (inference)**.**

- A $\mathfrak{v}$-task **policy** is a possible thing $\pi \in L_\mathfrak{v}$. It constrains how we complete inputs.
- π is a **correct policy** for a task α iff the correct outputs O_α of α are all and only the completions π' of π for which π' is also a completion of an input.
- The set of all correct policies for a task α is denoted Π_α.

Assume $\mathfrak{v}$-task ω and a policy $\pi \in L_\mathfrak{v}$. Inference proceeds as follows:

1. we are presented with an input $i \in I_\omega$, and
2. when $\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi) \neq \emptyset$, we select an output $e \in \mathrm{Ext}(i) \cap \mathrm{Ext}(\pi)$. If this set is empty, inference under π is undefined on i. For a correct policy $\pi \in \Pi_\omega$, the set is nonempty for every $i \in I_\omega$, because every input has a correct output.

3. If $e \in O_\omega$, then e is correct and the task "complete". $\pi \in \Pi_\omega$ implies $e \in O_\omega$, but $e \in O_\omega$ doesn't imply $\pi \in \Pi_\omega$ (an incorrect policy can imply a correct output).

Definition 18 (λ-tasks)**.**
Fix a $\mathcal{P}$-task ρ with $I_\rho \neq \emptyset$. A **disembodied task**, or **λ-task**, associated with ρ is a function λ_ρ whose domain contains every finite vocabulary $\mathfrak{v}' \subseteq \mathcal{P}$ and may contain additional vocabularies when stated. For every vocabulary in its domain, $\lambda_\rho(\mathfrak{v}')$ is a selected $\mathfrak{v}'$-task satisfying $\lambda_\rho(\mathfrak{v}') \sqsubseteq \rho$ and

$$\ell_{\mathfrak{v}'}(\lambda_\rho(\mathfrak{v}')) = \max\Big\{\ell_{\mathfrak{v}'}(\omega) \;\Big|\; \omega \text{ is a } \mathfrak{v}'\text{-task and } \omega \sqsubseteq \rho\Big\}.$$

Thus $\lambda_\rho(\mathfrak{v}')$ is an expressible child of ρ with greatest task height in $\mathfrak{v}'$. This set of expressible children is nonempty because $I_\rho \neq \emptyset$ implies that $\langle \emptyset, \emptyset \rangle$ is a strict child of ρ. When several maximisers exist, the λ-task includes a choice among them. When $\mathfrak{v}'$ is finite, $L_{\mathfrak{v}'}$ and the induced task space are finite, so the maximum exists. To compare λ-tasks, I fix a reference vocabulary $\mathfrak{v}_{\text{ref}}$ in their common domain and write

$$\lambda_1 \sqsubset \lambda_2 \quad \Longleftrightarrow \quad \lambda_1(\mathfrak{v}_{\text{ref}}) \sqsubset \lambda_2(\mathfrak{v}_{\text{ref}}).$$

Adjacent tasks are ordered after the higher task is grounded into the lower language. Thus the stack condition at adjacent layers is

$$g_i^\chi\Big(\lambda^{i+1}(\mathfrak{v}^{i+1})\Big) \sqsubseteq \lambda^i(\mathfrak{v}^i),$$

where $i+1$ is more abstract than i and g_i^χ is the one-step grounding of Definition 7. This compares the tasks instantiated by the stack.

Definition 19 (learning and selectors)**.**
A **selector** $<$ is a binary relation on possible things. I read $l_1 < l_2$ as "l_2 is ranked ahead of l_1".

A **learner** is a system represented by an abstraction layer with vocabulary $\mathfrak{v}$. Let α be a $\mathfrak{v}$-task with $\Pi_\alpha \neq \emptyset$, let ω be a $\mathfrak{v}$-task with $\alpha \sqsubset \omega$, and let $\mathcal{F}_\alpha \subseteq \Pi_\alpha$ be the nonempty set of correct policies that the learner has found. This set is all of Π_α when the learner has found every correct policy. The learner restricts attention to $\mathcal{F}_\alpha$ and applies a selector $<$ to it. The learner **learns** ω under $<$ if it outputs a $<$-maximal policy of $\mathcal{F}_\alpha$ that is correct for ω, meaning some $\pi^\star \in \mathcal{F}_\alpha \cap \Pi_\omega$ such that

$$\forall\, \pi \in \mathcal{F}_\alpha : \quad \neg(\pi^\star < \pi).$$

The learner, and the chosen $\pi^\star$, **generalise** to ω if the learner learns ω.

Two selectors are used throughout this appendix.

- $<_{\text{ff}}$ is the **freedom** selector, given by the ordering of Definition 5. Its maximal elements have maximal freedom $\text{ff}(l) = |\,\text{Ext}(l)|$. A freest correct policy was called a weakest correct policy in earlier work.
- $<_s$ is the **simplicity** selector. Writing $|l|$ for the number of possible facts in l, we have $l_1 <_s l_2$ iff $|l_1| > |l_2|$. Its maximal elements have minimal description length in the chosen vocabulary.

Definition 20 (Generalisation probability)**.**
Let P_α be a probability distribution over demand sets S, meaning finite sets of future demands for a task α. Let B_π be the set of future demands compatible with a policy π that is correct for α. Define

$$G_{P_\alpha}(\pi) := P_\alpha(S \subseteq B_\pi)\,.$$

A parent task generated by S includes the requirements of α and adds every member of S as an input and as a correct output. For a policy correct for α, the unseen region and buffer are $U = L_{\mathfrak{v}} \setminus \text{Ext}(I_\alpha)$ and $B_\pi = \text{Ext}(\pi) \cap U$. The identity $\text{Ext}(I_\alpha \cup S) \cap \text{Ext}(\pi) = O_\alpha \cup (\text{Ext}(S) \cap B_\pi)$ shows that the policy remains correct for that parent if and only if $S \subseteq B_\pi$ and S is closed under completion inside B_π, meaning S contains every completion of a member of S that lies in B_π.

7 Freest-correct generalisation

Proposition 10 (Uniform generalisation equivalence)**.**
Let $\mathfrak{v}$ be finite and let α be a $\mathfrak{v}$-task with $\Pi_\alpha \neq \emptyset$. I define an unseen region $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$. A demand set is a set $S \subseteq U$, and each member of S is a demand. A policy $\pi \in \Pi_\alpha$ generalises to S when

$$S \subseteq B_\pi, \qquad B_\pi := \mathrm{Ext}(\pi) \cap U.$$

If S is drawn uniformly from 2^U, then every correct policy has generalisation probability $G_{P_\alpha}(\pi) = 2^{|B_\pi|-|U|}$, so generalisation probability is a strictly increasing function of freedom on Π_α. Hence a correct policy maximises generalisation probability if and only if it is $<_{\mathrm{ff}}$-maximal in Π_α. Every non-$<_{\mathrm{ff}}$-maximal correct policy is strictly suboptimal.

Proof. By Lemma 9 with μ as cardinality, maximising $|\mathrm{Ext}(\pi)|$ over Π_α is equivalent to maximising $|B_\pi|$. Under a uniform distribution of demands, π generalises to S when S is a subset of B_π. There are $2^{|B_\pi|}$ such demands out of $2^{|U|}$ possible demands. Therefore

$$G_{P_\alpha}(\pi) = 2^{|B_\pi|-|U|},$$

which is strictly increasing in $|B_\pi|$, and hence in $|\mathrm{Ext}(\pi)|$ over Π_α. Therefore the policies with maximal generalisation probability are the policies with maximal freedom. □

Theorem 2 (Freest-correct is Bayes optimal under exchangeable task growth)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a child task α with $\Pi_\alpha \neq \emptyset$. Let $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$ be the unseen region. For each $\pi \in \Pi_\alpha$, I write $B_\pi := \mathrm{Ext}(\pi) \cap U$ for the unseen buffer of π. Let a later task add future demands $S \subseteq U$, and assume the distribution of S is exchangeable, meaning $P(S)$ depends only on $|S|$. A policy generalises to the demand set when $S \subseteq B_\pi$. Then for all $\pi, \pi' \in \Pi_\alpha$ with $|\mathrm{Ext}(\pi)| \leq |\mathrm{Ext}(\pi')|$, $P(S \subseteq B_\pi) \leq P(S \subseteq B_{\pi'})$. For any such pair with $|\mathrm{Ext}(\pi)| < |\mathrm{Ext}(\pi')|$, this inequality is strict if and only if $P(1 \leq |S| \leq |B_{\pi'}|) > 0$. In particular every freest correct policy

$$\pi^* \in \arg\max_{\pi \in \Pi_\alpha} |\mathrm{Ext}(\pi)|$$

maximises $P(S \subseteq B_\pi)$. Either every correct policy has generalisation probability $P(S = \emptyset)$ or the maximisers of $P(S \subseteq B_\pi)$ are the freest correct policies and no others. Under a randomised selector, a policy is drawn at random according to some probability distribution. Among all randomised selectors supported on Π_α, no selector has higher expected generalisation probability than a freest correct policy.

Proof. By Lemma 9 with μ as cardinality, maximising $|\mathrm{Ext}(\pi)|$ over Π_α is equivalent to maximising $|B_\pi|$. Let $b = |B_\pi|$. Since the distribution of S is exchangeable, there are nonnegative numbers r_m such that each subset of U with size m has probability r_m. Therefore

$$P(S \subseteq B_\pi) = \sum_{m=0}^{b} \binom{b}{m} r_m.$$

Probability $P(S \subseteq B_\pi) = \sum_{m=0}^{b} \binom{b}{m} r_m$ is nondecreasing in b, because every summand is nonnegative and $\binom{b}{m}$ is nondecreasing in b for fixed m. Hence any policy with maximal buffer, equivalently maximal freedom, maximises generalisation probability. Since $|\mathrm{Ext}(\pi)| \leq |\mathrm{Ext}(\pi')|$ if and only if $|B_\pi| \leq |B_{\pi'}|$ by Lemma 9, a correct policy with no less freedom than another has no lower generalisation probability. For $b < b'$ and $1 \leq m \leq b'$, Pascal's rule gives $\binom{b'}{m} = \binom{b'-1}{m} + \binom{b'-1}{m-1} \geq \binom{b}{m} + 1$, so

$$\sum_{m=0}^{b'} \binom{b'}{m} r_m - \sum_{m=0}^{b} \binom{b}{m} r_m = \sum_{m=1}^{b'} \Big[\binom{b'}{m} - \binom{b}{m} \Big] r_m$$

is strictly positive if and only if $r_m > 0$ for some $1 \leq m \leq b'$, that is, if and only if $P(1 \leq |S| \leq b') > 0$. Let b^* be the maximal buffer size over Π_α. If $P(1 \leq |S| \leq b^*) = 0$, then $r_m = 0$ for $1 \leq m \leq b^*$ and every correct policy has generalisation probability $r_0 = P(S = \emptyset)$. Otherwise every correct policy with a buffer smaller than b^* has strictly lower generalisation probability than a freest correct policy. Under a randomised selector, expected generalisation probability is a convex combination of the generalisation probabilities of correct policies. A convex combination cannot exceed the largest of its terms. □

Corollary 5 (Optimality does not hinge upon a uniform prior)**.**
Proposition 10, which assumes that every set of demands is equally likely, was the first optimality result I proved, and is I think the easiest to understand. However the optimality claims do not depend on that uniform prior. Under exchangeable demands, maximising freedom is optimal. Under independent demands with unequal probabilities, generalisation probability is a strictly increasing function of weighted freedom, so maximising weighted freedom is optimal. In atomless continuous domains, suppose equally measured regions have equal survival probability. Then survival probability never falls as buffer measure grows.

Proof. The uniform distribution is the exchangeable distribution with $r_m = 2^{-|U|}$ for every subset. For independent demands with unequal probabilities, suppose each $u \in U$ is demanded with probability $p_u \in [0, 1)$. Then

$$P(S \subseteq B_\pi) = \prod_{u \notin B_\pi} (1 - p_u),$$

so $P(S \subseteq B_\pi) = \prod_{u \in U}(1 - p_u) \exp(\sum_{u \in B_\pi} - \log(1 - p_u))$, with $\prod_{u \in U}(1 - p_u) > 0$, is a strictly increasing function of $\sum_{u \in B_\pi} - \log(1 - p_u)$, which is weighted freedom. For continuous domains, I put a reference measure μ on U and assume that equally measured regions have equal survival probability. If measurable $A, B \subseteq U$ satisfy $\mu(A) \leq \mu(B)$, by atomlessness there is a measurable $B' \subseteq B$ with $\mu(B') = \mu(A)$. Therefore $P(S \subseteq A) = P(S \subseteq B') \leq P(S \subseteq B)$. □

Lemma 7 (The freest correct policy permits every completion of every correct policy)**.**
Let α be a $\mathfrak{v}$-task with $I_\alpha \neq \emptyset$ and $\Pi_\alpha \neq \emptyset$, and let $C_\alpha := \bigcap_{i \in I_\alpha} i$ be the set of possible facts common to every input. Then there is a policy $R_\alpha \in \Pi_\alpha$ with $R_\alpha \cap C_\alpha = \emptyset$ and

$$\Pi_\alpha = \{R_\alpha \cup X \mid X \subseteq C_\alpha\}.$$

Hence R_α is the unique freest correct policy, every other correct policy contains more possible facts than R_α, and $\mathrm{Ext}(\pi) \subseteq \mathrm{Ext}(R_\alpha)$ for every $\pi \in \Pi_\alpha$.

Proof. I fix $\pi \in \Pi_\alpha$ and $i \in I_\alpha$. Since $\mathrm{Ext}(i) \subseteq \mathrm{Ext}(I_\alpha)$, the correctness equation $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$ gives $\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi) = \mathrm{Ext}(i) \cap O_\alpha$. The input i has a correct output, so this set is nonempty. Any member of it contains $i \cup \pi$, and every subset of a possible thing is a possible thing, so $i \cup \pi \in L_\mathfrak{v}$ and $\mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i) \cap O_\alpha$. The same is true of any $\pi' \in \Pi_\alpha$, so $\mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i \cup \pi')$. A possible thing is the smallest member of its own extension, so possible things with equal extensions are equal, and $i \cup \pi = i \cup \pi'$. This is true for every input, so every possible fact in which π and π' differ lies in every input, which is to say in C_α. Hence $R_\alpha := \pi \setminus C_\alpha$ is the same for every $\pi \in \Pi_\alpha$, and every correct policy is $R_\alpha \cup X$ with $X = \pi \cap C_\alpha$. Conversely, let $X \subseteq C_\alpha$. For every input i, $C_\alpha \subseteq i$ gives $i \cup R_\alpha \cup X = i \cup \pi$. So $R_\alpha \cup X$ is a subset of the possible thing $i \cup \pi$, hence a possible thing, and $\mathrm{Ext}(i) \cap \mathrm{Ext}(R_\alpha \cup X) = \mathrm{Ext}(i \cup \pi) = \mathrm{Ext}(i) \cap O_\alpha$. The union over the inputs gives $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(R_\alpha \cup X) = O_\alpha$, so $R_\alpha \cup X \in \Pi_\alpha$. Since $R_\alpha \subseteq R_\alpha \cup X$, $\mathrm{Ext}(R_\alpha \cup X) \subseteq \mathrm{Ext}(R_\alpha)$, with equality only when $X = \emptyset$, because possible things with equal extensions are equal. Since $R_\alpha \cap C_\alpha = \emptyset$, $|R_\alpha \cup X| = |R_\alpha| + |X|$, which exceeds $|R_\alpha|$ whenever $X \neq \emptyset$. □

Corollary 6 (Every demand set that a correct policy survives is also survived by the freest correct policy)**.**
Under the hypotheses of Lemma 7, with $U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$ and $B_\pi := \mathrm{Ext}(\pi) \cap U$,

1. $B_\pi \subseteq B_{R_\alpha}$ for every $\pi \in \Pi_\alpha$, so $G_{P_\alpha}(R_\alpha) \geq G_{P_\alpha}(\pi)$ under every probability distribution P_α of demands, exchangeable or not;

2. R_α is correct for every parent generated by a demand set in the sense of Definition 20, whenever some correct policy of α is correct for that parent;

3. if $\mathfrak{v}$ is finite, then R_α is correct for at least as many parents of α as any other correct policy.

Proof. By Lemma 7, $\mathrm{Ext}(\pi) \subseteq \mathrm{Ext}(R_\alpha)$, so $B_\pi \subseteq B_{R_\alpha}$. Every demand set inside B_π is then inside B_{R_α}, so $P_\alpha(S \subseteq B_\pi) \leq P_\alpha(S \subseteq B_{R_\alpha})$ for every distribution P_α. For the second item, let π be correct for the parent generated by S. By Definition 20, $S \subseteq B_\pi$ and S is closed under completion inside B_π. Then $S \subseteq B_{R_\alpha}$. A completion of a member of S contains a member of $\mathrm{Ext}(\pi)$, so it lies in $\mathrm{Ext}(\pi) \subseteq \mathrm{Ext}(R_\alpha)$. Such a completion lies in B_{R_α} if and only if it lies in U, and hence if and only if it lies in B_π. So S is closed under completion inside B_{R_α}, and R_α is correct for that parent. For the third item, let ω be a parent of α for which π is correct. By the correctness equation, $O_\omega = \mathrm{Ext}(I_\omega) \cap \mathrm{Ext}(\pi)$, so ω is determined by I_ω. Every input j of ω has a correct output, which contains $j \cup \pi$, so $j \cup \pi$ is a possible thing and so is its subset $j \cup R_\alpha$. In the task $\omega' := \langle I_\omega, \mathrm{Ext}(I_\omega) \cap \mathrm{Ext}(R_\alpha) \rangle$, every input j therefore has the correct output $j \cup R_\alpha$. The task ω' also has $I_\alpha \subsetneq I_\omega$, its correct outputs include $O_\alpha = \mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(R_\alpha)$, and R_α is one of its correct policies, so ω' is a parent of α for which R_α is correct. If π is correct for two distinct parents ω, then those parents have distinct sets of inputs and so give distinct parents ω'. Hence R_α is correct for at least as many parents as π. □

Over the full correct-policy set, both freedom and simplicity measured by policy cardinality select R_α; their choices can diverge on a searched subset that excludes R_α. Dominance under every distribution of demands concerns parents generated by demand sets. It does not assert correctness on every arbitrary parent on which a rival succeeds. For a fixed nested sequence of children with uniform demand law at each step, freedom selection maximises generalisation probability at every step and reaches any attainable probability threshold with no more examples. With the same positive additive acquisition and processing cost per example for every selector, the corresponding cost is no greater.

8 Product vocabularies and simplicity

Definition 21 (product vocabulary)**.**
Let $B_1, \ldots, B_m$ be pairwise disjoint finite sets of possible facts with $|B_i| \geq 2$. Assume there is a nonempty $A \subseteq \Phi$ such that each B_i partitions A, and that

$$\bigcap_{i=1}^{m} p_i \neq \varnothing \qquad \text{for every } (p_1, \ldots, p_m) \in B_1 \times \cdots \times B_m.$$

The product vocabulary is

$$\mathfrak{v}_{\mathrm{prod}} := \bigcup_{i=1}^{m} B_i.$$

For $l \in L_{\mathfrak{v}_{\mathrm{prod}}}$, I define

$$I(l) := \{i \in \{1, \ldots, m\} : l \cap B_i \neq \varnothing\}.$$

By the definition of an embodied language, every member of $L_{\mathfrak{v}_{\mathrm{prod}}}$ is a possible thing. I use simplicity selector $<_s$ and freedom selector $<_{\mathrm{ff}}$ from Definition 19.

Lemma 8 (product extension count)**.**
For every $l \in L_{\mathfrak{v}_{\text{prod}}}$,

$$|\operatorname{Ext}(l)| = \prod_{i \notin I(l)} (|B_i| + 1).$$

Proof. A possible thing contains at most one possible fact from each block because each B_i partitions A. Conversely, every selection of at most one possible fact from each block is satisfiable. To show this, I select one possible fact from each block unrepresented by the selection. The resulting full selection has nonempty intersection by assumption, so its subset does too. By definition, a completion y of l satisfies $l \subseteq y$, so it contains every possible fact from l and therefore represents every block represented by that possible thing. For each block $i \notin I(l)$, a completion either adds no possible fact from B_i or adds one of its $|B_i|$ possible facts. These choices are independent across blocks. Hence $|\operatorname{Ext}(l)| = \prod_{i \notin I(l)} (|B_i| + 1)$. □

Theorem 3 (when simplicity and freedom agree)**.**
On $L_{\mathfrak{v}_{\text{prod}}}$, simplicity selector $<_s$ and freedom selector $<_{\text{ff}}$ agree on all comparisons and ties if and only if

$$|B_1| = |B_2| = \cdots = |B_m|.$$

Proof. Assume all block sizes equal d. Since l represents exactly $|l|$ blocks, by Lemma 8,

$$|\operatorname{Ext}(l)| = (d+1)^{m-|l|}.$$

Function $(d+1)^{m-|l|}$ is strictly decreasing in $|l|$, so $l <_s l'$ if and only if $l <_{\text{ff}} l'$, with identical ties.

Conversely, assume two blocks have different sizes. I choose $i \neq j$ with $|B_i| \neq |B_j|$, and $p_i \in B_i$, $p_j \in B_j$. Let $l_i = \{p_i\}$ and $l_j = \{p_j\}$. Then $|l_i| = |l_j| = 1$, so simplicity selector $<_s$ ties them. By Lemma 8,

$$|\operatorname{Ext}(l_i)| = \prod_{k \neq i} (|B_k| + 1), \qquad |\operatorname{Ext}(l_j)| = \prod_{k \neq j} (|B_k| + 1).$$

Values $\prod_{k \neq i} (|B_k| + 1)$ and $\prod_{k \neq j} (|B_k| + 1)$ differ because $|B_i| \neq |B_j|$. Freedom selector $<_{\text{ff}}$ separates two possible things that simplicity selector $<_s$ ties. Therefore $<_s$ and $<_{\text{ff}}$ cannot agree on all comparisons and ties unless all block sizes are equal. □

Proposition 11 (simplicity sub-optimality)**.**
Here simplicity means minimum description length, measured as the cardinality of the smallest possible thing that implies a particular extension. In a task, that possible thing is a shortest correct policy. When a learner chooses between two correct policies, simplicity in this sense is neither necessary nor sufficient for choosing the one with the greater generalisation probability, under the uniform distribution of demands, or under any exchangeable distribution of demands that gives positive probability to some demand set of size between one and the larger of their two buffer sizes.

Proof. Let

$$\Phi = \{\phi_1, \phi_2, \phi_3, \phi_4\}$$

and let the vocabulary be $\mathfrak{v} = \{a, b, c, d, e, f\}$, where the six distinct possible facts are

$$\begin{aligned} a &= \{\phi_1\}, & b &= \{\phi_2\}, & c &= \{\phi_1, \phi_3\}, \\ d &= \{\phi_1, \phi_2, \phi_3\}, & e &= \{\phi_1, \phi_2, \phi_4\}, & f &= \{\phi_3, \phi_4\}. \end{aligned}$$

A subset of $\mathfrak{v}$ is a possible thing when its possible facts have nonempty intersection. I define

$$I_\alpha = \{\{b, d, e\}\}, \qquad O_\alpha = \{\{b, d, e\}\}.$$

The input $\{b, d, e\}$ is its own correct output. Its truth set is $b \cap d \cap e = \{\phi_2\}$, and no possible fact outside $\{b, d, e\}$ is true given ϕ_2, so $\mathrm{Ext}(\{b, d, e\}) = \{\{b, d, e\}\}$ and every subset of $\{b, d, e\}$ is a correct policy. I compare two of them, $\pi_s = \{b\}$ and $\pi_{\mathrm{ff}} = \{d, e\}$. Since π_s has fewer possible facts, $<_s$ ranks π_s ahead of π_{ff}. The extension of π_s is

$$\mathrm{Ext}(\pi_s) = \{\{b\}, \{b, d\}, \{b, e\}, \{b, d, e\}\},$$

so $|\mathrm{Ext}(\pi_s)| = 4$. The extension of π_{ff} is

$$\mathrm{Ext}(\pi_{\mathrm{ff}}) = \{\{d, e\}, \{a, d, e\}, \{b, d, e\}, \{c, d, e\}, \{a, c, d, e\}\},$$

so $|\mathrm{Ext}(\pi_{\mathrm{ff}})| = 5$, and $<_{\mathrm{ff}}$ ranks π_{ff} ahead of π_s. By Lemma 9 their buffers have sizes $|\mathrm{Ext}(\pi_s)| - |O_\alpha| = 3$ and $|\mathrm{Ext}(\pi_{\mathrm{ff}})| - |O_\alpha| = 4$. By Proposition 10, π_{ff} then has strictly greater generalisation probability than π_s under the uniform distribution over unseen demands, and by Theorem 2 the same is true under every exchangeable distribution of demands with $P(1 \leq |S| \leq 4) > 0$. Simplicity is insufficient because $<_s$ chooses π_s. It is unnecessary because the policy with the greater generalisation probability, π_{ff}, has the longer description. Simplicity also cannot choose between π_s and $\{d\}$, since $\{d\}$ contains one possible fact, as π_s does, while $|\mathrm{Ext}(\{d\})| = 12$. As Lemma 7 states, the empty thing is both the shortest and the freest correct policy. Hence the two selectors can part only when a learner has found correct policies but not the empty thing. □

Corollary 7 (Simplicity and freedom usually diverge on bounded machine vocabularies)**.**
In the bounded Turing configuration vocabulary of Definition 10, suppose $|\Gamma| \geq 2$ and each family[8]

$$\{Q_q : q \in Q\}, \quad \{H_i : 1 \leq i \leq S\}, \quad \{C_{j,a} : a \in \Gamma\}\ (1 \leq j \leq S)$$

partitions $\mathcal{M}_{M,S}$. Suppose also that the implementation is configuration-complete, meaning the map

$$\phi \longmapsto (q(\phi), i(\phi), T_\phi)$$

from $\mathcal{M}_{M,S}$ onto $Q \times \{1, \ldots, S\} \times \Gamma^S$ is surjective. Then simplicity selector $<_s$ and freedom selector $<_{\mathrm{ff}}$ coincide on $L_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}$ if and only if $|Q| = S = |\Gamma|$.

Proof. The partition assumptions specify one block for control, one for head position, and one for each tape cell. Every block has at least two possible facts. By configuration completeness, every selection of one possible fact from each block is jointly satisfiable. Suppose one possible fact occurred in two blocks. I select it from one block, a different possible fact from the other, and arbitrary possible facts from all remaining blocks. The shared possible fact also lies in the block from which the different possible fact was selected. The block partitions $\mathcal{M}_{M,S}$, so these two possible facts have empty intersection, contradicting configuration completeness. Thus $\mathfrak{v}^{\mathrm{conf}}_{M,S}$ is a product vocabulary whose block sizes are $|Q|$, S, and $|\Gamma|$ for each tape-cell block. By Theorem 3, the simplicity and freedom selectors agree if and only if all block sizes are equal. □

9 Measured languages and continuous demands

Definition 22 (measured embodied language and μ-freedom)**.**
A **measured embodied language** is a triple $(\mathfrak{v}, \mathcal{A}_L, \mu)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary, $\mathcal{A}_L$ is a σ-algebra on $L_\mathfrak{v}$, and μ is a σ-finite measure on $(L_\mathfrak{v}, \mathcal{A}_L)$. Assume $\mathrm{Ext}(l) \in \mathcal{A}_L$ for every $l \in L_\mathfrak{v}$. The **μ-freedom** of l is

$$\mathrm{ff}_\mu(l) := \mu(\mathrm{Ext}(l)).$$

[8] In mathematics a family is a set, called a family when its members are themselves sets or functions. Here it is a set of possible facts.

For a task α, I define the unseen region and the buffer of a correct policy by

$$U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha), \qquad B_\pi := \mathrm{Ext}(\pi) \cap U.$$

For every continuous task considered below, assume $\mathrm{Ext}(I_\alpha)$ and O_α are measurable. Assume also that U is measurable. For the PAC-Bayes certificates of Section 10, assume in addition that $0 < \mu(U) < \infty$.

Definition 23 (μ-exchangeable distribution of demands)**.**
A probability distribution of the demand set S over the measurable subsets of U is **μ-exchangeable** when, for every measurable $B \subseteq U$, the generalisation probability

$$G(B) := P(S \subseteq B)$$

depends on B only through $\mu(B)$ and is nondecreasing in $\mu(B)$. When that dependence is strictly increasing across the buffer measures attained by the policies under comparison, a larger attained buffer has strictly larger generalisation probability.

Theorem 4 (continuous freest correct optimality)**.**
Fix a measured embodied language and a task α with $\Pi_\alpha \neq \emptyset$ and $\mu(O_\alpha) < \infty$. A correct policy's generalisation shortfall under a distribution of demands is the amount by which its generalisation probability falls below the highest generalisation probability among correct policies. Under any μ-exchangeable distribution of demands, correct policies π, π' with $\mathrm{ff}_\mu(\pi) \leq \mathrm{ff}_\mu(\pi')$ satisfy $P(S \subseteq B_\pi) \leq P(S \subseteq B_{\pi'})$, so every correct policy maximising $\mathrm{ff}_\mu(\pi)$ also maximises $P(S \subseteq B_\pi)$. If generalisation probability is strictly increasing in buffer measure across the attained buffer measures, $\mathrm{ff}_\mu(\pi) < \mathrm{ff}_\mu(\pi')$ implies $P(S \subseteq B_\pi) < P(S \subseteq B_{\pi'})$, and every correct policy with less than maximal μ-freedom is strictly suboptimal. Hence the worst-case generalisation shortfall over the μ-exchangeable distributions of demands is minimised by μ-freest correct selection.

Proof. By Lemma 9, μ-freedom and buffer measure induce the same ordering on correct policies. By Definition 23, generalisation probability is a nondecreasing function of that buffer measure. When generalisation probability is strictly increasing across the attained buffer measures, the ordering on the attained values is strict. Suppose a policy is pointwise optimal for every distribution in the class. Then it has zero shortfall for every distribution, so its worst-case shortfall is minimal. □

Proposition 12 (measured-language reparameterisation invariance)**.**
Let $(L_{\mathfrak{v}}, \mathcal{A}_L, \mu)$ and $(L_{\mathfrak{v}'}, \mathcal{A}_{L'}, \mu')$ be measured embodied languages. Let π be a policy and U an unseen region in $L_{\mathfrak{v}}$, and let π' and U' be a policy and an unseen region in $L_{\mathfrak{v}'}$. Let $h : L_{\mathfrak{v}} \to L_{\mathfrak{v}'}$ be an injective measurable map satisfying $h_\# \mu = \mu'$, $h(\mathrm{Ext}(\pi)) = \mathrm{Ext}(\pi')$, and $h(U) = U'$. Then

$$\mathrm{ff}_{\mu'}(\pi') = \mathrm{ff}_\mu(\pi) \qquad \text{and} \qquad \mu'(B_{\pi'}) = \mu(B_\pi).$$

Proof. Since $h_\# \mu = \mu'$, $\mu'(E) = \mu(h^{-1}(E))$ for every measurable $E \subseteq L_{\mathfrak{v}'}$, and $h^{-1}(h(A)) = A$ for every $A \subseteq L_{\mathfrak{v}}$ by injectivity. Hence $\mathrm{ff}_{\mu'}(\pi') = \mu'(h(\mathrm{Ext}(\pi))) = \mu(\mathrm{Ext}(\pi)) = \mathrm{ff}_\mu(\pi)$, and since $B_{\pi'} = h(\mathrm{Ext}(\pi)) \cap h(U) = h(\mathrm{Ext}(\pi) \cap U)$ by injectivity, $\mu'(B_{\pi'}) = \mu(B_\pi)$. □

Proposition 13 (function-level invariance in one language)**.**
Let $F(\theta)$ and $F(\theta')$ be the same input-output function, and fix a vocabulary $\mathfrak{v}$ whose possible facts describe input-output functions. Then

$$\mathrm{Enc}_{\mathfrak{v}}(F(\theta)) = \mathrm{Enc}_{\mathfrak{v}}(F(\theta')).$$

Consequently their extensions, finite freedom, and μ-freedom are equal whenever measured in that same language.

Proof. By definition $\mathrm{Enc}_{\mathfrak{v}}(F)$ is the set of possible facts in $\mathfrak{v}$ that are true of the function F. Since $F(\theta)$ and $F(\theta')$ are one function, a possible fact is true of the first if and only if it is true of the second, so the two encodings are the same set of possible facts. A possible thing determines its extension, so equal encodings have equal extensions. Finite freedom is the cardinality of the extension, and μ-freedom is its measure under μ. Each is fixed by the extension alone, so each has the same value for the two encodings. □

Lemma 9 (fixed seen-region decomposition)**.**
Let α be a task. Each correct policy $\pi \in \Pi_\alpha$ satisfies

$$\mathrm{Ext}(\pi) = O_\alpha \dot\cup B_\pi$$

and

$$\mathrm{ff}_\mu(\pi) = \mu(O_\alpha) + \mu(B_\pi).$$

If $\mu(O_\alpha) < \infty$, μ-freedom and buffer measure induce the same ordering on correct policies. In particular, when $\mathfrak{v}$ is finite and μ is cardinality, correct policies $\pi, \pi' \in \Pi_\alpha$ satisfy $|\mathrm{Ext}(\pi)| \leq |\mathrm{Ext}(\pi')|$ if and only if $|B_\pi| \leq |B_{\pi'}|$.

Proof. By correctness, $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$. The seen region $\mathrm{Ext}(I_\alpha)$ and unseen region U partition $L_{\mathfrak{v}}$, so O_α and B_π are disjoint and satisfy $\mathrm{Ext}(\pi) = O_\alpha \dot\cup B_\pi$. The measure identity follows by countable additivity. The finite value $\mu(O_\alpha)$ is fixed across Π_α, so adding it preserves the ordering, with cardinality as a special case. □

I use Definitions 22 and 23. Throughout this section, U is measurable, $0 < \mu(U) < \infty$, $\mu(O_\alpha) < \infty$, and $\Pi_\alpha \neq \emptyset$. I use $G(\pi, P) := P(S \subseteq B_\pi)$ for a distribution P of future demands.

Theorem 5 (Continuous sufficiency under Poisson demands)**.**
If future demands are generated by a Poisson point process on U with intensity $\lambda\mu$ with $\lambda > 0$, then for every correct policy $\pi \in \Pi_\alpha$,

$$G(\pi, P) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

In particular, $G(\pi, P)$ is strictly increasing in $\mu(B_\pi)$ and therefore, by Lemma 9, strictly increasing in $\mathrm{ff}_\mu(\pi)$ on Π_α.

Proof. Under a Poisson point process with intensity measure $\lambda\mu$, the event $S \subseteq B_\pi$ is exactly the event that no points fall in the measurable complement $U \setminus B_\pi$. The Poisson void formula therefore gives

$$G(\pi, P) = P(S \subseteq B_\pi) = \exp\big(-\lambda\mu(U \setminus B_\pi)\big) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

This is strictly increasing in $\mu(B_\pi)$. By Lemma 9, on correct policies this is equivalent to strict increase in $\mathrm{ff}_\mu(\pi)$. □

Definition 24 (Nondegeneracy)**.**
A μ-exchangeable distribution P of demands is **nondegenerate** if for every measurable $A \subseteq U$ with $\mu(A) > 0$,

$$P(\emptyset \neq S \subseteq A) > 0.$$

Theorem 6 (Continuous strict necessity)**.**
Assume that $(U, \mathcal{A}_L|_U, \mu|_U)$ is atomless. If P is μ-exchangeable and nondegenerate, and if two correct policies satisfy

$$\mu(B_{\pi_1}) > \mu(B_{\pi_2}),$$

then

$$G(\pi_1, P) > G(\pi_2, P).$$

By Lemma 9, this is equivalent to strict ordering by μ-freedom on Π_α.

Proof. Choose measurable $B' \subseteq B_{\pi_1}$ with $\mu(B') = \mu(B_{\pi_2})$. Such a set exists by atomlessness. By Definition 23,

$$P(S \subseteq B') = P(S \subseteq B_{\pi_2}).$$

Set $A := B_{\pi_1} \setminus B'$, which has positive measure. Nondegeneracy gives $P(\emptyset \neq S \subseteq A) > 0$. This event is disjoint from $\{S \subseteq B'\}$ and implies $S \subseteq B_{\pi_1}$, so

$$P(S \subseteq B_{\pi_1}) \geq P(S \subseteq B') + P(\emptyset \neq S \subseteq A) > P(S \subseteq B_{\pi_2}).$$

□

Theorem 7 (Minimax optimality of μ-freedom)**.**
Let $\mathcal{E}$ be the class of all μ-exchangeable distributions of demands on U. If a μ-freest correct policy $\pi^* \in \Pi_\alpha$ exists, then

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P) \;=\; \inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P),$$

and π^* attains every policy maximum in this equality.

Proof. Let π^* be a correct policy maximising ff_μ on Π_α. By Theorem 4,

$$G(\pi^*, P) \geq G(\pi, P)$$

for all $\pi \in \Pi_\alpha$ and all $P \in \mathcal{E}$. So

$$\max_{\pi \in \Pi_\alpha} G(\pi, P) = G(\pi^*, P) \qquad \text{for every } P \in \mathcal{E},$$

which gives

$$\inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P) = \inf_{P \in \mathcal{E}} G(\pi^*, P).$$

Again by Theorem 4,

$$G(\pi^*, P) \geq G(\pi, P) \qquad \text{for all } \pi \in \Pi_\alpha \text{ and all } P \in \mathcal{E},$$

so

$$\inf_{P \in \mathcal{E}} G(\pi^*, P) \geq \inf_{P \in \mathcal{E}} G(\pi, P) \qquad \text{for all } \pi \in \Pi_\alpha.$$

Therefore π^* achieves $\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P)$. The minimax equality follows from the chain

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P) \geq \inf_{P \in \mathcal{E}} G(\pi^*, P) = \inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P) \geq \max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P).$$

□

10 PAC-Bayes survival certificates

The set of candidates $\mathcal{C}_0$ is measurable and fixed, and $S_1, \ldots, S_n \sim P$ are independently sampled demand sets. The jointly measurable score $u(c, S) \in [0, 1]$ is fixed before sampling. For Stack Theory, $\mathcal{C}_0 = \Pi_\alpha$ and $u(\pi, S) = \mathbf{1}\{S \subseteq B_\pi\}$. The relevant expectations and divergences are

$$\widehat{G}_{n,u}(Q) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}_{c \sim Q} u(c, S_i), \qquad G_u(Q, P) = \mathbb{E}_{c \sim Q, S \sim P} u(c, S),$$

$$\mathrm{kl}(a \| b) = a \log \frac{a}{b} + (1 - a) \log \frac{1 - a}{1 - b}, \qquad \mathrm{KL}(Q \| P_0) = \int \log \frac{dQ}{dP_0} \, dQ$$

when $Q \ll P_0$, and $\mathrm{KL}(Q \| P_0) = \infty$ otherwise. The usual endpoint conventions and natural logarithms apply. The prior may depend on the fixed child task but is fixed before the demand sets are drawn.

Theorem 8 (PAC-Bayes future-demand certificate)**.**
Let $n \geq 2$, let $\delta \in (0,1)$, and let P_0 be a prior on $\mathcal{C}_0$ fixed before $S_1, \ldots, S_n$ are drawn. With probability at least $1-\delta$, simultaneously for every posterior Q,

$$\mathrm{kl}\Big(\widehat{G}_{n,u}(Q)\Big\|G_u(Q,P)\Big) \leq \frac{\mathrm{KL}(Q\|P_0) + \log(2\sqrt{n}/\delta)}{n}.$$

Proof. I apply PAC-Bayes-kl to the bounded loss $1-u(c,S)$ and use

$$\mathrm{kl}(1-a\|1-b) = \mathrm{kl}(a\|b)$$

[Seeger, 2002, Maurer, 2004, Chugg et al., 2023]. □

For measurable A with $0 < \mu(A) < \infty$, the normalised restriction is

$$\mu(E \mid A) := \frac{\mu(E \cap A)}{\mu(A)}.$$

Theorem 9 (The closest distribution on a buffer)**.**
Assume

$$0 < \mu(B_\pi) \leq \mu(U) < \infty.$$

For every probability measure $Q \ll \mu(\,\cdot \mid U)$ supported on B_π,

$$\mathrm{KL}(Q\|\mu(\,\cdot \mid U)) = \mathrm{KL}(Q\|\mu(\,\cdot \mid B_\pi)) + \log\frac{\mu(U)}{\mu(B_\pi)}.$$

Consequently,

$$\inf_{\substack{Q \ll \mu(\,\cdot\,|U) \\ Q(B_\pi)=1}} \mathrm{KL}(Q\|\mu(\,\cdot \mid U)) = \log\frac{\mu(U)}{\mu(B_\pi)},$$

and the unique minimiser up to null sets is $\mu(\,\cdot \mid B_\pi)$.

Proof. On B_π,

$$\frac{d\mu(\,\cdot \mid B_\pi)}{d\mu(\,\cdot \mid U)} = \frac{\mu(U)}{\mu(B_\pi)}.$$

For every admissible Q,

$$\log\frac{dQ}{d\mu(\,\cdot \mid U)} = \log\frac{dQ}{d\mu(\,\cdot \mid B_\pi)} + \log\frac{\mu(U)}{\mu(B_\pi)}.$$

Integrating with respect to Q, I get $\mathrm{KL}(Q\|\mu(\,\cdot \mid U)) = \mathrm{KL}(Q\|\mu(\,\cdot \mid B_\pi)) + \log(\mu(U)/\mu(B_\pi))$. Since $\mathrm{KL}(Q\|\mu(\,\cdot \mid B_\pi)) \geq 0$, the infimum of $\mathrm{KL}(Q\|\mu(\,\cdot \mid U))$ is $\log(\mu(U)/\mu(B_\pi))$, which is attained at $Q = \mu(\,\cdot \mid B_\pi)$, where the first term vanishes. Since $\mathrm{KL}(Q\|\mu(\,\cdot \mid B_\pi)) = 0$ only when $Q = \mu(\,\cdot \mid B_\pi)$ up to null sets, the minimiser is unique. □

Corollary 8 (completion-restriction survival certificate)**.**
Assume $n \geq 2$, $\delta \in (0,1)$, and $0 < \mu(U) < \infty$. Let $S_1, \ldots, S_n$ be independent future-demand sets from P. Assume $(y,S) \mapsto \mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\}$ is jointly measurable. I define

$$\widehat{C}_n(\pi) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{y\sim\mu(\,\cdot\,|B_\pi)}\mathbf{1}\{S_i \subseteq \mathrm{Ext}(y) \cap U\},$$

$$C(\pi,P) := \mathbb{E}_{y\sim\mu(\,\cdot\,|B_\pi)}P(S \subseteq \mathrm{Ext}(y) \cap U).$$

With probability at least $1-\delta$, simultaneously for every policy $\pi \in L_{\mathfrak{v}}$ with $\mu(B_\pi) > 0$, where $B_\pi := \mathrm{Ext}(\pi) \cap U$,

$$\mathrm{kl}\Big(\widehat{C}_n(\pi)\Big\|C(\pi,P)\Big) \le \frac{\log[\mu(U)/\mu(B_\pi)] + \log(2\sqrt{n}/\delta)}{n}.$$

Moreover,

$$P(S \subseteq B_\pi) \ge C(\pi, P).$$

Proof. I apply Theorem 8 with the set of candidates $\mathcal{C}_0 = U$, prior $\mu(\cdot \mid U)$, posterior $\mu(\cdot \mid B_\pi)$, and survival score $u(y,S) = \mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\}$. Theorem 9 states the KL term. If $y \in B_\pi \subseteq \mathrm{Ext}(\pi)$, then $\mathrm{Ext}(y) \subseteq \mathrm{Ext}(\pi)$. Therefore

$$\mathbf{1}\{S \subseteq \mathrm{Ext}(y) \cap U\} \le \mathbf{1}\{S \subseteq B_\pi\}.$$

I take expectations over y and S. □

The buffer KL identity alone gives no classification-error guarantee: a sampled completion need not be child-correct or inherit its policy's loss. The certificates concern the specified future-demand score. They do not identify arbitrary parameter-space PAC-Bayes posteriors with freedom.

Proposition 14 (Both certificates survive measure-preserving renaming)**.**
Let $h : U \to U'$ be a bimeasurable bijection[9] with $h_\#\mu = \mu'$ and $h(B_\pi) = B_{\pi'}$. Then

$$h_\#\mu(\cdot \mid U) = \mu'(\cdot \mid U'), \qquad h_\#\mu(\cdot \mid B_\pi) = \mu'(\cdot \mid B_{\pi'}),$$

$$\mu'(B_{\pi'}) = \mu(B_\pi),$$

and

$$\mathrm{KL}(\mu(\cdot \mid B_\pi)\|\mu(\cdot \mid U)) = \mathrm{KL}(\mu'(\cdot \mid B_{\pi'})\|\mu'(\cdot \mid U'))\,.$$

If $S' = h(S)$, then survival of the policy is preserved. Survival of sampled completions is also preserved when

$$h(\mathrm{Ext}(y) \cap U) = \mathrm{Ext}(h(y))' \cap U'$$

for $\mu(\cdot \mid B_\pi)$-almost every $y \in B_\pi$.

Proof. Normalisation commutes with the pushforward because $h_\#\mu = \mu'$ and h maps the buffer to the primed buffer. KL is invariant when both distributions are pushed through the same measurable bijection. Direct image under a bijection preserves containment. Under the final extension condition, h maps each completion-survival event to its primed counterpart. □

11 Description length and priors

Proposition 15 (subjectivity of description length)**.**
Suppose there is no abstraction, in the sense that the vocabulary equals the full possible fact set ($\mathfrak{v} = \mathcal{P} = 2^\Phi$). Then the tautological truth set Φ has minimum description length 0, and every other truth set of a possible thing has minimum description length 1, meaning that it is the truth set of some one-fact thing and of no shorter thing. More generally this is true for every vocabulary $\mathfrak{v}$ with $T(l) \in \mathfrak{v}$ for every nonempty $l \in L_{\mathfrak{v}}$. Consequently, description length is highly representation dependent. In the extreme case $\mathfrak{v} = \mathcal{P}$, minimising $|l|$ does not meaningfully distinguish among policies by truth set.

[9] Primed symbols U', μ', π', $B_{\pi'}$ and $\mathrm{Ext}(\cdot)'$ are the counterparts of U, μ, π, B_π and $\mathrm{Ext}(\cdot)$ in a second measured embodied language, as in Proposition 12 in Section 9. The pushforward $h_\#\mu$ gives each measurable set $E \subseteq U'$ the weight $\mu(h^{-1}(E))$, so $h_\#\mu = \mu'$ means every set weighs the same before and after renaming by h.

Proof. If $\mathfrak{v} = \mathcal{P}$, then for every $l \in L_{\mathfrak{v}}$ the truth set $p := T(l)$ is itself a member of $\mathfrak{v}$. If $p = \Phi$, the empty thing already has truth set p and length 0. If $p \neq \Phi$, the singleton thing $\{p\}$ has length 1 and satisfies

$$T(\{p\}) = p = T(l).$$

The empty thing is the only possible thing of length 0 and its truth set is Φ, so no shorter thing has truth set p. Thus every truth set expressible by any possible thing is also expressible by either the empty thing or a one-fact thing when $\mathfrak{v} = \mathcal{P}$. I use $\mathfrak{v} = \mathcal{P}$ only to place p in $\mathfrak{v}$ when $p \neq \Phi$, where l is nonempty. Hence the argument is valid for every vocabulary with $T(l) \in \mathfrak{v}$ for every nonempty $l \in L_{\mathfrak{v}}$. Equality $T(\{T(l)\}) = T(l)$ concerns semantic representability, not equality of extensions inside an arbitrary vocabulary. One-fact representation in $\mathfrak{v} = \mathcal{P}$ shows that description length can be collapsed by changing vocabulary, so description length cannot be the invariant that explains generalisation. □

Lemma 10 (In degenerate vocabularies, simplicity is freedom)**.**
Assume $\mathfrak{v}$ is finite. Then

$$\mathrm{ff}(l) \leq 2^{|\mathfrak{v}|-|l|}$$

for every $l \in L_{\mathfrak{v}}$. Equality is attained for one $l \in L_{\mathfrak{v}}$, equivalently for every $l \in L_{\mathfrak{v}}$, if and only if $\mathfrak{v}$ is degenerate, with $T(\mathfrak{v}) \neq \emptyset$, every subset of $\mathfrak{v}$ satisfiable and $L_{\mathfrak{v}} = 2^{\mathfrak{v}}$. In this special case, freedom is a strictly decreasing function of description length (simplicity).

Proof. A completion of l is any satisfiable superset y with $l \subseteq y \subseteq \mathfrak{v}$. There are $2^{|\mathfrak{v}|-|l|}$ supersets before imposing satisfiability, one for each subset of the remaining possible facts $\mathfrak{v} \setminus l$. Hence $\mathrm{ff}(l) \leq 2^{|\mathfrak{v}|-|l|}$, with equality if and only if every superset of l in $\mathfrak{v}$ is satisfiable. A subset of a satisfiable set is satisfiable, since $y \subseteq y'$ gives $T(y') \subseteq T(y)$. The vocabulary $\mathfrak{v}$ is a superset of every l. Therefore every superset of l in $\mathfrak{v}$ is satisfiable if and only if $T(\mathfrak{v}) \neq \emptyset$, a condition independent of l. □

Corollary 9 (complexity confounding)**.**
Let $\mathfrak{v}$ be finite. At most $|\mathfrak{v}| + 1$ truth sets have minimum description length at most one in $L_{\mathfrak{v}}$, meaning that each is the truth set of the empty thing or of a thing of one possible fact. So when the possible things of $L_{\mathfrak{v}}$ have more truth sets than that, some truth set has minimum description length at least two, unlike in Proposition 15. In every degenerate vocabulary (Lemma 10) and in every product vocabulary whose blocks have one size (Theorem 3), freedom is a strictly decreasing function of description length. In such a vocabulary, let $\pi_1, \pi_2 \in \Pi_\alpha$ be correct policies for a $\mathfrak{v}$-task α with $|\pi_1| < |\pi_2|$. Then $G_{P_\alpha}(\pi_1) \geq G_{P_\alpha}(\pi_2)$ under every exchangeable distribution P_α of further demands as in Theorem 2, and $G_{P_\alpha}(\pi_1) > G_{P_\alpha}(\pi_2)$ under the uniform distribution of Proposition 10. Freedom therefore *confounds* simplicity and generalisation in such a vocabulary.

Proof. The empty thing is the one possible thing of length zero, and there are at most $|\mathfrak{v}|$ possible things of length one, so at most $|\mathfrak{v}| + 1$ truth sets have minimum description length at most one. For example, let $\Phi = \{1, 2, 3, 4\}$, $a = \{1, 2, 4\}$, $b = \{1, 3, 4\}$ and $\mathfrak{v} = \{a, b\}$. Then $L_{\mathfrak{v}} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ has four truth sets, so the truth set $\{1, 4\}$ of $\{a, b\}$ has minimum description length two. This vocabulary is degenerate, and its four possible things have freedoms 4, 2, 2 and 1. In a degenerate vocabulary $\mathrm{ff}(l) = 2^{|\mathfrak{v}|-|l|}$ by Lemma 10, and in a product vocabulary of m blocks of size d, $\mathrm{ff}(l) = (d+1)^{m-|l|}$ by Lemma 8. Both strictly decrease in $|l|$, so $|\pi_1| < |\pi_2|$ gives $\mathrm{ff}(\pi_1) > \mathrm{ff}(\pi_2)$. Theorem 2 then gives $G_{P_\alpha}(\pi_1) \geq G_{P_\alpha}(\pi_2)$ under every exchangeable P_α, and Proposition 10 gives $G_{P_\alpha}(\pi_1) > G_{P_\alpha}(\pi_2)$ under the uniform distribution. Neither result mentions description length, so the correlation between description length and generalisation is inherited from freedom. □

Corollary 10 (Weak constraints take simple forms under usage bias)**.**
Let $\mathfrak{v}$ be finite and let $\pi \in L_{\mathfrak{v}}$ be the policy a body realises. Let P be a usage distribution over

the completions of π, meaning that P assigns each $o \in \mathrm{Ext}(\pi)$ its probability of being used. Assume P satisfies the usage-bias condition that $\mathrm{ff}(o_1) > \mathrm{ff}(o_2)$ implies $P(o_1) \geq P(o_2)$. The condition is true, with strict inequality, when demand sets are drawn uniformly from the subsets of $L_{\mathfrak{v}}$, as in Proposition 10 with $L_{\mathfrak{v}}$ in place of U, and $P(o)$ is proportional to the probability that o generalises to the demand set, meaning that the demand set is a subset of $\mathrm{Ext}(o)$. A **layer above** π is a vocabulary $\mathfrak{w} \subseteq \mathfrak{f}(\mathfrak{v}, \pi)$ of truth sets of completions of π (Definition 6), with a grounding choice $\chi : \mathfrak{w} \to \mathrm{Ext}(\pi)$ such that $T(\chi(q)) = q$ (Definition 7). The completion $\chi(q)$ is the grounding of q. In a stack (Definition 26) whose layer i has $\mathfrak{v}^i = \mathfrak{v}$ and $\pi^i = \pi$, the vocabulary $\mathfrak{v}^{i+1}$ with a grounding choice is a layer above π. A completion o has a **form** of k possible facts in the layer when some $a \subseteq \mathfrak{w}$ with $|a| = k$ has $T(a) = T(o)$, and its minimum description length in the layer is the least such k. The **usage** of the layer is $\sum_{q \in \mathfrak{w}} P(\chi(q))$, the usage of the groundings of its possible facts. Fix $m \geq 1$. Among layers of at most m possible facts:

1. A layer of greatest usage exists. One such layer, $\mathfrak{w}^*$ with grounding choice χ^*, results when I take the completions of π in order of decreasing freedom, with ties broken by decreasing usage, and take each as the grounding of its truth set unless a completion taken earlier has the same truth set, until m completions are taken or no completions remain.

2. In every layer of greatest usage, the grounding of each possible fact is used at least as often as every completion whose truth set is not a possible fact of the layer, and at least as often as every other completion that has the same truth set as that grounding.

3. Under $\mathfrak{w}^*$, if $o_2 = \chi^*(q)$ for some $q \in \mathfrak{w}^*$ and $\mathrm{ff}(o_1) > \mathrm{ff}(o_2)$, then $T(o_1) \in \mathfrak{w}^*$, so o_1 has a form of one possible fact in the layer.

4. At most $\sum_{j=0}^{k} \binom{m}{j}$ truth sets have minimum description length at most k in a layer of m possible facts. So if the completions of π have more truth sets than that, some completion has no form of k or fewer possible facts in any such layer.

Proof. For a truth set $q \in \mathfrak{f}(\mathfrak{v}, \pi)$, let $M(q)$ be the greatest usage of a completion with truth set q. A layer $\mathfrak{w}$ has usage at most $\sum_{q \in \mathfrak{w}} M(q)$, with equality when the grounding of each possible fact is a completion of greatest usage among the completions with the same truth set. So the greatest usage of a layer of at most m possible facts is the sum of the m largest values of M, or of all of them when there are fewer than m truth sets, and some layer attains it, since there are finitely many layers and grounding choices. By the usage-bias condition, a list of the completions in order of decreasing freedom, with ties broken by decreasing usage, is in order of nonincreasing usage. The first completion listed with a given truth set q therefore has usage $M(q)$, and the truth sets appear, by their first completions, in order of nonincreasing M. In the first part I take the first m truth sets so listed, each with a grounding of usage $M(q)$, so $\mathfrak{w}^*$ attains the greatest usage. For the second part, let $\mathfrak{w}$ with grounding choice χ have greatest usage, and let $w = \chi(q)$ be the grounding of some $q \in \mathfrak{w}$. If some completion o with $T(o) \notin \mathfrak{w}$ had $P(o) > P(w)$, then q replaced by $T(o)$ and w by o would give a layer of at most m possible facts with greater usage. If some o with $T(o) = q$ had $P(o) > P(w)$, then w replaced by o would do the same. For the third part, let $o_2 = \chi^*(q)$ for some $q \in \mathfrak{w}^*$ and $\mathrm{ff}(o_1) > \mathrm{ff}(o_2)$. Then o_1 is listed before o_2, so the truth set of o_1 first appears before o_2. Since o_2 is a grounding under χ^*, it is the first completion listed with its truth set, and that truth set is among the m taken. Every truth set that first appears earlier is also among them, so $T(o_1) \in \mathfrak{w}^*$ and $\{T(o_1)\}$ is a form of one possible fact. For the last part, a layer of m possible facts has at most $\sum_{j=0}^{k} \binom{m}{j}$ things of at most k possible facts, and each has one truth set. □

Corollary 11 (Simple-general correlation is freedom induced)**.**
Under the conditions of Corollary 10, let $\pi_1, \pi_2 \in \Pi_\alpha$ be correct policies for a $\mathfrak{v}$-task α that are completions of π. If $\pi_2 = \chi^*(q)$ for some $q \in \mathfrak{w}^*$ and $\mathrm{ff}(\pi_1) > \mathrm{ff}(\pi_2)$, then π_1 has a form of

one possible fact in $\mathfrak{w}^*$. If the distribution P_α of further demands for α is exchangeable as in Theorem 2, and $\pi_1 = \chi^*(q)$ for some $q \in \mathfrak{w}^*$ while π_2 has no form of one possible fact in $\mathfrak{w}^*$, then $G_{P_\alpha}(\pi_1) \geq G_{P_\alpha}(\pi_2)$. A simple form above appears to predict generalisation. Without the usage-bias condition, that prediction is no longer guaranteed.

Proof. The first claim is the third part of Corollary 10. For the second, if $\mathrm{ff}(\pi_2) > \mathrm{ff}(\pi_1)$ then that part would give π_2 a form of one possible fact, so $\mathrm{ff}(\pi_2) \leq \mathrm{ff}(\pi_1)$. Under exchangeable demands, Theorem 2 gives $G_{P_\alpha}(\pi_2) \leq G_{P_\alpha}(\pi_1)$ whenever $|\operatorname{Ext}(\pi_2)| \leq |\operatorname{Ext}(\pi_1)|$. The apparent link between a simple form above and generalisation is therefore inherited from freedom through the usage distribution. Without the usage-bias condition, the completions of greatest freedom need not be those of greatest usage, so they need not be groundings in a layer of greatest usage. □

Corollary 10 places the simple forms in a layer above the body, whose possible facts are truth sets of the body's completions. A memory is different. Its vocabulary $\mathfrak{m}$ can be any finite vocabulary, and the possible thing of $\mathfrak{m}$ that stands for a possible thing l need not share a truth set with l. I call it a codeword. A codeword whose truth set is not inside the truth set of the thing it encodes can be true while that thing is false, which is what Section 20 calls an inert representation. A memory whose codewords are all of this kind is inert memory. The corollary below is Corollary 10 for codewords. It ranks every codeword by cardinality, because a code can give any possible thing any codeword, while the forms of a layer are tied to truth sets.

Corollary 12 (Weak constraints compress under usage bias)**.**
Let P be a usage distribution over the finite set $L_\mathfrak{v}$, meaning that P assigns each possible thing its probability of being used. Assume P satisfies the usage-bias condition that $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies $P(l_1) \geq P(l_2)$. The condition is true, with strict inequality, when demand sets are drawn uniformly from the subsets of $L_\mathfrak{v}$, as in Proposition 10 with $L_\mathfrak{v}$ in place of U, and $P(l)$ is proportional to the probability that l generalises to the demand set, meaning that the demand set is a subset of $\operatorname{Ext}(l)$. Let $\mathfrak{m}$ be a finite vocabulary of forms, such as the vocabulary of a memory. A **code** assigns each possible thing $l \in L_\mathfrak{v}$ its own possible thing $c(l) \in L_\mathfrak{m}$, its **codeword**, so that $c(l_1) \neq c(l_2)$ whenever $l_1 \neq l_2$. The cardinality $|c(l)|$ of a codeword is the number of possible facts in it. Codewords in the usual sense are codewords in this sense. On a tape of S cells whose contents all occur, over an alphabet Γ of at least two symbols, the word $a_1 \cdots a_k$ with $k \leq S$ is the possible thing $\{C_{1,a_1}, \ldots, C_{k,a_k}\}$ of the vocabulary of cell facts $C_{j,a}$ of Definition 10, of cardinality k. The expected cardinality of a code is $\sum_{l \in L_\mathfrak{v}} P(l)\,|c(l)|$. Assume a code exists. Then there exists a code with minimum expected cardinality whose codewords $c^*(l)$ satisfy

$$\mathrm{ff}(l_1) > \mathrm{ff}(l_2) \quad \Longrightarrow \quad |c^*(l_1)| \leq |c^*(l_2)|.$$

Every code with minimum expected cardinality has codewords $c(l)$ that satisfy $P(l_1) > P(l_2) \Rightarrow |c(l_1)| \leq |c(l_2)|$. Therefore every code with minimum expected cardinality satisfies the displayed ordering at every pair with $P(l_1) \neq P(l_2)$, and under uniform demands it satisfies the displayed ordering at every pair. No equality claim is made for equal-freedom ties. For each k, let N_k be the number of possible things of $L_\mathfrak{m}$ with at most k possible facts. Then $N_k \leq \sum_{j=0}^{k} \binom{|\mathfrak{m}|}{j}$, every code assigns codewords of more than k possible facts to at least $|L_\mathfrak{v}| - N_k$ possible things, and under c^* none of those possible things is freer than a possible thing whose codeword has at most k possible facts. If $\mathfrak{m}$ is a product vocabulary (Definition 21) whose blocks all have one size, then $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies $\mathrm{ff}(c^*(l_1)) \geq \mathrm{ff}(c^*(l_2))$, where the freedom of a codeword is its freedom in $L_\mathfrak{m}$.

Proof. A code exists by assumption. Since $L_\mathfrak{v}$ and $L_\mathfrak{m}$ are finite, there are finitely many codes, so a code with minimum expected cardinality exists. I take any minimum expected cardinality code and fix its set of codewords. If $P(l_1) > P(l_2)$ but $|c(l_1)| > |c(l_2)|$, I exchange $c(l_1)$ and $c(l_2)$. The

set of codewords is unchanged, so the result is a code. After the exchange the cardinalities of the codewords of l_1 and l_2 are swapped and expected cardinality is strictly lower, a contradiction. Thus no optimal code assigns a codeword of greater cardinality to a higher-probability possible thing than to a lower-probability possible thing. For each equal-probability class, I reorder the codewords assigned to that class by such exchanges, so that higher-freedom possible things receive codewords of no greater cardinality than those of lower-freedom possible things. Expected cardinality is unchanged because the probabilities in the class are equal. The reordered code, with codewords $c^*(l)$, still has minimum expected cardinality. By the usage-bias condition, a possible thing with strictly greater freedom never has lower probability. So $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies $|c^*(l_1)| \leq |c^*(l_2)|$. Under uniform demands, $2^{\mathrm{ff}(l)}$ of the $2^{|L_\mathfrak{v}|}$ subsets of $L_\mathfrak{v}$ are demands to which l generalises, namely the subsets of $\mathrm{Ext}(l)$. So l generalises to a uniformly drawn demand set with probability $2^{\mathrm{ff}(l)-|L_\mathfrak{v}|}$, and $P(l)$ is proportional to $2^{\mathrm{ff}(l)}$, which strictly increases with $\mathrm{ff}(l)$. Hence $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ implies $P(l_1) > P(l_2)$, and every code with minimum expected cardinality satisfies the displayed ordering. Each possible thing of $L_\mathfrak{m}$ is a subset of $\mathfrak{m}$, so N_k is at most the number of subsets of $\mathfrak{m}$ with at most k members. The codewords of a code are distinct, so at most N_k possible things have codewords of at most k possible facts. If $|c^*(l_1)| > k \geq |c^*(l_2)|$, then $\mathrm{ff}(l_1) > \mathrm{ff}(l_2)$ would contradict the displayed ordering, so $\mathrm{ff}(l_1) \leq \mathrm{ff}(l_2)$. If $\mathfrak{m}$ is a product vocabulary whose r blocks all have size d, then each $y \in L_\mathfrak{m}$ has freedom $(d+1)^{r-|y|}$ in $L_\mathfrak{m}$ by Lemma 8, as in the proof of Theorem 3, and this is a strictly decreasing function of $|y|$. So $|c^*(l_1)| \leq |c^*(l_2)|$ implies $\mathrm{ff}(c^*(l_1)) \geq \mathrm{ff}(c^*(l_2))$. □

The consequence for generalisation is as in Corollary 11. Under c^*, a correct policy with a codeword of smaller cardinality than another correct policy for the same task is no less free, so under exchangeable demands it has no lower generalisation probability by Theorem 2. Section 20 is about what holding any codeword costs, whatever its cardinality. Each codeword held across ticks of objective time is a commitment beyond the contents it stands for, and Proposition 38 shows that commitment costs in tenure and in viable completions.

Lemma 11 (Decoder keys are hidden entropy)**.**
Let $\mathcal{M}$ be a **finite** microstate set with $|\mathcal{M}| = 2^S$, where S is the number of bits needed to index the microstate set. Let $\mathcal{D}$ be a finite family of operationally distinct decoders.

For each $D \in \mathcal{D}$, let $\mathcal{M}^{(D)} \subseteq \mathcal{M}$ be the set of microstates on which D succeeds uniformly at some fixed accuracy. Assume a uniform coverage bound

$$|\mathcal{M}^{(D)}| \leq d_{\max} \qquad \text{for every } D \in \mathcal{D}.$$

If the protocol must succeed for every microstate, meaning $\bigcup_{D\in\mathcal{D}} \mathcal{M}^{(D)} = \mathcal{M}$, then

$$|\mathcal{D}| \geq \frac{|\mathcal{M}|}{d_{\max}} = \frac{2^S}{d_{\max}}.$$

Therefore any control register that must select among $\mathcal{D}$ at verification time must support at least $|\mathcal{D}|$ reliably distinguishable internal states. Equivalently any key vocabulary $\mathfrak{v}_K$ representing that register must satisfy

$$\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}| \geq S - \log_2 d_{\max}.$$

Proof. The covering condition gives

$$|\mathcal{M}| \leq \sum_{D\in\mathcal{D}} |\mathcal{M}^{(D)}| \leq |\mathcal{D}|\, d_{\max}.$$

Rearrange to get the lower bound on $|\mathcal{D}|$. A selector register that can implement any $D \in \mathcal{D}$ needs at least one reliably distinguishable internal state per operationally distinct decoder. So the register has at least $|\mathcal{D}|$ internal states and any representing vocabulary satisfies $\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}|$. □

Remark 1 (Freedom does not imply inclusion)**.**
Freedom compares only *cardinalities*. In general, $\mathrm{ff}(l_1) \geq \mathrm{ff}(l_2)$ does *not* imply $l_1 \subseteq l_2$, nor does it imply $\mathrm{Ext}(l_1) \supseteq \mathrm{Ext}(l_2)$. However, if $l_1 \subseteq l_2$ (i.e. l_2 is a completion of l_1), then $\mathrm{Ext}(l_2) \subseteq \mathrm{Ext}(l_1)$ and hence $\mathrm{ff}(l_1) \geq \mathrm{ff}(l_2)$.

Corollary 13 (Occam priors collapse under no abstraction)**.**
Fix a finite candidate family $H \subseteq L_{\mathcal{P}}$, where $\mathcal{P} = 2^{\Phi}$. If $\mathfrak{v} = \mathcal{P}$ (no abstraction), then every $h \in H$ has a singleton semantic representation of length 1 by Proposition 15. Therefore any prior or selector on H that depends only on this semantic length proxy is uniform on H. In particular, the semantic-length Occam prior carries no discriminatory information on a no-abstraction candidate family.

Proof. By Proposition 15, when $\mathfrak{v} = \mathcal{P}$ every candidate $h \in H$ has a singleton semantic representation $\{T(h)\}$ with description length

$$|\{T(h)\}| = 1.$$

So any prior or selector on the finite family H that depends only on this semantic length proxy assigns the same score to every candidate. After normalisation, the induced prior is therefore uniform on H. Hence the semantic-length Occam prior carries no discriminatory information on a no-abstraction candidate family. □

12 Stacks and delegation

Definition 25 (unseen-buffer utility of intelligence)**.**
For any task $\gamma = \langle I_\gamma, O_\gamma \rangle$ whose vocabulary $\mathfrak{v}$ is finite, I define

$$U_\gamma := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\gamma), \qquad B_\pi^\gamma := \mathrm{Ext}(\pi) \cap U_\gamma.$$

For each correct policy $\pi \in \Pi_\gamma$, I define its unseen-buffer value by

$$\beta_\gamma(\pi) := |B_\pi^\gamma|.$$

The **utility of intelligence** of γ is the largest unseen-buffer value achievable by a correct policy:

$$\epsilon(\gamma) = \begin{cases} 0, & \Pi_\gamma = \emptyset, \\ \max_{\pi \in \Pi_\gamma} \beta_\gamma(\pi), & \Pi_\gamma \neq \emptyset. \end{cases}$$

For $\pi \in \Pi_\gamma$, by Lemma 9 with μ as cardinality, $\beta_\gamma(\pi) = |\,\mathrm{Ext}(\pi)| - |O_\gamma| = |\,\mathrm{Ext}(\pi) \setminus O_\gamma|$. The maximum exists because the vocabulary is finite.

Definition 26 (stack)**.**
A **stack** [Bennett, 2024] of depth n is specified by the following data.

- A sequence of disembodied tasks $\langle \lambda^0, \lambda^1, \ldots, \lambda^n \rangle$ such that $\lambda^{i+1} \sqsubset \lambda^i$.
- A sequence of vocabularies $\langle \mathfrak{v}^0, \mathfrak{v}^1, \ldots, \mathfrak{v}^n \rangle$.
- A sequence of policies $\langle \pi^0, \pi^1, \ldots, \pi^n \rangle$ with $\pi^i \in L_{\mathfrak{v}^i}$ for each i.

For each $i = 0, \ldots, n-1$, $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i)$, so each higher layer embodies some or all of the truth sets of the completions of the policy beneath it. I call a stack **over-constrained** if there exists some layer i with $\Pi_{\lambda^i(\mathfrak{v}^i)} = \varnothing$.

Theorem 10 (Time in a stack)**.**
Let $\langle \mathfrak{v}^0, \dots, \mathfrak{v}^n \rangle$ be the finite vocabularies of a stack, and let $\tau : I \to \Phi$ be a timeline. For every adjacent pair of layers and all distinctions $\phi, \phi' \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}^i}(\phi) = \mathrm{Enc}_{\mathfrak{v}^i}(\phi')$ implies $\mathrm{Enc}_{\mathfrak{v}^{i+1}}(\phi) = \mathrm{Enc}_{\mathfrak{v}^{i+1}}(\phi')$. Hence for every finite timeline fragment h of τ,

$$\kappa_{\mathfrak{v}^{i+1}}(h) \le \kappa_{\mathfrak{v}^i}(h).$$

Every tick of subjective time in layer $i{+}1$ is a tick of subjective time in layer i, so for $u \le u'$ in I,

$$\kappa^\tau_{\mathfrak{v}^i}(u) = \kappa^\tau_{\mathfrak{v}^i}(u') \quad \Longrightarrow \quad \kappa^\tau_{\mathfrak{v}^{i+1}}(u) = \kappa^\tau_{\mathfrak{v}^{i+1}}(u'),$$

and every uninterrupted block of constant lower encoding lies within one block of constant higher encoding. Consequently, for every $t \in I$,

$$\kappa^\tau_{\mathfrak{v}^{i+1}}(t) \le \kappa^\tau_{\mathfrak{v}^i}(t),$$

and hence

$$0 \le \kappa^\tau_{\mathfrak{v}^n}(t) \le \cdots \le \kappa^\tau_{\mathfrak{v}^0}(t) \le t.$$

For any vocabulary $\mathfrak{v}$, equality $\kappa^\tau_{\mathfrak{v}}(t) = t$ is attained if and only if its encoding changes at every adjacent transition from 0 through t.

Proof. By the stack and abstractor definitions, every $q \in \mathfrak{v}^{i+1}$ is $T(o)$ for some $o \in \mathrm{Ext}(\pi^i)$ with $o \subseteq \mathfrak{v}^i$. Therefore $\phi \in q$ if and only if $o \subseteq \mathrm{Enc}_{\mathfrak{v}^i}(\phi)$. Equal lower encodings of two distinctions therefore give equal higher encodings. Suppose tick u of objective time is a tick of subjective time in layer $i{+}1$. Then some possible fact $q \in \mathfrak{v}^{i+1}$ changes truth value between $\tau(u-1)$ and $\tau(u)$. By the stack and abstractor definitions, $q = T(o)$ for some $o \in \mathrm{Ext}(\pi^i)$. If every possible fact $p \in o$ had the same truth value given both environmental distinctions, then their intersection $T(o) = q$ would also have the same truth value. Therefore some $p \in o \subseteq \mathfrak{v}^i$ changes truth value. The lower encoding changes, so layer i records a tick of subjective time at the same tick of objective time. Every tick of subjective time counted at layer $i{+}1$ is therefore counted at layer i. Thus $\kappa_{\mathfrak{v}^{i+1}}(h) \le \kappa_{\mathfrak{v}^i}(h)$. Taking $h = \tau_{\le t}$, I obtain $\kappa^\tau_{\mathfrak{v}^{i+1}}(t) \le \kappa^\tau_{\mathfrak{v}^i}(t)$. Applying this adjacent-layer inequality for $i = 0, \dots, n-1$ shows that $\kappa^\tau_{\mathfrak{v}^n}(t) \le \cdots \le \kappa^\tau_{\mathfrak{v}^0}(t)$. A layer clock increments by one at each change of its encoding and otherwise stays constant. Hence if a lower clock is constant between u and u', its higher clock is constant over that interval. Each of t ticks of objective time advances any layer clock by at most one, so its upper bound is t. Equality is attained if and only if all t increments are one. □

Proposition 16 (The Law of the Stack)**.**
In a stack with adjacent layers i and $i{+}1$, suppose $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i) = \{T(o) \mid o \in \mathrm{Ext}(\pi^i)\}$. Then the achievable behavioural flexibility at layer $i{+}1$ is bottlenecked by the freedom of the realised policy at layer i in the following cardinal sense:

$$\sup_{\pi \in L_{\mathfrak{v}^{i+1}}} |\mathrm{Ext}(\pi)| \;\le\; 2^{|\,\mathrm{Ext}(\pi^i)|}.$$

When $\mathfrak{v}^{i+1}$ is finite, let γ^{i+1} be any $\mathfrak{v}^{i+1}$-task with $\Pi_{\gamma^{i+1}} \neq \emptyset$, such as the induced layer-$(i{+}1)$ task $\lambda^{i+1}(\mathfrak{v}^{i+1})$ when it has a correct policy. Then $\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \ge 1$ and

$$|\,\mathrm{Ext}(\pi^i)| \;\ge\; \log_2\big(\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|\big).$$

Higher targets for the utility of intelligence at layer $i{+}1$ imply larger lower bounds on the freedom of the layer-i policy. The present freedom of the layer-i policy sets a bound, and a target above that bound cannot be reached without increasing that freedom.

Proof. A layer i policy π^i has a set of completions $\mathrm{Ext}(\pi^i)$. Every possible fact of the next layer vocabulary $\mathfrak{v}^{i+1}$ is the truth set of one of those completions, so $|\mathfrak{v}^{i+1}| \leq |\mathrm{Ext}(\pi^i)|$. Any possible thing in layer $i{+}1$ is a subset of $\mathfrak{v}^{i+1}$, so the entire layer-$i{+}1$ language has size at most $2^{|\mathfrak{v}^{i+1}|}$, which is at most $2^{|\mathrm{Ext}(\pi^i)|}$. Every extension set $\mathrm{Ext}(\pi)$ at layer $i{+}1$ is a subset of that language, so $|\mathrm{Ext}(\pi)| \leq 2^{|\mathrm{Ext}(\pi^i)|}$. When $\mathfrak{v}^{i+1}$ is finite, by the definition of ϵ there is some correct layer-$(i{+}1)$ policy with $|\mathrm{Ext}(\pi)| = \epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|$, because $\Pi_{\gamma^{i+1}} \neq \emptyset$ and the set of correct policies is finite. This cardinality is at least one, because a correct policy is a possible thing and is a member of its own extension. Applying $|\mathrm{Ext}(\pi)| \leq 2^{|\mathrm{Ext}(\pi^i)|}$ to that policy and taking base-2 logarithms, I get $|\mathrm{Ext}(\pi^i)| \geq \log_2(\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|)$. □

Corollary 14 (Delegation is necessary for lowering F_2)**.**
Let $\mathfrak{v}^i$ be finite and $\mathfrak{v}^{i+1} \subseteq \mathfrak{f}(\mathfrak{v}^i, \pi^i)$ as in Proposition 16. Fix a possible thing $\mu \in L_{\mathfrak{v}^{i+1}}$ in layer $i{+}1$, viability, whose extension $\mathrm{Ext}(\mu)$ represents viable completions. For any policy $\pi \in L_{\mathfrak{v}^{i+1}}$ with $\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu) \neq \emptyset$, let p_π be uniform on Ω_π and p_μ uniform on $\mathrm{Ext}(\mu)$. I use $D_{\mathrm{KL},2}$ for Kullback–Leibler divergence with base-2 logarithms. I define

$$F_2(\pi) := D_{\mathrm{KL},2}(p_\pi \| p_\mu) = \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|.$$

This is a divergence in bits, not thermodynamic free energy. The physical identification follows below. Then

$$|\Omega_\pi| \leq |\mathrm{Ext}(\pi)| \leq 2^{|\mathrm{Ext}(\pi^i)|}.$$

Consequently, if at least N viable completions must remain open to the layer-$(i{+}1)$ policy, then

$$|\mathrm{Ext}(\pi^i)| \geq \log_2 N.$$

Lower F_2 targets therefore imply larger lower bounds on the freedom of the layer-i policy [Bennett, 2024].

Proof. The vocabulary $\mathfrak{v}^{i+1} \subseteq \{T(o) \mid o \in \mathrm{Ext}(\pi^i)\}$ has at most $|\mathrm{Ext}(\pi^i)|$ possible facts. Every possible thing of $L_{\mathfrak{v}^{i+1}}$ is a subset of this vocabulary, and every completion of π is such a possible thing. Therefore $|\mathrm{Ext}(\pi)| \leq 2^{|\mathrm{Ext}(\pi^i)|}$, which is the bound of Proposition 16. Since $\Omega_\pi \subseteq \mathrm{Ext}(\pi)$, $|\Omega_\pi| \leq 2^{|\mathrm{Ext}(\pi^i)|}$. If $N \leq |\Omega_\pi|$ then $N \leq 2^{|\mathrm{Ext}(\pi^i)|}$, and taking base-2 logarithms gives $|\mathrm{Ext}(\pi^i)| \geq \log_2 N$. □

The F_2 ordering uses the stated uniform p_π and p_μ. Lower-layer restrictions may make viability μ inexpressible or leave no correct upper-layer policy. The premise $\mu \in L_{\mathfrak{v}^{i+1}}$ presupposes its representability. Optimisation confined to the upper layer cannot remove either obstruction without changing lower constraints. A physical identification is also required to relate F_2 to physical free energy. F_2 is defined in Corollary 14 from uniform distributions over viable completions. Let $F_{\mathrm{phys}}[p]$ be the thermodynamic nonequilibrium free energy of physical microstate distribution p. In a physical application, suppose I identify the completions in $\mathrm{Ext}(\mu)$ bijectively with the full physical microstate space being modelled. Suppose I also use a fixed Hamiltonian, meaning an energy function over microstates. With that Hamiltonian, the canonical equilibrium distribution at temperature Θ_0 assigns each microstate of energy E a probability proportional to its Boltzmann weight $\mathrm{e}^{-E/(k_B\Theta_0)}$. Suppose finally that I take the resulting distribution p_μ to be this canonical equilibrium distribution. Then, by the standard nonequilibrium identity,

$$F_{\mathrm{phys}}[p_\pi] - F_{\mathrm{phys}}[p_\mu] = k_B\Theta_0 \ln 2\, F_2(\pi).$$

The factor $\ln 2$ appears because F_2 is defined with base-two logarithms. Uniform p_μ can be a canonical equilibrium distribution only when mapped microstates have equal Boltzmann weights. Without that physical identification, F_2 remains an information quantity in bits.

The identification above requires equal weights and one microstate per completion. Both requirements can be dropped. Let $\mathcal{X} \subseteq \Phi$ be a finite set of distinctions, the microstates of the

physical model. Let q be a probability distribution on $\mathcal{X}$ with $q_x > 0$ for every $x \in \mathcal{X}$, and for $A \subseteq \Phi$ write $q(A) := \sum_{x \in A \cap \mathcal{X}} q_x$. Let $q_{\mathfrak{v}} := (\mathrm{Enc}_{\mathfrak{v}})_{\#} q$ be the pushforward of q through the encoding map, as in Proposition 12. It is the measure on $L_{\mathfrak{v}}$ with

$$q_{\mathfrak{v}}(D) = q(\{x \in \mathcal{X} \mid \mathrm{Enc}_{\mathfrak{v}}(x) \in D\}) \qquad \text{for } D \subseteq L_{\mathfrak{v}}.$$

The triple $(\mathfrak{v}, 2^{L_{\mathfrak{v}}}, q_{\mathfrak{v}})$ is then a measured embodied language in the sense of Definition 22, with $q_{\mathfrak{v}}$ as the measure. For the equilibrium distribution, each microstate x has energy E_x and contains an amount $m_{j,x}$ of each material constituent j that the system exchanges with its surroundings. The reference environment[10] has temperature Θ_0 and chemical potential η_j for constituent j. The equilibrium distribution at temperature Θ_0, also called the Gibbs distribution, is

$$q_x := \frac{\mathrm{e}^{-h_x/(k_B\Theta_0)}}{\sum_{y \in \mathcal{X}} \mathrm{e}^{-h_y/(k_B\Theta_0)}}, \qquad h_x := E_x - \sum_j \eta_j m_{j,x}.$$

The nonequilibrium free energy of a distribution p on $\mathcal{X}$ is

$$F_{\mathrm{phys}}[p] := \sum_{x \in \mathcal{X}} p_x h_x + k_B\Theta_0 \sum_{x \in \mathcal{X}} p_x \ln p_x,$$

with $0 \ln 0 := 0$. I call $F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q]$ the excess free energy of p, meaning its free energy above that of the equilibrium distribution. Without exchanged constituents, $h_x = E_x$ and F_{phys} is the Helmholtz free energy. With them, it is a grand potential. For $A \subseteq \Phi$ with $q(A) > 0$, I write $q|_A$ for q conditioned on A, meaning $(q|_A)_x = q_x/q(A)$ for $x \in A \cap \mathcal{X}$ and $(q|_A)_x = 0$ otherwise, and $q(B \mid A) := q(A \cap B)/q(A)$ for the conditional probability of $B \subseteq \Phi$ given A. For distributions p and p' on $\mathcal{X}$, the Kullback–Leibler divergence is $\mathrm{KL}(p\|p') := \sum_x p_x \ln(p_x/p'_x)$, with natural logarithms. The same formula, summed over possible things, defines KL for two distributions on $L_{\mathfrak{v}}$.

Proposition 17 (Free energy of a commitment)**.**
Let q be the equilibrium distribution above and $q_{\mathfrak{v}} := (\mathrm{Enc}_{\mathfrak{v}})_{\#} q$. For every $l \in L_{\mathfrak{v}}$, $\mathrm{ff}_{q_{\mathfrak{v}}}(l) = q_{\mathfrak{v}}(\mathrm{Ext}(l)) = q(T(l))$, so the $q_{\mathfrak{v}}$-freedom of l is the equilibrium probability that l is true. Let $\mu, \pi \in L_{\mathfrak{v}}$ with $q(T(\pi) \cap T(\mu)) > 0$. Then $q_{\mathfrak{v}}(\Omega_\pi) = q(T(\pi) \cap T(\mu))$, where Ω_π is the set of viable completions of π (Corollary 14). Every distribution p on $\mathcal{X}$ with $p(T(\pi) \cap T(\mu)) = 1$ satisfies

$$F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q|_{T(\mu)}] = k_B\Theta_0\Big(-\ln q(T(\pi) \mid T(\mu)) + \mathrm{KL}\big(p\|q|_{T(\pi)\cap T(\mu)}\big)\Big).$$

Hence, among distributions confined to $T(\pi) \cap T(\mu)$, the least free energy exceeds $F_{\mathrm{phys}}[q|_{T(\mu)}]$ by $-k_B\Theta_0 \ln q(T(\pi) \mid T(\mu))$, and only $q|_{T(\pi)\cap T(\mu)}$ attains it. If q is uniform and $\mathrm{Enc}_{\mathfrak{v}}$ maps $\mathcal{X}$ one to one onto $\mathrm{Ext}(\mu)$, this least excess is $k_B\Theta_0 \ln 2\, F_2(\pi)$. Suppose a process takes the system from a distribution p_0 with $p_0(T(\mu)) = 1$ to a distribution p_1 with $p_1(T(\pi) \cap T(\mu)) = 1$. Suppose also that the physical model satisfies

$$W = F_{\mathrm{phys}}[p_1] - F_{\mathrm{phys}}[p_0] + \Theta_0 S_{\mathrm{gen}}, \qquad S_{\mathrm{gen}} \geq 0,$$

where W is the work done on the system and S_{gen} is the entropy generated. Then

$$W + \big(F_{\mathrm{phys}}[p_0] - F_{\mathrm{phys}}[q|_{T(\mu)}]\big) \geq -k_B\Theta_0 \ln q(T(\pi) \mid T(\mu)).$$

[10] A reference environment is the surroundings of a system, idealised as inexhaustible. However much energy or matter the system exchanges with such surroundings, their temperature, pressure and chemical composition stay fixed. A system left alone comes to equilibrium with its surroundings, and the most work that can be extracted from it before it reaches equilibrium is its excess free energy. Equilibrium and free energy therefore both depend upon the surroundings, which is why each is defined relative to a reference environment.

Proof. For $l \in L_{\mathfrak{v}}$ and $x \in \mathcal{X}$, $\mathrm{Enc}_{\mathfrak{v}}(x) \in \mathrm{Ext}(l)$ if and only if $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(x)$, which holds if and only if every possible fact in l contains x, meaning $x \in T(l)$. So $q_{\mathfrak{v}}(\mathrm{Ext}(l)) = q(T(l))$. Since $\Omega_\pi = \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu)$, $\mathrm{Enc}_{\mathfrak{v}}(x) \in \Omega_\pi$ if and only if $x \in T(\pi) \cap T(\mu)$, so $q_{\mathfrak{v}}(\Omega_\pi) = q(T(\pi) \cap T(\mu))$.

Write $A := T(\mu)$ and $C := T(\pi) \cap T(\mu)$, so that $q(C) > 0$ and $C \subseteq A$. Since $\ln q_x = -h_x/(k_B\Theta_0) - \ln \sum_y \mathrm{e}^{-h_y/(k_B\Theta_0)}$, every distribution p satisfies $F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q] = k_B\Theta_0 \,\mathrm{KL}(p\|q)$ [Esposito and Van den Broeck, 2011]. For p with $p(A) = 1$, $\mathrm{KL}(p\|q|_A) = \mathrm{KL}(p\|q) + \ln q(A)$, and with $p = q|_A$ this gives $\mathrm{KL}(q|_A\|q) = -\ln q(A)$. Subtracting, $F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q|_A] = k_B\Theta_0 \,\mathrm{KL}(p\|q|_A)$ for every p with $p(A) = 1$. For p with $p(C) = 1$, the ratio of $q|_C$ to $q|_A$ is $q(A)/q(C)$ at every microstate in C, so $\mathrm{KL}(p\|q|_A) = \mathrm{KL}(p\|q|_C) + \ln(q(A)/q(C))$. This is Theorem 9 with microstates in place of possible things, and since $\ln(q(A)/q(C)) = -\ln q(C \mid A) = -\ln q(T(\pi) \mid T(\mu))$, it gives the displayed identity. The Kullback–Leibler divergence is nonnegative and zero only when the two distributions are equal, so the least value is attained at $p = q|_C$ alone. When q is uniform and $\mathrm{Enc}_{\mathfrak{v}}$ maps $\mathcal{X}$ one to one onto $\mathrm{Ext}(\mu)$, every microstate is in A and $|C \cap \mathcal{X}| = |\Omega_\pi|$. So $q(A) = 1$ and $q(C) = |\Omega_\pi|/|\,\mathrm{Ext}(\mu)|$, and $-\ln q(C \mid A) = \ln 2 \,\log_2(|\,\mathrm{Ext}(\mu)|/|\Omega_\pi|) = \ln 2 \, F_2(\pi)$ by Corollary 14. For the last claim, substitute the identity for p_1 and $F_{\mathrm{phys}}[p_0] - F_{\mathrm{phys}}[q|_A] = k_B\Theta_0 \,\mathrm{KL}(p_0\|q|_A)$ into the balance, then drop the nonnegative terms $k_B\Theta_0 \,\mathrm{KL}(p_1\|q|_C)$ and $\Theta_0 S_{\mathrm{gen}}$. □

Proposition 18 (Excess free energy seen from a layer)**.**
Let q be the equilibrium distribution above and p any distribution on $\mathcal{X}$, with pushforwards $p_{\mathfrak{v}} := (\mathrm{Enc}_{\mathfrak{v}})_{\#}p$ and $q_{\mathfrak{v}} := (\mathrm{Enc}_{\mathfrak{v}})_{\#}q$ on $L_{\mathfrak{v}}$. For a possible thing l with $p_{\mathfrak{v}}(\{l\}) > 0$, let $p(\cdot \mid l)$ and $q(\cdot \mid l)$ be p and q conditioned on the microstates whose encoding is l. Then

$$F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q] = k_B\Theta_0 \,\mathrm{KL}(p_{\mathfrak{v}}\|q_{\mathfrak{v}}) + k_B\Theta_0 \sum_{l:\, p_{\mathfrak{v}}(\{l\})>0} p_{\mathfrak{v}}(\{l\}) \,\mathrm{KL}(p(\cdot \mid l)\|q(\cdot \mid l)).$$

Both terms are nonnegative, so $k_B\Theta_0 \,\mathrm{KL}(p_{\mathfrak{v}}\|q_{\mathfrak{v}}) \le F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q]$, with equality if and only if $p(\cdot \mid l) = q(\cdot \mid l)$ for every l with $p_{\mathfrak{v}}(\{l\}) > 0$. In a stack (Definition 26) with layers $\mathfrak{v}^i$ and $\mathfrak{v}^{i+1}$,

$$\mathrm{KL}(p_{\mathfrak{v}^{i+1}}\|q_{\mathfrak{v}^{i+1}}) \le \mathrm{KL}(p_{\mathfrak{v}^i}\|q_{\mathfrak{v}^i}).$$

Equality holds if and only if the following is true for every layer-$(i{+}1)$ encoding with positive probability under p. Among the layer-i encodings that are mapped to it, p and q induce the same conditional distribution.

Proof. Take $x \in \mathcal{X}$ with $p_x > 0$ and let $l := \mathrm{Enc}_{\mathfrak{v}}(x)$. Then $p_{\mathfrak{v}}(\{l\}) \ge p_x > 0$ and $q_{\mathfrak{v}}(\{l\}) \ge q_x > 0$, so $p_x = p_{\mathfrak{v}}(\{l\})\, p(x \mid l)$ and $q_x = q_{\mathfrak{v}}(\{l\})\, q(x \mid l)$, and

$$\ln \frac{p_x}{q_x} = \ln \frac{p_{\mathfrak{v}}(\{l\})}{q_{\mathfrak{v}}(\{l\})} + \ln \frac{p(x \mid l)}{q(x \mid l)}.$$

Multiplying by p_x and summing over the microstates with $p_x > 0$, grouped by encoding, gives $\mathrm{KL}(p\|q) = \mathrm{KL}(p_{\mathfrak{v}}\|q_{\mathfrak{v}}) + \sum_l p_{\mathfrak{v}}(\{l\}) \,\mathrm{KL}(p(\cdot \mid l)\|q(\cdot \mid l))$, since $\sum_{x:\mathrm{Enc}_{\mathfrak{v}}(x)=l} p_x = p_{\mathfrak{v}}(\{l\})$ and $p_x/p_{\mathfrak{v}}(\{l\}) = p(x \mid l)$. The proof of Proposition 17 gives $F_{\mathrm{phys}}[p] - F_{\mathrm{phys}}[q] = k_B\Theta_0 \,\mathrm{KL}(p\|q)$. Each Kullback–Leibler divergence is nonnegative and is zero only when its two distributions are equal, which gives the inequality and its equality case.

For the stack, every possible fact of $\mathfrak{v}^{i+1}$ is $T(o)$ for some $o \subseteq \mathfrak{v}^i$, and $x \in T(o)$ if and only if $o \subseteq \mathrm{Enc}_{\mathfrak{v}^i}(x)$, as in the proof of Theorem 10. So the layer-$(i{+}1)$ encoding of x is a function g of its layer-i encoding, and $p_{\mathfrak{v}^{i+1}} = g_{\#}p_{\mathfrak{v}^i}$ and $q_{\mathfrak{v}^{i+1}} = g_{\#}q_{\mathfrak{v}^i}$. The identity above requires only that $\mathrm{Enc}_{\mathfrak{v}}$ is a function on $\mathcal{X}$ and that q is positive. So it applies with the set $\mathrm{Enc}_{\mathfrak{v}^i}(\mathcal{X})$ of encodings of microstates in place of $\mathcal{X}$, on which $q_{\mathfrak{v}^i}$ is positive, with $p_{\mathfrak{v}^i}$ and $q_{\mathfrak{v}^i}$ in place of p and q, and with g in place of $\mathrm{Enc}_{\mathfrak{v}}$. This gives the inequality between layers, and its equality case is the equality case of that identity. □

13 Irreducible regret

Theorem 11 (Regret is freedom lost)**.**
Fix a finite list Z of history-task pairs $z = (h, Q)$, where h is an interaction record and Q is a binary prediction task. Here an interaction record is any finite record used by the prediction setup. It need not be a timeline or a finite timeline fragment. Each z has evaluation weight $\rho_z \geq 0$ with $\sum_{z \in Z} \rho_z = 1$. Let $p_z \in [0, 1]$ be its conditional probability that test Q returns 1, given record h. The optimal answer is $a^*(z) = 1$ when $p_z > 1/2$, $a^*(z) = 0$ when $p_z < 1/2$, and either answer at a tie. I define the normalised decision margin for this history-task pair

$$m_z := \frac{|2p_z - 1|}{\max\{p_z, 1 - p_z\}} \in [0, 1].$$

This is the normalised-regret weight $\rho_i = 2|p_i - \frac{1}{2}|/(\frac{1}{2} + |p_i - \frac{1}{2}|)$ in Bennett [2026c], where $|p_i - \frac{1}{2}|$ is called the margin. Here m_z denotes its normalised-regret weight, rather than its unnormalised margin. The weight is zero at a tie. A representation M maps each interaction record to a memory state. This map partitions Z into groups

$$C_{s,Q_0} = \{(h, Q) \in Z \mid M(h) = s,\ Q = Q_0\}.$$

An M-based policy gives one answer $a_C \in \{0, 1\}$ on each group. I define the group weights

$$W_C(a) := \sum_{z \in C\,:\, a^*(z) = a} \rho_z m_z.$$

Normalised regret at z is one minus the ratio of the policy's actual success probability to the optimal success probability $\max\{p_z, 1 - p_z\}$. Its average over Z weighted by ρ is

$$R^M_\rho(a_C) := \sum_C W_C(1 - a_C).$$

Finally I define the irreducible regret of M

$$K_\rho(M) := \sum_C \min\{W_C(0), W_C(1)\}.$$

Then

$$\min_{M\text{-based policies}} R^M_\rho(a_C) = K_\rho(M).$$

Moreover, the regret of every M-based policy decomposes as

$$R^M_\rho(a_C) = K_\rho(M) + \sum_C \Big(W_C(1 - a_C) - \min\{W_C(0), W_C(1)\}\Big).$$

Term $K_\rho(M)$ is the regret unavoidable under memory M. Sum $\sum_C \big(W_C(1 - a_C) - \min\{W_C(0), W_C(1)\}\big)$ is avoidable prediction error given representation M. An M-based policy attains $K_\rho(M)$ if and only if $a_C \in \arg\max_{a \in \{0,1\}} W_C(a)$ for every group C.

Proof. Choosing a worse answer at z reduces success probability by $|2p_z - 1|$. Normalising by $\max\{p_z, 1 - p_z\}$, I recover m_z. Groups are disjoint, and an M-based policy chooses one answer per group. If it answers 0 on group C, it incurs cost equal to the weighted mass of points whose optimal answer is 1, namely $W_C(1)$. If it answers 1, it incurs $W_C(0)$. The smallest possible cost on that group is therefore $\min\{W_C(0), W_C(1)\}$. Summing over groups, I get $K_\rho(M)$. For a policy with answers a_C, I add and subtract this groupwise minimum:

$$\begin{aligned} R^M_\rho(a_C) = &\sum_C \min\{W_C(0), W_C(1)\} \\ &+ \sum_C \Big(W_C(1 - a_C) - \min\{W_C(0), W_C(1)\}\Big). \end{aligned}$$

The first sum is the irreducible regret, since $\sum_C \min\{W_C(0), W_C(1)\} = K_\rho(M)$. The second sum, $\sum_C (W_C(1 - a_C) - \min\{W_C(0), W_C(1)\})$, is nonnegative. A sum of nonnegative terms vanishes if and only if every term does, and the term for C vanishes if and only if $W_C(1 - a_C) = \min\{W_C(0), W_C(1)\}$, meaning $a_C \in \arg\max_{a \in \{0,1\}} W_C(a)$. □

Weighted freedom in the degenerate vocabulary. The degenerate vocabulary contains two distinct possible facts $p_{z,0}, p_{z,1}$ for each $z \in Z$, all true given one common distinction, so every thing is a possible thing. With no task inputs every policy is correct and every possible thing unseen. For an M-based decision rule $a = (a_C)$, put

$$\pi_a = \{p_{z,a_{C(z)}} : z \in Z\}, \qquad o_z = \{p_{z,a^*(z)}\} \cup \{p_{z',0}, p_{z',1} : z' \neq z\}.$$

The weight of o_z is $\rho_z m_z$, and every other possible thing has weight zero. Then $\pi_a \subseteq o_z$ iff $a_{C(z)} = a^*(z)$. The weighted freedom of π_a is $W_Z - R^M_\rho(a)$, where $W_Z := \sum_z \rho_z m_z$. With $W^*(M)$ the maximum weighted freedom among M-based policies,

$$W_Z - W^*(M) = \min_a R^M_\rho(a) = K_\rho(M).$$

If the demands o_z occur independently with probabilities $1 - \exp(-\rho_z m_z)$, optimal M-based generalisation is $\exp[-K_\rho(M)]$, since the product of probabilities of avoiding excluded demands is the exponential of their negative total weight. This distribution of demands is μ-exchangeable in the sense of Definition 23, for the measure that gives each unseen output o_z the weight $\rho_z m_z$, since the probability of generalising depends upon a buffer only through its weight and rises with that weight. For any nonempty admissible policy family permitting outputs from at most one answer per group,

$$W_Z - W^*(M) \geq K_\rho(M),$$

with equality iff a single admissible policy attains the larger of $W_C(0)$ and $W_C(1)$ in every group simultaneously. Within each test, the coarsest zero-regret grouping of positive-weight points identifies points with the same optimal answer. Representation convergence up to renaming additionally requires this grouping to be realisable by one memory code $M(h)$ across all tests. Weighted freedom uses a fixed demand prior; cross-vocabulary comparisons require a correspondence of weights, as well as outputs.

I use indexed notation for these extensions of Theorem 11. Evaluation points are $x_i = (h_i, T_i)$ with masses $\mu_i > 0$ summing to 1, conditional probabilities p_i, and optimal predictions $y_i \in \{0, 1\}$. Here μ_i corresponds to ρ_{z_i} in that theorem. I set

$$d_i := |p_i - \tfrac{1}{2}|, \qquad r_i := \frac{2d_i}{\frac{1}{2} + d_i},$$

so r_i corresponds to its m_{z_i}. A representation M groups points with equal $M(h_i)$ and equal T_i. I denote this partition by $\mathcal{C}_M$. On each group C, a randomised policy predicts 1 with probability $q_C \in [0, 1]$. Normalised regret at x_i is $\delta_i(\pi)$.

For this section I abbreviate $K_\rho(M)$ as $K(M)$. With $A_C := \sum_{i \in C,\, y_i = 1} \mu_i r_i$ and $B_C := \sum_{i \in C,\, y_i = 0} \mu_i r_i$, its equivalent deletion form is

$$K(M) = \sum_C \min\{A_C, B_C\} = \min\left\{ \sum_{i \in B} \mu_i r_i \;\middle|\; y \text{ is constant on } C \setminus B \text{ for every } C \in \mathcal{C}_M \right\}.$$

Each group becomes constant after deleting one prediction class. Deleting its lighter class minimises cost.

Lemma 12 (Pointwise regret decomposition)**.**
Let π be any M-based policy with report probability q_C on cell C. For $i \in C$,

$$\delta_i(\pi) = r_i\big((1 - q_C)\mathbf{1}\{y_i = 1\} + q_C \mathbf{1}\{y_i = 0\}\big).$$

Proof. If $y_i = 1$, then $p_i \geq \frac{1}{2}$ and $V_i^\star = p_i = \frac{1}{2} + d_i$. The success probability under report q_C is $V_i^\pi = q_C p_i + (1 - q_C)(1 - p_i)$. So $\delta_i = 1 - V_i^\pi / V_i^\star = 2d_i(1 - q_C)/(\frac{1}{2} + d_i) = r_i(1 - q_C)$. The $y_i = 0$ case is symmetric, giving $\delta_i = r_i q_C$. □

Corollary 15 (Policy-wise decomposition)**.**
For each cell $C \in \mathcal{C}_M$, let $A_C := \sum_{i \in C,\, y_i = 1} \mu_i r_i$, $B_C := \sum_{i \in C,\, y_i = 0} \mu_i r_i$, and choose any

$$q_C^\star \in \arg\min_{q \in [0,1]} \big(A_C(1 - q) + B_C q\big).$$

Then every M-based policy π with report probabilities q_C satisfies

$$\sum_{i=1}^{n} \mu_i \delta_i(\pi) = K(M) + \sum_{C \in \mathcal{C}_M} |A_C - B_C|\, |q_C - q_C^\star|.$$

In particular,

$$K(M) \leq \sum_{i=1}^{n} \mu_i \delta_i(\pi),$$

with equality exactly when the policy is cellwise optimal on every non-tied cell.

Proof. On each cell C, the policy contributes $A_C(1 - q_C) + B_C q_C$. The minimum is $\min\{A_C, B_C\}$, achieved at $q_C^\star$. If $A_C > B_C$, then $q_C^\star = 1$ and

$$A_C(1 - q_C) + B_C q_C = B_C + (A_C - B_C)(1 - q_C) = \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|.$$

If $A_C < B_C$, then $q_C^\star = 0$ and

$$A_C(1 - q_C) + B_C q_C = A_C + (B_C - A_C) q_C = \min\{A_C, B_C\} + |A_C - B_C|\, |q_C - q_C^\star|.$$

If $A_C = B_C$, both sides reduce to A_C for every q_C. Summing over cells and using Theorem 11 gives the decomposition. The equality criterion is immediate from the absolute-value term. □

Theorem 12 (Multi-class exact reduction)**.**
Assume now a K-class setting in which each support point has an optimal label $y_i \in \{1, \ldots, K\}$ and, for $i \in C$, pointwise regret takes the weighted misreport form

$$\delta_i(\pi) = r_i(1 - q_C^{(y_i)}),$$

where $q_C \in \Delta^{K-1}$ is the class-distribution reported on cell C. Define

$$A_C^{(k)} := \sum_{i \in C,\, y_i = k} \mu_i r_i$$

for each cell C and class k. Then

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \Big(\sum_{k=1}^{K} A_C^{(k)} - \max_k A_C^{(k)} \Big).$$

Proof. An M-based policy assigns a distribution $q_C \in \Delta^{K-1}$ over the K classes for each cell C. Under the pointwise assumption in the theorem statement, its expected regret on cell C is

$$\sum_{k=1}^{K} A_C^{(k)}(1 - q_C^{(k)}) = \Big(\sum_k A_C^{(k)} \Big) - \sum_k A_C^{(k)} q_C^{(k)}.$$

To minimise this over $q_C \in \Delta^{K-1}$, maximise the inner product $\sum_k A_C^{(k)} q_C^{(k)}$. This is a linear function on the simplex, so its maximum is achieved at a vertex, namely all mass on the class $k^\star = \arg\max_k A_C^{(k)}$. The minimum per-cell regret is therefore $(\sum_k A_C^{(k)}) - \max_k A_C^{(k)}$. To make y single-valued on C, one must delete every point not in the dominant class, costing the same quantity. Summing over cells gives both sides. □

Proposition 19 (Lipschitz stability of K)**.**
For fixed optimal labels, if $\hat{r}_i$ are perturbed margin weights, then $|\widehat{K}(M) - K(M)| \le \sum_{i=1}^n \mu_i |\hat{r}_i - r_i|$. Since r is Lipschitz with constant 4 on $(0,1)$ (the one-sided derivatives satisfy $|dr/dp| \le 4$ on $(0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$), estimation error in conditional probabilities, including a change of optimal label, satisfies $|\widehat{K}(M) - K(M)| \le 4 \sum_{i=1}^n \mu_i |\hat{p}_i - p_i|$.

Proof. For each cell C, $|\min\{\hat{A}_C, \hat{B}_C\} - \min\{A_C, B_C\}| \le |\hat{A}_C - A_C| + |\hat{B}_C - B_C| \le \sum_{i \in C} \mu_i |\hat{r}_i - r_i|$. Sum over cells. For the second bound, on $(0, \frac{1}{2})$ we have $r = (1-2p)/(1-p)$ with $|dr/dp| = 1/(1-p)^2 \le 4$, and on $(\frac{1}{2}, 1)$ we have $r = (2p-1)/p$ with $|dr/dp| = 1/p^2 \le 4$. At $p = \frac{1}{2}$, r has a kink but both one-sided derivatives have magnitude 4. So r is Lipschitz with constant 4 on $(0,1)$, giving $|\hat{r}_i - r_i| \le 4|\hat{p}_i - p_i|$ when optimal labels stay fixed. If an estimated probability crosses $1/2$, each fixed prediction's pointwise regret remains 4-Lipschitz. For prediction 0 it is 0 below $1/2$ and $(2p-1)/p$ above $1/2$. For prediction 1 it is $(1-2p)/(1-p)$ below $1/2$ and 0 above $1/2$. Both are continuous at $1/2$. Every fixed policy's expected regret therefore changes by at most $4 \sum_i \mu_i |\hat{p}_i - p_i|$, and taking minima over the same policies preserves this bound. □

Proposition 20 (Measurable exact reduction)**.**
Let $(\mathcal{X}, \Sigma, \mu)$ be a probability space. Let $(\mathcal{Z}, \mathcal{B})$ be a standard Borel space and let $\zeta : \mathcal{X} \to \mathcal{Z}$, $y : \mathcal{X} \to \{0,1\}$, and $r : \mathcal{X} \to [0,1]$ be measurable.

For a measurable decision rule $q : \mathcal{Z} \to [0,1]$, define

$$R(q) := \mathbb{E}\Big[r\Big((1 - q(\zeta))\mathbf{1}\{y = 1\} + q(\zeta)\mathbf{1}\{y = 0\}\Big)\Big].$$

Let $a, b : \mathcal{Z} \to [0,1]$ be measurable versions satisfying

$$a(\zeta) = \mathbb{E}[r\mathbf{1}\{y = 1\} \mid \sigma(\zeta)] \quad \text{and} \quad b(\zeta) = \mathbb{E}[r\mathbf{1}\{y = 0\} \mid \sigma(\zeta)] \qquad \text{a.s.}$$

Then

$$\inf_{q : \mathcal{Z} \to [0,1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}].$$

Moreover any measurable selector

$$q^\star(z) \in \arg\min_{q \in [0,1]} \big(a(z)(1-q) + b(z)q\big)$$

attains the infimum; for example $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$.

Proof. Because $(\mathcal{Z}, \mathcal{B})$ is standard Borel, the conditional expectations with respect to $\sigma(\zeta)$ admit measurable versions as functions of ζ; those are the displayed functions a and b. For any measurable $q : \mathcal{Z} \to [0,1]$, the random variable $q(\zeta)$ is $\sigma(\zeta)$-measurable, so the tower property gives

$$R(q) = \mathbb{E}[a(\zeta)(1 - q(\zeta)) + b(\zeta)q(\zeta)]\,.$$

For each fixed z, the integrand

$$a(z)(1-q) + b(z)q$$

is affine in $q \in [0,1]$, so its minimum is $\min\{a(z), b(z)\}$, attained by any selector $q^\star(z) \in \arg\min_{q \in [0,1]}(a(z)(1-q) + b(z)q)$. The explicit choice $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$ is measurable because a and b are measurable.

Substituting this pointwise minimiser into the previous display yields

$$\inf_{q : \mathcal{Z} \to [0,1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}],$$

which is finite because $0 \le a, b \le 1$. □

Theorem 13 (Representation convergence)**.**
Let $I_+ = \{i : \mu_i r_i > 0\}$ be the informative support. Define the coarsest pure diagnostic partition $\mathcal{P}^\star_+$ on I_+ by grouping points that share the same test value T_i and optimal label y_i. Then $K(M) = 0$ iff $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}^\star_+$.

Assume that an admissible representation $M(h)$ induces $\mathcal{P}^\star_+$ on I_+. This requires a single code for each history, used across every test. Call a representation M **informative-minimal zero-regret** if $K(M) = 0$ and there is no representation M' with $K(M') = 0$ such that $\mathcal{C}_{M'}|_{I_+}$ is strictly coarser than $\mathcal{C}_M|_{I_+}$. Then M is informative-minimal zero-regret iff

$$\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+.$$

Consequently, any two informative-minimal zero-regret representations induce the same diagnostic partition on I_+, up to labels for its groups.

Proof. By Theorem 11, $K(M) = 0$ iff $\min\{A_C, B_C\} = 0$ for every cell $C \in \mathcal{C}_M$. On the informative support I_+, this is equivalent to saying that each cell contains only one optimal label at its fixed test value. That is exactly the statement that $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}^\star_+$. This proves the first claim.

For the minimality claim, note that by assumption the partition $\mathcal{P}^\star_+$ itself is representable, and it is zero-regret because every one of its cells is pure by construction. So any zero-regret representation whose induced partition on I_+ is strictly finer than $\mathcal{P}^\star_+$ cannot be informative-minimal: coarsening it down to $\mathcal{P}^\star_+$ preserves zero regret. Conversely, if $\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+$, then no strictly coarser zero-regret partition exists, because any strict coarsening would merge two groups with different optimal predictions at one test and would therefore create an impure informative group. Hence M is informative-minimal zero-regret iff $\mathcal{C}_M|_{I_+} = \mathcal{P}^\star_+$.

Finally, if M_1 and M_2 are both informative-minimal zero-regret, then they induce the same partition $\mathcal{P}^\star_+$ on I_+. Their diagnostic group labels may differ, but matching equal groups gives a bijection between those labels. □

Proposition 21 (Coordination cost of shared representations)**.**
Let N subsystems each face a diagnostic family on the same informative support I_+ with the same single test T. Let $\mathcal{P}^\star_j$ be the coarsest pure partition for subsystem j on I_+, with $|\mathcal{P}^\star_j| = k_j$ cells. Then every shared representation that achieves zero regret for all N subsystems simultaneously must have at least

$$|\mathcal{P}^\star_1 \vee \cdots \vee \mathcal{P}^\star_N|$$

cells on I_+, where $\vee$ denotes the common refinement. This lower bound is tight in the unrestricted class of shared representations. More generally, it is tight whenever the class contains a representation whose informative quotient is exactly the common refinement. Hence in that case the minimum number of cells in a shared zero-regret representation is exactly

$$|\mathcal{P}^\star_1 \vee \cdots \vee \mathcal{P}^\star_N|.$$

Moreover,

$$|\mathcal{P}^\star_1 \vee \cdots \vee \mathcal{P}^\star_N| \;\le\; \min\Big\{|I_+|,\; \prod_{j=1}^{N} k_j\Big\}.$$

The product bound is achievable when $|I_+| \ge \prod_{j=1}^{N} k_j$ and every tuple of one cell from each partition has nonempty intersection.

Proof. By Theorem 13, zero regret for subsystem j requires the shared partition to refine $\mathcal{P}^\star_j$. The coarsest partition refining all N partitions is the common refinement $\mathcal{P}^\star_1 \vee \cdots \vee \mathcal{P}^\star_N$, so any shared zero-regret representation needs at least that many cells.

This lower bound is tight whenever the shared representation class contains a representation whose informative quotient is the common refinement. In particular, this is automatic in the unrestricted quotient class. Such a representation has induced partition exactly $\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star$. Because each refinement cell lies inside a pure cell of every subsystem partition, it has zero regret for every subsystem simultaneously. So in that case the minimum possible cell count is exactly the size of the common refinement.

Each refinement cell is an intersection of one cell from each subsystem partition. There are at most $\prod_{j=1}^N k_j$ such tuples, and of course no partition of I_+ can have more than $|I_+|$ nonempty cells. Hence

$$|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star| \le \min\Big\{|I_+|,\ \prod_{j=1}^N k_j\Big\}.$$

If $|I_+| \ge \prod_{j=1}^N k_j$ and every tuple of subsystem cells has nonempty intersection, then every possible intersection occurs and the refinement has exactly $\prod_j k_j$ cells. □

Remark 2 (Splintering on the diagnostic quotient)**.**
Take three support points x_1, x_2, x_3 under one common test T.

Subsystem A has optimal labels $y^A(x_1) = 1$, $y^A(x_2) = 0$, $y^A(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{A,+}^\star = \{\{x_1\}, \{x_2, x_3\}\}$.

Subsystem B has labels $y^B(x_1) = 0$, $y^B(x_2) = 1$, $y^B(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{B,+}^\star = \{\{x_2\}, \{x_1, x_3\}\}$.

These minimal zero-regret quotients do not coincide. If each subsystem separately minimises its own irreducible regret, they converge to different partitions (Theorem 13). If they are forced to share one common zero-regret quotient, they must use the finer common refinement $\{\{x_1\}, \{x_2\}, \{x_3\}\}$.

Proposition 22 (Regret ordering)**.**
Define $M_1 \preceq_\rho M_2$ iff $K(M_1) \ge K(M_2)$. This is a total preorder on representations, and it becomes a partial order on the equivalence classes under

$$M_1 \sim_\rho M_2 \iff K(M_1) = K(M_2).$$

It is consistent with the partition refinement lattice. If $\mathcal{C}_{M_2}|_{I_+}$ refines $\mathcal{C}_{M_1}|_{I_+}$, then $K(M_2) \le K(M_1)$. More sharply, let a parent cell C_1 split into subcells $C_2^{(j)}$ and define $\Delta_j := A_{C_2^{(j)}} - B_{C_2^{(j)}}$. Then

$$\min\{A_{C_1}, B_{C_1}\} - \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{1}{2}\left(\sum_j |\Delta_j| - \left|\sum_j \Delta_j\right|\right).$$

So the inequality is strict iff at least one mixed parent cell is split into subcells with opposite-signed imbalances. Equivalently, strict improvement occurs iff some mixed parent cell is split into subcells with different dominant labels.

Proof. If $\mathcal{C}_{M_2}$ refines $\mathcal{C}_{M_1}$, every cell $C_1 \in \mathcal{C}_{M_1}$ is partitioned into subcells $C_2^{(j)} \in \mathcal{C}_{M_2}$. The total weights therefore satisfy

$$A_{C_1} = \sum_j A_{C_2^{(j)}}, \qquad B_{C_1} = \sum_j B_{C_2^{(j)}}.$$

Using $\min\{x, y\} = \frac{x+y-|x-y|}{2}$, I get

$$\min\{A_{C_1}, B_{C_1}\} = \frac{\sum_j (A_{C_2^{(j)}} + B_{C_2^{(j)}}) - \left|\sum_j (A_{C_2^{(j)}} - B_{C_2^{(j)}})\right|}{2},$$

while

$$\sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{\sum_j (A_{C_2^{(j)}} + B_{C_2^{(j)}}) - \sum_j |A_{C_2^{(j)}} - B_{C_2^{(j)}}|}{2}.$$

Subtracting gives the displayed identity

$$\min\{A_{C_1}, B_{C_1}\} - \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\} = \frac{1}{2}\left(\sum_j |\Delta_j| - \left|\sum_j \Delta_j\right|\right) \geq 0,$$

by the triangle inequality. Summing over parent cells yields $K(M_1) \geq K(M_2)$.

The relation $\preceq_\rho$ is reflexive, transitive, and total because it is induced by the scalar score K. It need not be antisymmetric on representations themselves, since distinct representations can have the same regret floor. Quotienting by $\sim_\rho$ restores antisymmetry, so the induced relation on equivalence classes is a partial order.

For strictness, the triangle inequality is strict exactly when the summands are not all of the same sign. So the improvement on a given parent cell is strict iff the subcell imbalances Δ_j take both positive and negative values. That is equivalent to saying the refined mixed parent cell contains subcells with different dominant labels. Therefore the global inequality is strict iff this happens for at least one refined mixed parent cell. □

Corollary 16 (Unweighted irreducible regret on the informative region)**.**
Fix $\gamma \in (0, \frac{1}{2}]$ and let $\mathcal{X}_\gamma := \{i : d_i \geq \gamma\}$. The unweighted irreducible regret on the informative region, $K_\gamma(M)$, is defined by replacing $\mu_i r_i$ with μ_i in Theorem 11 and restricting to $\mathcal{X}_\gamma$. Then with $c(\gamma) = 4\gamma/(1+2\gamma)$,

$$c(\gamma) K_\gamma(M) \;\leq\; \inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \delta_i(\pi) \;\leq\; K_\gamma(M).$$

Proof. Let

$$s_\gamma := \sum_{i \in \mathcal{X}_\gamma} \mu_i.$$

If $s_\gamma = 0$, then both $K_\gamma(M)$ and the restricted regret are zero, so the claim is trivial. Assume $s_\gamma > 0$ and define normalised masses

$$\tilde{\mu}_i := \frac{\mu_i}{s_\gamma} \qquad \text{for } i \in \mathcal{X}_\gamma.$$

Apply Theorem 11 to the restricted support $\mathcal{X}_\gamma$ with masses $\tilde{\mu}_i$. This gives

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \tilde{\mu}_i \delta_i(\pi) = \widetilde{K}_{r,\gamma}(M),$$

where $\widetilde{K}_{r,\gamma}(M)$ is the corresponding weighted deletion cost on the restricted support. Multiplying by s_γ yields the identity for the original masses on the restricted support

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \delta_i(\pi) = K_{r,\gamma}(M),$$

with

$$K_{r,\gamma}(M) := \min\Big\{ \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i r_i \;\Big|\; B \subseteq \{1, \ldots, n\},\ y \text{ is constant on } (C \cap \mathcal{X}_\gamma) \setminus B \text{ for every } C \in \mathcal{C}_M \Big\}.$$

Now on $\mathcal{X}_\gamma$ I have $r_i \in [c(\gamma), 1]$. Therefore, for every admissible deletion set B,

$$c(\gamma) \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i \leq \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i r_i \leq \sum_{i \in B \cap \mathcal{X}_\gamma} \mu_i.$$

Taking minima over the same admissible family gives

$$c(\gamma)K_\gamma(M) \le K_{r,\gamma}(M) \le K_\gamma(M).$$

Combining this with the restricted exact reduction proves the claim. □

I use the independent binary prediction construction associated with Theorem 11. Each evaluation point corresponds to an unseen output u_i with weight $w_i := \mu_i r_i$. For an admissible policy ϑ,

$$W(\vartheta) := \sum_{u_i \in \mathrm{Ext}(\vartheta)\cap U} w_i, \qquad Z := \sum_i w_i, \qquad W^*(M) := \max_\vartheta W(\vartheta).$$

Independent choices across prediction groups give $W^*(M) = Z - K(M)$. Under these assumptions, the following corollary relates regret to a weighted KL floor.

Corollary 17 (Free-energy floor from regret)**.**
Under the independent prediction construction above, assume $Z = \sum_{i=1}^n w_i > 0$. Define the finite measure

$$\nu(A) := \sum_{u_i \in A} w_i \qquad \text{on} \qquad U = \{u_1, \dots, u_n\}.$$

For any admissible policy ϑ with $W(\vartheta) > 0$, define the weighted prior and posterior

$$p_U(u_i) := \frac{w_i}{Z}, \qquad q_\vartheta(u_i) := \frac{w_i\, \mathbf{1}\{u_i \in \mathrm{Ext}(\vartheta) \cap U\}}{W(\vartheta)}.$$

Let

$$F_2^\nu(\vartheta) := D_{\mathrm{KL},2}(q_\vartheta \| p_U).$$

Then

$$F_2^\nu(\vartheta) = \log_2 \frac{Z}{W(\vartheta)}.$$

If $W^\star(M) > 0$ (equivalently $K(M) < Z$), then the minimum weighted free-energy achievable by an admissible policy is

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K(M)}.$$

If $W^\star(M) = 0$ (equivalently $K(M) = Z$), then no admissible policy retains positive weighted unseen mass, so there is no admissible policy with finite weighted free-energy relative to p_U. In that case the attainable floor is $+\infty$. In particular, whenever $0 < K(M) < Z$, the weighted free-energy floor is strictly above the zero-regret value 0 and is strictly increasing in $K(M)$.

Proof. By construction, q_ϑ is the normalised restriction of the measure ν to the retained unseen region $\mathrm{Ext}(\vartheta) \cap U$, while p_U is the normalised version of ν on all of U. Therefore, for q_ϑ-almost every u_i,

$$\frac{q_\vartheta(u_i)}{p_U(u_i)} = \frac{w_i/W(\vartheta)}{w_i/Z} = \frac{Z}{W(\vartheta)}.$$

The ratio is constant on the support of q_ϑ. Hence

$$F_2^\nu(\vartheta) = D_{\mathrm{KL},2}(q_\vartheta \| p_U) = \sum_{u_i} q_\vartheta(u_i) \log_2 \frac{q_\vartheta(u_i)}{p_U(u_i)} = \log_2 \frac{Z}{W(\vartheta)}.$$

This is strictly decreasing in $W(\vartheta)$. Therefore, if $W^\star(M) > 0$, the smallest attainable weighted free-energy is achieved by maximising $W(\vartheta)$, i.e. by $W^\star(M)$. Theorem 11 gives $W^\star(M) = Z - K(M)$, so

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K(M)}.$$

If instead $W^\star(M) = 0$, then there is no admissible policy with $W(\vartheta) > 0$, so the domain on which F_2^ν is finite is empty. In the extended reals the attainable floor is therefore $+\infty$. If $K(M) = 0$, the finite formula gives 0. If $0 < K(M) < Z$, the denominator decreases and the logarithm increases strictly. □

14 Bagi and Unagi

For $\Pi_\alpha \neq \emptyset$, Bagi selects

$$\mathrm{Bagi}_{\mathfrak{v}}(\alpha) \in \arg\max_{\pi \in \Pi_\alpha} |\operatorname{Ext}(\pi)|.$$

Operational Unagi first maximises the specified utility of intelligence over admissible embodiments, then minimises K_ρ among utility ties, then applies Bagi within the selected embodiment. Semantic Unagi instead compares the distinct unseen truth sets specified below. These optimisations presume that the required optima are attained.

Bagi compares policies within one vocabulary by freedom, which is the number of completions of a policy. Unagi compares pairs of a vocabulary and a policy. This is because two vocabularies can describe one way the world might be with different numbers of possible things. So semantic Unagi tallies something that does not depend upon how many descriptions a vocabulary has for each thing. It tallies truth sets. A truth set is the set of distinctions given which a possible thing is true, so two completions with the same truth set are true in the same situations and are tallied once. The score of a pair is the number of distinct truth sets of the policy's completions, less the truth sets of the outputs the task specifies as correct. The maximal vocabulary $\mathcal{P}$ would give the broadest comparison, but every result in this section is stated for finite vocabularies in the domain of the λ-task.

The score cannot be inflated by padding the vocabulary. Suppose m possible facts are added, each of which can be included in a completion or left out of it without changing the truth set $T(y)$ of any completion y under comparison. Each old completion then has 2^m written forms, so the freedom $|\operatorname{Ext}(\pi)|$ of every policy is multiplied by 2^m. Bagi's ordering within the vocabulary is unchanged, and so is the set of truth sets in Definition 27, so the semantic Unagi score is unchanged. A recoding that preserves truth sets likewise preserves the score. Freedom is thus a property of what a policy permits, while any calculation of it takes place inside some embodied language.

Definition 27 (Semantic Unagi score for a disembodied task)**.**
Fix a disembodied task λ_ρ and a nonempty class V of finite vocabularies in its domain. I define the set of admissible pairs

$$\mathcal{A}(\lambda_\rho, V) := \{(\mathfrak{v}, \pi) \mid \mathfrak{v} \in V,\ \pi \in \Pi_{\lambda_\rho(\mathfrak{v})}\}.$$

Assume $\mathcal{A}(\lambda_\rho, V) \neq \emptyset$. For an admissible pair, I define

$$\beta_{\lambda_\rho, V}(\mathfrak{v}, \pi) := \Big|\{T(y) \mid y \in \operatorname{Ext}(\pi)\} \setminus \{T(o) \mid o \in O_{\lambda_\rho(\mathfrak{v})}\}\Big|,$$

where the extension is computed in the paired vocabulary $\mathfrak{v}$. Possible things with the same truth set are tallied once. The number $\beta_{\lambda_\rho, V}(\mathfrak{v}, \pi)$ is the **semantic Unagi score** of the pair. Semantic Unagi ranks admissible pairs by

$$(\mathfrak{v}_1, \pi_1) \preceq_{\lambda_\rho, V} (\mathfrak{v}_2, \pi_2) \quad \Longleftrightarrow \quad \beta_{\lambda_\rho, V}(\mathfrak{v}_1, \pi_1) \leq \beta_{\lambda_\rho, V}(\mathfrak{v}_2, \pi_2).$$

Proposition 23 (Semantic Unagi optimisation)**.**
Fix λ_ρ and V as above. Let

$$V_{\mathrm{ad}} := \{\mathfrak{v} \in V \mid \Pi_{\lambda_\rho(\mathfrak{v})} \neq \emptyset\},$$

and assume the following maximum is attained. I define

$$V^\star := \arg\max_{\mathfrak{v}\in V_{\mathrm{ad}}} \max_{\pi\in\Pi_{\lambda_\rho(\mathfrak{v})}} \beta_{\lambda_\rho,V}(\mathfrak{v},\pi).$$

An admissible pair $(\mathfrak{v}^\star,\pi^\star)$ maximises $\preceq_{\lambda_\rho,V}$ if and only if

$$\mathfrak{v}^\star \in V^\star$$

and

$$\beta_{\lambda_\rho,V}(\mathfrak{v}^\star,\pi^\star) = \max_{\pi\in\Pi_{\lambda_\rho(\mathfrak{v}^\star)}} \beta_{\lambda_\rho,V}(\mathfrak{v}^\star,\pi).$$

Equivalently, its score is the maximum over all admissible pairs.

Proof. For each admissible pair, its policy lies in the class associated with its vocabulary in the definition of $V^\star$. The pair's score cannot exceed the maximum over that class, and hence cannot exceed the largest of those maxima. A pair attains the global maximum if its vocabulary lies in $V^\star$ and its policy attains the maximum for that vocabulary. Conversely, any global maximiser must attain the maximum for its vocabulary, and that vocabulary must lie in $V^\star$. □

Semantic Unagi chooses, from the class V, an admissible pair of greatest score. Bagi compares policies inside one body by freedom, and semantic Unagi compares bodies by the distinct truth sets their policies leave available. When several policies in the chosen body attain the greatest score, Bagi chooses the freest among them.

Future demands give a lower bound on the score. Suppose a later task adds future demands, as in Theorem 2, and let $\mathcal{T}$ be the finite set of their distinct truth sets. I write

$$\widehat{\beta}_{\mathcal{T}}(\mathfrak{v},\pi) := \Big|\mathcal{T} \cap \Big(\{T(y) \mid y \in \mathrm{Ext}(\pi)\} \setminus \{T(o) \mid o \in O_{\lambda_\rho(\mathfrak{v})}\}\Big)\Big|$$

for the number of future demands, distinct in truth set, that the policy's completions express and the task does not specify as correct. Every such truth set is tallied by $\beta_{\lambda_\rho,V}$ too, so

$$\widehat{\beta}_{\mathcal{T}}(\mathfrak{v},\pi) \le \beta_{\lambda_\rho,V}(\mathfrak{v},\pi).$$

Operational Unagi maximises $\widehat{\beta}_{\mathcal{T}}$ first, as its utility of intelligence. If every truth set has the same positive weight, the total weight of the truth sets tallied by $\widehat{\beta}_{\mathcal{T}}$ is at most the total weight of those tallied by $\beta_{\lambda_\rho,V}$, with equality only when the future demands exhaust the truth sets that $\beta_{\lambda_\rho,V}$ tallies. In Section 13 I relate the regret tie-break to weighted freedom.

In the absence of abstraction, when $\mathfrak{v} = \mathcal{P}$, the semantic Unagi score is the number of further truth sets compatible with the policy, less those of the correct outputs. When that number is infinite, comparison needs a measure or a restriction to a finite class. Across physical embodiments, Unagi compares a stated admissible class. To build a system that attains this upper bound, one would need to engineer abstraction layers that express the freest correct policies for the task.

Corollary 18 (Semantic and operational Unagi)**.**
Fix one list of history-task pairs Z with total weight $W_Z := \sum_{z\in Z} \rho_z m_z$, and let each admissible vocabulary $\mathfrak{v}$ induce a representation $M_{\mathfrak{v}}$. In the degenerate vocabulary of Section 13, every assignment of one answer per group is an $M_{\mathfrak{v}}$-based policy, and $W^*(M_{\mathfrak{v}})$ is the largest weighted freedom among all of them. The answer in one group can be chosen without regard to the answers in the other groups, so the best policy takes, in every group, the answer with the larger of $W_C(0)$ and $W_C(1)$, and

$$W^*(M_{\mathfrak{v}}) = W_Z - K_\rho(M_{\mathfrak{v}}).$$

Hence minimising K_ρ maximises weighted freedom $W^*(M_\mathfrak{v})$ for the task in this degenerate vocabulary. In an embodied language other than the degenerate one, a task built in the same way may permit only some of those policies, because a choice of answers in different groups need not form a possible thing. Take any nonempty family[11] of permitted policies, each admitting outputs from at most one prediction per group, and let $W^*(M_\mathfrak{v})$ be the largest weighted freedom in that family. Then

$$W^*(M_\mathfrak{v}) \leq W_Z - K_\rho(M_\mathfrak{v}).$$

Thus K_ρ lower-bounds the weighted freedom lost, $W_Z - W^*(M_\mathfrak{v})$. Irreducible regret K_ρ gives no lower bound on semantic Unagi's score itself. Suppose each $M_\mathfrak{v}$-based policy is paired with a correct policy π of $\lambda_\rho(\mathfrak{v})$. The pairing is assumed to match the unseen outputs in the buffer of the $M_\mathfrak{v}$-based policy one to one with the distinct unseen truth sets of π, all with equal weight. Then the weighted freedom of the $M_\mathfrak{v}$-based policy is proportional to semantic Unagi's score $\beta_{\lambda_\rho,V}(\mathfrak{v},\pi)$ in Definition 27, so the two scores give one ranking.

Now fix a representation $M = M_\mathfrak{v}$, and let $\mathcal{A}$ be the nonempty finite family of available policies, each admitting outputs from at most one prediction per group. For a policy $\pi \in \mathcal{A}$ with unseen buffer $B_\pi = \mathrm{Ext}(\pi) \cap U$, I write $s_\pi(C)$ for the weight of the unseen outputs in group C that π admits,

$$s_\pi(C) := \sum_{\substack{z \in C \\ o_z \in B_\pi}} \rho_z m_z.$$

Since π admits outputs from at most one prediction in each group, its admitted weight in a group is at most the larger of $W_C(0)$ and $W_C(1)$,

$$0 \leq s_\pi(C) \leq \max\{W_C(0), W_C(1)\}.$$

The largest weighted freedom available within $\mathcal{A}$ is

$$W^*_\mathcal{A}(M) := \max_{\pi \in \mathcal{A}} \sum_C s_\pi(C).$$

The loss from restriction to $\mathcal{A}$ is the smallest total shortfall, over the policies in $\mathcal{A}$, of the admitted weights below the group maxima,

$$\Delta_\mathcal{A}(M) := \min_{\pi \in \mathcal{A}} \sum_C \big(\max\{W_C(0), W_C(1)\} - s_\pi(C)\big).$$

These quantities satisfy

$$W^*_\mathcal{A}(M) = W_Z - K_\rho(M) - \Delta_\mathcal{A}(M).$$

The loss $\Delta_\mathcal{A}(M)$ is nonnegative, and it is zero if and only if some one policy in $\mathcal{A}$ attains the group maximum in every group at once. Enlarging $\mathcal{A}$ cannot increase this loss, provided that the groups, the weights and the scores $s_\pi(C)$ of the policies in the old $\mathcal{A}$ are unchanged. In particular, $\Delta_\mathcal{A}(M) = 0$ when $\mathcal{A}$ contains every M-based policy of the degenerate construction, since the policy that takes the answer with the larger weight in every group is then available. Suppose, as in Section 13, that each unseen output o_z is demanded independently with probability $1 - \exp(-\rho_z m_z)$. Then policy π generalises, meaning every demanded output is in its buffer, with probability $\exp\big(-(W_Z - \sum_C s_\pi(C))\big)$. The best policy in $\mathcal{A}$ generalises with probability $\exp\big(-K_\rho(M) - \Delta_\mathcal{A}(M)\big)$.

Proof. The task in this degenerate vocabulary has independent predictions for each group. So I can use Section 13's identity $\min R^M_\rho = W_Z - W^*(M)$ over M-based policies. Theorem 11 states that $\min R^M_\rho = K_\rho(M)$. Equating the two expressions for minimum regret, I get $W_Z - W^*(M_\mathfrak{v}) =$

[11] In mathematics a family is a set, called a family when its members are themselves sets or functions. Here it is a set of policies, each of which is a set of possible facts.

$K_\rho(M_\mathfrak{v})$. Since W_Z is fixed across the compared representations, minimising K_ρ is equivalent to maximising W^*. Each admissible policy has weight at most $\max\{W_C(0), W_C(1)\}$ in each group C. Summing this upper bound and subtracting it from total weight, I get $W_Z - W^*(M_\mathfrak{v}) \geq K_\rho(M_\mathfrak{v})$, which rearranges to $W^*(M_\mathfrak{v}) \leq W_Z - K_\rho(M_\mathfrak{v})$. Under equal weights, the weighted mass of a policy's unseen buffer is proportional to the number of unseen outputs in that buffer. Under the pairing, these outputs correspond one to one with the distinct unseen truth sets of the paired policy. Therefore the two scores give one ranking.

For the decomposition, take any policy $\pi \in \mathcal{A}$. Its admitted weight $\sum_C s_\pi(C)$ plus its total shortfall $\sum_C(\max\{W_C(0), W_C(1)\} - s_\pi(C))$ equals

$$\sum_C \max\{W_C(0), W_C(1)\} = W_Z - K_\rho(M).$$

This quantity does not depend on π. So the policy with the largest admitted weight is the policy with the smallest total shortfall, and $W^*_\mathcal{A}(M) = W_Z - K_\rho(M) - \Delta_\mathcal{A}(M)$. Both the maximum and the minimum are attained because $\mathcal{A}$ is finite and nonempty. Each group's shortfall is nonnegative, so the total shortfall of a policy is zero if and only if that policy attains the maximum in every group. Enlarging $\mathcal{A}$ adds candidates to the minimum, so the minimum shortfall cannot increase.

A policy generalises when no output outside its buffer B_π is demanded. Under independent demands this has probability $\prod_{z:o_z \notin B_\pi} \exp(-\rho_z m_z) = \exp\big(-(W_Z - \sum_C s_\pi(C))\big)$, which rises with the admitted weight. So the best policy in $\mathcal{A}$ generalises with probability $\exp\big(-W_Z + W^*_\mathcal{A}(M)\big) = \exp\big(-K_\rho(M) - \Delta_\mathcal{A}(M)\big)$. □

So weighted freedom is lost in two ways. Records that call for different predictions can share one memory state, and that loss is $K_\rho(M)$. With the memory fixed, the available policies may be unable to take the best prediction in every group at once, and that loss is $\Delta_\mathcal{A}(M)$. Together they are the weighted freedom lost by the best policy in $\mathcal{A}$. A comparison across embodiments still requires the correspondence, stated above, between the demands evaluated in each embodiment and their weights.

Proposition 24 (When a coarsest zero-regret memory exists)**.**
In the setting of Theorem 11, let H_+ be the set of interaction records h that appear in some pair $z = (h, Q)$ with positive regret weight $\rho_z m_z$. Every partition below is a partition of H_+. If the same pair (h, Q) appears more than once in Z, its conditional probability p_z is the same at every occurrence. Two records h and h' are **compatible** when they have the same optimal prediction on every test Q for which both (h, Q) and (h', Q) have positive regret weight. A memory M has $K_\rho(M) = 0$ if and only if any two records with the same memory state under M are compatible.

One partition is **coarser** than another when each of its groups is a union of groups of the other. If the partition of a memory M is coarser than the partition of a memory M', then $K_\rho(M') \leq K_\rho(M)$. A **coarsest zero-regret memory** is a memory with $K_\rho(M) = 0$ whose partition of H_+ is coarser than the partition of every other memory with $K_\rho(M) = 0$. Such a memory exists if and only if compatibility is transitive. When it exists, its groups are the classes of mutually compatible records, so it is unique up to renaming the memory states. Transitivity is guaranteed when every record in H_+ has positive regret weight on the same tests, and is sometimes true without that condition.

Proof. Each group C is the set of pairs with one memory state and one test, and it contributes $\min\{W_C(0), W_C(1)\}$ to $K_\rho(M)$. That contribution is zero if and only if one of $W_C(0)$ and $W_C(1)$ is zero, which is to say that every pair in C with positive weight has the same optimal prediction. Two records with the same memory state are in the same group for every test. So $K_\rho(M) = 0$ if and only if any two records with the same memory state have the same optimal prediction on every test for which both have positive weight, which is compatibility.

Suppose the partition of M is coarser than the partition of M'. Fix a test Q and a group C of M for that test. Let $C'_1, \ldots, C'_k$ be the groups of M' for test Q whose records have, under M, the memory state of C. The pairs in C with positive weight are the pairs with positive weight in $C'_1, \ldots, C'_k$, so $W_C(a) = \sum_j W_{C'_j}(a)$ for each answer a. The minimum of two sums is at least the sum of the minima, so $\min\{W_C(0), W_C(1)\} \geq \sum_j \min\{W_{C'_j}(0), W_{C'_j}(1)\}$. Summing over the groups of M gives $K_\rho(M) \geq K_\rho(M')$.

Compatibility is reflexive and symmetric. If it is also transitive, it is an equivalence relation, and the memory whose states are the equivalence classes has $K_\rho(M) = 0$ by the first paragraph. In every other zero-regret partition, records that share a group are compatible, so each group is a subset of one equivalence class, and the class partition is coarser than it. Conversely, suppose that a coarsest zero-regret partition exists. Take any compatible pair of records, and consider the memory under which those two records share one state and every other record has a state of its own. This memory has $K_\rho(M) = 0$, so that pair must share a group of the coarsest partition. No incompatible pair shares a group of it, since it is zero-regret. So two records are in the same group of the coarsest partition if and only if they are compatible. Being in the same group is transitive, so compatibility is transitive. The groups are then the compatibility classes, which fixes the partition up to the names of the states.

When every record in H_+ has positive regret weight on the same tests, two records are compatible if and only if they have the same list of optimal predictions over those tests, and equality of lists is transitive. Transitivity can also be true without common coverage. For example, two records that agree on every test that both of them face are compatible even when one of them faces a further test that the other does not. □

15 Novelty, damage and autopoiesis

Definition 28 (Viable policy buffers)**.**
Fix a finite vocabulary $\mathfrak{v}$ and a child task $\alpha = \langle I_\alpha, O_\alpha \rangle$ with a finite nonempty subset $\mathcal{V} \subseteq \Pi_\alpha$ of correct policies under comparison. The unseen region is

$$U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha).$$

This represents the demands an unknown future might present us with. For each policy $\pi \in \mathcal{V}$, I define its unseen buffer as

$$B_\pi := \mathrm{Ext}(\pi) \cap U.$$

This is the part of the unseen region covered by the policy.

Definition 29 (Timeline continuations and demand sets)**.**
Fix a timeline $\tau : I \to \Phi$ and $t \in I$. Fix its prefix $\tau_{\leq t}$ and a finite future test interval $J_t = \{t+1, \ldots, t+r\}$ with $r \in \mathbb{N}$ and $r \geq 1$. For each distinction $\phi \in \Phi$, the maximal possible thing seen through $\mathfrak{v}$ is the encoding $\mathrm{Enc}_\mathfrak{v}(\phi)$ from Definition 8. Since $\phi \in \bigcap \mathrm{Enc}_\mathfrak{v}(\phi)$, the encoding is an element of $L_\mathfrak{v}$. An infinite timeline $c : \mathbb{N} \to \Phi$ is admissible for α on J_t when $c(s) = \tau(s)$ for every $s \leq t$, and

$$\mathrm{Enc}_\mathfrak{v}(c(s)) \in \mathrm{Ext}(I_\alpha) \quad \Longrightarrow \quad \mathrm{Enc}_\mathfrak{v}(c(s)) \in O_\alpha$$

for every $s \in J_t$. I call c a timeline continuation, because it continues a particular past. Let $\mathcal{C}_{\tau,t}$ be a finite nonempty set of such admissible timeline continuations, equipped with probability measure $P_\mathcal{C}$. In a physical application, $\mathcal{C}_{\tau,t}$ is restricted to timeline continuations permitted by the chosen physical model. For $c \in \mathcal{C}_{\tau,t}$ and a thing $x \subseteq \mathfrak{v}$, $\rho_{c,J_t}(x)$ is the tenure[12] of x, the

[12]Tenure is the period of objective time for which a thing persists, measured in ticks of objective time. A tick is a position in a timeline. The timeline assigns one distinction to each tick, and a thing is true at a tick when it is true given that distinction (Definition 2).

number of ticks for which x persists in the timeline c, defined as

$$\rho_{c,J_t}(x) := |\{s \in J_t \mid c(s) \in T(x)\}|.$$

For $c \in \mathcal{C}_{\tau,t}$, define the demand set

$$\Delta(c) := \{\, \mathrm{Enc}_{\mathfrak{v}}(c(s)) \mid s \in J_t,\ \mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U \,\}.$$

Thus $\Delta(c)$ is the set of unseen maximal outputs that are true during the future test interval of timeline c. Note that every member of $\Delta(c)$ lies in the unseen region U, which contains every unseen buffer. Now remember how I defined a probability measure $P_{\mathcal{C}}$ over these timeline continuations. Each continuation determines a demand set, so $P_{\mathcal{C}}$ determines a probability measure over demand sets. The induced novelty distribution on 2^U is

$$P_\tau(S) := P_{\mathcal{C}}(\{c \in \mathcal{C}_{\tau,t} : \Delta(c) = S\}).$$

The novelty distribution P_τ is exchangeable when $P_\tau(S) = P_\tau(S')$ whenever S and S' are sets of unseen encodings with $|S| = |S'|$.[13]

Lemma 13 (A policy survives a future when, and only when, that future's demands are inside its buffer)**.**
Under Definition 29, for every $\pi \in \mathcal{V}$ and every $c \in \mathcal{C}_{\tau,t}$,

$$\rho_{c,J_t}(\pi) = |J_t| \quad \Longleftrightarrow \quad \Delta(c) \subseteq B_\pi.$$

Moreover,

$$\rho_{c,J_t}(\pi) = |J_t| - \big|\{s \in J_t : \mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U \setminus B_\pi\}\big|.$$

Consequently,

$$P_{\mathcal{C}}(\rho_{c,J_t}(\pi) = |J_t|) = P_\tau(\{S \subseteq U : S \subseteq B_\pi\}).$$

Proof. For any $s \in J_t$, the possible thing π is true given $c(s)$ when

$$\pi \subseteq \mathrm{Enc}_{\mathfrak{v}}(c(s)),$$

which is when $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in \mathrm{Ext}(\pi)$.

If $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U$, then π is true given $c(s)$ if and only if that encoding lies in $B_\pi = \mathrm{Ext}(\pi) \cap U$. If $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in \mathrm{Ext}(I_\alpha)$, by admissibility it lies in O_α. Since $\pi \in \Pi_\alpha$, by correctness $O_\alpha \subseteq \mathrm{Ext}(\pi)$. The sets U and $\mathrm{Ext}(I_\alpha)$ partition $L_{\mathfrak{v}}$, so π is false given $c(s)$ iff $\mathrm{Enc}_{\mathfrak{v}}(c(s)) \in U \setminus B_\pi$. Counting the ticks of J_t gives the second display, and π is true at every tick in J_t iff every unseen encoding in the continuation lies in B_π, meaning $\Delta(c) \subseteq B_\pi$.

The probability equality follows because P_τ is the pushforward of $P_{\mathcal{C}}$ under Δ. □

Proposition 25 (Survival under novelty)**.**
Let $S \subseteq U$ be drawn from a probability distribution P on 2^U that is exchangeable, meaning $P(S)$ depends only on $|S|$. Then $P(\{S \subseteq U : S \subseteq B_\pi\})$ is a nondecreasing function of $|B_\pi|$, and when $|B_{\pi'}| > |B_\pi|$, $P(\{S \subseteq U : S \subseteq B_{\pi'}\}) > P(\{S \subseteq U : S \subseteq B_\pi\})$ if and only if $P(1 \le |S| \le |B_{\pi'}|) > 0$. Under the uniform P where every $S \subseteq U$ is equally likely,

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = 2^{|B_\pi| - |U|}.$$

[13] Only encodings occur in $\Delta(c)$, so P_τ is concentrated on sets of unseen encodings and exchangeability is a condition on those sets. If exchangeability were imposed on every subset of U, it would force $P_\tau(\emptyset) = 1$ whenever U contains a possible thing that is no distinction's encoding. In any product vocabulary over Φ, U contains such a possible thing, because every encoding has one possible fact per block and a nonempty U contains the empty thing.

Proof. Let $b = |B_\pi|$ and $N = |U|$. Exchangeability means there are nonnegative numbers r_m such that every subset of U with size m has probability r_m. Hence

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = \sum_{m=0}^{b} \binom{b}{m} r_m.$$

For fixed m, the term $\binom{b}{m}$ is nondecreasing in b, and every coefficient r_m is nonnegative. So the whole sum is nondecreasing in b. For $b < b'$, the difference between the sums at b' and at b is $\sum_{m=1}^{b'} [\binom{b'}{m} - \binom{b}{m}] r_m$ with $\binom{b}{m} = 0$ for $m > b$. Every bracket is positive, since $\binom{b'}{m} > \binom{b}{m}$ for $1 \le m \le b'$. Hence the difference is positive if and only if $r_m > 0$ for some $1 \le m \le b'$, which is $P(1 \le |S| \le b') > 0$. Under the uniform distribution, $r_m = 2^{-N}$ for every subset of size m, so

$$P(\{S \subseteq U : S \subseteq B_\pi\}) = \sum_{m=0}^{b} \binom{b}{m} 2^{-N} = 2^{b-N}.$$

□

For the finite viable-policy set $\mathcal{V}$, put

$$F(\pi) := P_\tau\{S \subseteq U : S \subseteq B_\pi\}, \qquad M_\pi := \{\pi' \in \mathcal{V} : F(\pi') \ge F(\pi)\}, \qquad I(\pi) := -\log_2 \frac{|M_\pi|}{|\mathcal{V}|}.$$

Thus functional information is the survival-rank statistic on this fixed $\mathcal{V}$, in bits [Hazen et al., 2007, Szostak, 2003].

Theorem 14 (The Law of Increasing Functional Information)**.**
Fix the objects in Definitions 28 and 29. Let π be drawn from $\mathcal{V}$ by a probability distribution w, uniform or otherwise, and a continuation c from $P_\mathcal{C}$ independently of π. If the full persistence event $\rho_{c,J_t}(\pi) = |J_t|$ has positive probability, then

$$\mathbb{E}[I(\pi) \mid \rho_{c,J_t}(\pi) = |J_t|] \ge \mathbb{E}[I(\pi)]\,,$$

with strict inequality if and only if $F(\pi)$ is not constant on the policies to which w gives positive probability.

Proof. By Lemma 13, conditioning on full persistence through the continuation is conditioning on the event $S \subseteq B_\pi$. For each $\pi \in \mathcal{V}$, this event has probability $F(\pi)$.

I list the policies in $\mathcal{V}$ as $\pi_1, \dots, \pi_k$ in nondecreasing order of F.

$$F(\pi_1) \le F(\pi_2) \le \cdots \le F(\pi_k).$$

Let $q_j := F(\pi_j)$ and $I_j := I(\pi_j)$. From the definition of M_π, the sequence (I_j) is nondecreasing in the same order as (q_j). If $q_i < q_j$, then $M_{\pi_j} \subsetneq M_{\pi_i}$, so $I_j > I_i$.

Before selection, π is π_j with probability $w_j := w(\pi_j)$, so

$$\mathbb{E}[I(\pi)] = \sum_{j=1}^{k} w_j I_j.$$

After conditioning on survival, policy π_j is weighted by $w_j q_j$. Since the survival event has positive probability, $\sum_j w_j q_j > 0$, and

$$\mathbb{E}[I(\pi) \mid \rho_{c,J_t}(\pi) = |J_t|] = \frac{\sum_{j=1}^{k} w_j q_j I_j}{\sum_{j=1}^{k} w_j q_j}.$$

By the covariance identity, since $\sum_j w_j = 1$,

$$\sum_{j=1}^{k} w_j q_j I_j - \left(\sum_{j=1}^{k} w_j q_j\right)\left(\sum_{j=1}^{k} w_j I_j\right) = \frac{1}{2}\sum_{i,j} w_i w_j (q_i - q_j)(I_i - I_j).$$

Every term on the right is nonnegative because (q_j) and (I_j) are ordered the same way. Therefore

$$\frac{\sum_{j=1}^{k} w_j q_j I_j}{\sum_{j=1}^{k} w_j q_j} \geq \sum_{j=1}^{k} w_j I_j.$$

If F is not constant on the policies with $w_j > 0$, then some i, j with $w_i, w_j > 0$ satisfy $q_i < q_j$ and $I_i < I_j$, so $w_i w_j (q_i - q_j)(I_i - I_j) > 0$. At least one summand in the covariance identity is then positive, so the inequality is strict. If F is constant on those policies, then every q_j with $w_j > 0$ equals one positive number, so the two expectations coincide. The uniform draw, $w_j = 1/k$, is one case of the distribution w in the theorem. □

For repeated independent continuations under one novelty law, selection updates w_j to $w'_j = w_j F(\pi_j)/\sum_i w_i F(\pi_i)$. Keeping $\mathcal{V}$ and its rank statistic I fixed, Theorem 14 applies at every round; expected functional information is nondecreasing.

Definition 30 (Damage history, mend, and closure)**.**
Fix a finite embodied language $L_{\mathfrak{v}}$, viability $\mu \in L_{\mathfrak{v}}$, a policy $\pi \in L_{\mathfrak{v}}$, a timeline prefix $\tau_{\leq t}$, a horizon $H \in \mathbb{N}$, and transition constraints that determine which timelines are **permitted**. The prefix $\tau_{\leq t}$ is the history of timeline τ from tick 0 through the present tick t (Definition 2). In a physical application, transition constraints include conservation laws and thermodynamic requirements on energy and material flows (Section 27). Damage can occur at ticks $t+1, \ldots, t+H$. A **damage history** is a set

$$\delta \subseteq \{1, \ldots, H\} \times \mathfrak{v}.$$

A pair $(j, r) \in \delta$ records a **disruption** of possible fact r at tick $t + j$. A timeline c **continues** $\tau_{\leq t}$ when $c_{\leq t} = \tau_{\leq t}$. For a timeline c that continues $\tau_{\leq t}$, damage history δ **occurs in timeline** c when each of its disruptions occurs in timeline c. Disruption (j, r) occurs in timeline c when possible fact r is false at tick $t + j$, meaning $c(t + j) \notin r$. Basically, a disruption (j, r) occurs when r is not true j ticks of objective time after the present tick t. All damage histories form a finite set $\mathbb{D}_H := 2^{\{1,\ldots,H\}\times\mathfrak{v}}$. A **damage distribution** is a probability measure Q_H on $\mathbb{D}_H$.

A possible fact $r \in \pi$ is **damage-sensitive** when Q_H gives positive probability to damage histories that contain a disruption of r. It is **essential** for μ when viability ends if r is never restored. Formally, take any permitted timeline c on which r is false at every tick from some tick s onward, meaning $c(t') \notin r$ for every tick $t' \geq s$. Then μ is false at some tick $u \geq s$, meaning $c(u) \notin T(\mu)$, so viability has ended by tick u. A possible fact that is both damage-sensitive and essential is a **point of failure**. $k(\pi)$ denotes the number of points of failure for π.

A damage history δ is **non-fatal** to π after timeline prefix $\tau_{\leq t}$ when δ occurs in some permitted timeline c that continues $\tau_{\leq t}$ and has $c(u) \in T(\mu) \cap T(\pi)$ at some objective time $u > t + H$. I call such c a **timeline in which δ is mended**, and such u an **objective time at which δ is mended**. Damage histories that are non-fatal to π after timeline prefix $\tau_{\leq t}$ form π's **mend** over horizon H,

$$R_\pi(H) := \{\delta \in \mathbb{D}_H \mid \delta \text{ is non-fatal to } \pi \text{ after } \tau_{\leq t}\}.$$

Damage drawn from damage distribution Q_H lies in this mend with probability

$$q_\pi(H) := Q_H(R_\pi(H)),$$

which is π's **probability of mending**. It is the chance that π can mend the damage it meets over horizon H.

A policy π has **finite capacity to mend damage** $m_\pi \in \mathbb{N}$ when, for every horizon H, each damage history in $R_\pi(H)$ contains at most m_π disruptions of points of failure in π.

The same definition applies with any timeline prefix h in place of $\tau_{\leq t}$, with the final tick of h as the present, and I then say δ is **non-fatal to π after timeline prefix h.** A policy π is **organisationally closed** relative to (μ, Q_H, H) when after each repair the system can be ready for the next. Formally, there is a set $\mathcal{E}$ of timeline prefixes that contains $\tau_{\leq t}$. From each $h \in \mathcal{E}$, each $\delta \in R_\pi(H)$ with $Q_H(\{\delta\}) > 0$ is non-fatal to π after timeline prefix h. A timeline c in which δ is mended can be chosen so that its history $c_{\leq u}$ up to an objective time u at which δ is mended lies in $\mathcal{E}$. I call $\mathcal{E}$ the prefix set. I call π **organisationally closed** when it is organisationally closed relative to (μ, Q_H, H) for every $H \in \mathbb{N}$.

Definition 31 (Joint persistence)**.**
Let U be an unseen region and let $B_\pi = \mathrm{Ext}(\pi) \cap U$ be the unseen buffer of π. Let a demand set $S \subseteq U$ be drawn from a novelty distribution P_N on 2^U and a damage history $\delta \in \mathbb{D}_H$ from Q_H, independently. Policy π persists through the joint test when $S \subseteq B_\pi$ and $\delta \in R_\pi(H)$. Its persistence probability is

$$P_H(\pi) = P_N(S \subseteq B_\pi)\, Q_H(\delta \in R_\pi(H)).$$

Under the uniform novelty distribution on U, define log persistence by

$$\Lambda_H(\pi) := \log_2 P_H(\pi) = |B_\pi| - |U| + \log_2 q_\pi(H),$$

whenever $q_\pi(H) > 0$, and $\Lambda_H(\pi) := -\infty$ when $q_\pi(H) = 0$.

Physical feasibility also restricts repair histories. Under Definition 30, a damage history lies in $R_\pi(H)$ only when a repair continuation satisfies the balances of Section 27, remains within the available energy, power, and material bounds, restores viability, and restores the conditions under which π remains true. Physical feasibility therefore partly determines $q_\pi(H)$ and, under Definition 31, $P_H(\pi)$, so persistence contains no separate factor for it. In my account, which has a single resource, these physical constraints determine which acquisitions, expenditures and storage states are possible. Repair, reproduction and other activities share that resource budget.

Lemma 14 (Complex passivity must die)**.**
Assume a policy π that cannot mend damage has $k(\pi) = k > 0$ points of failure. In the independent damage model, each such possible fact is disrupted at each tick with probability $p \in (0, 1)$, independently across possible facts and ticks. Suppose the policy survives if and only if no such possible fact is disrupted. Then

$$q_\pi(H) = (1-p)^{kH},$$

and therefore

$$\lim_{H \to \infty} q_\pi(H) = 0.$$

Under uniform novelty,

$$\Lambda_H(\pi) = |B_\pi| - |U| + kH \log_2(1-p),$$

which tends to $-\infty$ as $H \to \infty$.

Proof. There are kH independent opportunities for a point of failure to be disrupted. The probability that none of them is disrupted is $(1-p)^{kH}$. Since $0 < 1 - p < 1$ and $k > 0$, this probability tends to zero with H. Substituting $q_\pi(H) = (1-p)^{kH}$ into Definition 31, I obtain $\Lambda_H(\pi) = |B_\pi| - |U| + kH \log_2(1-p)$. Because $\log_2(1-p) < 0$, the final term tends to $-\infty$. □

Lemma 15 (Repair without closure is murdered by damage)**.**
Assume a policy π has $k(\pi) = k > 0$ points of failure and finite capacity to mend damage $m_\pi < \infty$. In the independent damage model of Lemma 14, let $N_H \sim \text{Binomial}(kH, p)$ count essential disruptions over horizon H. Then

$$q_\pi(H) \leq \Pr[N_H \leq m_\pi],$$

and hence

$$\lim_{H\to\infty} q_\pi(H) = 0.$$

Proof. Under the independent damage model, N_H is the number of essential disruptions among the kH independent opportunities. A policy with finite capacity to mend damage m_π cannot survive histories with $N_H > m_\pi$, so $q_\pi(H) \leq \Pr[N_H \leq m_\pi]$. Since $p > 0$ and $k > 0$, the mean of N_H is $kHp \to \infty$ while m_π is fixed. Therefore $\Pr[N_H \leq m_\pi] \to 0$, for example by the law of large numbers. Thus $q_\pi(H) \to 0$. □

Proposition 26 (Closure provideth an abiding chance)**.**
Let π be organisationally closed relative to (μ, Q_W, W) with prefix set $\mathcal{E}$. The horizon is divided into **periods**. For $i \geq 1$, period i occupies positions $(i-1)(W+1)+1$ to $(i-1)(W+1)+W$ of a damage history, and position $i(W+1)$ has no damage. Suppose π **mends each period's damage immediately after that period**. This means that, from each $h \in \mathcal{E}$ with final tick s, each $\delta \in R_\pi(W)$ with $Q_W(\{\delta\}) > 0$ is mended at objective time $s + W + 1$ in some timeline whose history up to that objective time lies in $\mathcal{E}$. The damage in period i is read as a set $\delta_i \subseteq \{1, \ldots, W\} \times \mathfrak{v}$, which is drawn from a distribution $Q_W^{(i)}$ that gives positive probability only to histories to which Q_W gives positive probability. The draws for different periods are independent. For each horizon H, the damage distribution Q_H is the distribution of the resulting history restricted to positions $1, \ldots, H$. Let

$$\varepsilon_i := Q_W^{(i)}(\mathbb{D}_W \setminus R_\pi(W))$$

be the chance that the damage in period i lies outside the mend. Then for every horizon H, with $n := \lfloor H/(W+1) \rfloor + 1$,

$$q_\pi(H) \;\geq\; \prod_{i=1}^{n}(1-\varepsilon_i).$$

If $\varepsilon_i < 1$ for every i and $\sum_{i\geq 1} \varepsilon_i < \infty$, then for every horizon H

$$q_\pi(H) \;\geq\; q_c := \prod_{i\geq 1}(1-\varepsilon_i) \;>\; 0,$$

so q_c is an abiding chance for π.

Proof. Fix a horizon H and let $n := \lfloor H/(W+1) \rfloor + 1$, so that periods $1, \ldots, n$ cover every position up to H and $n(W+1) > H$. I draw the period histories $\delta_1, \ldots, \delta_n$ and let δ be the horizon-H history they form. Suppose every δ_i lies in $R_\pi(W)$. Each drawn δ_i has positive probability under $Q_W^{(i)}$, hence under Q_W, so closure, with mending immediately after each period, applies to it from every prefix in $\mathcal{E}$. I set $h_0 := \tau_{\leq t}$, which lies in $\mathcal{E}$ and has final tick $s_0 := t$. For $i = 1, \ldots, n$ in turn, I write $s_i := s_{i-1} + W + 1$. By closure, with mending immediately after each period, there is a permitted timeline c^i that continues h_{i-1} and in which δ_i occurs at ticks $s_{i-1}+1, \ldots, s_{i-1}+W$. This timeline satisfies $c^i(s_i) \in T(\mu) \cap T(\pi)$ and has $h_i := c^i_{\leq s_i} \in \mathcal{E}$. Then $s_i = t + i(W+1)$ for every i.

The final timeline c^n continues $\tau_{\leq t}$, because c^n agrees with h_{n-1} up to tick s_{n-1}, h_{n-1} agrees with h_{n-2} up to tick s_{n-2}, and so on down to $h_0 = \tau_{\leq t}$. Every disruption of δ occurs in timeline c^n. A disruption at position $(i-1)(W+1)+j$ with $1 \leq j \leq W$ is a disruption of δ_i at tick

$s_{i-1} + j \le s_i$, which occurs in timeline c^i. The timeline c^n agrees with c^i up to tick s_i, so c^n preserves this disruption. Positions $i(W+1)$ have no damage. Finally $c^n(s_n) \in T(\mu) \cap T(\pi)$ with $s_n = t + n(W+1) > t + H$. So c^n is a timeline in which δ is mended after timeline prefix $\tau_{\le t}$, and $\delta \in R_\pi(H)$.

Hence $\delta \in R_\pi(H)$ whenever $\delta_1, \ldots, \delta_n \in R_\pi(W)$, and by independence of the periods,

$$\begin{aligned} q_\pi(H) = Q_H(R_\pi(H)) &\ge \prod_{i=1}^{n} Q_W^{(i)}(R_\pi(W)) = \prod_{i=1}^{n}(1-\varepsilon_i) \\ &\ge \prod_{i\ge 1}(1-\varepsilon_i) = q_c. \end{aligned}$$

Because every $\varepsilon_i < 1$ and $\sum_i \varepsilon_i < \infty$, only finitely many ε_i exceed $1/2$. Also, $\log(1-\varepsilon) \ge -2\varepsilon$ for $0 \le \varepsilon \le 1/2$. So $\sum_i \log(1-\varepsilon_i)$ is finite and $q_c > 0$. □

Lemma 16 (An abiding chance means probability of persistence is greater than zero)**.**
Let π_c be a policy satisfying $\liminf_{H\to\infty} q_{\pi_c}(H) \ge q_c > 0$. Then

$$\liminf_{H\to\infty} P_H(\pi_c) \ge P_N(S \subseteq B_{\pi_c}) q_c.$$

Under the uniform novelty distribution this becomes

$$\liminf_{H\to\infty} P_H(\pi_c) \ge 2^{|B_{\pi_c}|-|U|} q_c.$$

Let π_o be another policy and set $\Delta b := |B_{\pi_o}| - |B_{\pi_c}|$, which may be negative. Suppose both probabilities of mending are positive at horizon H. Then π_c has greater log persistence than π_o if and only if

$$\log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)} > \Delta b.$$

If $q_{\pi_o}(H) = 0 < q_{\pi_c}(H)$, then π_c wins automatically.

Proof. By independence, $P_H(\pi_c) = P_N(S \subseteq B_{\pi_c}) q_{\pi_c}(H)$. Taking lim inf and using $\liminf q_{\pi_c}(H) \ge q_c$, I obtain $\liminf_{H\to\infty} P_H(\pi_c) \ge P_N(S \subseteq B_{\pi_c}) q_c$. Under uniform novelty, $P_N(S \subseteq B_{\pi_c}) = 2^{|B_{\pi_c}|-|U|}$ by Proposition 25. For the comparison, I subtract log persistences:

$$\begin{aligned} \Lambda_H(\pi_c) - \Lambda_H(\pi_o) &= (|B_{\pi_c}| - |B_{\pi_o}|) + \log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)} \\ &= -\Delta b + \log_2 \frac{q_{\pi_c}(H)}{q_{\pi_o}(H)}. \end{aligned}$$

Difference $-\Delta b + \log_2\big(q_{\pi_c}(H)/q_{\pi_o}(H)\big)$ is positive if and only if $\log_2\big(q_{\pi_c}(H)/q_{\pi_o}(H)\big) > \Delta b$, when both probabilities of mending are positive. If $q_{\pi_o}(H) = 0 < q_{\pi_c}(H)$, then $\Lambda_H(\pi_o) = -\infty$ while $\Lambda_H(\pi_c)$ is finite, so π_c wins automatically. □

Relative to bounds b_s and k_s on static policies, a policy is **non-static** when $|B_\pi| > b_s$ and **complex** when $k(\pi) > k_s$.

Theorem 15 (Closure Necessity Theorem)**.**
Fix a finite embodied language, an unseen region U, a novelty distribution P_N on 2^U, and the independent damage model of Lemma 14 with $p \in (0,1)$. Let $\mathcal{S}$ be a static policy class whose unseen buffers are bounded by b_s and whose points of failure number at most k_s, so $|B_\pi| \le b_s$ and $k(\pi) \le k_s$ for all $\pi \in \mathcal{S}$. Suppose $P_N(|S| > b_s) > 0$, so static policies face positive-probability demands they cannot contain. Let $\mathcal{N}$ be a class of complex policies with finite capacity to mend damage. Then:

1. no policy in $\mathcal{S}$ survives any demand set S with $|S| > b_s$, so every $\pi \in \mathcal{S}$ has $P_H(\pi) \leq 1 - P_N(|S| > b_s) < 1$ for every H;
2. every policy in $\mathcal{N}$ has asymptotic persistence probability zero;
3. any policy π_c with $\liminf_H q_{\pi_c}(H) > 0$ and $P_N(S \subseteq B_{\pi_c}) > 0$ has positive long-horizon persistence.

Thus, once novelty with positive probability exceeds the static bound b_s, and non-closed complex policies have only finite capacity to mend damage, closure is how complex policies remain viable under ongoing damage over long horizons.

Proof. If $|S| > b_s$ and $|B_\pi| \leq b_s$, then $S \not\subseteq B_\pi$, so no static policy with buffer at most b_s survives that demand set. Hence for every $\pi \in \mathcal{S}$, $P_N(S \subseteq B_\pi) \leq P_N(|S| \leq b_s) = 1 - P_N(|S| > b_s) < 1$, and $P_H(\pi) = P_N(S \subseteq B_\pi)\, q_\pi(H) \leq 1 - P_N(|S| > b_s)$ for every H. Each $\pi \in \mathcal{N}$ has $k(\pi) > k_s \geq 0$ and finite capacity to mend damage, so by Lemma 15, $q_\pi(H) \to 0$ and hence $P_H(\pi) \leq q_\pi(H) \to 0$. Let $q_c := \liminf_H q_{\pi_c}(H) > 0$. Lemma 16 shows that $\liminf_H P_H(\pi_c) \geq P_N(S \subseteq B_{\pi_c}) q_c$, and positive $P_N(S \subseteq B_{\pi_c})$ then implies $\liminf_H P_H(\pi_c) > 0$. Buffer inequality proves item 1 of Theorem 15, limit $P_H(\pi) \to 0$ proves item 2, and $\liminf_H P_H(\pi_c) > 0$ proves item 3. □

Theorem 16 (Void of the Unviable)**.**
Assume a finite embodied language, an unseen region U, uniform novelty, and the independent damage model of Lemma 14 with $p \in (0, 1)$, and fix bounds b_s and k_s. Let π_c be a policy with a buffer of size b_c and an abiding chance q_c, meaning $\liminf_{H\to\infty} q_{\pi_c}(H) \geq q_c > 0$. Let $\mathcal{M}$ be any finite set of complex policies with finite capacity to mend damage. Let π_s be a static policy with a buffer of size b_s and at most k_s points of failure. Suppose π_s satisfies one of the following.

1. $q_{\pi_s}(H) = q_s > 0$ for every H, and $b_c - b_s + \log_2(q_c/q_s) > 0$.
2. π_s cannot mend damage, and survives if and only if none of its $k(\pi_s) \geq 1$ points of failure is disrupted.

Then there exists H_0 such that for every $H \geq H_0$,

$$\Lambda_H(\pi_c) > \Lambda_H(\pi_s) > \max_{\pi\in\mathcal{M}} \Lambda_H(\pi).$$

Proof. For $\varepsilon \in (0, q_c)$, the bound $\liminf_{H\to\infty} q_{\pi_c}(H) \geq q_c$ gives H_1 with $q_{\pi_c}(H) \geq q_c - \varepsilon$ for all $H \geq H_1$. It follows that $\Lambda_H(\pi_c) \geq b_c - |U| + \log_2(q_c - \varepsilon)$ for all $H \geq H_1$. In case 1, premise $b_c - b_s + \log_2(q_c/q_s) > 0$ is strict, so I choose ε with $b_c - b_s + \log_2((q_c - \varepsilon)/q_s) > 0$. Then for all $H \geq H_1$,

$$\Lambda_H(\pi_c) \geq (b_c - |U|) + \log_2(q_c - \varepsilon) > (b_s - |U|) + \log_2 q_s = \Lambda_H(\pi_s).$$

For every $\pi \in \mathcal{M}$, Lemma 15 shows that $q_\pi(H) \to 0$. Because $L_{\mathfrak{v}}$ is finite and $\mathcal{M}$ is finite, the buffers $|B_\pi|$ are bounded above on $\mathcal{M}$. Therefore

$$\Lambda_H(\pi) = |B_\pi| - |U| + \log_2 q_\pi(H)$$

tends to $-\infty$ uniformly over $\mathcal{M}$. Hence for sufficiently large H, every policy in $\mathcal{M}$ has lower log persistence than π_s, while $b_c - b_s + \log_2(q_c/q_s) > 0$ implies higher log persistence for π_c than for π_s. Hence $\Lambda_H(\pi_c) > \Lambda_H(\pi_s) > \max_{\pi\in\mathcal{M}} \Lambda_H(\pi)$ for all sufficiently large H.

In case 2, by the argument of Lemma 14,

$$\Lambda_H(\pi_s) = b_s - |U| + k(\pi_s) H \log_2(1 - p),$$

which tends to $-\infty$ because $k(\pi_s) \geq 1$, while $\Lambda_H(\pi_c) \geq b_c - |U| + \log_2(q_c - \varepsilon)$ for all $H \geq H_1$. Hence $\Lambda_H(\pi_c) > \Lambda_H(\pi_s)$ for all sufficiently large H. Now I take $\pi \in \mathcal{M}$ with $k := k(\pi)$ and $m := m_\pi$, and write $n := kH$. Whenever $n \geq \max\{1, m\}$, by Lemma 15,

$$q_\pi(H) \leq \sum_{i=0}^{m} \binom{n}{i} p^i (1-p)^{n-i} \leq (m+1)\, n^m (1-p)^{n-m},$$

because $\binom{n}{i} \leq n^m$, $p^i \leq 1$ and $(1-p)^{n-i} \leq (1-p)^{n-m}$ for every $i \leq m$. Therefore

$$\begin{aligned} \Lambda_H(\pi_s) - \Lambda_H(\pi) \geq b_s - |B_\pi| - \log_2(m+1) \\ - m \log_2(kH) + m \log_2(1-p) \\ + (k - k(\pi_s))H \left|\log_2(1-p)\right|. \end{aligned}$$

Since $k(\pi_s) \leq k_s < k$ implies $k - k(\pi_s) \geq 1$, the final term grows linearly in H while $m \log_2(kH)$ grows logarithmically. Hence $\Lambda_H(\pi_s) - \Lambda_H(\pi) \to \infty$. Buffers are bounded because $L_{\mathfrak{v}}$ is finite, and $\mathcal{M}$ is finite, so one H_0 serves every $\pi \in \mathcal{M}$. □

Full-support exploration means that each unseen completion in the current buffer has positive probability of being expressed over repeated operation; it can occur in a fixed finite buffer. For a fixed countable set $\mathcal{P}_*$ of higher-layer policies, local finiteness means that only finitely many policies have been realised at each finite time. Open-ended policy generation means that the realised policies are not contained in any fixed finite subset. These are distinct conditions.

Theorem 17 (Autopoietic Theorem)**.**
At every finite time t, let the system have a finite embodied vocabulary $\mathfrak{v}_t$ and a finite set of realised policies drawn from one fixed countable set $\mathcal{P}_*$ of possible higher-layer policies. The vocabulary may change with time. Suppose the system is exposed indefinitely to damage and to viability-relevant demands, meaning demands that end viability when unmet. Damage and demand opportunities may overlap, and no independence is assumed. Assume the following.

1. Along every timeline that remains viable, there are infinitely many successive designated damage opportunities. At each opportunity, conditional upon the full preceding timeline prefix and viability to that point, a damaging event has probability at least $p_0 > 0$.

2. For every viable finite timeline prefix h, there are finite capacities $m_0(h), m_1(h) \in \mathbb{N}$, determined by h. On every continuation of h with no further response to damage, viability cannot continue through its $(m_0(h)+1)$st subsequent damaging event. A renewal restores the lower-layer conditions under which the same response policy or process is available again. On every continuation of h with no further renewal, viability cannot continue through its $(m_1(h)+1)$st subsequent damaging event. These capacities may grow across successive prefixes.

3. For every fixed finite class $\mathcal{K} \subseteq \mathcal{P}_*$ of higher-layer policies, there is $\varepsilon_{\mathcal{K}} > 0$ and an infinite sequence of designated demand opportunities along every timeline that remains viable while confined to $\mathcal{K}$. At each opportunity, conditional upon the full preceding timeline prefix and viability to that point, a viability-relevant demand that no policy in $\mathcal{K}$ can meet has probability at least $\varepsilon_{\mathcal{K}}$.

4. The system has positive probability of remaining viable indefinitely.

Then, under Assumptions 1 to 3, the probability is zero that a timeline remains viable indefinitely with only finitely many responses to damage, or only finitely many renewals of the conditions for those responses, or with every realised higher-layer policy inside one fixed finite subset of $\mathcal{P}_*$. In particular, under Assumption 4, conditional on indefinite viability, almost every timeline

contains infinitely many responses to damage, infinitely many renewals of the conditions for those responses, and open-ended policy generation. By almost every timeline I mean all but a set of probability zero. Those renewals constitute organisational closure. If they are generated through processes internal to the system's specified organisational boundary, they also meet autopoiesis' self-production condition.

Proof. Let D_j indicate damage at the jth designated damage opportunity and let V_j be the event of viability through that opportunity. For $r \geq 0$, let $N_{r,n} := \sum_{j=r+1}^{r+n} D_j$. By Assumption 1 and induction over successive opportunities,

$$\Pr(V_{r+n} \cap \{N_{r,n} \leq m\}) \leq \sum_{i=0}^{m} \binom{n}{i} p_0^i (1-p_0)^{n-i} \longrightarrow 0$$

for every r and m. For $r, m \geq 0$, let $E_{r,m}$ denote indefinite viability with no response to damage after opportunity r and $N_{r,n} \leq m$ for every $n \geq 1$. At $r = 0$, the prefix at opportunity r is the initial prefix. For every n,

$$\Pr(E_{r,m}) \leq \Pr(V_{r+n} \cap \{N_{r,n} \leq m\}) \longrightarrow 0,$$

so $\Pr(E_{r,m}) = 0$. In an indefinitely viable timeline with finitely many responses, I can choose an opportunity r after responses have ceased, using $r = 0$ if none occur. Let h be the timeline's prefix at opportunity r. Assumption 2 then implies that this timeline lies in $E_{r,m}$ with $m = m_0(h) < \infty$. Thus $\bigcup_{r,m \geq 0} E_{r,m}$ contains every indefinitely viable timeline with finitely many responses, and this countable union has probability zero.

Replacing response with renewal and $m_0(h)$ with $m_1(h)$ proves that indefinitely viable timelines with finitely many renewals also have probability zero. Each renewal restores lower-layer conditions for the response policy, so these renewals constitute organisational closure.

Fix any finite class $\mathcal{K}$ of higher-layer policies. By Assumption 3 and successive conditioning, the system has probability at most $(1 - \varepsilon_{\mathcal{K}})^n$ of remaining viable through n designated demand opportunities while confined to $\mathcal{K}$. This bound tends to zero. Thus indefinite viability while confined to $\mathcal{K}$ has probability zero. The set $\mathcal{P}_*$ is countable, so its finite subsets form a countable family. Indefinite viability while confined to some fixed finite class is therefore a countable union of probability-zero events. By Assumption 4, conditioning on indefinite viability is well defined. Consequently, assuming indefinite viability, almost every timeline has infinitely many damage responses, infinitely many renewals, and policies outside every fixed finite class. The vocabulary and the set of realised policies are finite at each time, while their unions over all times need not be finite. □

Corollary 19 (A widening mend provideth an abiding chance)**.**
With the periods and the period distributions $Q_W^{(i)}$ of Proposition 26, let $B_1, B_2, \ldots \subseteq \mathbb{D}_W$ be sets of damage histories of one period. Each B_i is a set of damage histories the system can repair in period i. Let $\mathcal{E}_0, \mathcal{E}_1, \ldots$ be sets of timeline prefixes with $\tau_{\leq t} \in \mathcal{E}_0$[14]. Each $\mathcal{E}_i$ contains histories the system can have once period i has been repaired. Suppose that for every period $i \geq 1$ and every prefix $h \in \mathcal{E}_{i-1}$ with final tick s, each damage history $\delta \in B_i$ with $Q_W^{(i)}(\{\delta\}) > 0$ is mended by π at objective time $s + W + 1$ in some timeline whose history up to that objective time lies in $\mathcal{E}_i$. Let $\varepsilon_i := Q_W^{(i)}(\mathbb{D}_W \setminus B_i)$ be the chance that period i's damage lies outside B_i. If $\varepsilon_i < 1$ for every i and $\sum_{i \geq 1} \varepsilon_i < \infty$, then

$$q_\pi(H) \;\geq\; \prod_{i \geq 1} (1 - \varepsilon_i) \;>\; 0$$

[14] $\tau_{\leq t}$ is the present, meaning the history of the timeline from tick 0 to the current tick t, as in Definition 30. Repairs are only assumed from prefixes in the sets $\mathcal{E}_i$. So the chain of period repairs can only begin from a prefix in $\mathcal{E}_0$. The probability of mending $q_\pi(H)$ measures how likely damage is to be non-fatal after the present prefix, so the present $\tau_{\leq t}$ must lie in $\mathcal{E}_0$.

for every horizon H. Now I take the independent damage model of Lemma 14, in which each of π's k points of failure is disrupted at each tick with probability p. So a period has kW opportunities for such a disruption. Suppose that in period i the system can repair any damage that disrupts at most m_i of those points. So B_i contains every such period history. Then

$$\varepsilon_i \leq \Pr\big[\text{Binomial}(kW, p) > m_i\big].$$

If $m_i \geq kW$ from some period i^* onward, no later period can be fatal, so $\varepsilon_i = 0$ for every $i \geq i^*$ and

$$q_c = \prod_{i<i^*} (1 - \varepsilon_i)$$

is a positive abiding chance.

Proof. At step i, the construction in the proof of Proposition 26 rests only on δ_i being mended immediately after its period, from a prefix in one set into a prefix in the next. So it applies with $\mathcal{E}_{i-1}$ and B_i in place of $\mathcal{E}$ and $R_\pi(W)$, starting from $h_0 = \tau_{\leq t} \in \mathcal{E}_0$. Hence $\delta \in R_\pi(H)$ whenever $\delta_1 \in B_1, \ldots, \delta_n \in B_n$, so $q_\pi(H) \geq \prod_{i\leq n}(1 - \varepsilon_i) \geq \prod_{i\geq 1}(1 - \varepsilon_i)$, and the product is positive as before. Under the independent damage model, a period history outside B_i disrupts more than m_i of the k points of failure, and the number of such disruptions in a period is Binomial(kW, p). So $\varepsilon_i \leq \Pr[\text{Binomial}(kW, p) > m_i]$. This bound is zero when $m_i \geq kW$ and at most $1 - (1 - p)^{kW} < 1$ otherwise. So when $m_i \geq kW$ for every $i \geq i^*$, the sum $\sum_i \varepsilon_i$ has finitely many nonzero terms, each below one. Hence $\prod_{i<i^*}(1 - \varepsilon_i)$ is positive. □

A resource surplus can sustain the loop. I now give an explicit resource balance for a process run by the system itself. When the reserve covers a cycle's cost, the cycle restores the conditions needed for further acquisition and repair.

Proposition 27 (Surplus sustains repair)**.**
Fix a finite embodied language, viability μ, and policy π as in Definition 30. Let $\mathcal{E}$ be a set of permitted timeline prefixes whose final distinction is in $T(\mu) \cap T(\pi)$, so that μ and π are both true at the final tick of each. I fix a process that generates permitted timelines by acquiring resource and repairing damage, using events observed so far. The process checks viability after repair at cycle boundaries.

For a fixed positive integer W, suppose that from every prefix in $\mathcal{E}$ with at least $a > 0$ units of resource, this process completes one cycle of W ticks of objective time, and that for every cycle outcome of positive probability the prefix at the end of that cycle is again in $\mathcal{E}$. The operating requirement a covers every resource requirement within a cycle, including costs incurred before acquisition succeeds. Storage permits every finite reserve.

From each such prefix, resource gain and cost of acquisition and repair have a common bounded joint distribution (G, C), with $G, C \geq 0$. Cycle outcomes are independent of earlier cycles. Gain and cost within one cycle may be dependent. Reserve changes by $G - C$ at cycle completion.

Let an initial prefix $\tau_{\leq t} \in \mathcal{E}$ have reserve $A_0 = a + r$, where $r > 0$. For a comparison, I draw independent cycle outcomes (G_n, C_n), each with the distribution of (G, C), for all $n \geq 1$, and define

$$X_n := G_n - C_n, \qquad \gamma := \mathbb{E}[X_1] > 0,$$

$$A_n := a + r + \sum_{j=1}^{n} X_j, \qquad E_\infty := \{A_n \geq a \text{ for every } n \geq 0\}.$$

Then some $\theta > 0$ has $\mathbb{E}[e^{-\theta X_1}] < 1$, and every $\theta > 0$ with $\mathbb{E}[e^{-\theta X_1}] \leq 1$ satisfies

$$\Pr(E_\infty) \geq 1 - e^{-\theta r} > 0.$$

On E_∞, the process reaches these reserves, every cycle completes, and μ and π are true at every cycle boundary. For damage distributions Q_H induced by the process,

$$q_\pi(H) \geq \Pr(E_\infty) \geq 1 - e^{-\theta r} \qquad \text{for every such } \theta \text{ and every } H \in \mathbb{N}.$$

Suppose a permitted finite initial phase has probability $p_{\text{start}} > 0$ of meeting its viability checks and reaching $\mathcal{E}$ with reserve at least $a+r$. Then that initial phase followed by perpetual operation has probability at least $p_{\text{start}}(1 - e^{-\theta r})$. Probabilities of mending from the initial prefix of that phase have this lower bound as well.

Proof. Boundedness permits differentiation at zero, giving

$$\frac{d}{d\theta}\mathbb{E}[e^{-\theta X_1}]\Big|_{\theta=0} = -\gamma < 0.$$

Since this expectation equals one at zero, some $\theta > 0$ has $\mathbb{E}[e^{-\theta X_1}] < 1$. Fix any $\theta > 0$ with $\mathbb{E}[e^{-\theta X_1}] \leq 1$.

Let $S_n := A_n - a$ and let $\sigma := \inf\{n \geq 1 : S_n < 0\}$, with $\sigma = \infty$ if this set is empty. Independence of these cycle outcomes gives, by conditioning successively,

$$\mathbb{E}[e^{-\theta S_{\min\{\sigma,n\}}}] \leq e^{-\theta r}.$$

Indeed, after σ the quantity inside the expectation remains fixed, while before σ its conditional expectation at the next step is multiplied by $\mathbb{E}[e^{-\theta X_1}] \leq 1$. On $\{\sigma \leq n\}$ that quantity exceeds one, so

$$\Pr(\sigma \leq n) \leq e^{-\theta r}.$$

Taking $n \to \infty$ proves $\Pr(E_\infty) \geq 1 - e^{-\theta r} > 0$.

I use identical random cycle outcomes for this comparison and for the process itself until reserve falls below a. Initially, $\tau_{\leq t} \in \mathcal{E}$ and $A_0 \geq a$. Whenever the prefix at the end of a completed cycle is in $\mathcal{E}$ and the reserve is at least a, the specified process executes another cycle, whose gain and cost are those of its fresh cycle outcome. So reserve changes by X_n, and the cycle's final prefix is again in $\mathcal{E}$. Induction establishes actual completion of every cycle on E_∞.

Fix a horizon H. Every realised history on E_∞ continues $\tau_{\leq t}$, is a timeline in which its own damage history through H occurs, and reaches a cycle boundary $u > t + H$ at which μ and π are true. So the actual continuation is a timeline in which that damage history is mended, in the sense of Definition 30. Hence on E_∞ the realised horizon-H damage history lies in $R_\pi(H)$. This proves $q_\pi(H) \geq \Pr(E_\infty)$. I prove the final claim by conditioning on each prefix in which the initial phase is successful. □

Corollary 20 (Surplus accumulates)**.**
Under Proposition 27, conditional on E_∞,

$$\frac{A_n}{n} \longrightarrow \gamma > 0 \qquad \text{almost surely.}$$

Consequently, every fixed finite reserve is eventually exceeded at every later cycle boundary.

Proof. Before conditioning, the bounded independent increments obey the strong law of large numbers,

$$\frac{1}{n}\sum_{j=1}^{n} X_j \longrightarrow \gamma \qquad \text{almost surely.}$$

This probability-one event also has conditional probability one given E_∞, because $\Pr(E_\infty) > 0$. On E_∞, actual reserve equals $a + r + \sum_{j=1}^{n} X_j$, so $A_n/n \to \gamma$. Positive γ implies $A_n \to \infty$. □

A learner can spend surplus on improving how it learns. I represent learning by a task from Definition 16. Its inputs encode child tasks, and its correct outputs encode policies that generalise to their parent tasks. A modification trial records the learning procedure before that trial and the resources then available. A correct output adds a record of a replacement procedure, now in use, with greater performance on a fixed learning task, together with restored conditions for acquisition and repair. The successor performs subsequent learning and modification trials. The task encoding identifies success on the demands added by a trial with those demands lying in the buffer of the procedure in use, as in Definition 20.

Proposition 28 (Surplus funds learner improvement)**.**
Assume the cycle process, operating requirement a, and bounded independent and identically distributed net changes X_i of Proposition 27, with $\gamma := \mathbb{E}[X_1] > 0$. Assume the distribution of the X_i, the cycle timing and the operating requirement a all continue to apply during every trial and installation described below. Let the operating reserve at the start be $A_0 = a + r$, where $r > 0$. I define $X_i^+ := \max\{X_i, 0\}$ and choose

$$0 < \lambda < \frac{\gamma}{\mathbb{E}[X_1^+]}.$$

After cycle i, λX_i^+ units of resource are allocated to trials. The operating reserve then changes by $Y_i := X_i - \lambda X_i^+$. Since $\gamma \le \mathbb{E}[X_1^+]$, this choice has $\lambda < 1$. Deductions occur after repair and reduce only positive net gains, so the operating requirement a remains sufficient. If K_n is the total expenditure on trials through cycle n, the resource available for trials is

$$J_n := \lambda \sum_{i=1}^{n} X_i^+ - K_n \ge 0.$$

Every trial, including evaluation and installation, has a total cost of at most $c > 0$ and terminates after finitely many cycles. Trials run sequentially, another starting after a finite delay whenever c units are available. All additional operating costs are included in that trial cost.

Fix a finite vocabulary $\mathfrak{v}$, a learning task α, a finite nonempty set $\mathcal{V} \subseteq \Pi_\alpha$ of learning procedures, and a novelty distribution P_N on 2^U, where $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$. I use the existing buffers and performance

$$B_\ell := \mathrm{Ext}(\ell) \cap U, \qquad F(\ell) := P_N(S \subseteq B_\ell).$$

The current procedure $\ell \in \mathcal{V}$ performs each trial. A proposed successor $\ell' \in \mathcal{V}$ is adopted only when $F(\ell') > F(\ell)$, with every comparison correct. Installation preserves the cycle process of Proposition 27 and replaces the procedure that performs subsequent learning and trials. Whenever the current performance is below $\max_{\eta \in \mathcal{V}} F(\eta)$, assume that each funded trial has probability at least $p \in (0, 1]$ of producing and installing an F-maximiser, conditional upon the full viable timeline prefix at which that trial starts. One fixed p applies to every such prefix.

Let E_∞ be continued operation with $A_n \ge a$ for every n under this allocation. Then some $\theta > 0$ has $\mathbb{E}[e^{-\theta Y_1}] \le 1$. The bound $\Pr(E_\infty) \ge 1 - e^{-\theta r} > 0$ is true for every such θ. Moreover, almost every timeline in E_∞ reaches $\max_{\eta \in \mathcal{V}} F(\eta)$ after finitely many trials and remains at it thereafter. For every k, the probability of remaining viable through k trials without reaching that maximum is at most $(1-p)^k$. If P_N is uniform on 2^U, the procedure reached also maximises freedom among procedures in $\mathcal{V}$.

Proof. The operating changes Y_i are bounded, independent and identically distributed, and

$$\mathbb{E}[Y_1] = \gamma - \lambda \mathbb{E}[X_1^+] > 0.$$

Proposition 27 applied to Y_i gives $\Pr(E_\infty) \geq 1 - e^{-\theta r} > 0$ for every $\theta > 0$ with $\mathbb{E}[e^{-\theta Y_1}] \leq 1$. By the strong law of large numbers, almost surely on E_∞,

$$\frac{A_n}{n} \longrightarrow \gamma - \lambda \mathbb{E}[X_1^+] > 0, \qquad \frac{\lambda \sum_{i=1}^n X_i^+}{n} \longrightarrow \lambda \mathbb{E}[X_1^+] > 0.$$

The resource allocated to trials therefore grows without bound. If only finitely many trials occurred, their total expenditure would be finite, so J_n would eventually exceed c and another trial would start. Hence trials continue indefinitely on E_∞. Successive conditioning gives

$$\Pr\left(\begin{array}{c} \text{viable through } k \text{ trials and} \\ \text{maximal performance not yet reached} \end{array} \right) \leq (1-p)^k \longrightarrow 0.$$

Thus timelines in E_∞ that never reach maximal performance have probability zero. The comparisons preserve maximal performance after it is reached. Under uniform P_N, Proposition 25 gives $F(\ell) = 2^{|B_\ell| - |U|}$, and correctness for the common task α gives $|\operatorname{Ext}(\ell)| = |O_\alpha| + |B_\ell|$. Maximising F therefore maximises freedom on $\mathcal{V}$. □

Further rounds require further attainable improvements. A round is one run of the construction above with its own finite family[15] of procedures. Continuing beyond a finite $\mathcal{V}$ requires a comparison of learning performance that remains meaningful across rounds, resources for every trial and installation, and a constant positive lower bound on discovery probability per trial within each round. If acquisition or repair changes between rounds, the initial reserves must support those changed requirements. An indefinitely repeated construction can bound the probability of failure to complete round j, conditional on completing previous rounds, by $0 \leq \varepsilon_j < 1$ with $\sum_j \varepsilon_j < \infty$. Completion of all rounds then has a positive product bound. These conditions permit continuing improvement. They impose no increase in its rate.

Proposition 29 (Repeated conditional opportunities)**.**
Consider a process observed at successive opportunities $n = 1, 2, \ldots$, and let $\mathcal{F}_n$ be its full history through opportunity n. Let E_n be an event determined by that history, for example that damage occurs at opportunity n, and let q_n be its probability conditional on the history before opportunity n,

$$q_n := \Pr(E_n \mid \mathcal{F}_{n-1}) \in [0,1], \qquad n \geq 1.$$

Then, except on a set of timelines of probability zero, the events occur infinitely often whenever the conditional probabilities sum to infinity,

$$\sum_{n=1}^{\infty} q_n = \infty \quad \Longrightarrow \quad E_n \text{ occurs infinitely often.}$$

The same implication is true on any event of continued operation, with each q_n computed under the original law of the process rather than conditioned on that event. The result requires neither independence nor a fixed positive lower bound on q_n.

Proof. Fix an integer $r \geq 0$ and, for $n \geq r$, consider the random variable

$$Z_n^{(r)} := \mathbf{1}_{\bigcap_{j=r+1}^n E_j^c} \exp\left(\sum_{j=r+1}^{n} q_j \right).$$

The indicator equals one when none of the events $E_{r+1}, \ldots, E_n$ occurs, and zero otherwise, so $Z_n^{(r)}$ is zero from the first event after opportunity r onwards. At $n = r$ the sum is empty and

[15] In mathematics a family is a set, called a family when its members are themselves sets or functions. Here it is a set of learning procedures.

$Z_r^{(r)} = 1$. Each $Z_n^{(r)}$ is nonnegative and at most e^{n-r}, so its expectation is finite. For $n > r$, the indicator for n is the indicator for $n-1$ times $\mathbf{1}_{E_n^c}$, and q_n is determined by the history before opportunity n, so

$$\begin{aligned}\mathbb{E}[Z_n^{(r)} \mid \mathcal{F}_{n-1}] &= Z_{n-1}^{(r)} e^{q_n}(1-q_n) \\ &\leq Z_{n-1}^{(r)}.\end{aligned}$$

The inequality is $e^q(1-q) \leq 1$ for $q \in [0,1]$. Taking expectations, $\mathbb{E}Z_n^{(r)} \leq \mathbb{E}Z_{n-1}^{(r)} \leq \cdots \leq \mathbb{E}Z_r^{(r)} = 1$, and Fatou's lemma gives

$$\mathbb{E}\Big[\liminf_{n\to\infty} Z_n^{(r)}\Big] \leq 1.$$

So $\liminf_n Z_n^{(r)}$ is finite except on a set of probability zero. Now consider the timelines on which no event occurs after opportunity r and the conditional probabilities sum to infinity. On these timelines $Z_n^{(r)} = \exp(\sum_{j=r+1}^{n} q_j)$, which tends to infinity, so they form a set of probability zero. A timeline with a divergent sum and only finitely many events is such a timeline for some r. There are countably many r, so the set of all such timelines has probability zero, and so does its intersection with any event of continued operation. □

The proposition applies to exploration as follows. Fix a finite buffer B and let X_n be the completion expressed at opportunity n. Once operation ends, every later X_n is assigned a fixed value that is not in B. Suppose that for each completion $x \in B$ the conditional probabilities of expressing x sum to infinity,

$$\sum_{n=1}^{\infty} \Pr(X_n = x \mid \mathcal{F}_{n-1}) = \infty \qquad \text{for each } x \in B$$

almost surely on the event of continued operation, with each conditional probability computed under the original law of the process. Then, by the proposition applied to the events $X_n = x$, every completion in B is expressed infinitely often, almost surely on that event. Since B is finite, there is a finite time, which may depend on the timeline, by which every completion in B has been expressed at least once.

Corollary 21 (Declining and adaptive risks)**.**
The conclusions of Theorem 17 remain true when the fixed bounds p_0 and $\varepsilon_{\mathcal{K}}$ of its Assumptions 1 and 3 are replaced by the following two conditions.

1. At each designated damage opportunity there is a conditional probability of a damaging event. On almost every timeline that remains viable indefinitely, these conditional probabilities sum to infinity.

2. Fix a finite class $\mathcal{K} \subseteq \mathcal{P}_*$. At each designated demand opportunity there is a conditional probability of a viability-relevant demand that no policy in $\mathcal{K}$ can meet. On almost every timeline that remains viable indefinitely while confined to $\mathcal{K}$, these conditional probabilities sum to infinity.

The other assumptions of the theorem are unchanged. Each relevant timeline has infinitely many designated opportunities, the capacities $m_0(h)$ and $m_1(h)$ after a final response or renewal are finite, $\mathcal{P}_*$ is countable, and indefinite viability has positive probability.

Each designated opportunity is identified from the history as the process unfolds. Each probability of damage, and each probability of a demand of the kind specified, is conditioned on the full history before its opportunity, under the original law of the process. The sequence of damage outcomes stops when viability ends. For each $\mathcal{K}$, the sequence of demand outcomes stops when viability ends or when a realised policy outside $\mathcal{K}$ first appears. After a sequence stops, every later term of it is assigned one fixed outcome that marks the stop.

Proof. By Proposition 29 applied to the damage events, almost every timeline that remains viable indefinitely has infinitely many damaging events. Suppose that on such a timeline the responses to damage stop after some finite prefix h. By Assumption 2, viability cannot continue through the $(m_0(h)+1)$st damaging event after h, yet infinitely many damaging events follow h. So the responses continue without end. The same argument with $m_1(h)$ in place of $m_0(h)$ shows the same of the renewals, including the case in which no renewal ever occurs.

Now fix a finite class $\mathcal{K}$ and consider the timelines that remain viable indefinitely while confined to $\mathcal{K}$. By the proposition applied to the demand events, almost every one of them faces infinitely many demands that no policy in $\mathcal{K}$ can meet. Such a demand is unmet while the process is confined to $\mathcal{K}$, and an unmet viability-relevant demand ends viability. So the event of remaining viable indefinitely while confined to $\mathcal{K}$ has probability zero. The finite subsets of the countable set $\mathcal{P}_*$ form a countable family, and a countable union of events of probability zero has probability zero. So the probability that an indefinitely viable timeline is confined to some fixed finite class is zero. Since indefinite viability has positive probability, conditioning on it gives the conclusion of the theorem. □

The fixed positive bounds of the theorem imply these conditions, since infinitely many terms each at least p_0 sum to infinity. The conditions also allow probabilities that approach zero, such as $q_n = 1/(n+1)$, whose sum is still infinite.

The Leaning Glade is an illustration that I use in Chapter IX of the book. It is an engineered ecology that evolves a pump, and the pump draws work from a gravity gradient. I now bound the power of that pump. Let a mass stream with rate $\dot{m}$ cross a gravitational potential drop $\Delta\phi_g > 0$[16]. Let P_{grav} be the power extracted from the gravitational drop. Then

$$P_{\text{grav}} \leq \dot{m}\,\Delta\phi_g.$$

When gravitational acceleration is nearly constant over height difference Δh, I write $\Delta\phi_g = g_{\text{grav}}\Delta h$. Then

$$P_{\text{grav}} \leq \dot{m}\,g_{\text{grav}}\Delta h.$$

A static gravitational field gives zero net potential drop around a closed cycle[17]. Resetting the mass consumes at least the recovered gravitational work in the reversible limit[18]. Sustained output therefore requires an incoming high-potential mass stream, an outgoing low-potential stream, or another process that performs the reset[19]. In the Leaning Glade example, the possible thing[20] g in the lower layer remains true only while the mass flow, reset process, waste export, and physical balances remain satisfied[21]. A physical model determines associated energy, power,

[16] A mass stream is a flow of matter, and its rate $\dot{m}$ is how many kilograms of it pass each second. The drop $\Delta\phi_g$ is how much energy each kilogram releases as it falls. Power is how fast energy is delivered. So the inequality below means that the pump can deliver no more energy each second than the falling matter releases each second.

[17] A static gravitational field is gravity that does not change over time. A closed cycle is a round trip, and anything that goes around a round trip climbs just as far as it falls.

[18] Resetting the mass means bringing it back to the top so that it can fall again. The recovered gravitational work is the energy put to use as it falls. The reversible limit is the ideal case with no friction or other waste. Even in that case, lifting the mass back up takes all the energy its fall released, and in practice it takes more.

[19] Sustained output is a steady stream of energy that can be put to use. It needs fresh matter flowing in from high up, matter flowing out at the bottom so that it never has to be lifted back, or something else to do the lifting with energy from elsewhere. A hydroelectric dam works the third way, since the sun lifts its water back up as rain.

[20] In my earlier papers a distinction is a state, a possible fact a program and a possible thing a statement or an aspect.

[21] The mass flow is the mass stream above, and the reset process is whatever brings its matter back to the top. Waste export is the removal of spent matter and heat from the Glade. Each physical balance is an equation for one quantity, such as energy or a kind of matter. It states that what flows into the Glade equals what flows out, plus the change in how much of that quantity the Glade contains. These equations are true because energy and matter are never created or destroyed (the Physical quantities section).

and material flows[22].

For the Leaning Glade I use the following corollary of Theorem 17.

Corollary 22 (Artificial worlds can grow life)**.**
Let g be a possible thing in a lower layer such that g remains true along every viable timeline fragment under consideration. Suppose the higher-layer systems conditional on g satisfy the hypotheses of Theorem 17. Then, conditional on indefinite viability, almost every timeline of each such higher-layer system contains infinitely many responses to damage, infinitely many renewals of the conditions for those responses, and open-ended policy generation. Those renewals constitute organisational closure. Response to damage without end, renewal, and policy generation beyond every fixed finite class follow whether g arose without design or is maintained by engineered machinery.

Proof. I restrict the higher-layer process to the part of the environment $T(g)$ in which the lower-layer condition is true. By assumption, the damage, restoration, novelty, and indefinite-viability premises of Theorem 17 are true on that restriction. Theorem 17 therefore implies response to damage without end, organisational closure, and generation beyond every fixed finite policy class. This proof depends on the persistence of g alone. Response to damage without end, closure, and open-ended policy generation therefore follow whether geology, prior life, or engineering produced that lower layer. □

Corollary 23 (Repairs per period)**.**
Take the physical model of Proposition 17 and its symbols, with equilibrium distribution q at temperature Θ_0 and viability μ. Suppose that during one period the system is repaired n times in sequence. Repair j takes the system from a distribution p_j with $p_j(T(\mu)) = 1$ to a distribution p'_j with $p'_j(T(\pi_j) \cap T(\mu)) = 1$ and $q(T(\pi_j) \cap T(\mu)) > 0$. Here π_j is the possible thing whose truth set consists of the distinctions in which the jth repaired part is whole. Let W_j be the work done upon the system during repair j, and let $E_j := F_{\text{phys}}[p_j] - F_{\text{phys}}[q|_{T(\mu)}]$ be the excess free energy of the system when repair j begins. Suppose that each repair satisfies the work balance of Proposition 17. Then

$$\sum_{j=1}^{n} \Big(-k_B \Theta_0 \ln q(T(\pi_j) \mid T(\mu)) - E_j \Big) \leq \sum_{j=1}^{n} W_j .$$

If every repair begins from the equilibrium distribution given viability, so that $E_j = 0$ for every j, if $q(T(\pi_j) \mid T(\mu)) < 1$ for every j, and if the work done upon the system during the repairs sums to at most W, then

$$n \leq \frac{W}{\min_j \left(-k_B \Theta_0 \ln q(T(\pi_j) \mid T(\mu)) \right)} .$$

In the Leaning Glade, if a mass stream with constant rate $\dot{m}$ crossing a gravitational drop $\Delta\phi_g$ is the only source of work upon the Glade, then the work done upon the Glade during the repairs of a period of length Δt sums to at most $\dot{m}\,\Delta\phi_g\,\Delta t$, so W can be taken to be that amount.

Proof. For each j, Proposition 17 applied to repair j gives $W_j + E_j \geq -k_B \Theta_0 \ln q(T(\pi_j) \mid T(\mu))$. Summing over j gives the first display. With $E_j = 0$ for every j, each term on the left is at least the minimum over j, so n times that minimum is at most $\sum_j W_j$, which is at most W. The minimum is positive because $q(T(\pi_j) \mid T(\mu)) < 1$ for every j. The bound upon W is the power bound above, $P_{\text{grav}} \leq \dot{m}\,\Delta\phi_g$, multiplied by the length of the period. □

[22] A physical model is a description of a system in which time is measured in units such as seconds and every change obeys the laws of physics, such as the law that energy is never created or destroyed (Section 1). Material flows are movements of matter into or out of a system.

16 Causality and organisms

Definition 32 (intervention)**.**
If *int* and *obs* are "events" which have happened, then we say that *int* has **caused** *obs* if *obs* happens whenever *int* does. In our formalism, an **event** is a possible thing in $L_{\mathfrak{v}}$, and an event **happens** or is **observed** if and only if it is an existing thing. If $obs \in L_{\mathfrak{v}}$ is sensorimotor activity we interpret as an "observed event", and $int \in L_{\mathfrak{v}}$ is an **intervention** (by an acting system or other agency, in the sense described by Pearl [Pearl and Mackenzie, 2018]) that causes that event, then the intervention entails the observation:

$$T(int) \subseteq T(obs).$$

The syntactic condition $obs \subseteq int$ is always sufficient for $T(int) \subseteq T(obs)$. The converse requires a vocabulary in which semantic entailment is reflected by inclusion of possible things. I use inclusion of truth sets here because the causal definition should not depend on redundant vocabulary items.

Definition 33 (causal identities)**.**
Fix an embodied language $L_{\mathfrak{v}}$. Let $INT \subseteq L_{\mathfrak{v}}$ be a set of possible things, the (putative) *interventions*, and $OBS \subseteq L_{\mathfrak{v}}$ a set of matched *non-intervention observations*, meaning observations of the same outcomes occurring without the intervention. I define the set of causal identities

$$\mathcal{C}(INT, OBS) := \left\{ c \in L_{\mathfrak{v}} \setminus \{\emptyset\} \;\middle|\; \begin{array}{l} (\forall\, int \in INT)\ T(int) \subseteq T(c) \text{ and} \\ (\forall\, obs \in OBS)\ T(obs) \nsubseteq T(c) \end{array} \right\}.$$

Each $c \in \mathcal{C}(INT, OBS)$ is a **causal identity** for whatever is common to the interventions but not present in the observations. When $L_{\mathfrak{v}}$ is finite and $\mathcal{C}(INT, OBS) \neq \emptyset$, a **freedom maximising causal identity** is any $c^\star \in \mathcal{C}(INT, OBS)$ that maximises freedom over $\mathcal{C}(INT, OBS)$: $|\mathrm{Ext}(c^\star)| = \max\{|\mathrm{Ext}(c)| \mid c \in \mathcal{C}(INT, OBS)\}$. If a policy π is a causal identity, then the associated task is to classify interventions: given an event $e \in L_{\mathfrak{v}}$, decide "source present" iff $T(e) \subseteq T(\pi)$.

Definition 34 (purpose, goal or intent)**.**
We consider a policy c which is a causal identity corresponding to INT and OBS to be the **intent**, **purpose** or **goal** ascribed to the interventions. c is what the interventions share in common, meaning the "name" or "identity" of behaviour *is* the "intent", "goal" or "purpose" of behaviour. Just as an intervention caused an observation, the intervention was caused by the intent which motivated the agency undertaking the intervention. Correct intent inference requires inferring a causal identity that implies subsequent interventions.

Theorem 18 (The Psychophysical Principle of Causality)**.**
Fix a finite vocabulary, and let a body have parts $j = 1, \ldots, m$, such as its cells, its organs and the whole body. For a part j, let $\mu := \mu_j$ be a possible thing encoding the most basic valence partition, viable versus not viable, which I call the viability of part j. For any policy π, define the set of viable completions

$$\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu).$$

I use the viability divergence F_2 of Corollary 14,

$$F_2(\pi) := \log_2 |\,\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|.$$

Let π be any policy with $\Omega_\pi \neq \emptyset$. Let the added commitment $c \subseteq \mathfrak{v}$ be a thing. Commitment c is **not required** by viability inside π when there exists some $o \in \Omega_\pi$ with $T(o) \nsubseteq T(c)$. If c is

not required, then either $\Omega_{\pi\cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$, where $\Omega_{\pi\cup c}$ is taken to be empty when $\pi \cup c \notin L_{\mathfrak{v}}$. In fact $\Omega_{\pi\cup c} = \{o \in \Omega_\pi : c \subseteq o\}$. Hence $\Omega_{\pi\cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$ if and only if some $o \in \Omega_\pi$ omits a possible fact of c, which holds for every commitment not required by viability.

Proof. If $\pi \cup c \notin L_{\mathfrak{v}}$, then $\mathrm{Ext}(\pi \cup c) = \emptyset$ and hence $\Omega_{\pi\cup c} = \emptyset$. In this case, no $o \in \Omega_\pi$ contains c, since $\pi \cup c \subseteq o$ would give $\emptyset \neq T(o) \subseteq T(\pi \cup c) = \emptyset$. Otherwise $o \in \Omega_{\pi\cup c}$ iff $o \in L_{\mathfrak{v}}$ contains π, c and μ, iff $o \in \Omega_\pi$ and $c \subseteq o$. In either case $\Omega_{\pi\cup c} = \{o \in \Omega_\pi : c \subseteq o\} \subseteq \Omega_\pi$. If every $o \in \Omega_\pi$ contains c, then $\Omega_{\pi\cup c} = \Omega_\pi$ and $F_2(\pi \cup c) = F_2(\pi)$. If some $o \in \Omega_\pi$ omits a possible fact of c, then $o \notin \Omega_{\pi\cup c}$, so $\Omega_{\pi\cup c} \subsetneq \Omega_\pi$. If $\Omega_{\pi\cup c}$ is empty, the first disjunct holds. Otherwise $|\Omega_{\pi\cup c}| < |\Omega_\pi|$, which implies $F_2(\pi \cup c) > F_2(\pi)$. Finally, if c is not required, choose $o \in \Omega_\pi$ with $T(o) \not\subseteq T(c)$. If $c \subseteq o$, then $T(o) \subseteq T(c)$. This contradicts the choice of o. Hence o omits a possible fact of c. □

Theorem 19 (Bottom-up Psychophysical Principle from persistence)**.**
Fix a timeline $\tau : I \to \Phi$, an interval $[u_1, u_2]$ with $u_1, u_2 \in I$, and things $\pi, \mu, c \subseteq \mathfrak{v}$, where an impossible thing has tenure zero by Definition 14[23]. Here $\mu := \mu_j$ is the viability of a part j and π the current policy. Then

$$\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c) \leq \rho_{\tau,[u_1,u_2]}(\pi \cup \mu).$$

Strict inequality $\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c) < \rho_{\tau,[u_1,u_2]}(\pi \cup \mu)$ holds if and only if $\pi \cup \mu$ is true and c is false given $\tau(u)$ for some $u \in \{u_1, \ldots, u_2\}$. Thus adding a commitment shortens tenure over the interval if and only if the timeline visits a certain kind of distinction during the interval. At such a distinction, policy and viability are true and the commitment is false.

Proof. I apply Corollary 4 with $l = \pi \cup \mu$. The extra commitment c refines l to $l \cup c$, so c cannot extend tenure. The two values differ by $|\{u \in \{u_1, \ldots, u_2\} : \tau(u) \in T(l) \setminus T(c)\}|$, the number of ticks at which l is true and c is false. Hence $\rho_{\tau,[u_1,u_2]}(l \cup c) < \rho_{\tau,[u_1,u_2]}(l)$ if and only if this set is nonempty. □

Corollary 24 (Representation-evaluation collapse)**.**
In the setting of Theorems 18 and 19, an added commitment c is **zero-cost inside** π for part j when it does not empty the set of viable completions, does not raise F_2 and does not shorten tenure in the timelines being considered. If c is not entailed by the viability partition inside π, then c is not zero-cost inside π. Indeed c is zero-cost inside π if and only if $c \subseteq \pi \cup \mu$. A primitive commitment is one that the organism has from the start rather than builds out of others. Therefore, if a primitive commitment c is guaranteed zero-cost across all viable timelines inside π, then $c \subseteq \pi \cup \mu$. Each possible fact of c is then a possible fact of π or of μ, so the commitment is true wherever $\pi \cup \mu$ is true. For the primitive policy $\pi_0 = \emptyset$, each possible fact of c is a possible fact of μ, so the commitment is part of viability. Since viability is an evaluative partition, every such guaranteed primitive representation is evaluative in this formal sense.

Proof. If c is not required by viability inside π, then by Theorem 18 adding c either empties the set of viable completions or raises F_2, and neither depends on the timeline. If the bottom-up timeline visits a viable distinction excluded by c, then by Theorem 19 adding c shortens tenure. Such a timeline exists, because a timeline may consist of a single distinction. Let ϕ be a distinction of which $\pi \cup \mu$ is true and c is false. The timeline consisting of ϕ alone visits a viable distinction excluded by c. Hence a commitment is not zero-cost in any timeline if it is not entailed by viability. Thus any primitive commitment with that guarantee is required by viability inside π, so $T(o) \subseteq T(c)$ for every $o \in \Omega_\pi$. Since $\Omega_\pi \neq \emptyset$, some possible thing contains $\pi \cup \mu$. Hence $\pi \cup \mu \in L_{\mathfrak{v}}$ and $\pi \cup \mu \in \Omega_\pi$, which gives $T(\pi \cup \mu) \subseteq T(c)$. For the equivalence, if $c \subseteq \pi \cup \mu$ then $\pi \cup \mu \cup c = \pi \cup \mu$, so tenure is unchanged in every timeline. In that case every $o \in \Omega_\pi$ contains c,

[23]Tenure is the period of objective time for which a thing persists.

so $\Omega_{\pi\cup c} = \Omega_\pi$ by Theorem 18 and F_2 is unchanged. Conversely, if $\Omega_{\pi\cup c} \neq \emptyset$ and F_2 is not raised, then by Theorem 18 every $o \in \Omega_\pi$ contains c, and $\pi \cup \mu \in \Omega_\pi$, so $c \subseteq \pi \cup \mu$. □

Corollary 25 (Value-first construction order)**.**
Let the possible thing $\mu := \mu_j$ be the viability of a part j, and let $\pi_0 := \emptyset$ be the primitive current policy. A primitive commitment c is **viability-entailing** when $T(\mu) \subseteq T(c)$ at the level being considered. Let $\mathcal{Q}$ be a primitive representational partition of distinctions, meaning a block in the sense of Definition 21, so every distinction satisfies one and only one possible fact in $\mathcal{Q}$. Partition $\mathcal{Q}$ **separates** two distinctions when they satisfy different possible facts of $\mathcal{Q}$, and a possible fact separates two distinctions when it is true given one of them and false given the other. Committing to $\mathcal{Q}$ given a distinction means adding to the current policy the possible fact of $\mathcal{Q}$ true when that distinction is made. Partition $\mathcal{Q}$ is **zero-cost** for part j when committing to it given any viable distinction is zero-cost inside π_0 in the sense of Corollary 24. If $\mathcal{Q}$ separates two viable distinctions, then $\mathcal{Q}$ is not zero-cost for part j. Therefore any guaranteed zero-cost primitive partition must be evaluative. The possible fact it adds given a viable distinction must follow from attraction or repulsion relative to viability. The primitive layer is layer 0 of the stack. If each possible fact of the primitive layer is added by a partition zero-cost for some part, given a distinction viable for that part, then each possible fact of the primitive layer is evaluative for that part. When each possible fact of the primitive layer is evaluative for some part, value neutral object partitions can appear at later layers of a stack only as patterns over those evaluative possible facts. In other words, the system will be blind to any objects and properties to which the valence of every part is indifferent.

Proof. Suppose $\mathcal{Q}$ separates two viable distinctions ϕ and ϕ'. Committing to $\mathcal{Q}$ given ϕ adds the possible fact $q \in \mathcal{Q}$ true given ϕ, and q is false given ϕ', so $T(\mu) \nsubseteq q$ and q is not viability-entailing. Then q is an added commitment above and beyond viability inside π_0, since $\mu \in \Omega_{\pi_0}$ and $T(\mu) \nsubseteq q$. By Corollary 24 with $\pi = \pi_0$, q is not zero-cost. Hence $\mathcal{Q}$ is not zero-cost for part j. By contraposition, a partition separates no two viable distinctions if it is guaranteed zero-cost for part j at the primitive stage. So, given a viable distinction, such a partition adds a possible fact entailed by viability. Those possible facts are evaluative by definition. For the remaining claims, I apply the argument above to each part. Given a distinction viable for some part, a partition zero-cost for that part therefore adds a possible fact evaluative for that part. By hypothesis every possible fact of the primitive layer is added in this way, so each is evaluative for some part. Suppose each later layer is built on the one beneath as in Definition 26. Every possible fact of a later layer is the truth set $T(o)$ of a possible thing o of the layer beneath. Hence, if two distinctions satisfy the same possible facts beneath, they satisfy the same possible facts above. By induction over the layers, if no evaluative possible fact separates two distinctions, then no possible fact of any later layer separates them. Every later partition is therefore a pattern over evaluative possible facts, and distinctions between which the valence of every part is indifferent remain indistinguishable at every layer. □

Definition 35 (organism)**.**
I describe the circumstances of an organism $\mathfrak{o}$ by the four-tuple

$$\mathfrak{o} = \langle \mathfrak{v}_\mathfrak{o}, \mu_\mathfrak{o}, \mathfrak{p}_\mathfrak{o}, <_\mathfrak{o} \rangle,$$

where

- $\mathfrak{v}_\mathfrak{o}$ is the organism's embodied vocabulary.
- $\mu_\mathfrak{o}$ is its viability task, and $O_{\mu_\mathfrak{o}}$ contains the outputs counted as fit.
- $\mathfrak{p}_\mathfrak{o}$ is the set of policies currently retained by the organism.

- $<_{\mathfrak{o}}$ is a binary relation over $\Gamma_{\mathfrak{v}_{\mathfrak{o}}}$ called the organism's **preferences**.

An organism stores only so many policies, and there are two ways of coming by them. Let $\mathfrak{p}_{n.s.} \subseteq L_{\mathfrak{v}_{\mathfrak{o}}}$ be the reflexes fixed by natural selection, which an organism has without having learnt them. Let $\mathcal{H}_{<t_{\mathfrak{o}}} \subseteq \Gamma_{\mathfrak{v}_{\mathfrak{o}}}$ be a finite nonempty family of history tasks recording what it has undergone, and I write

$$\mathfrak{p}_{\mathcal{H}_{<t_{\mathfrak{o}}}} := \bigcup_{\zeta \in \mathcal{H}_{<t_{\mathfrak{o}}}} \Pi_{\zeta}$$

for the policies correct on at least one of those tasks. Then

$$\mathfrak{p}_{\mathfrak{o}} \subseteq \mathfrak{p}_{n.s.} \cup \mathfrak{p}_{\mathcal{H}_{<t_{\mathfrak{o}}}}.$$

Inheritance and experience together bound what an organism can have at its disposal. Where a later argument needs a history task, $\mathfrak{h}_{<t_{\mathfrak{o}}} \in \mathcal{H}_{<t_{\mathfrak{o}}}$ denotes the one I've chosen. It is shorthand, and the organism remains a four-tuple. For $\zeta \in \mathcal{H}_{<t_{\mathfrak{o}}}$, I write $\hat{\zeta}$ for the task holding every interaction the organism underwent in the stretch of history that ζ summarises. The organism has **selective memory** when some ζ omits inputs of $\hat{\zeta}$, so that $\zeta \sqsubset \hat{\zeta}$ with $O_{\zeta} = O_{\hat{\zeta}} \cap \mathrm{Ext}(I_{\zeta})$. Dropping an input removes a demand, so $\Pi_{\hat{\zeta}} \subseteq \Pi_{\zeta}$ and the set of correct policies grows. One inconsistent interaction can leave $\Pi_{\hat{\zeta}}$ empty, and discarding that interaction leaves a task with correct policies [Bennett, 2023b].

Definition 36 (preconditions)**.**
If $\mathfrak{o}$ is an organism, and c is a causal identity:

- the **differentiation** precondition is met iff $c \in L_{\mathfrak{v}_{\mathfrak{o}}}$, and
- the **alignment** precondition is met iff $\mathfrak{o}$ must learn c to remain "fit".

Proposition 30 (Causal identities are minimal commitments)**.**
Fix a finite vocabulary, viability μ, a policy π, and sets INT, OBS with $\mathcal{C}(INT, OBS) \neq \emptyset$. Let $c \subsetneq c'$ be members of $\mathcal{C}(INT, OBS)$ with $\Omega_{\pi \cup c} \neq \emptyset$, and read $\Omega_{\pi \cup c'} = \emptyset$ when $\pi \cup c' \notin L_{\mathfrak{v}}$. Then

1. $\rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c') \leq \rho_{\tau,[u_1,u_2]}(\pi \cup \mu \cup c)$ for every timeline τ and interval $[u_1, u_2]$, with strict inequality if and only if the timeline visits a distinction of which $\pi \cup \mu \cup c$ is true and c' is false;
2. $\Omega_{\pi \cup c'} \subseteq \Omega_{\pi \cup c}$, so $\Omega_{\pi \cup c'} = \emptyset$ or $F_2(\pi \cup c') \geq F_2(\pi \cup c)$, and if $c' \setminus c$ is not required by viability inside $\pi \cup c$, then $\Omega_{\pi \cup c'} = \emptyset$ or $F_2(\pi \cup c') > F_2(\pi \cup c)$;
3. $\mathrm{Ext}(\pi \cup c') \subseteq \mathrm{Ext}(\pi \cup c)$, strictly if and only if $c' \not\subseteq \pi \cup c$, and $\mathrm{Ext}(c') \subsetneq \mathrm{Ext}(c)$.

Hence a $\subseteq$-minimal member of $\mathcal{C}(INT, OBS)$ is never ranked below a member that properly contains it when rankings are based on tenure or on F_2. The minimal member is ranked strictly above that member by freedom. Therefore the freedom maximising causal identity of Definition 33 is $\subseteq$-minimal.

Proof. For the first part apply Theorem 19 with policy $\pi \cup c$ and added commitment $c' \setminus c$, so that $(\pi \cup c) \cup \mu \cup (c' \setminus c) = \pi \cup \mu \cup c'$. At a tick at which $\pi \cup \mu \cup c$ is true, c' is false if and only if $c' \setminus c$ is false. Hence the theorem's strict case holds if and only if the timeline visits a distinction of which $\pi \cup \mu \cup c$ is true and c' is false. For the second part, every completion of $\pi \cup c'$ completes $\pi \cup c$, so $\Omega_{\pi \cup c'} \subseteq \Omega_{\pi \cup c}$. This gives $F_2(\pi \cup c') \geq F_2(\pi \cup c)$ when $\Omega_{\pi \cup c'} \neq \emptyset$. The strict case follows from Theorem 18 with policy $\pi \cup c$ and added commitment $c' \setminus c$. For the third part, $\pi \cup c \subseteq \pi \cup c'$, so every completion of $\pi \cup c'$ completes $\pi \cup c$. When $c' \not\subseteq \pi \cup c$ the possible thing $\pi \cup c$ completes $\pi \cup c$ but not $\pi \cup c'$, and when $c' \subseteq \pi \cup c$ the two extensions coincide since $\pi \cup c' = \pi \cup c$. Likewise $c \in \mathrm{Ext}(c) \setminus \mathrm{Ext}(c')$. A freedom maximising causal identity therefore has no proper subset in $\mathcal{C}(INT, OBS)$, since such a subset would have strictly more completions. □

Proposition 31 (Do-operator as conditioning on an intervention indicator)**.**
Let X and Y be random variables and let $do(X = x)$ denote the intervention that sets X to the value x. Introduce an auxiliary binary variable $A \in \{0, 1\}$ indicating the regime, where $A = 1$ means "intervene and set $X = x$" and $A = 0$ means "observe" (no intervention). Consider an augmented model that preserves each regime's mechanisms and whose regime choice is independent of its exogenous variables. Assume $\mathbb{P}(A = 1) > 0$. In this model, X is generated by

$$X := \begin{cases} x, & A = 1, \\ (\text{its original mechanism}), & A = 0. \end{cases}$$

Then the interventional distribution is an ordinary conditional in the augmented model:

$$\mathbb{P}(Y \in B \mid do(X = x)) \;=\; \mathbb{P}(Y \in B \mid A = 1) \qquad \text{for all measurable } B.$$

Moreover, whenever $\mathbb{P}(X = x) > 0$,

$$\begin{aligned} \mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1)\, \mathbb{P}(A = 1 \mid X = x) \\ + \mathbb{P}(Y \in B \mid A = 0, X = x)\, \mathbb{P}(A = 0 \mid X = x). \end{aligned}$$

Proof. Fix x. Let $P_{\text{obs}}(X, Y)$ denote the observational joint distribution and let $P_{\text{do}(x)}(Y)$ denote the interventional distribution of Y under $do(X = x)$. I can always construct a joint distribution on (A, X, Y) as follows:

- Sample $A \in \{0, 1\}$ according to any distribution with $\mathbb{P}(A = 1) > 0$.
- If $A = 0$, sample (X, Y) from the observational distribution $P_{\text{obs}}(X, Y)$.
- If $A = 1$, set $X := x$ deterministically and sample Y from $P_{\text{do}(x)}(Y)$.

By construction, the conditional law of Y given $A = 1$ is exactly the interventional law of Y under $do(X = x)$, so $\mathbb{P}(Y \in B \mid A = 1) = \mathbb{P}(Y \in B \mid do(X = x))$ for all measurable B. Because $A = 1$ implies $X = x$, I also have $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$.

Finally, assuming $\mathbb{P}(X = x) > 0$, apply the law of total probability conditioned on $X = x$:

$$\begin{aligned} \mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1, X = x)\, \mathbb{P}(A = 1 \mid X = x) \\ + \mathbb{P}(Y \in B \mid A = 0, X = x)\, \mathbb{P}(A = 0 \mid X = x). \end{aligned}$$

and substitute $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$ to obtain the stated decomposition. □

17 Orders of self

Definition 37 (1ST-order-self)**.**
Let $INT_{\mathfrak{o}} \subseteq L_{\mathfrak{v}_{\mathfrak{o}}}$ denote idealised self-interventions, possible things, for an organism $\mathfrak{o}$, and let $OBS_{\mathfrak{o}} \subseteq L_{\mathfrak{v}_{\mathfrak{o}}}$ denote matched non-self observations. Any freedom maximising causal identity

$$\mathfrak{o}^1 \in \arg \max_{c \in \mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}})} |\operatorname{Ext}(c)|$$

is called a **1ST-order-self** for $\mathfrak{o}$. Existence requires the differentiation precondition $\mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}}) \neq \emptyset$ and the alignment precondition that viability depends on distinguishing self-intervention from matched observation.

An intervention includes the context needed for its specified consequence; an action label alone need not determine an invariant effect. Alignment together with differentiation is a sufficient precondition for learning the identity, not a necessary one: an organism may construct an identity that viability does not require.

Theorem 20 (Intervention control requires a represented distinction)**.**
Let a finite decision problem have cases that are pairs (e, r) of a possible thing $e \in L_{\mathfrak{v}}$ and a value $r \in \{\text{int}, \text{obs}\}$ recording whether the organism brought e about or only observed it. A policy chooses its action from an internal state that represents the case. Suppose the cases (e, int) and (e, obs) have no action in common that maximises the utility of intelligence. Then no policy whose internal state omits r is optimal on both, so any optimal policy encodes a distinction between self-intervention and matched observation on which its action depends.

Proof. If the internal state omits r, the two cases are represented identically, so any policy reading that state chooses the same action in both. No action is optimal in both, so the policy is suboptimal in at least one. □

Definition 38 (chain notation)**.**
Suppose I have two organisms, $\mathfrak{a}$ (Alice) and $\mathfrak{b}$ (Bob). A causal identity $c_{\mathfrak{a}}^{\mathfrak{b}}$ denotes the causal identity that $\mathfrak{a}$ constructs for $\mathfrak{b}$: informally, what Alice takes Bob to intend or cause.

More generally, for a finite chain of agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_k$ I write

$$c_{\mathfrak{a}}^{\mathfrak{x}_1\mathfrak{x}_2\cdots\mathfrak{x}_k}$$

for Alice's nested causal identity in which $\mathfrak{x}_1$ models $\mathfrak{x}_2$ models … models $\mathfrak{x}_k$. For example, $c_{\mathfrak{a}}^{\mathfrak{ba}}$ denotes Alice's model of Bob's model of Alice, what Alice thinks Bob thinks Alice intends. The maximal chain depth that $\mathfrak{a}$ can represent is constrained by $\mathfrak{a}$'s vocabulary $\mathfrak{v}_{\mathfrak{a}}$. Finally, Bob need not be a biological organism. Bob can be anything for which Alice constructs a causal identity.

Definition 39 ($\mathrm{n}^{\mathbf{th}}$ order self)**.**
Fix an agent $\mathfrak{a}$. An $n^{\mathbf{th}}$-order self of $\mathfrak{a}$ is any causal identity whose chain ends at $\mathfrak{a}$ after interleaving $\mathfrak{a}$ with $n-1$ models of other agents.

Formally, let $c_{\mathfrak{a}}^{\mathfrak{a}}$ denote the 1ST-order-self from Definition 37, and set

$$\mathfrak{a}^1 := c_{\mathfrak{a}}^{\mathfrak{a}}.$$

For $n \geq 2$, choose agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_{n-1}$ (not necessarily distinct) and set

$$\mathfrak{a}^n := c_{\mathfrak{a}}^{\mathfrak{x}_1\mathfrak{a}\mathfrak{x}_2\mathfrak{a}\cdots\mathfrak{x}_{n-1}\mathfrak{a}}.$$

For example, relative to another agent $\mathfrak{b}$, I have

$$\mathfrak{a}^1 = c_{\mathfrak{a}}^{\mathfrak{a}}, \qquad \mathfrak{a}^2 = c_{\mathfrak{a}}^{\mathfrak{ba}}, \qquad \mathfrak{a}^3 = c_{\mathfrak{a}}^{\mathfrak{baba}}.$$

I use $\mathfrak{a}^n$ to denote an arbitrary $n^{\mathbf{th}}$-order self, and chain notation to denote a specific one, for example $c_{\mathfrak{a}}^{\mathfrak{ba}}$.

A 3RD-order-self predicts how another agent reads present commitments. The term also denotes the policy that the causal identity licenses to constrain future policies on that prediction, represented by Bind in Definition 44.

Proposition 32 ($\mathrm{n}^{\mathbf{th}}$ order self convergence)**.**
An organism $\mathfrak{o}$ has current history task $\mathfrak{h}_{<t_{\mathfrak{o}}}$ and viability task $\mu_{\mathfrak{o}}$ with

$$\mathfrak{h}_{<t_{\mathfrak{o}}} \sqsubset \mu_{\mathfrak{o}}.$$

Successful learning of $\mu_{\mathfrak{o}}$ from $\mathfrak{h}_{<t_{\mathfrak{o}}}$ yields a retained correct policy,

$$\pi^\star \in \mathfrak{p}_{\mathfrak{o}} \cap \Pi_{\mu_{\mathfrak{o}}}.$$

An $n^{\mathbf{th}}$-order self $\mathfrak{o}^n$ is expressible in $L_{\mathfrak{v}_{\mathfrak{o}}}$. The alignment precondition is that every correct policy for $\mu_{\mathfrak{o}}$ learnable from $\mathfrak{h}_{<t_{\mathfrak{o}}}$ entails that self:

$$T(\pi) \subseteq T(\mathfrak{o}^n) \qquad \text{for every such policy } \pi.$$

The learned policy therefore entails the self,

$$T(\pi^\star) \subseteq T(\mathfrak{o}^n).$$

If $\mathfrak{p}_{\mathfrak{o}}$ is closed under entailed commitments, meaning

$$\pi \in \mathfrak{p}_{\mathfrak{o}}, \quad c \in L_{\mathfrak{v}_{\mathfrak{o}}}, \quad T(\pi) \subseteq T(c) \quad \Longrightarrow \quad c \in \mathfrak{p}_{\mathfrak{o}},$$

then $\mathfrak{o}^n \in \mathfrak{p}_{\mathfrak{o}}$.

Proof. The learned policy $\pi^\star$ is one of the correct policies learnable from $\mathfrak{h}_{<t_{\mathfrak{o}}}$. The alignment precondition therefore gives $T(\pi^\star) \subseteq T(\mathfrak{o}^n)$. Since $\pi^\star \in \mathfrak{p}_{\mathfrak{o}}$ and $\mathfrak{o}^n \in L_{\mathfrak{v}_{\mathfrak{o}}}$, closure under entailed commitments gives $\mathfrak{o}^n \in \mathfrak{p}_{\mathfrak{o}}$. This implication holds for any such policy selected by freedom maximisation. □

Proposition 33 (Agreement of 1ST-order-selves)**.**
Let $c, c' \in \mathcal{C}(INT_{\mathfrak{o}}, OBS_{\mathfrak{o}})$. Then for every possible thing $e \in INT_{\mathfrak{o}} \cup OBS_{\mathfrak{o}}$,

$$T(e) \subseteq T(c) \quad \Longleftrightarrow \quad T(e) \subseteq T(c').$$

Thus every 1ST-order-self classifies the organism's interventions and its matched observations the same way.

Proof. If $e \in INT_{\mathfrak{o}}$, both inclusions are true by Definition 33. If $e \in OBS_{\mathfrak{o}}$, both inclusions are false. The two possible things c and c' therefore agree on every possible thing that defines the causal identity. □

Decoder mismatch (hypothesis). When every available mind hypothesis gives an observed trace a much smaller likelihood than the noise model, Bayes' rule lowers the posterior odds of each available mind hypothesis against noise. The suggested application to the Fermi paradox is a hypothesis about recognition across incompatible stacks [Bennett, 2026a].

18 Communication, meaning and commitment

Theorem 21 (A speaker needs a model of the listener)**.**
Let U be a set of utterances and M a set of meanings. A listener is characterised by a decoder $d : U \to M$, the function taking what it hears to what it understands. Let D be the set of decoders the speaker admits as possible. The speaker wants the listener to arrive at $m \in M$, and an utterance u works for decoder d when $d(u) = m$. For a speaker whose utterance is fixed before d is known, every utterance fails for some $d \in D$ if and only if no $u \in U$ has $d(u) = m$ for every $d \in D$. A speaker whose utterance can depend on d has, for every $d \in D$, an utterance that works for d if and only if every $d \in D$ has some u_d with $d(u_d) = m$.

Now take the binary case. Let $m \in \{0, 1\}$ be drawn from any distribution, and let $D = \{d_0, d_1\}$ where d_0 returns the bit it was sent and d_1 flips it. The listener's decoder is drawn uniformly from D, independently of m, and hidden from the speaker. The speaker sends one probe and

observes the listener's response before its final utterance, and the response is the bit the listener's decoder returns for the probe. A speaker whose final utterance is independent of that response succeeds with probability $\frac{1}{2}$, which is chance. A speaker whose final utterance depends on that response succeeds with probability 1 [Bennett, 2026a].

The state separating the two speakers represents how this listener maps utterances to meanings, and that is the functional role of a 2ND-order-self.

Proof. If a speaker's utterance is fixed before d is known, the speaker sends the same u to every listener. If no member of U works for every $d \in D$, then $d(u) \neq m$ for some $d \in D$, so that utterance fails for at least one member of D. Conversely, if some member of U works for every $d \in D$, a speaker that fixes it fails for no member of D. Suppose every $d \in D$ has some u_d with $d(u_d) = m$. If a speaker's utterance depends on d, the speaker sends u_d, and $d(u_d) = m$ by construction. Hence that speaker's utterance works for every member of D. Conversely, if no utterance works for some $d \in D$, then no speaker has an utterance that works for that d.

For the binary case, take a speaker whose final utterance s depends on m alone. Since $d_1(s) = 1 \oplus d_0(s)$, the occasions on which $d_0(s) = m$ are those on which $d_1(s) \neq m$. The decoder is drawn uniformly and independently of m, so, averaged over d_0 and d_1, the success probability is $\frac{1}{2}$ whatever distribution the speaker uses. Now let the speaker send probe $p = 0$ and read the response, which is 0 under d_0 and 1 under d_1. Hence the speaker identifies the decoder. The speaker sends m when the decoder is d_0 and $1 \oplus m$ when it is d_1, and the listener arrives at m in both cases. Telling the members of D apart is therefore what separates chance from certainty, and a state that does so represents how the listener will take the speaker's utterance. □

Corollary 26 (Deception requires the same model of the listener as truthful report)**.**
In either case of Theorem 21, suppose its conditions hold for a meaning m' that the speaker holds to be false, in place of m. Causing the listener to arrive at m' then requires the state that separates the two speakers in Theorem 21. Deception and report differ in which meaning the speaker wants the listener to arrive at, and call for the same model of the listener.

Proof. I replace m by m' throughout. If a speaker's utterance is fixed before d is known, the speaker fails for some $d \in D$ when no utterance works for every member of D. By contrast, if a speaker discriminates between the members of D, it chooses u_d with $d(u_d) = m'$. In the binary case a speaker plants m' with probability $\frac{1}{2}$ if it ignores the probe response, and with probability 1 if it reads that response. □

Corollary 27 (Human-like report requires 1ST and 2ND-order-selves)**.**
A speaker has **human-like report** when it conveys its intended meaning m with probability 1 whichever member of D is the listener's decoder. It must do so in every setting of Theorem 21 that admits such a speaker, including settings in which the listener also responds to other speakers with the same decoder. **Human-like access consciousness** is availability for reasoning and for human-like report.

1. In each such setting in which no utterance conveys m under every member of D, the utterance of a speaker with human-like report depends upon a state that takes different values for two members of D. Such a state represents how the listener takes the speaker's utterances, which is the functional role of the 2ND-order-self $c^{\mathfrak{ba}}_{\mathfrak{a}}$ of Definitions 38 and 39, with $\mathfrak{a}$ the speaker and $\mathfrak{b}$ the listener.

2. In the binary case, suppose instead that on some occasions the speaker sends its probe 0 and registers the listener's response to it, and on the other occasions the speaker is silent and registers the listener's response to another speaker's utterance 1. Suppose also that the kind of occasion is drawn independently of m and of the decoder, and that each kind has positive probability. Then some speaker conveys m with probability 1, and the

final utterance of every such speaker depends upon whether it sent its probe, which is the distinction between self-intervention and matched observation of Theorem 20. The probe is a possible thing of the speaker's embodied language that is true whenever the speaker sends it and false whenever the speaker is silent, so the speaker meets the differentiation precondition of the 1ST-order-self (Definition 37).

Hence human-like access consciousness requires a 1ST-order-self and a 2ND-order-self.

Proof. For the first item, suppose a speaker's utterance does not vary with the listener's decoder. The speaker then sends one utterance u to every listener, and since no utterance conveys m under every member of D, some $d \in D$ has $d(u) \neq m$ (Theorem 21). A speaker with human-like report therefore sends different utterances to two members of D, so the state from which it chooses its utterance takes different values for them.

For the second item, let y be the response the speaker registers and s its final utterance. After its own probe, $y = d(0)$, which is 1 if and only if the decoder flips, so s conveys m if and only if $s = m \oplus y$. After the other speaker's utterance, $y = d(1)$, which is 0 if and only if the decoder flips, so s conveys m if and only if $s = m \oplus y \oplus 1$. The decoder is uniform and independent of m and of the kind of occasion, so for every value of m with positive probability, each value of y has positive probability on each kind of occasion. For such m and y, the two kinds of occasion form a matched pair of cases of Theorem 20, with no final utterance that conveys m in both. Hence a speaker whose final utterance does not depend upon whether it sent its probe fails with positive probability. A speaker that sends $m \oplus y$ after its own probe and $m \oplus y \oplus 1$ otherwise conveys m with probability 1. A speaker whose viability depends upon human-like report therefore meets the alignment precondition of Definition 37.

Let p be the probe, let INT be the set of the speaker's interventions, each of which contains p, and let OBS be the set of matched observations, each of which is true given some distinction in which the speaker is silent. Then $T(int) \subseteq T(p)$ for every $int \in INT$. The probe is false whenever the speaker is silent, so $T(obs) \nsubseteq T(p)$ for every $obs \in OBS$, and p is nonempty because the empty possible thing is true given every distinction. The probe is an output of the speaker, so it is a possible thing of the speaker's embodied language. Hence $p \in \mathcal{C}(INT, OBS)$ by Definition 33, so $\mathcal{C}(INT, OBS) \neq \emptyset$, which is the differentiation precondition of Definition 37. □

Definition 40 (protosymbol system)**.**
Assume an organism $\mathfrak{o}$. For each policy $p \in \mathfrak{p}_\mathfrak{o}$ there exists a set $\mathfrak{s}_p = \{\alpha \in \Gamma_{\mathfrak{v}_\mathfrak{o}} : p \in \Pi_\alpha\}$ of all tasks for which p is a correct policy. The union of all such sets is

$$\mathfrak{s}_\mathfrak{o} = \bigcup_{p \in \mathfrak{p}_\mathfrak{o}} \{\alpha \in \Gamma_{\mathfrak{v}_\mathfrak{o}} : p \in \Pi_\alpha\}$$

We call $\mathfrak{s}_\mathfrak{o}$ a "protosymbol system". A $\mathfrak{v}$-task $\alpha \in \mathfrak{s}_\mathfrak{o}$ is called a "protosymbol".

Definition 41 (interpretation)**.**
For each input $i \in L_{\mathfrak{v}_\mathfrak{o}}$, define the protosymbols signified by i as

$$\mathfrak{S}_\mathfrak{o}(i) := \{\, \alpha \in \mathfrak{s}_\mathfrak{o} \mid i \in I_\alpha \,\}.$$

Whenever $\mathfrak{S}_\mathfrak{o}(i)$ is finite and nonempty, restrict $<_\mathfrak{o}$ to a total preorder on this set and fix a tie-breaker for its maximal elements. The organism selects

$$\alpha^\star \in \max_{<_\mathfrak{o}} \mathfrak{S}_\mathfrak{o}(i),$$

chooses a policy known to be correct, $p_{\alpha^\star} \in \mathfrak{p}_\mathfrak{o} \cap \Pi_{\alpha^\star}$, and infers an output

$$o \in \mathrm{Ext}(i) \cap \mathrm{Ext}(p_{\alpha^\star}).$$

If $\mathfrak{S}_{\mathfrak{o}}(i) = \emptyset$, no retained correct policy exists, or $\mathrm{Ext}(i) \cap \mathrm{Ext}(p_{\alpha^\star}) = \emptyset$, interpretation fails on input i. Otherwise $\alpha^\star$ is the **interpretation** of i and o is the interpreted output.

Definition 42 (affect)**.**
Suppose organisms $\mathfrak{a}$ and $\mathfrak{b}$ inhabit the same environment. Suppose $\mathfrak{a}$ interprets input $i \in L_{\mathfrak{v}_{\mathfrak{a}}}$ as output o. For a part $v \subseteq i$ of the input, I write $i \setminus v$ for the input obtained by deleting the possible facts in v. If $\mathfrak{b}$ makes output k, let $k^{\mathfrak{a}} \in L_{\mathfrak{v}_{\mathfrak{a}}}$ be the part of k registered in $\mathfrak{a}$'s vocabulary.

- A **possible thing** $v \subseteq i$ affects $\mathfrak{a}$ on i when interpreting $i \setminus v$ produces an output different from o.
- The **organism** $\mathfrak{b}$ affects $\mathfrak{a}$ on i by making k when some $v \subseteq i \cap k^{\mathfrak{a}}$ affects $\mathfrak{a}$ on i.

Definition 43 (meaning)**.**
The **meaning** an organism $\mathfrak{o}$ ascribes to an input i is the protosymbol $\alpha^\star \in \mathfrak{s}_{\mathfrak{o}}$ selected in Definition 41.

Assume organism $\mathfrak{a}$ chooses output u on input $i_{\mathfrak{a}}$, and organism $\mathfrak{b}$ subsequently interprets input $i_{\mathfrak{b}}$ and produces output $o_{\mathfrak{b}}$. Let $o_{\mathfrak{b}}^{\mathfrak{a}} \in L_{\mathfrak{v}_{\mathfrak{a}}}$ be the part of Bob's output expressible in Alice's vocabulary. Alice **means** protosymbol $\alpha \in \mathfrak{s}_{\mathfrak{a}}$ by choosing u when she intends all three conditions below.

1. Bob interprets his input and $o_{\mathfrak{b}}^{\mathfrak{a}} \in O_\alpha$.
2. Bob recognises Alice's intention in the operational sense that Alice's output affects his input and hence his interpretation.
3. Bob produces an output registered in O_α because of that recognition. Had Alice not chosen u, Bob would not have produced an output whose expressible part is in O_α.

Meaningful communication therefore requires the organisms to affect one another, Bob's response to be legible in Alice's vocabulary, and their learned protosymbol systems to permit the intended response. With similar bodies, experiences, and preferences, these conditions are easier to meet.

Proposition 34 (Cooperation with advance detection)**.**
An infinitely repeated Prisoner's Dilemma has payoffs $T > R > P > S$, discount factor $\beta \in [0, 1)$ and probability $q \in [0, 1]$ that planned betrayal is inferred before the other player cooperates. Both players cooperate until betrayal or exploitative intent is detected and then defect forever, with no false detection of cooperative intent. Cooperation forever is a best response when

$$R - P \geq (1 - \beta)(1 - q)(T - P).$$

Proof. The discounted payoff from cooperation is $V_C = R/(1 - \beta)$. A first planned defection yields expected payoff

$$V_D = qP + (1 - q)T + \frac{\beta P}{1 - \beta} = \frac{P}{1 - \beta} + (1 - q)(T - P).$$

Thus $V_C \geq V_D$ is the stated inequality. At a later first defection, the same payoff difference is discounted, and mutual defection is a best response after detection because $P > S$. □

Mirror Symbol Hypothesis. An organism uses its own preferences and protosymbol system as a prior for another organism's causal identity, then adjusts that prediction through interaction. Shared embodied preferences make this reuse a proposed basis for empathy and social learning [Bennett and Maruyama, 2022].

Definition 44 (A two-stage trust/commitment game)**.**
Fix a commitment cost c with $0 < c < 1$. Two agents $\mathfrak{a}$ (Alice, the actor) and $\mathfrak{b}$ (Bob, the audience) play the following extensive-form game.

1. **Stage 0 (self-binding).** $\mathfrak{a}$ chooses either *Bind* (B) or *Free* (F). Choosing B incurs an immediate cost c to $\mathfrak{a}$ and removes $\mathfrak{a}$'s ability to *exploit* in Stage 2. Choosing F incurs no cost and leaves Stage 2 unconstrained.

2. **Stage 1 (audience response).** $\mathfrak{b}$ observes $\mathfrak{a}$'s choice (B or F) and chooses either *Trust* (Y) or *Do not trust* (N). If $\mathfrak{b}$ chooses N, the game ends with payoff $(0, 0)$ after F and $(-c, 0)$ after B, because the binding cost was already paid.

3. **Stage 2 (act).** If $\mathfrak{b}$ chose Y, then if $\mathfrak{a}$ chose B in Stage 0, $\mathfrak{a}$ must play *Honour* (K). If $\mathfrak{a}$ chose F in Stage 0, $\mathfrak{a}$ chooses either K or *Exploit* (E). Payoffs before subtracting the binding cost are $u_{\mathfrak{a}}(Y, K) = 1$, $u_{\mathfrak{b}}(Y, K) = 1$, $u_{\mathfrak{a}}(Y, E) = 2$, $u_{\mathfrak{b}}(Y, E) = -1$. If $\mathfrak{a}$ chose B in Stage 0, then c is subtracted from $\mathfrak{a}$'s payoff.

Theorem 22 (Separating audience modelling from self-binding)**.**
A subgame-perfect equilibrium is a choice of strategy for each player such that, from every stage of the game onward, no player can gain by changing strategy alone. In the game of Definition 44, the following hold.

1. If the self-binding move B is not available ($\mathfrak{a}$ can only play F), then the unique subgame-perfect equilibrium outcome is N with payoff $(0, 0)$.

2. If B is available and $0 < c < 1$, then in the unique subgame-perfect equilibrium $\mathfrak{a}$ plays B, $\mathfrak{b}$ plays Y after observing B (and N after observing F), and the Stage 2 outcome is necessarily K, yielding payoff $(1 - c, 1)$.

Proof. (1) If B is unavailable, then after $\mathfrak{b}$ chooses Y the actor $\mathfrak{a}$ chooses between K and E and strictly prefers E because $2 > 1$. Anticipating this, $\mathfrak{b}$ strictly prefers N to Y because $0 > -1$. Hence the unique subgame-perfect equilibrium outcome is N.

(2) Now allow B. In the subgame following (F, Y), $\mathfrak{a}$ plays E because $2 > 1$, so $\mathfrak{b}$'s best response after observing F is N. In the subgame following $(B, \cdot)$, $\mathfrak{a}$ must play K, so $\mathfrak{b}$'s best response after observing B is Y because $1 > 0$. Anticipating these best responses, $\mathfrak{a}$ compares Stage 0 payoffs. Choosing F yields 0 (since $\mathfrak{b}$ chooses N), whereas choosing B yields $1 - c$. Because $0 < c < 1$, we have $1 - c > 0$, so $\mathfrak{a}$ strictly prefers B. Every best response in this induction is strict, so the subgame-perfect equilibrium is unique. □ □

19 Collective identity and embodied goals

The parts of a collective have $\mathfrak{v}$-tasks $\omega_1, \ldots, \omega_m$ and correct-policy sets $\Pi_{\omega_j} \subseteq L_{\mathfrak{v}}$.

Definition 45 (Restricted policies, over-constraint, and splinters)**.**
The **collective policies** are $\Pi_\omega := \bigcap_{j=1}^m \Pi_{\omega_j}$. The collective has an identity at $\mathfrak{v}$ iff $\Pi_\omega \neq \emptyset$. To model top-down boundary conditions, fix restricted policy sets $\Pi'_{\omega_j} \subseteq \Pi_{\omega_j}$ and define the restricted collective policies $\Pi'_\omega := \bigcap_{j=1}^m \Pi'_{\omega_j}$. The system is **over-constrained at $\mathfrak{v}$** if $\Pi'_\omega = \emptyset$. A **splinter** is a nonempty index set $S \subseteq \{1, \ldots, m\}$ such that $\bigcap_{j \in S} \Pi'_{\omega_j} \neq \emptyset$.

Proposition 35 (Over-constraint implies splintering)**.**
Assume at least one part still has a correct policy ($\Pi'_{\omega_j} \neq \emptyset$ for some j). A policy is feasible for part j when it lies in Π'_{ω_j}. If the system is over-constrained ($\Pi'_\omega = \emptyset$), then no policy is feasible for every part. At least one proper splinter exists, meaning a splinter that excludes at least one part. For any realised policy $\pi \in L_{\mathfrak{v}}$ with $S(\pi) := \{j : \pi \in \Pi'_{\omega_j}\} \neq \emptyset$, the set $S(\pi)$ is a proper splinter.

Proof. If some π were feasible for all parts then $\pi \in \bigcap_{j=1}^m \Pi'_{\omega_j} = \Pi'_\omega$, contradicting $\Pi'_\omega = \emptyset$. Thus no realised policy is feasible for every part. Since at least one part still has a correct policy,

choose j with $\Pi'_{\omega_j} \neq \emptyset$ and choose $\pi \in \Pi'_{\omega_j}$. Then $S(\pi)$ is nonempty and $\pi \in \bigcap_{j \in S(\pi)} \Pi'_{\omega_j}$, so its restricted policy intersection is nonempty. Since no policy is feasible for every part, $S(\pi)$ is a proper subset of $\{1, \ldots, m\}$. Hence $S(\pi)$ is a proper splinter. □

Corollary 28 (Incompatible maximal splinters)**.**
Suppose the parts of a collective share no policy after restriction, so $\Pi'_\omega = \emptyset$, and every part still has at least one policy of its own, so $\Pi'_{\omega_j} \neq \emptyset$ for each j. Then every part lies in a maximal splinter that excludes at least one part of the former collective. This means there are at least two maximal splinters, and no policy is shared across the members of any two maximal splinters, so for distinct maximal splinters S and R,

$$\bigcap_{j \in S \cup R} \Pi'_{\omega_j} = \emptyset.$$

Proof. Each part has a policy of its own, so every singleton $\{j\}$ is a splinter. I begin with any singleton and add parts for as long as the members still share a policy. There are finitely many parts, so this stops, and it stops at a maximal splinter. The maximal splinter leaves someone out, since a splinter containing every part would require a policy in Π'_ω, which is empty. Were there only one maximal splinter, every part would lie in it, so it would contain every part, which has just been ruled out. Finally, if two different maximal splinters S and R shared a policy across all their members, then $S \cup R$ would be a splinter larger than both, and neither could have been maximal. □

Proposition 36 (Shared representations need every distinction)**.**
Let parts $1, \ldots, m$ of a collective receive the same inputs Z, and suppose each part's task has one correct output for each input. A single representation of those inputs serves every part, each interpreting using a decoder of its own. A part must discriminate between two inputs when they are associated with different correct outputs. If part j is correct on inputs z and z' and must discriminate between them, then the representation distinguishes z from z'. There are decoders under which each and every part is correct on every input if and only if the representation is sufficient to distinguish between any two inputs that one or more parts must discriminate between.

Proof. Suppose the representation failed to distinguish between two inputs z and z' that part j must discriminate between. Part j's decoder has only the representation to go on, so it produces the same output for z and z', and the resulting output can be correct for at most one of them. Part j is therefore wrong on z or on z'. If every part is correct on every input, the representation distinguishes every pair that some part must discriminate between. Conversely, suppose the representation distinguishes every pair that some part must discriminate between. Give part j the decoder that returns, for each representation, the correct output of any input having that representation. If two inputs with the same representation had different correct outputs for part j, part j would have to discriminate between them and the representation would distinguish them. So the decoder is well defined, and it is correct on every input. □

Proposition 37 (Embodied notions of success are at odds with strong orthogonality)**.**
Let $\mathcal{C}$ be a nonempty set of software states, let Γ be a nonempty set of behaviours, let $f_1 \in \mathcal{C}$ be a software state, let $f_2 : \mathcal{C} \to \Gamma$ be the body or interpreter, and let $f_3 : \Gamma \to \{0, 1\}$ be the goal environment. Assume f_3 is nontrivial, so some behaviour succeeds and some behaviour fails. Then success is a function neither of f_1 alone, nor of (f_1, f_3), nor of (f_1, f_2). It is a property of the whole triple (f_1, f_2, f_3).

Proof. I choose behaviours γ_+ and γ_- with $f_3(\gamma_+) = 1$ and $f_3(\gamma_-) = 0$. There is a body f_2^+ with $f_2^+(c) = \gamma_+$ for every $c \in \mathcal{C}$, so $f_3(f_2^+(f_1)) = 1$. There is also a body f_2^- with $f_2^-(c) = \gamma_-$ for every $c \in \mathcal{C}$, so $f_3(f_2^-(f_1)) = 0$. Neither the software state f_1 nor the goal environment f_3 has

changed, but success has changed because the body changed. Thus success is a function neither of f_1 alone nor of (f_1, f_3). Conversely, let $\gamma = f_2(f_1)$ be any realised behaviour. This behaviour succeeds under a goal environment $f : \Gamma \to \{0, 1\}$ if and only if $f(\gamma) = 1$. The admissible goals therefore depend on the realised behaviour. Goal environments with $f(\gamma) = 1$ and with $f(\gamma) = 0$ both exist, so with f_1 and f_2 fixed, success changes with the goal environment and is not a function of (f_1, f_2). Success is determined by software, body, and goal environment together. □

No selves, no alignment. The necessity claims concern conduct whose correctness depends on causal responsibility, audience-relative interpretation, or commitments maintained across change. Theorems 20, 21 and 22 establish the respective separations. The embodied orthogonality proposition addresses the strong factorisation of successful goals from bodily feasibility.

Recursive self impairment (conjecture). Self-binding in an unhealthy collective may propagate corrupted shared meanings and higher-order identities into its splinters and their parts, recursively impairing coordination.

20 Consciousness and temporal conditions

Definition 46 (levels of consciousness)**.**

1. an organism that acts but does not learn, meaning $\mathfrak{p}_\mathfrak{o}$ is fixed from birth.
2. an organism that learns, but $\mathfrak{o}^1 \notin \mathfrak{p}_\mathfrak{o}$ either because $\mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o}) = \emptyset$ (failing the "differentiation precondition") or because the organism's viability does not hinge upon $\mathfrak{o}^1$ (failing the "alignment precondition").
3. reafference and phenomenal or core consciousness are achieved when $\mathfrak{o}^1 \in \mathfrak{p}_\mathfrak{o}$ is learned by an organism as a consequence of attraction to and repulsion from possible things in $L_{\mathfrak{v}_\mathfrak{o}}$.
4. (a) access or self reflexive consciousness is achieved when $\mathfrak{o}^2 \in \mathfrak{p}_\mathfrak{o}$.
   (b) hard consciousness [Boltuc, 2012] is achieved when a phenomenally conscious organism learns a 2ND-order-self (an organism is consciously aware of the contents of 2ND-order-selves, which must have quality if learned through phenomenal consciousness).
5. meta self reflexive consciousness (human level hard consciousness) is achieved when $\mathfrak{o}^3 \in \mathfrak{p}_\mathfrak{o}$.

Proposition 38 (Arpeggio shortens tenure, and thus threatens viability)**.**
Fix a finite vocabulary $\mathfrak{v}$, viability μ, a policy $\pi \in L_\mathfrak{v}$ with $\Omega_\pi \neq \emptyset$, and a moment $\ell = \{p_1, \ldots, p_n\} \in L_\mathfrak{v}$ with $n \geq 2$. A **record** of a content p_j is a possible fact $r_j \in \mathfrak{v}$ with $r_j \not\subseteq p_j$, so that r_j can be true while p_j is false. A chordal realisation of ℓ is a time at which every content is true. When the contents become true at the same time, I take it to add no possible fact to π, because they are true together and nothing need be held across ticks. An arpeggiated realisation of ℓ over a fragment $\tau_{[t_0,t_1]}$ is one in which no distinction of the fragment satisfies every content. The system therefore holds a record r_j of each content p_j other than the last to occur. Each record is held from the tick at which p_j occurs to the tick at which the last content occurs. Let $R = \{r_1, \ldots, r_{n-1}\}$ and let $\pi_A = \pi \cup R$ be the arpeggiating policy, read with $T(\pi_A) = \emptyset$ when $\pi_A \notin L_\mathfrak{v}$. Then, with R the records or any other finite subset of $\mathfrak{v}$,

1. for every timeline τ and interval $[u_1, u_2]$, $\rho_{\tau,[u_1,u_2]}(\pi_A) \leq \rho_{\tau,[u_1,u_2]}(\pi)$[24], with strict inequality if and only if the timeline visits a distinction in $T(\pi) \setminus T(\pi_A)$ during the interval;
2. $\Omega_{\pi_A} \subseteq \Omega_\pi$, so $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) \geq F_2(\pi)$. If some $r \in R$ is not required by viability inside π, meaning some $o \in \Omega_\pi$ has $T(o) \not\subseteq r$, then $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) > F_2(\pi)$;

[24]Here ρ is tenure, the period of objective time for which a thing persists.

3. the inequality of the first part is unchanged when π and R are replaced by their groundings at any lower layer of a finite chain.

Hence a system able to hold every content of ℓ at the same time must have concurrency capacity of at least n (Proposition 40). Such a system is never worse off realising ℓ as a chord whose contents become true at the same time, and is strictly better off whenever either inequality is strict.

Proof. The first part is Corollary 4 with $l = \pi$ and $c = R$, since $\pi_A = \pi \cup R$. For the second part, the set Ω_{π_A} is empty when $\pi_A \notin L_{\mathfrak{v}}$. Every member of this set is a possible thing containing π and μ, so $\Omega_{\pi_A} \subseteq \Omega_\pi$ and $F_2(\pi_A) \geq F_2(\pi)$ whenever $\Omega_{\pi_A} \neq \emptyset$. For every $r \in R$, $T(R) \subseteq r$, so the $o \in \Omega_\pi$ with $T(o) \nsubseteq r$ supposed in the second part has $T(o) \nsubseteq T(R)$, which is to say R is not required by viability inside π. Hence, by Theorem 18 with the thing $c = R$, $\Omega_{\pi_A} = \emptyset$ or $F_2(\pi_A) > F_2(\pi)$. The third part is Corollary 2 with $l = \pi$ and the thing $c = R$. For the final claim, a chordal realisation adds nothing to π when the contents of ℓ become true at the same time. Proposition 40 shows that a capacity below n forbids such a realisation. So a system has capacity at least n when every content can be true in it at the same time, and then nothing forces the records. The first two parts then rank π at least as high as π_A by tenure and by F_2, and strictly higher whenever either inequality is strict. □

Definition 47 (Arpeggio and Chord)**.**
Let $\ell = \{p_1, \ldots, p_n\} \subseteq \mathfrak{v}$ be a moment with $n \geq 1$, let $\tau : I \to \Phi$ be a timeline, and let $\tau_{[t_0,t_1]}$ be a finite timeline fragment of τ. Arpeggio holds when

$$\forall i \in \{1, \ldots, n\}\ \exists u_i \in \{t_0, \ldots, t_1\} \quad \tau(u_i) \in p_i.$$

I call ℓ **co-instantiated** in $\tau_{[t_0,t_1]}$ when

$$\exists u \in \{t_0, \ldots, t_1\} \quad \tau(u) \in T(\ell).$$

Co-instantiation requires $T(\ell) \neq \emptyset$, so a co-instantiated moment is a possible thing, $\ell \in L_{\mathfrak{v}}$, while an Arpeggio need not be. Chord holds when the following three conditions are true:

1. The moment ℓ is co-instantiated in $\tau_{[t_0,t_1]}$.
2. The size of ℓ satisfies $|\ell| \leq c_{\text{conc}}$, where c_{conc} is the concurrency capacity.
3. The sites of ℓ interact within the time window.

Concurrency capacity and interaction within the time window are defined in Definition 48 below.

Definition 48 (Sites and interaction)**.**
Let $\ell \subseteq \mathfrak{v}$ be a **moment**, a thing. Sites_ℓ is the nonempty finite set of physical sites that instantiate the contents of ℓ at an arbitrarily chosen abstraction layer, the one at which the test is made. The **interaction graph** $\text{Graph}_\ell = (\text{Sites}_\ell, \text{Edges}_\ell)$ has vertex set Sites_ℓ. An edge identifies two sites that the tested binding mechanism requires to **interact**, each sending a signal that the other receives[25]. Let $\tau_{[t_0,t_1]}$ be a finite timeline fragment in the sense of Definition 2, the restriction of the timeline τ to the ticks $t_0 \leq u \leq t_1$. A **physical model** is a layer with the clock of Theorem 10 and a unit for that clock, in the sense of Section 27. The model assigns a duration to each span of its own clock, while a tick of objective time is a position in the order of changes, and thus has no duration. The **time window** of the fragment is the span of that layer time across $\tau_{[t_0,t_1]}$. The window's duration is θ, calculated by multiplying the number of ticks

[25]This is a minimum standard. They must at least be able to interact. The nature of that interaction is unclear. I merely present a bound to which conscious systems must necessarily conform, but necessity is an easier standard to meet than sufficiency in this case. Co-instantiation and interaction of contents suggests synchronisation.

of the model's subjective time across the fragment by the unit. A required edge is completed within the time window when both of its signals are sent and received inside it. The sites of ℓ **interact within the time window** when Graph_ℓ is connected and every required edge is completed within the time window.

The **concurrency capacity** c_{conc} is the largest number of contents the physical model can instantiate together at that abstraction layer.

Proposition 39 (Temporal Gap separation)**.**
For every ℓ, τ and $\tau_{[t_0,t_1]}$, co-instantiation implies Arpeggio. Some ℓ, τ and $\tau_{[t_0,t_1]}$ satisfy Arpeggio without co-instantiation, and so without Chord.

Proof. If some tick u satisfies every p_i, I choose $u_i = u$ for every content, and Arpeggio follows. For the second claim, Theorem 23 gives a moment that satisfies Arpeggio without co-instantiation in any fragment that visits two different distinctions and misses a third. The fragment (a, b) with $p_1 = \{a, c\}$ and $p_2 = \{b, c\}$ is that theorem's case $n = 2$. □

Proposition 40 (Persistence and capacity)**.**
Let $\ell = \{p_1, \ldots, p_n\} \subseteq \mathfrak{v}$ be a moment, let $\tau : I \to \Phi$ be a timeline, and let $\tau_{[t_0,t_1]}$ be a finite timeline fragment of τ.

1. Suppose each p_i is true at some tick $u_i \in \{t_0, \ldots, t_1\}$ and, once true, remains true through $u^* = \max_i u_i$, the last of these ticks. Then every content is true at u^*.

2. If at each time of the fragment at most $c < n$ contents of ℓ are true, then there is no time of the fragment at which every content is true.

Proof. For persistent contents, let $u^* = \max_i u_i$. Every content has become true by u^*, and under persistence each one remains true through u^*, so $\tau(u^*) \in T(\ell)$. Under concurrency limit $c < n$, any time satisfying all n contents would violate that limit, so no such time exists. □

For a moment $g \subseteq \mathfrak{v}$ and a present possible thing $r \in L_{\mathfrak{v}}$, if $T(r) \subseteq T(g)$, then every content of g is presently true. If $T(r) \nsubseteq T(g)$, some distinction satisfies r without g. An inert record given such a distinction instantiates a different moment. If $T(g) = \emptyset$, no possible thing r can entail g. Persistence of the contents themselves, as in Proposition 40, can close the temporal gap; coexistence of inert records alone does not.

Corollary 29 (Subject and moment)**.**
Fix a timeline $\tau : I \to \Phi$ and a finite timeline fragment $\tau_{[t_0,t_1]}$. Fix a finite chain as in Theorem 1. Let g_s be the grounding of a 1ST-order-self at the lowest layer of the chain and let g_c be the grounding of a moment at that layer. Then

$$T(g_s \cup g_c) = T(g_s) \cap T(g_c).$$

The joint claim that this subject currently undergoes this moment is co-instantiated in $\tau_{[t_0,t_1]}$ when

$$\exists u \in \{t_0, \ldots, t_1\} \quad \tau(u) \in T(g_s) \cap T(g_c).$$

Separate occurrence of g_s and g_c at different times does not imply $\exists u \in \{t_0, \ldots, t_1\} : \tau(u) \in T(g_s) \cap T(g_c)$. Let r be the grounding at that layer of an inert representation of the moment. The distinction at a tick $u \in \{t_0, \ldots, t_1\}$ can then instantiate the subject together with the representation, while failing to instantiate the subject together with the original moment. This happens if and only if

$$\tau(u) \in T(g_s) \cap T(r) \setminus T(g_c).$$

Neither condition changes its truth value if g_s, g_c and r are replaced by the self, the moment and the representation, or by their groundings at any layer of the chain.

Proof. By Definition 4, $T(g_s \cup g_c) = T(g_s) \cap T(g_c)$ for the things g_s and g_c, so co-instantiation means that some tick $u \in \{t_0, \ldots, t_1\}$ has $\tau(u) \in T(g_s) \cap T(g_c)$. By Proposition 39, separate occurrence is insufficient. Membership $\tau(u) \in T(g_s) \cap T(r) \setminus T(g_c)$ means subject and representation occur at u while the original moment does not. By Theorem 1, the self, the moment and the representation have the truth sets $T(g_s)$, $T(g_c)$ and $T(r)$, as do their groundings at every layer of the chain. Hence both conditions have the same truth value at every layer. □

Theorem 23 (Arpeggio collapse)**.**
Let a finite timeline fragment $\tau_{[t_0,t_1]}$ visit $n \geq 2$ different distinctions $x_1, \ldots, x_n$, suppose there is a distinction x_* that it does not visit, and let the vocabulary contain the possible facts $p_i = \{x_i, x_*\}$. Then the moment $\ell = \{p_1, \ldots, p_n\}$ has $T(\ell) = \{x_*\}$, so $\ell \in L_{\mathfrak{v}}$, and satisfies Arpeggio in $\tau_{[t_0,t_1]}$ while co-instantiation fails. So if Arpeggio suffices, ℓ qualifies as an Arpeggiated consciousness, the contents of which are not instantiated by any one tick within the fragment.

Proof. For each visited distinction x_i, the possible fact $p_i = \{x_i, x_*\}$ is in the vocabulary by hypothesis, so ℓ is a moment. At the tick at which the fragment visits x_i, possible fact p_i is true, so every content of ℓ is true at some time of the fragment, which is Arpeggio. Because the visited distinctions are distinct and $n \geq 2$, no visited distinction lies in all possible facts p_i, while x_* lies in all of them, so

$$\bigcap_{i=1}^{n} p_i = \{x_*\}.$$

So $T(\ell) = \{x_*\}$ and $\ell \in L_{\mathfrak{v}}$. Since the fragment does not visit x_*, no distinction of the fragment satisfies every content, so co-instantiation is false in this fragment. Yet if Arpeggio suffices, ℓ qualifies as an Arpeggiated consciousness. □

Layer A has finite vocabulary $\mathfrak{v}_A$, intervention and matched-observation sets $INT_A, OBS_A \subseteq L_{\mathfrak{v}_A}$, and $\mathcal{C}_A := \mathcal{C}(INT_A, OBS_A)$ as in Definition 33. Enlarging to $\mathfrak{v} = \mathfrak{v}_A \cup \{q_X\}$ preserves old possible things and causal identities, because satisfiability and the identity conditions depend only on truth sets. Freedom is recomputed in the enlarged language, so its maximising identity may change.

Theorem 24 (Arpeggiated inclusion drift)**.**
Let A have a causal identity $c_A \in \mathcal{C}_A$, let $q_X \in \mathcal{P}$ be a possible fact of another part X, and let $\mathfrak{v} := \mathfrak{v}_A \cup \{q_X\}$. Choose

$$c^\star \in \arg \max_{c \in \mathcal{C}(INT_A, OBS_A)} |\operatorname{Ext}(c)|,$$

with the causal identities and extensions taken in $L_{\mathfrak{v}}$. Under the differentiation and alignment preconditions of Definition 37, $c^\star$ is a 1ST-order-self. Let a timeline τ visit $T(int_A)$ for some $int_A \in INT_A$ and visit q_X. Then for every $c \in \mathcal{C}(INT_A, OBS_A)$, and in particular for $c^\star$, the moment

$$\ell := c \cup \{q_X\} \subseteq \mathfrak{v}$$

satisfies Arpeggio in every finite fragment of τ containing both visits. Hence, if Arpeggio suffices, X is included in an Arpeggiated consciousness, although X need not have a 1ST-order-self and q_X need never be true at a time at which c is true.

Proof. Since $L_{\mathfrak{v}_A} \subseteq L_{\mathfrak{v}}$ and the conditions of Definition 33 depend only on truth sets, c_A is a causal identity in $L_{\mathfrak{v}}$. The set of causal identities is therefore nonempty. It is finite because $\mathfrak{v}$ is finite, so $c^\star$ exists. By the first condition of Definition 33, $T(int_A) \subseteq T(c)$ for every $c \in \mathcal{C}(INT_A, OBS_A)$. So every possible fact of c is true at the tick u_1 at which τ visits $T(int_A)$, and the possible fact q_X is true at the tick u_2 at which τ visits q_X. Any finite fragment $\tau_{[t_0,t_1]}$ with $t_0 \leq \min\{u_1, u_2\}$ and $t_1 \geq \max\{u_1, u_2\}$ contains both ticks, so every content of ℓ is true at some time of the fragment, which is Arpeggio. If Arpeggio suffices, ℓ qualifies as an Arpeggiated consciousness.

No step depends on a causal identity for X, and no step depends on a time at which c and q_X are true together. □

Definition 49 (Distances and signal speed)**.**
Let ℓ be a moment with sites Sites_ℓ and interaction graph $\mathrm{Graph}_\ell = (\mathrm{Sites}_\ell, \mathrm{Edges}_\ell)$ from Definition 48. Let $d(x, y)$ be the physical distance between sites. The diameter of ℓ is

$$D(\ell) = \max_{x,y \in \mathrm{Sites}_\ell} d(x, y).$$

Let $v > 0$ be the maximum signal speed. If Graph_ℓ is connected, let $\mathrm{hops}(x, y)$ be the length in edges of a shortest path from x to y. The hop diameter is

$$\mathrm{hop}(\mathrm{Graph}_\ell) = \max_{x,y \in \mathrm{Sites}_\ell} \mathrm{hops}(x, y).$$

The length of the longest required edge is

$$L_{\max}(\mathrm{Graph}_\ell, d) := \begin{cases} 0, & \mathrm{Edges}_\ell = \emptyset, \\ \max_{\{x,y\} \in \mathrm{Edges}_\ell} d(x, y), & \mathrm{Edges}_\ell \neq \emptyset. \end{cases}$$

Theorem 25 (Edge length and spacetime diameter bounds)**.**
Let ℓ be a moment over a finite timeline fragment $\tau_{[t_0,t_1]}$, and let θ be the duration of the fragment's window in the chosen physical model. Suppose the following conditions are true:

1. Every required edge in Graph_ℓ supports a signal in each direction, each sent and received inside the fragment's window.
2. The physical model registers the emission and receipt of each such signal.
3. No relevant signal travels faster than v in the model's units.

Then

$$L_{\max}(\mathrm{Graph}_\ell, d) \leq v\theta,$$

equivalently

$$\theta \geq \frac{L_{\max}(\mathrm{Graph}_\ell, d)}{v}.$$

If Graph_ℓ is connected, then using only the shape of the interaction graph I obtain

$$D(\ell) \leq \mathrm{hop}(\mathrm{Graph}_\ell)\, v\theta.$$

Proof. For any edge $\{a, b\} \in \mathrm{Edges}_\ell$, inside the time window a sends a signal that b receives. So its travel time $t_{a\to b} \geq 0$ satisfies

$$t_{a\to b} \leq \theta.$$

Since no signal travels faster than v, the trip takes at least $d(a, b)/v$. Hence

$$d(a, b)/v \leq t_{a\to b} \leq \theta,$$

so the edge length satisfies

$$d(a, b) \leq v\theta.$$

Thus $L_{\max}(\mathrm{Graph}_\ell, d) \leq v\theta$ whenever $\mathrm{Edges}_\ell \neq \emptyset$, and trivially otherwise, since then $L_{\max}(\mathrm{Graph}_\ell, d) = 0$. Now suppose Graph_ℓ is connected. For any two sites $x, y \in \mathrm{Sites}_\ell$, let

$$x = x_0, x_1, \ldots, x_k = y$$

be a shortest path in Graph_ℓ, so $k = \mathrm{hops}(x, y) \leq \mathrm{hop}(\mathrm{Graph}_\ell)$. By the triangle inequality,

$$d(x, y) \leq \sum_{j=0}^{k-1} d(x_j, x_{j+1}) \leq k\, v\theta \leq \mathrm{hop}(\mathrm{Graph}_\ell)\, v\theta.$$

Taking the maximum over $x, y \in \mathrm{Sites}_\ell$, I obtain

$$D(\ell) \leq \mathrm{hop}(\mathrm{Graph}_\ell)\, v\theta.$$

□

For a complete interaction graph with at least two sites, $\mathrm{hop}(\mathrm{Graph}_\ell) = 1$ and $D(\ell) \leq v\theta$. For a hub graph with at least three sites, $\mathrm{hop}(\mathrm{Graph}_\ell) = 2$ and $D(\ell) \leq 2v\theta$. A Chordal moment's required sites must belong to one connected interaction component within the window.

Corollary 30 (Feasible window)**.**
Let ℓ be a moment over a finite timeline fragment $\tau_{[t_0,t_1]}$ in a chosen physical model, and let θ be the duration of the fragment's window. Let $\mathcal{R}$ be a set of permitted timings, each recording when every content first becomes true and how long it stays true. Suppose Chord is required for every $k \in \mathcal{R}$. Suppose $\theta_{\mathrm{pers}}(\ell)$ is a guaranteed persistence duration for every content of ℓ once it becomes true. Assume $\mathcal{R}$ includes a timing in which one content first becomes true at window start, remains true through $\theta_{\mathrm{pers}}(\ell)$, and is false afterwards, while another first becomes true at window end. Then the window must satisfy

$$\theta_{\min}(\ell) \leq \theta \leq \theta_{\mathrm{pers}}(\ell),$$

where $\theta_{\min}(\ell) = L_{\max}(\mathrm{Graph}_\ell, d)/v$. Conversely, suppose the time window lasts no longer than $\theta_{\mathrm{pers}}(\ell)$. This means a content that becomes true in the time window will persist through to the end of the time window. Hence by the end of the window, all of the contents are true, so ℓ is co-instantiated by the final tick.

Proof. By Theorem 25, $\theta \geq L_{\max}(\mathrm{Graph}_\ell, d)/v$. To prove $\theta \leq \theta_{\mathrm{pers}}(\ell)$, I choose a permitted timing with onsets and expiry as specified. If $\theta > \theta_{\mathrm{pers}}(\ell)$, the early content in that timing is false by the time the late content becomes true, while the late content is false at every earlier tick of the fragment. No tick of the fragment contains both. Chord is false in this permitted timing, contradicting the requirement that Chord be true for every $k \in \mathcal{R}$. Hence $\theta \leq \theta_{\mathrm{pers}}(\ell)$. When $\theta \leq \theta_{\mathrm{pers}}(\ell)$, suppose a content has its onset inside the window. Then the content remains true for at least $\theta_{\mathrm{pers}}(\ell) \geq \theta$ after its onset, hence through the end of the window. So every earlier content with its onset inside the window remains true when the last one becomes true. Under a permitted timing in which every content first becomes true inside the window, Proposition 40 then gives co-instantiation at the tick at which the last content becomes true. □

Corollary 31 (Same possible fact above, different control below)**.**
Let π^i be the policy at layer i and let $R \in \mathfrak{v}^{i+1}$ be a possible fact at layer $i+1$. Suppose $T(\pi^i) \setminus R$ has at least two and finitely many distinctions, let k be their number, and suppose $\mathfrak{v}^i$ contains the possible fact R and the possible fact $\Phi \setminus \{s\}$ for each distinction $s \in T(\pi^i) \setminus R$. For the distinctions $s_1, \ldots, s_k$ of $T(\pi^i) \setminus R$, let $p_j = \Phi \setminus \{s_j\}$. Then the possible things $x = \pi^i \cup \{R\}$ and $y = \pi^i \cup \{p_1, \ldots, p_k\}$ in layer i are completions of π^i with

$$T(x) = T(y) = R,$$

such that removing any p_j from y gives a possible thing with truth set strictly larger than R, and for each j the distinction s_j satisfies every possible fact of y except p_j. Either possible thing can be the grounding of R in the sense of Definition 7, and $x \neq y$. The two groundings therefore agree on R at layer $i+1$. At layer i, the truth of R in the grounding x depends on the one further possible fact R alone, because $R \subseteq T(\pi^i)$. In the grounding y, the truth of R depends on π^i and each of k further possible facts.

Proof. Since $R \in \mathfrak{v}^{i+1}$, there is a completion o of π^i with $T(o) = R$, so R is nonempty and $R \subseteq T(\pi^i)$. Hence $T(x) = T(\pi^i) \cap R = R$. The distinctions $s_1, \ldots, s_k$ are the distinctions of $T(\pi^i)$ outside R, so $T(y) = T(\pi^i) \cap \bigcap_j p_j = T(\pi^i) \setminus \{s_1, \ldots, s_k\} = R$. Both x and y are finite, contain π^i and have truth set R. Since R is nonempty, both are completions of π^i with truth set R. Either therefore satisfies the condition of Definition 7 for the grounding of R. Removing p_j from y gives a possible thing with truth set $T(\pi^i) \setminus \{s_m : m \neq j\} = R \cup \{s_j\}$, which strictly contains R because $s_j \in T(\pi^i)$. The distinction s_j satisfies π^i and satisfies p_m for every $m \neq j$, while falsifying p_j. Since $k \geq 2$, y has more possible facts than x, so $x \neq y$. At layer $i+1$ both groundings give the possible fact R with the same truth set. At layer i, R is true under the grounding x if and only if the possible fact R is true, because $R \subseteq T(\pi^i)$ leaves π^i redundant. Under the grounding y, R is true if and only if π^i and all of $p_1, \ldots, p_k$ are true. The distinction s_j falsifies p_j alone. □

Proposition 41 (Occurrence does not determine order)**.**
Fix a finite vocabulary $\mathfrak{v}$ and an admissible finite fragment $h = (\phi_0, \ldots, \phi_N)$, meaning one whose consecutive distinctions differ and which the transition constraints permit. Let r permute $\{0, \ldots, N\}$, and suppose $h' = (\phi_{r(0)}, \ldots, \phi_{r(N)})$ is also admissible. For every $p \in \mathfrak{v}$,

$$\exists j \in \{0, \ldots, N\}\ \phi_j \in p \quad \Longleftrightarrow \quad \exists j \in \{0, \ldots, N\}\ \phi_{r(j)} \in p.$$

Consequently Arpeggio has identical truth value for every thing $\ell \subseteq \mathfrak{v}$ on h and h'. Co-instantiation considered alone also has identical truth value. Nevertheless, the order of events can differ. Which contents occur need not determine their order. For an actual timeline, layer clocks instead satisfy $u \leq u' \Rightarrow \kappa^{\tau}_{\mathfrak{v}}(u) \leq \kappa^{\tau}_{\mathfrak{v}}(u')$.

Proof. Since r is bijective, every visited distinction appears at a position before and after permutation. This proves equivalence for each p, and hence for Arpeggio, the conjunction over contents of these occurrence conditions. Applying that observation to $T(\ell)$ proves invariance of co-instantiation alone.

For event order, let $\Phi = \{a, b, c\}$, $p = \{a, c\}$, $q = \{b, c\}$ and $\mathfrak{v} = \{p, q\}$. Let both fragments (a, b) and (b, a) be permitted. Each content occurs in one fragment if and only if it occurs in the other, and both p and q are true given c. Yet first occurrence of p precedes first occurrence of q in only one fragment. Suppose a function's inputs specify only which contents occur in each fragment. Then the function receives identical arguments on both fragments and cannot reproduce that difference. Finally, each increment of a layer clock is zero or one, so its value cannot decrease along objective time. Admissibility of the permuted fragment h' is a premise. □

Constitutive explanation. The Psychophysical Principle identifies represented objects and properties with causes of evaluation, so evaluation precedes representation. My explanation of experience is constitutive, identifying experience with embodied evaluation organised around a causal self. Complex feelings are composed of less complex ones, down to one-dimensional attraction and repulsion, and the subject is the learned 1ST-order-self. Conscious attention is identified with the contents of active 2ND-order-selves, and 3RD-order-selves support projected social interactions and an impelling narrative whose anticipated evaluations are themselves felt. The levels of Definition 46 assume representations built from evaluations, and assignments of biological taxa to those levels remain conjectural.

Some might try to allege there is still a gap between embodied evaluation and experience, but this mistakes a constitutive explanation for a claim about two separate events. My argument explains qualitative character through patterns of embodied evaluation, the subject through the 1ST-order-self, and reflective awareness through higher orders of self. Their organisation and relations account for the features constituting experience. My conclusion is that this organised activity is experience, because it accounts for everything that constitutes feeling. Feeling is

therefore not a further effect that must be added to the completed account. Demanding an additional explanation of why it "feels like something" assumes that experience is something beyond the very features and relations whose constitution has been explained. Such an unjustified assumption can be dismissed out of hand. I have, by any reasonable standard, provided a formal explanation of everything that constitutes feeling. My identification of qualia with chords of valence is the conclusion of this cumulative argument from persistence through evaluation to selves. It is backed by proofs, but the individual mathematical results need not independently establish that entire conclusion. Anyone hoping to offer a substantive objection must identify a feature I've left unexplained, an unsupported inference, or a counterexample. Simply repeating that the account explains function but not feeling does no such thing! It presupposes the separation the argument refutes.

Human-like access. Here access requires audience-relative Gricean report. By Corollary 27, human-like access consciousness requires a 1ST-order-self and a 2ND-order-self. Generic availability for reasoning or broadcast alone is not this report condition.

Multilayer functionalism. Equivalence of an experience requires preservation of its constitutive evaluations and causal relations under grounding. By Proposition 7, one finer-layer constitutive condition on which implementations differ refutes the proposed equivalence. Whole-stack equality is stronger than equality of one experience. Since an experience is composed of these evaluations, a system preserving them but lacking that experience is contradictory. Matching only higher-layer reports does not preserve them.

Conscious contraptions. A conscious machine must necessarily possess:

1. A freedom-maximising, delegated, collective, self-organising architecture, with the persistent structure of a solid brain, in which evaluation precedes representation.

2. The hierarchy of first-, second- and third-order selves, specifically satisfying the differentiation and alignment preconditions for such selves to arise.

3. Sufficient concurrency to support co-instantiation of the desired conscious contents in time. Co-instantiation then follows from freedom-maximisation and the precedence of evaluation over representation (Proposition 38).

4. Physical interaction. The contents of an experience must physically connect, interacting in order to form a single unified moment of subjective experience. Without interaction, a mind could unite contents whose sites no signal can connect within a moment. Such unity would require information to travel faster than light, in violation of special relativity. I justify physical interaction with special relativity, as I justify finite embodied vocabularies with the Bekenstein bound, since neither follows from change alone. Contents instantiated at different sites must interact within the time window of the moment (Corollary 30). By Theorem 25, that time window must satisfy $\theta \geq L_{\max}/v$.

A system may also need the capacities of an artificial scientist to be sufficient for human-like consciousness. These are representation of a hypothesis space and a means of ordering hypotheses, inference, causal learning, communication, planning, and physical embedding within the environment. The sufficiency claim is a hypothesis. The signal-speed and capacity bounds are conditional mathematical results.

Corollary 32 (Global broadcast is neither necessary nor sufficient for human-like access consciousness)**.**

A **global workspace** is a workspace of limited capacity for which contents compete, with the winning content broadcast to processes for reasoning, memory, action and speech [Baars, 1997, Mashour et al., 2020].

1. Some system with a global workspace lacks human-like access consciousness.

2. Some system without a global workspace has human-like access consciousness.

Hence global broadcast is neither necessary nor sufficient for human-like access consciousness.

Proof. For the first item, let a global workspace broadcast a meaning m to a speech process that chooses its utterance from m alone. The utterance of that process takes the same value whichever member of D is the listener's decoder, so in the binary case of Theorem 21 it conveys m with probability $\frac{1}{2}$. By Corollary 27 the system lacks human-like report, and so lacks human-like access consciousness, although m is available for reasoning, memory, action and speech.

For the second item, let each content reach processes for reasoning, memory, action and speech by dedicated channels, with no shared workspace and no competition between contents. Let the speech process also receive the listener's responses by a channel of its own. Some settings of Theorem 21 admit a speaker who conveys m with probability 1 whichever member of D is the listener's decoder. In each such setting, let the speech process choose its utterances by the policy of such a speaker. The speech process then has human-like report, and each content is available for reasoning by its own channel. Hence the system has human-like access consciousness without a global workspace. □

IIT: necessity and choice of units. The sets H and P contain, respectively, hosts of experience and systems satisfying proposed physical postulates. The premises $H \subseteq P$ and that every experience has a host do not imply $P \subseteq H$: adjoining $x \in P \setminus H$ preserves those premises. An identity or sufficiency premise is additional [Bennett, 2026k]. The book's two-switch example has $A' = A$, $B' = A \mathrm{XOR} B$. Units (A, B) and $(B, A \mathrm{XOR} B)$ preserve the whole-system transition structure but specify different independent interventions. Under the 2023 IIT calculation, the pair's system integrated information is respectively 0 and 2 ibits; the 2026 deterministic-zero revision removes this deterministic example. The cited software study gives a noisy variant with distinct unit verdicts and a corresponding m-switch construction. A no-overlap rule relative to selected micro-units does not itself choose those units [Bennett, 2026k, Albantakis et al., 2023, Mayner et al., 2026, Marshall et al., 2026].

21 Vocabulary-relative computation

Proposition 42 (Every task is a satisfiability problem)**.**
Let $i, \pi \in L_{\mathfrak{v}}$. Then

$$\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi) = \{o \in L_{\mathfrak{v}} : i \cup \pi \subseteq o\},$$

which is $\mathrm{Ext}(i \cup \pi)$ when $i \cup \pi \in L_{\mathfrak{v}}$ and empty otherwise. In particular, for a $\mathfrak{v}$-task α and $i \in I_\alpha$, inference under π on i selects a possible thing containing the input and the policy, and it is defined if and only if the input and the policy together are a possible thing.

Proof. By Definition 5, $\mathrm{Ext}(i) \cap \mathrm{Ext}(\pi)$ is the set of $o \in L_{\mathfrak{v}}$ with $i \subseteq o$ and $\pi \subseteq o$. If $i \cup \pi \in L_{\mathfrak{v}}$ then the set is $\mathrm{Ext}(i \cup \pi)$ by the same definition. If some o lies in the set then $T(i \cup \pi) \supseteq T(o) \neq \varnothing$, and $i \cup \pi$ is finite, so $i \cup \pi \in L_{\mathfrak{v}}$ by Definition 4. □

Definition 50 (run on an input)**.**
Let α be a $\mathfrak{v}$-task, let τ be a timeline and let u be a tick at which an input $i \in I_\alpha$ is true, meaning $\tau(u) \in T(i)$. The **run** of α on i along τ from u is the finite timeline fragment $\tau_{[u,u+n]}$ for the least $n \geq 0$ with $u + n \in I$ such that some correct output $o \in O_\alpha$ with $i \subseteq o$ is true given $\tau(u + n)$. If no such n exists, the run is undefined. Its **run time** is $\kappa_{\mathfrak{v}}(\tau_{[u,u+n]})$.

Definition 51 (complexity classes relative to a vocabulary)**.**
Let $\mathfrak{v}$ be a finite vocabulary and let $\mathcal{T}$ be a set of timelines, the **permitted timelines**, those a physical model permits from a presented input in the sense of Section 27. A thing $x \subseteq \mathfrak{v}$ is

presented in $\tau \in \mathcal{T}$ when $\tau(0) \in T(x)$, and $|x|$, its number of possible facts, is its description length. A $\mathfrak{v}$-task α is **completed within** $f : \mathbb{N} \to \mathbb{N}$ under $\mathcal{T}$ when both of the following conditions are true. Every input is presented in some permitted timeline. In every permitted timeline τ in which an input i is presented, the run of α on i along τ from tick 0 is defined with run time at most $f(|i|)$. A task is in **P relative to $\mathfrak{v}$ under $\mathcal{T}$** when it is completed within some polynomial. Let $a, r \in \mathfrak{v}$ be two possible facts with $T(a) \cap T(r) = \varnothing$ that a permitted timeline, once in either, does not leave, meaning that $\tau(t) \in T(a)$ implies $\tau(t') \in T(a)$ for every later tick t' of $\tau \in \mathcal{T}$, and likewise for r. These are the **verdicts**. A **decision task** is a $\mathfrak{v}$-task whose inputs are divided into yes-inputs and no-inputs, with no input a completion of another. Its correct outputs are the completions of yes-inputs that contain a and the completions of no-inputs that contain r. A decision task α is in **NP relative to $\mathfrak{v}$ under $\mathcal{T}$** when there is a polynomial p satisfying both of the following requirements. For every yes-input i, there is a thing $c \subseteq \mathfrak{v}$ with $|c| \leq p(|i|)$, called a **certificate**, such that $i \cup c$ is presented in some permitted timeline. For that certificate, the run of α on i along τ from tick 0 is defined with run time at most $p(|i|)$ in every permitted timeline τ in which $i \cup c$ is presented. For every no-input i, every thing c with $|c| \leq p(|i|)$ and every permitted timeline τ in which $i \cup c$ is presented, $\tau(t) \notin T(i \cup \{a\})$ at every tick t with $\kappa_{\mathfrak{v}}(\tau_{[0,t]}) \leq p(|i|)$. The vocabulary and the permitted timelines may form a family indexed by a tape bound $S \in \mathbb{N}$[26], as the configuration vocabularies of Proposition 43 do. A task is in P or NP relative to such a family when a single polynomial bound, or a single p in the NP case, works for every S large enough for the runs in question.

Proposition 43 (P and NP are the classes of the Turing machine)**.**
Let M be an embodied S-bounded machine as in Definition 9 that halts on every input and is **complete**, meaning that the map $\phi \mapsto (q(\phi), i(\phi), T_\phi, u_\phi)$ from $\mathcal{M}_{M,S}$ to $Q \times \{1, \ldots, S\} \times \Gamma^S \times \{0,1\}^S$ is surjective, which extends the configuration completeness of Corollary 7 to the usage facts. Let $\# \in \Sigma$ be a separator symbol. The classes **relative to** M are the classes relative to $\mathfrak{v}^{\text{conf}}_{M,S}$ under the runs of M that end at a halting state, with $Q_{q_{\text{acc}}}$ and $Q_{q_{\text{rej}}}$ as verdicts and S as large as the runs in question require, one polynomial serving for every such S as in Definition 51. A task is completed within f relative to M when it is completed within f under those runs. For $L \subseteq (\Sigma \setminus \{\#\})^*$ let α_L be the decision task whose inputs are the possible things

$$i_x := \{C_{j,x_j} : 1 \leq j \leq |x|\} \cup \{C_{|x|+1,\#}\},$$

that $x\#$ is written from cell 1, with i_x a yes-input when $x \in L$. Then

1. in a run of M that begins with $x\#$ on the tape, the run of α_L on i_x along it from tick 0 is defined if and only if the run of M reaches a halting state with the verdict of x and with $x\#$ still in cells 1 to $|x|+1$, and it is then the run of M itself, with run time the number of steps of M;

2. $L \in$ P if and only if α_L is in P relative to some M;

3. $L \in$ NP if and only if α_L is in NP relative to some M.

Proof. For the first item, i_x is true at tick 0. Every correct output contains $Q_{q_{\text{acc}}}$ or $Q_{q_{\text{rej}}}$, so none is true while the state is nonhalting. At the halting state a correct output is true if and only if i_x is true given the same distinction, which is $x\#$ in cells 1 to $|x|+1$, and the state is the decision on x. The run of α_L is then the whole run of M, and its run time is the number of steps by Proposition 1.

For the second, suppose α_L is completed within p relative to M. The input i_x is presented in the run of M on $x\#$, which ends at a halting state because M halts on every input. By the

[26] A family indexed by S is a function of S, one vocabulary and one set of permitted timelines for each tape bound. This is the other sense of family in mathematics, a list of objects indexed by a set.

first item, M halts on $x\#$ within $p(|x|+1)$ steps with the verdict of x. A machine decides L in polynomial time if it writes # after its input and then runs M. Conversely let M' decide L in polynomial time. Let M halt on every tape that does not begin with some $x\#$. On a tape beginning $x\#$, the machine writes a marker in cell $|x|+2$, copies x to the right of the marker and runs M' on the copy. The marker moves one cell to the right and a blank is written in the cell it leaves whenever M' needs a further cell. Thus cells 1 to $|x|+1$ are unchanged and nothing written after the separator affects the run. This costs polynomially many further steps, every i_x is presented in the run on $x\#$, and by the first item α_L is completed within a polynomial relative to M.

For the third, let $L \in$ NP with a polynomial q and a verifier V that decides within a polynomial in $|x|$ whether a string y over $\Sigma \setminus \{\#\}$ with $|y| \le q(|x|)$ is a valid certificate for x, where $x \in L$ if and only if some certificate is valid [Sipser, 2012]. Let M halt on every tape that does not begin with some $x\#$, and, on a tape beginning $x\#$, copy x and then the symbols after the separator up to the next separator or blank, at most $q(|x|)$ of them, behind a marker as above, run V on the copy and halt with its verdict. For $x \in L$ with a valid y, the thing $c := \{C_{|x|+1+j,y_j} : 1 \le j \le |y|\} \cup \{C_{|x|+2+|y|,\#}\}$ has at most $q(|x|)+1$ possible facts and is presented with i_x in the run on $x\#y\#$. In every run in which $i_x \cup c$ is presented, the symbols after the first separator begin $y\#$. Hence M accepts within polynomially many steps with $x\#$ in place. The run of α_L on i_x is defined by the first item. For $x \notin L$ no certificate is valid, so V rejects whatever M copies and M never accepts. So α_L is in NP relative to M. Conversely suppose α_L is in NP relative to M with polynomial p. On x and a string y with $|y| < p(|x|+1)$, let V run M on $x\#y$ for $p(|x|+1)$ steps and accept if M has accepted with $x\#$ still in cells 1 to $|x|+1$. If $x \in L$, some thing c with at most $p(|x|+1)$ possible facts is presented with i_x in some run of M, whose tape begins $x\#$. In that run a correct output is true within $p(|x|+1)$ steps, so M accepts with $x\#$ in place. In those steps the head reads at most cells 1 to $p(|x|+1)$. Let y be the symbols of that tape after the separator up to cell $p(|x|+1)$ and before the first blank. Then the run on $x\#y$ is the same for those steps, and V accepts on x and y. If $x \notin L$ and $|y| < p(|x|+1)$, the thing $\{C_{|x|+1+j,y_j} : 1 \le j \le |y|\} \cup \{C_{|x|+2+|y|,b}\}$ has at most $p(|x|+1)$ possible facts and is presented with i_x in the run on $x\#y$. The run on $x\#y$ ends at a halting state because M halts on every input. In that run, the no-input condition implies that no distinction within $p(|x|+1)$ steps has both $x\#$ in place and the accept state, so V rejects. So V is a polynomial-time verifier for L. □

Remark 3 (Finite-state machines as tasks)**.**
A deterministic finite-state machine (FSM) is a tuple

$$f = \langle Q, \Sigma, \delta, q_0, A \rangle$$

with finite state set Q, finite input alphabet Σ, transition function $\delta : Q \times \Sigma \to Q$, start state $q_0 \in Q$, and accepting set $A \subseteq Q$.

I can encode f as a finite $\mathfrak{v}_f$-task in Stack Theory by choosing the (finite) environment

$$\Phi_f := Q \times \Sigma \times Q,$$

whose microstates are triples (q, σ, q') interpreted as "current state q, read symbol σ, next state q'."

Define the following possible facts (subsets of Φ_f):

$$\mathrm{Curr}_q := \{(q,\sigma,q') : \sigma \in \Sigma,\ q' \in Q\}, \qquad \mathrm{Sym}_\sigma := \{(q,\sigma,q') : q \in Q,\ q' \in Q\},$$

$$\mathrm{Next}_{q'} := \{(q,\sigma,q') : q \in Q,\ \sigma \in \Sigma\}, \qquad \mathrm{Step}_f := \{(q,\sigma,q') : q' = \delta(q,\sigma)\}.$$

Let $\mathrm{Acc}_f := \{(q,\sigma,q') : q' \in A\}$ and $\mathrm{Rej}_f := \Phi_f \setminus \mathrm{Acc}_f$.

Let $\mathfrak{v}_f$ be the finite vocabulary consisting of all possible facts Curr_q ($q \in Q$), Sym_σ ($\sigma \in \Sigma$), $\mathrm{Next}_{q'}$ ($q' \in Q$), and the three global possible facts $\mathrm{Step}_f, \mathrm{Acc}_f, \mathrm{Rej}_f$.

For each pair $(q, \sigma) \in Q \times \Sigma$ define an input possible thing

$$i_{q,\sigma} := \{\mathrm{Curr}_q, \mathrm{Sym}_\sigma, \mathrm{Step}_f\}$$

and an output possible thing

$$o_{q,\sigma} := i_{q,\sigma} \cup \{\mathrm{Next}_{\delta(q,\sigma)}\} \cup \begin{cases} \{\mathrm{Acc}_f\} & \text{if } \delta(q,\sigma) \in A, \\ \{\mathrm{Rej}_f\} & \text{otherwise.} \end{cases}$$

Then $\rho_f := \langle I_f, O_f \rangle$ with $I_f := \{i_{q,\sigma} : (q,\sigma) \in Q \times \Sigma\}$ and $O_f := \{o_{q,\sigma} : (q,\sigma) \in Q \times \Sigma\}$ is a $\mathfrak{v}_f$-task: each input $i_{q,\sigma}$ has a completion in the outputs, and each output refines its corresponding input.

The task determines the transition relation encoded by Step_f and the accepting set encoded by Acc_f. The initial state q_0 must be specified separately.

Definition 52 (oracle fact)**.**
Let $p \subseteq \Phi$ be a possible fact, and let M have three further states $q_?$, q_{yes} and q_{no}, with δ defined on $(Q \setminus \{q_{\mathrm{acc}}, q_{\mathrm{rej}}, q_?\}) \times \Gamma$. A pair $\phi, \psi \in \mathcal{M}_{M,S}$ with $q(\phi) = q_?$ is a **query step on** p when $i(\psi) = i(\phi)$, $T_\psi = T_\phi$, $u_\psi = u_\phi$, and $q(\psi) = q_{\mathrm{yes}}$ if $\phi \in p$ while $q(\psi) = q_{\mathrm{no}}$ otherwise. For a set of strings $A \subseteq \Gamma^*$ and $\phi \in \mathcal{M}_{M,S}$, the **query string** of ϕ is the tape contents $T_\phi(1)T_\phi(2)\cdots$ up to the first blank, and the **oracle fact** of A is

$$O_A := \{\phi \in \mathcal{M}_{M,S} : \text{the query string of } \phi \text{ lies in } A\}.$$

A **run of** M **with oracle** A is a finite timeline fragment in $\mathcal{M}_{M,S}$ that begins as a run of M does and ends at a halting state, and whose consecutive pairs are machine steps or query steps on O_A. The **oracle vocabulary** is $\mathfrak{v}^{\mathrm{conf}}_{M,S} \cup \{O_A\}$. The usual convention gives the oracle a separate query tape. The two conventions simulate each other with polynomial overhead, so the classes below are the same under either.

Lemma 17 (oracle steps are layer time)**.**
Let $r = (\phi_0, \ldots, \phi_n)$ be a finite timeline fragment in $\mathcal{M}_{M,S}$ whose consecutive pairs are each a machine step or a query step on some possible fact, and let $\mathfrak{v} \supseteq \mathfrak{v}^{\mathrm{conf}}_{M,S}$ be a finite vocabulary. Then

$$\kappa_{\mathfrak{v}}(r) = n.$$

If r is a run of M with oracle A, the sequence $(q(\phi_t), i(\phi_t), T_{\phi_t})_{t \le n}$ is a computation of the oracle Turing machine M^A whose query string is the tape up to the first blank. Conversely, suppose the implementation is complete as in Proposition 43. Then every computation of M^A whose head stays within cells 1 to S is realised by such a run.

Proof. At an ordinary step the head moves, so $H_{i(\phi_{t-1})} \neq H_{i(\phi_t)}$ and the encoding changes, as in Proposition 1. At a query step the state fact changes from $Q_{q_?}$ to $Q_{q_{\mathrm{yes}}}$ or $Q_{q_{\mathrm{no}}}$, so the encoding changes. Each of the n consecutive pairs is therefore a tick of subjective time in $\mathfrak{v}^{\mathrm{conf}}_{M,S}$, so $\kappa_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(r) = n$. Hence $\kappa_{\mathfrak{v}}(r) = n$, since $\kappa_{\mathfrak{v}^{\mathrm{conf}}_{M,S}}(r) \le \kappa_{\mathfrak{v}}(r)$ by Proposition 3 and $\kappa_{\mathfrak{v}}(r) \le n$. In a run of M with oracle A, ordinary steps follow δ by Definition 9, and a query step moves to q_{yes} when the query string lies in A and to q_{no} otherwise, which is the transition of M^A at $q_?$. So the projected sequence is a computation of M^A. Conversely, fix a configuration of such a computation, with the usage facts produced by the computation's history. Under completeness, some distinction of $\mathcal{M}_{M,S}$ attains that configuration with those usage facts. Consecutive distinctions chosen this way satisfy the conditions of a machine step or a query step, so the computation is realised by a run of M with oracle A. □

The class P^A contains the languages decided by some M^A within polynomially many steps in the input length, and NP^A contains those for which some M^A decides within polynomially many steps whether a polynomially bounded certificate written beside the input is valid, which is the certificate form of the usual definition [Sipser, 2012]. Relative to M^A, meaning relative to $\mathfrak{v}^{\mathrm{conf}}_{M,S} \cup \{O_A\}$ under the runs of M with oracle A, Proposition 43 is proved by the same argument, since query steps are ticks of subjective time by Lemma 17. So with the tape bound S as large as the runs require, a language lies in P^A or NP^A if and only if its decision task lies in P or NP relative to some machine with a query step on O_A. I write P and NP relative to a vocabulary $\mathfrak{v}$ containing $\mathfrak{v}^{\mathrm{conf}}_{M,S}$ for the unions of these classes over the machines whose query steps, if any, are on oracle facts O_A in $\mathfrak{v}$.

Proposition 44 (Satisfiability on the tape)**.**
Let M be an embodied S-bounded machine as in Definition 9, complete as in Proposition 43, with a separator $\# \in \Sigma$. For $L \subseteq (\Sigma \setminus \{\#\})^*$ let α_L be the decision task of Proposition 43 and let O_L be the oracle fact of the set of strings beginning $x\#$ with $x \in L$. Let each formula F in conjunctive normal form be encoded as a string $\langle F \rangle \in (\Sigma \setminus \{\#\})^m$ by a standard encoding. This means that distinct formulas have distinct strings, that encoding and decoding are computable in polynomial time, and that the length of $\langle F \rangle$ is polynomial in the size of F. The task α_{SAT} and oracle O_{SAT} are α_L and O_L for the language of encodings of satisfiable formulas. The inputs of α_{SAT} include the possible things

$$i_F := \{C_{j,\langle F \rangle_j} : 1 \leq j \leq m\} \cup \{C_{m+1,\#}\},$$

that $\langle F \rangle\#$ is written from cell 1, and the task's correct outputs are the possible things $o \supseteq i_F$ with $Q_{q_{\mathrm{acc}}} \in o$ when F is satisfiable and $Q_{q_{\mathrm{rej}}} \in o$ when it is not. Then

1. for every L, every input of α_L has a correct output, and when α_L has a yes-input and a no-input, no possible thing of any vocabulary containing $\mathfrak{v}^{\mathrm{conf}}_{M,S}$ is a correct policy for it;

2. α_{SAT} is in NP relative to some M, and it is in P relative to some M if and only if satisfiability lies in P;

3. for every L, O_L is true given a distinction if and only if the input on its tape has a correct output of α_L containing $Q_{q_{\mathrm{acc}}}$, and α_L is completed within the constant 3 relative to a machine with a query step on O_L, under its runs with that oracle.

Proof. By completeness each of i_x, $i_x \cup \{Q_{q_{\mathrm{acc}}}\}$ and $i_x \cup \{Q_{q_{\mathrm{rej}}}\}$ is a possible thing, so every input of α_L has a correct output. Suppose $\pi \in L_{\mathfrak{v}}$ for a vocabulary $\mathfrak{v} \supseteq \mathfrak{v}^{\mathrm{conf}}_{M,S}$ satisfies $\mathrm{Ext}(I_{\alpha_L}) \cap \mathrm{Ext}(\pi) = O_{\alpha_L}$. Then every correct output contains π. Let i_x be a yes-input and $i_{x'}$ a no-input. The correct outputs $i_x \cup \{Q_{q_{\mathrm{acc}}}\}$ and $i_{x'} \cup \{Q_{q_{\mathrm{rej}}}\}$ both contain π, so $\pi \subseteq i_x \cap i_{x'}$. The possible thing $i_{x'} \cup \{Q_{q_{\mathrm{acc}}}\}$ contains $i_{x'}$ and π, so it lies in $\mathrm{Ext}(I_{\alpha_L}) \cap \mathrm{Ext}(\pi) = O_{\alpha_L}$. But the only input it contains is $i_{x'}$, since $i_{x''} \subseteq i_{x'} \cup \{Q_{q_{\mathrm{acc}}}\}$ puts the separator of $i_{x''}$ in the cell of the separator of $i_{x'}$ and then $x'' = x'$. Hence, as $x' \notin L$, the possible thing $i_{x'} \cup \{Q_{q_{\mathrm{acc}}}\}$ is not a correct output. This contradiction proves the first item.

The second item is Proposition 43 for the language of encodings of satisfiable formulas. This language lies in NP, since a machine given an assignment after the separator checks each clause against it in polynomially many steps [Sipser, 2012].

For the third, the query string of ϕ begins $x\#$ if and only if i_x is true given ϕ, since $x\#$ has no blank. Only one x has i_x true given ϕ, since the separator marks the end of x. So O_L is true given ϕ if and only if the string on its tape lies in L, which is when $i_x \cup \{Q_{q_{\mathrm{acc}}}\}$ is a correct output. Let $\delta(q_0, a) = (q_?, a, R)$, $\delta(q_{\mathrm{yes}}, a) = (q_{\mathrm{acc}}, a, R)$ and $\delta(q_{\mathrm{no}}, a) = (q_{\mathrm{rej}}, a, R)$ for every a. On every tape beginning $x\#$ the run is an ordinary step to $q_?$, a query step to q_{yes} or q_{no} according to O_L, and an ordinary step to q_{acc} or q_{rej}, the verdict of x. It leaves cells 1 to

$|x| + 1$ unchanged, so the run of α_L on i_x is defined. Each step changes the state fact or moves the head, so the run time is 3. □

Proposition 45 (A vocabulary in which P ≠ NP)**.**
There is a relation $R \subseteq \bigcup_n \{0,1\}^n \times \{0,1\}^n$ such that $L_R := \{x : (x, y) \in R \text{ for some } y \text{ with } |y| = |x|\}$ lies in NP^R and not in P^R. Relative to the vocabulary $\mathfrak{v}^{\mathrm{conf}}_{M,S} \cup \{O_R\}$, with O_R the oracle fact of the strings $x\#y$ with $(x, y) \in R$ and the tape bound large enough for the runs in question, P ≠ NP [Baker et al., 1975].

Proof. $L_R \in \mathrm{NP}^R$ for every R, since a certificate y is checked by one query. Let $(M_i, p_i)_{i \geq 1}$ enumerate every pair of an oracle machine and a polynomial, so that every polynomial-time oracle machine appears with a clock that bounds its running time. I build R in stages, and at each stage the membership of finitely many strings is fixed. At stage i I choose n_i larger than the length of every string fixed so far and with $2^{n_i} > p_i(n_i)$, and run M_i on 0^{n_i} for $p_i(n_i)$ steps, answering each query by the membership fixed so far and by no for every string not yet fixed. Fewer than 2^{n_i} of the queries have the form $0^{n_i}\#y$, so some $y^\star$ of length n_i is never queried. If the run rejects or fails to halt in time, I put $(0^{n_i}, y^\star)$ into R, and if it accepts I put nothing. I then fix the membership of every string queried and of every string of the form $0^{n_i}\#y$ as it now stands. Later stages add only pairs whose strings are longer than every string fixed before, so they change nothing fixed. Hence the simulated run is the run of M_i^R on 0^{n_i} within $p_i(n_i)$ steps. If some polynomial-time oracle machine decided L_R, it would be M_i with running time bounded by p_i for some i. Hence it would halt on 0^{n_i} within $p_i(n_i)$ steps, rejecting when $0^{n_i} \in L_R$ or accepting when $0^{n_i} \notin L_R$. So $L_R \notin \mathrm{P}^R$. By Lemma 17 the step counts are layer time in the oracle vocabulary, and for a complete machine that halts on every input and has a separator, Proposition 43 relative to M^R puts the decision task α_{L_R} in NP and not in P relative to that vocabulary, which is the last claim. □

In the absence of abstraction (when $\mathfrak{v} = \mathcal{P}$), with oracle querying allowed, O_L is available for every language L. Proposition 44 then gives a constant-time decision for each L, so P = NP = PSPACE is the class of all languages in this oracle model. This collapse and the oracle separation concern different permitted vocabularies. They do not settle P versus NP for the configuration vocabulary. Adding a PSPACE-complete oracle yields $\mathrm{P}^A = \mathrm{NP}^A$, so properties shared before and after that addition cannot by themselves prove separation. Finiteness, vocabulary-relative description length and layer time are insufficient. The conclusion is compatible with an extended Church–Turing simulation thesis [Baker et al., 1975, Bernstein and Vazirani, 1997].

The oracle separation and the collapse relative to a PSPACE-complete set extend to every vocabulary with countably many oracle facts. For sets of strings $A_1, A_2, \ldots$ I write $J := \{\langle k, x\rangle : x \in A_k\}$ for their **join**, where $\langle k, x\rangle$ is a pairing of a number and a string that is computable and decodable in polynomial time. I write PSPACE^J for the languages decided by some M^J within polynomially many cells of its tape, including the cells of each query string.

Corollary 33 (Either answer is one oracle fact away)**.**
Let $\mathfrak{v}$ be the configuration vocabulary $\mathfrak{v}^{\mathrm{conf}}_{M,S}$ with finitely or countably many oracle facts $O_{A_1}, O_{A_2}, \ldots$ added, and the tape bound as large as the runs require. Then

1. there is a relation R such that P ≠ NP relative to $\mathfrak{v} \cup \{O_R\}$, with O_R as in Proposition 45;
2. there is a set of strings C such that P = NP relative to $\mathfrak{v} \cup \{O_C\}$;
3. whichever of P = NP and P ≠ NP is true relative to $\mathfrak{v}$, the other is true relative to $\mathfrak{v}$ with one oracle fact added.

Proof. Relative to $\mathfrak{v} \cup \{O_R\}$, P and NP are the unions of the classes of the machines whose query steps, if any, are on oracle facts of that vocabulary, as I defined them above. Each such machine

has a finite description, and the vocabulary has countably many oracle facts, so the pairs of such a machine and a polynomial can be listed as $(M_i, p_i)_{i \geq 1}$, the same list for every R. I build R in stages as in the proof of Proposition 45, with this list in place of the list of oracle machines. At stage i I run M_i on 0^{n_i} for $p_i(n_i)$ steps, answering each query step on O_{A_k} by membership in A_k, and each query step on O_R by the membership fixed so far and by no for every string not yet fixed. The sets A_k are fixed in advance, so the argument of that proof is unchanged. The simulated run is the run of M_i, and M_i errs on 0^{n_i} whenever it halts within $p_i(n_i)$ steps. Hence L_R is not in P relative to $\mathfrak{v} \cup \{O_R\}$, while it is in NP relative to that vocabulary, since a certificate is checked by one query step on O_R. This proves the first item.

For the second item, let J be the join of the sets A_k, and let C be the set of strings $\langle N, x, 1^m \rangle$ such that N^J accepts x within m cells of its tape, for a machine N and a string x. Let L be in NP relative to $\mathfrak{v} \cup \{O_C\}$, so that some machine with query steps on oracle facts of that vocabulary checks a certificate of length at most $q(|x|)$ within $q(|x|)$ steps, for a polynomial q. A machine with oracle J decides L within polynomially many cells by trying every certificate in turn. It answers a query step on O_{A_k} by querying $\langle k, y \rangle$ in J. It answers a query step on O_C about $\langle N, y, 1^m \rangle$ by running N^J on y within m cells, stopped after as many steps as N has configurations within m cells. Here m is at most the length of the query string, and so polynomial in $|x|$. Hence $L \in \mathrm{PSPACE}^J$. Conversely, let $L \in \mathrm{PSPACE}^J$ be decided by N^J within $s(|x|)$ cells for a polynomial s. Then $x \in L$ if and only if $\langle N, x, 1^{s(|x|)} \rangle \in C$. A machine with a query step on O_C writes that string in polynomially many steps and queries it once, so L is in P relative to $\mathfrak{v} \cup \{O_C\}$. Hence every language in NP relative to $\mathfrak{v} \cup \{O_C\}$ lies in P relative to it, and the two classes coincide, since P lies within NP [Baker et al., 1975]. The third item follows from the first two. □

Whether P equals NP in the configuration vocabulary itself remains unknown. The argument of Corollary 33 is the same for every vocabulary of this kind, and among them are vocabularies with each answer.

22 Assembly observables

Two results pertain to assembly observables. In Proposition 46 the layer varies while the bearer is fixed, where a bearer is a physical object described by a layer. In Theorem 26 the assembly space, basic objects, joins, inventory, and stopping rule are fixed while pair choice varies.

Definition 53 (Assembly observables)**.**
Fix finite basic object types and permitted binary joins. A type produced by a join is added to the available types and may be reused in later joins. An object type's *assembly space* is the collection of recursive assembly pathways that produce it. For observed object type i, its *assembly index* a_i is the number of joins in a shortest pathway from the basic types. Its *copy number* n_i is the number of observed copies. If an ensemble contains N observed types and N_{T} objects in total, its ensemble assembly is

$$A = \sum_{i=1}^{N} \mathrm{e}^{a_i} \left(\frac{n_i - 1}{N_{\mathrm{T}}} \right),$$

where e is Euler's number [Sharma et al., 2023].

Interpretation. Assembly theory measures an object by the shortest sequence of joins that builds it and by the number of copies in which it appears. The assembly index is the number of joins in the shortest such sequence from the basic parts, and copy number is how many copies are observed. Ensemble assembly weights each kind of object by both, so a sample full of many copies of objects that are hard to build scores highly. The idea is that such a sample points to

selection, since chance seldom assembles the same intricate thing twice. Everything here concerns how an object was constructed and how common it is. How well it works, how it copes with new demands and whether it recovers from damage lie outside the measure.

Proposition 46 (Layer-relative assembly index)**.**
One physical bearer can have different assembly indices at two layers that have different distinctions and are related by a join-preserving map, meaning a map sending each join of two parts to a join of their images. Its ensemble assembly also differs at every copy number $n \geq 2$.

Proof. Let the fine layer consist of nonempty ordered strings of length at most four over the basic block types `a` and `b`. Let concatenation be the permitted join, and represent the bearer by

$$\texttt{aabb}.$$

The pathway

$$\texttt{a} + \texttt{a} = \texttt{aa}, \qquad \texttt{b} + \texttt{b} = \texttt{bb}, \qquad \texttt{aa} + \texttt{bb} = \texttt{aabb}$$

uses three joins. Two joins are too few, because after one join only one composite type of length two has been constructed. Reaching length four on the second join would require joining that type to itself, producing a string with two identical halves. The two halves of `aabb` are `aa` and `bb`. Its fine-layer assembly index is therefore three.

Let the coarse layer replace both fine block types `a` and `b` with one basic block type `x`, so block order survives while block type is lost. Map both `a` and `b` to `x` and extend the map through concatenation. The same bearer is then represented by `xxxx`. The pathway

$$\texttt{x} + \texttt{x} = \texttt{xx}, \qquad \texttt{xx} + \texttt{xx} = \texttt{xxxx}$$

uses two joins, while one join from basic blocks cannot produce length four. Its coarse-layer assembly index is two. For any $n \geq 2$ observed copies of the bearer, fine and coarse ensemble assemblies are

$$\mathrm{e}^3 \frac{n-1}{n} \qquad \text{and} \qquad \mathrm{e}^2 \frac{n-1}{n}.$$

□

Interpretation. An object's assembly index depends on what one agrees to call a basic part. Admit the wheel as a basic part and a bicycle is built in fewer joins, though the bicycle is unchanged. In the proof, a row of four blocks requires three joins when two kinds of block are distinguished and two joins when they are not. Renaming the same parts under the same rules changes nothing, so the dependence comes from which distinctions one draws. Assembly index is therefore relative to the layer at which one looks, in the same way that description length is.

I also show that assembly observables at the endpoint do not identify production probabilities. At a production state, a permitted pair is an unordered pair of distinct active occurrences whose types may be joined, where an active occurrence is an object present at that state and not yet consumed by a join. The production probabilities give each permitted pair a conditional probability of being joined next. Joining is *equiprobable* when every permitted pair has the same conditional probability. It is *preferential* when two permitted pairs have different positive conditional probabilities at a state reached with positive probability. A stopping rule states when production ends, and a production run's endpoint is its ensemble at that moment. Two sets of production probabilities are distinguishable from the endpoint only when they induce different probability distributions over endpoints.

Theorem 26 (Endpoint non-identification)**.**
For every pair of integers $a, n \geq 2$, there is one finite assembly space on which every choice of

production probabilities produces

$$N = 1, \qquad a_1 = a, \qquad n_1 = N_{\mathrm{T}} = n, \qquad A = \mathrm{e}^a \frac{n-1}{n}$$

with probability one. The inventory is n noninteracting compartments that each begin with 2^a monomer occurrences, and the stopping rule is that production continues while a permitted pair exists. Hence no property of the production probabilities is identifiable from the endpoint over any class of processes on this space. For each finite preference ratio $\nu > 0$[27], joining is equiprobable when $\nu = 1$ and preferential when $\nu \neq 1$. Production may also be restricted so that every chosen pair contains the newest aggregate. In particular neither ν nor whether $\nu \neq 1$ is identifiable from the endpoint.

Proof. I use unary chain types with lengths $1, \ldots, 2^a$. Length one is basic. Joining lengths r and s produces length $r + s$ whenever $r + s \leq 2^a$. Repeated doubling constructs length 2^a in a joins. After j type-level joins, no available type can have length greater than 2^j. The chain of length 2^a therefore has assembly index a.

A compartment begins with 2^a active monomer occurrences. Each production join consumes two active occurrences and inserts their joined chain. Total length remains 2^a, while each join reduces the number of active occurrences by one. Every active pair is permitted because its two positive lengths sum to at most the conserved total. Every history therefore ends after $2^a - 1$ joins with one chain of length 2^a.

The first pair is chosen uniformly. After each join, every pair containing the newest aggregate is given weight ν and every other pair weight one, and the weights are normalised. At $\nu = 1$ all pairs are equiprobable. After the first join there are $2^a - 1 \geq 3$ active occurrences. There is therefore both a pair containing the newest aggregate and a pair not containing it. For every finite $\nu > 0$ with $\nu \neq 1$, these pairs have different positive conditional probabilities while every history still has positive probability. Production may instead be restricted to pairs containing that aggregate. By conservation, every choice of production probabilities has the same terminal chain. Running n noninteracting compartments and observing their terminal chains, I obtain $a_1 = a$, $n_1 = N_{\mathrm{T}} = n$, and $A = \mathrm{e}^a(n-1)/n$ with probability one. □

Interpretation. Every process of construction ends in the same place. I begin with a quantity of unit blocks and join pieces two at a time. Whether the pieces are paired at random, paired with a preference, or always joined to the newest piece, the blocks end as one long chain with the same assembly index and the same copy number. The endpoint is identical, so it cannot reveal which process produced it, nor whether any preference was at work. Since assembly theory infers selection from the endpoint, the endpoint alone underdetermines the thing assembly theory is meant to detect. Distinguishing the processes requires observing them in progress.

When joining histories reconverge on one endpoint, the distribution over those histories cannot be recovered from that endpoint. Under equiprobable production with $n = 2$, I find $A = \mathrm{e}^a/2$, which grows without bound as a and the initial inventory 2^a grow. So no fixed threshold on A alone excludes equiprobable production over this class of processes [Bennett, 2026m]. This result concerns identification from the endpoint with no other evidence, and a specified chemical transition model, resource bound or temporal observation may exclude the construction.

[27]The first pair is chosen uniformly. After each join, each pair containing the newest aggregate has weight ν, while each other pair has weight one. The newest aggregate is the chain formed by that join. The next pair is chosen with probability proportional to its weight.

23 Macro descriptions and scale

A macro map is a map $q : \Phi \to Z$, with fibres $q^{-1}(\{z\})$. For a finitely supported probability mass function P, write $H(P) := -\sum_\phi P(\phi) \log_2 P(\phi)$, with $0 \log_2 0 := 0$.

Lemma 18 (Truth sets are unions of macro fibres)**.**
Fix an environment Φ and a finite vocabulary $\mathfrak{v} \subseteq 2^\Phi$. For any possible thing $l \in L_\mathfrak{v}$ and any $\phi_1, \phi_2 \in \Phi$, if $\mathrm{Enc}_\mathfrak{v}(\phi_1) = \mathrm{Enc}_\mathfrak{v}(\phi_2)$ then $\phi_1 \in T(l) \iff \phi_2 \in T(l)$.

Proof. By Definition 8, $\phi \in T(l) \iff l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi)$. If $\mathrm{Enc}_\mathfrak{v}(\phi_1) = \mathrm{Enc}_\mathfrak{v}(\phi_2)$, then $l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi_1) \iff l \subseteq \mathrm{Enc}_\mathfrak{v}(\phi_2)$. □

Proposition 47 (Truth sets are pullbacks of encoding images)**.**
For every possible thing $l \in L_\mathfrak{v}$, $T(l) = \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l))$, where $\mathcal{M}(l) = \{\mathrm{Enc}_\mathfrak{v}(\phi) \mid \phi \in T(l)\}$. If $\Phi_\star \subseteq \Phi$ is finite, write $\mathcal{M}_\star(l) := \{\mathrm{Enc}_\mathfrak{v}(\phi) \mid \phi \in T(l) \cap \Phi_\star\}$ and $S_{\mathrm{state}}(l) := \log_2 |T(l) \cap \Phi_\star|$. Then

$$|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\}) \cap \Phi_\star|.$$

If all fibres have the same size C, then $S_{\mathrm{state}}(l) = \log_2 |\mathcal{M}_\star(l)| + \log_2 C$.

Proof. Fix l. For any ϕ, $\phi \in \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l)) \iff \mathrm{Enc}_\mathfrak{v}(\phi) \in \mathcal{M}(l) \iff \exists\, \psi \in T(l) : \mathrm{Enc}_\mathfrak{v}(\psi) = \mathrm{Enc}_\mathfrak{v}(\phi)$. By Lemma 18, this implies $\phi \in T(l)$. Conversely, $\phi \in T(l)$ implies $\mathrm{Enc}_\mathfrak{v}(\phi) \in \mathcal{M}(l)$. This proves $T(l) = \mathrm{Enc}_\mathfrak{v}^{-1}(\mathcal{M}(l))$.

For the counting identity, the fibres $\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\})$ are disjoint, so $|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_\mathfrak{v}^{-1}(\{m\}) \cap \Phi_\star|$. If all fibre sizes equal C, then $|T(l) \cap \Phi_\star| = C \cdot |\mathcal{M}_\star(l)|$. Taking $\log_2$ gives the entropy identity. □

Proposition 48 (Entropy chain rule for a macro map)**.**
Let P be a probability mass function on Φ with finite support $\Phi_\star$. Let $q : \Phi \to Z$ be surjective and define

$$P_Z(z) := \sum_{\phi : q(\phi) = z} P(\phi).$$

For each z with $P_Z(z) > 0$, define the conditional distribution on the fibre by

$$P(\phi \mid z) := \frac{P(\phi)}{P_Z(z)}.$$

Then

$$H(P) = H(P_Z) + \sum_{z\,:\,P_Z(z) > 0} P_Z(z)\, H(P(\cdot \mid z)).$$

If for every z with $P_Z(z) > 0$ the conditional is uniform on $q^{-1}(\{z\})$ and the positive-mass fibres have sizes $C_z := |q^{-1}(\{z\})|$, then

$$H(P) = H(P_Z) + \sum_{z\,:\,P_Z(z) > 0} P_Z(z) \log_2 C_z.$$

In particular, if all positive-mass fibres have the same size C, then $H(P) = H(P_Z) + \log_2 C$.

Proof. For every ϕ with $P(\phi) > 0$, the macro state $z = q(\phi)$ satisfies $P_Z(z) > 0$, so

$$\log P(\phi) = \log P_Z(q(\phi)) + \log P(\phi \mid q(\phi)).$$

Multiply by $-P(\phi)$ and sum over $\Phi_\star$. Grouping terms by $z = q(\phi)$ gives

$$H(P) = H(P_Z) + \sum_{z\,:\,P_Z(z) > 0} P_Z(z)\, H(P(\cdot \mid z)).$$

If the conditional distribution on each fibre of positive mass is uniform on $q^{-1}(\{z\})$, then $H(P(\cdot \mid z)) = \log_2 C_z$ for every occupied fibre. Substituting yields the displayed weighted formula. If all occupied fibres have the same size C, the weighted average reduces to $\log_2 C$. □

Lemma 19 (Relabel-invariant vocabularies cannot separate relabelings)**.**
Let G be a finite group acting on Φ. Say $\mathfrak{v}$ is **relabel-invariant** if for every $p \in \mathfrak{v}$ and every $g \in G$, $g \cdot p := \{g(\phi) : \phi \in p\} \in \mathfrak{v}$. Then for every $g \in G$ and every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = g \cdot \mathrm{Enc}_{\mathfrak{v}}(\phi) := \{g \cdot p : p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)\}$.

If additionally $g \cdot p = p$ for all $p \in \mathfrak{v}$ and all $g \in G$ (the vocabulary is literally invariant, not just covariant), then $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$ for all g, ϕ. In this case, the entire orbit $G \cdot \phi$ maps to the same encoding. The encoding fibre has size at least $|G \cdot \phi|$.

Proof. $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff \phi \in p \iff g(\phi) \in g \cdot p$. So $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff g \cdot p \in \mathrm{Enc}_{\mathfrak{v}}(g(\phi))$. If $g \cdot p = p$ for all p, this gives $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Corollary 34 (The $N!$ term in the Gibbs paradox)**.**
Let Φ consist of N labelled particles in k boxes, with $k \geq N$, and let the symmetric group S_N act by permuting particle labels. If the vocabulary is literally invariant under S_N (it encodes only occupation numbers, not which particle is where), then for any configuration with all particles in distinct boxes, the encoding fibre has size $N!$. If moreover the microstate distribution is uniform on that orbit, then the within-fibre entropy contribution in Proposition 48 is $\log_2(N!)$. The Gibbs correction is the subtraction of this relabelling entropy.

Proof. If all N particles are in distinct boxes, every permutation of labels gives a different microstate but the same occupation-number encoding. The orbit under S_N has size $N!$ because the stabiliser is trivial. By Lemma 19, all $N!$ microstates in that orbit share the same encoding, so the fibre size is $N!$. If P is uniform on the orbit, then the positive-mass conditional on that fibre is uniform on $N!$ points. Proposition 48 therefore gives a within-fibre contribution of $\log_2(N!)$. □

Definition 54 (Covering and box dimension)**.**
Let $A \subseteq \mathbb{R}^d$ be bounded. For $\varepsilon > 0$, the **covering number** $N(A, \varepsilon)$ is the minimum number of closed ε-balls needed to cover A. The **(upper) box-counting dimension** is

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \to 0} \frac{\log N(A, \varepsilon)}{-\log \varepsilon}.$$

Proposition 49 (Scale as a macro map)**.**
Fix $\varepsilon > 0$ and let $\mathcal{G}_\varepsilon$ be the standard grid of half-open cubes of side length ε in $\mathbb{R}^d$. Let $\mathcal{G}_\varepsilon(A)$ be the finite set of grid cells that intersect A, and define

$$q_\varepsilon : A \to \mathcal{G}_\varepsilon(A)$$

by sending each point to the unique grid cell that contains it. This is a finite macro map on A. Two points that land in the same cell are indistinguishable at resolution ε. Write

$$N_\varepsilon(A) := |\mathcal{G}_\varepsilon(A)|$$

for the number of occupied cells. Then $\log_2 N_\varepsilon(A)$ is the scale entropy at resolution ε. Furthermore,

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \to 0} \frac{\log N_\varepsilon(A)}{-\log \varepsilon}.$$

Proof. Because A is bounded, only finitely many ε-grid cells meet A, so $\mathcal{G}_\varepsilon(A)$ is finite. The map q_ε sends each point of A to its cell, so it is a macro map on the bounded support A. Any two points in the same cell are at Euclidean distance at most $\sqrt{d}\,\varepsilon$, so those points are indistinguishable at that resolution. The number of occupied cells is exactly $N_\varepsilon(A)$, and the corresponding scale entropy is $\log_2 N_\varepsilon(A)$. The grid-cell count and the ball-cover count differ by at most dimension-dependent constants, so they yield the same upper box dimension. □

Proposition 50 (Scale entropy diverges on bounded infinite sets)**.**
If $A \subseteq \mathbb{R}^d$ is bounded and infinite, then

$$N_\varepsilon(A) \to \infty \qquad \text{and hence} \qquad \log_2 N_\varepsilon(A) \to \infty$$

as $\varepsilon \to 0$. In particular, if $\overline{\dim}_B(A) > 0$, then the same conclusion holds.

Proof. Suppose not. Then there exists $M \in \mathbb{N}$ and a sequence $\varepsilon_n \to 0$ such that

$$N_{\varepsilon_n}(A) \le M \qquad \text{for all } n.$$

For each n, choose at most M occupied ε_n-grid cells that cover A, and choose one point

$$x_{n,1}, \dots, x_{n,M}$$

from the closure of each chosen cell, repeating points if fewer than M cells are needed. Because A is bounded, after passing to a subsequence I may assume that for each $j = 1, \dots, M$,

$$x_{n,j} \to x_j \qquad \text{as } n \to \infty$$

for some points $x_1, \dots, x_M \in \mathbb{R}^d$.

Now fix any $a \in A$. For each n, the point a lies in one of the chosen cells, so

$$\|a - x_{n,j_n}\| \le \sqrt{d}\,\varepsilon_n$$

for some index $j_n \in \{1, \dots, M\}$. By passing to a further subsequence I may assume $j_n = j$ is constant. Then

$$\|a - x_j\| \le \|a - x_{n,j}\| + \|x_{n,j} - x_j\| \to 0,$$

so $a = x_j$. Since $a \in A$ was arbitrary, I get

$$A \subseteq \{x_1, \dots, x_M\},$$

which contradicts that A is infinite.

Therefore $N_\varepsilon(A) \to \infty$, and the logarithmic conclusion follows immediately. Positive upper box dimension implies that A is infinite, so the final claim is a corollary. □

24 Feature vocabularies

Definition 55 (Region-class vocabulary)**.**
Fix a finite set of regions $\mathcal{R} = \{r_1, \dots, r_R\}$ and a class set $C = \{0, \dots, K-1\}$. Define the environment

$$\Phi_{\mathcal{R}} := C^{\mathcal{R}},$$

the set of all total assignments of classes to regions. For each region $r \in \mathcal{R}$ and class $c \in C$, define the possible fact

$$p_{r,c} := \{g \in \Phi_{\mathcal{R}} : g(r) = c\}.$$

The resulting vocabulary is

$$\mathfrak{v}_{\text{reg}} := \{p_{r,c} : r \in \mathcal{R},\ c \in C\}.$$

Proposition 51 (Structure of the region-class language)**.**
$L_{\mathfrak{v}_{\text{reg}}}$ is exactly the set of partial functions from $\mathcal{R}$ to C, encoded as consistent subsets of $\mathfrak{v}_{\text{reg}}$. Consequently

$$|L_{\mathfrak{v}_{\text{reg}}}| = (K+1)^R,$$

because each region can be assigned one of K classes or left unspecified.

Proof. A subset $l \subseteq \mathfrak{v}_{\text{reg}}$ is consistent iff no region is assigned two different classes. That is exactly the condition for l to encode a partial function from $\mathcal{R}$ to C. Each of the R regions has $K+1$ independent options. It can be sent to one of the K classes or left unspecified. □

Definition 56 (Training-label policy and free regions)**.**
Fix training data with inputs $x_1, \ldots, x_n$. Let $\mathcal{R}_{\text{train}} \subseteq \mathcal{R}$ be the set of regions containing at least one training point. For each $r \in \mathcal{R}_{\text{train}}$, let y_r be a majority training label in that region, breaking ties arbitrarily. Define the **training-label policy**

$$\pi_{\text{train}} := \{p_{r,y_r} : r \in \mathcal{R}_{\text{train}}\}.$$

Let

$$\mathcal{R}_{\text{free}} := \mathcal{R} \setminus \mathcal{R}_{\text{train}}, \qquad k_{\text{free}} := |\mathcal{R}_{\text{free}}|.$$

Theorem 27 (Weakness of the training-label policy)**.**
Under $\mathfrak{v}_{\text{reg}}$,

$$\text{ff}(\pi_{\text{train}}) = |\operatorname{Ext}(\pi_{\text{train}})| = (K+1)^{k_{\text{free}}}.$$

In the usual ten-class case, this is $11^{k_{\text{free}}}$.

Proof. The extension $\operatorname{Ext}(\pi_{\text{train}})$ consists of all partial classifications that agree with π_{train} on the training regions. Every region in $\mathcal{R}_{\text{train}}$ is already fixed. Every region in $\mathcal{R}_{\text{free}}$ still has $K+1$ independent choices. It can be assigned one of the K classes or left unspecified. Multiplying those choices gives $(K+1)^{k_{\text{free}}}$. □

Remark 4 (Approximation and shared-weight caveat)**.**
This vocabulary is intentionally crude. Inside a fixed ReLU region, the final affine layer can still classify different inputs differently. So the one-class-per-region encoding is exact only when each relevant region contains at most one input. It also ignores the shared-weight constraint. A single final linear layer must realise all assignments at once. The next vocabulary repairs that.

Definition 57 (Feature-classifier vocabulary)**.**
Fix a finite evaluation set $S = \{x_1, \ldots, x_m\}$, a feature map $\varphi : S \to \mathbb{R}^d$, and a class set $C = \{0, \ldots, K-1\}$. Let the environment be the parameter space

$$\Phi_\varphi := \mathbb{R}^{Kd+K},$$

whose elements are the class weight vectors and biases (W, b). For each $x \in S$ and class $c \in C$, define

$$p_{x,c} := \{(W,b) \in \Phi_\varphi : W_c \cdot \varphi(x) + b_c > W_{c'} \cdot \varphi(x) + b_{c'} \text{ for all } c' \neq c\}.$$

The resulting vocabulary is

$$\mathfrak{v}_\varphi := \{p_{x,c} : x \in S,\ c \in C\}.$$

Proposition 52 (Consistency is linear feasibility)**.**
A subset $l \subseteq \mathfrak{v}_\varphi$ lies in $L_{\mathfrak{v}_\varphi}$ iff the strict linear system

$$W_{c_i} \cdot \varphi(x_i) + b_{c_i} > W_c \cdot \varphi(x_i) + b_c$$

for every possible fact $p_{x_i,c_i} \in l$ and every $c \neq c_i$ is feasible. Equivalently, $l \in L_{\mathfrak{v}_\varphi}$ iff for some $\varepsilon > 0$ the LP with constraints

$$(W_{c_i} - W_c) \cdot \varphi(x_i) + (b_{c_i} - b_c) \geq \varepsilon$$

for every such pair is feasible.

Proof. By Definition 57, $\bigcap_{p \in l} p$ is the set of all (W, b) satisfying the displayed strict inequalities. If l assigns two classes to the same input, the system contains a strict inequality and its reverse, so it is infeasible. Otherwise $l \in L_{\mathfrak{v}_\varphi}$ iff the strict system has a solution.

For the LP reformulation, suppose $(W^\star, b^\star)$ satisfies all strict inequalities. There are only finitely many of them, so the minimum positive margin

$$\varepsilon^\star := \min_{i,\, c \neq c_i} \left[(W^\star_{c_i} - W^\star_c) \cdot \varphi(x_i) + (b^\star_{c_i} - b^\star_c) \right]$$

is strictly positive. Then $(W^\star, b^\star)$ is feasible for every $0 < \varepsilon \leq \varepsilon^\star$. Conversely, any LP solution with $\varepsilon > 0$ satisfies the original strict system. □

Definition 58 (Training policy and pointwise pair proxy)**.**
Given training data $(x_1, y_1), \ldots, (x_n, y_n)$ with each $x_i \in S$, define

$$\pi := \{p_{x_i,y_i} : i = 1, \ldots, n\}.$$

Assume $\pi \in L_{\mathfrak{v}_\varphi}$. Let

$$S_{\text{train}} := \{x_1, \ldots, x_n\}, \qquad S_{\text{unseen}} := S \setminus S_{\text{train}}.$$

For each $x_j \in S_{\text{unseen}}$, define the **pointwise extension count**

$$\text{ext}(x_j) := |\{c \in C : \pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}\}|.$$

Define the **feature pair proxy**

$$\text{PP}(\varphi) := \sum_{x_j \in S_{\text{unseen}}} \text{ext}(x_j).$$

Theorem 28 (Feature-classifier reduction and affine invariance)**.**
Fix φ, S, and a training policy $\pi \in L_{\mathfrak{v}_\varphi}$.

1. For each unseen $x_j \in S_{\text{unseen}}$, $\text{ext}(x_j)$ equals the number of classes $c \in C$ for which $p_{x_j,c}$ appears in some completion $l \in \text{Ext}(\pi)$.
2. $\text{PP}(\varphi)$ is computable by solving $K \cdot |S_{\text{unseen}}|$ LP feasibility problems.
3. If $A \in \mathbb{R}^{d \times d}$ is invertible, $a \in \mathbb{R}^d$, and $\varphi'(x) = A\varphi(x) + a$, then $\text{PP}(\varphi') = \text{PP}(\varphi)$.
4. Every $l \in L_{\mathfrak{v}_\varphi}$ is realised by a single shared parameter vector (W, b).

Proof. For Part (1), suppose first that $\pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}$. Then this set is itself a completion of π, so $p_{x_j,c}$ appears in some $l \in \text{Ext}(\pi)$. Conversely, if $p_{x_j,c} \in l$ for some $l \in \text{Ext}(\pi)$, then $\pi \cup \{p_{x_j,c}\} \subseteq l$. Consistency is inherited by subsets because intersecting fewer possible facts cannot make a nonempty intersection empty. So $\pi \cup \{p_{x_j,c}\} \in L_{\mathfrak{v}_\varphi}$.

Part (2) is immediate from Proposition 52. For each unseen input there are K candidate classes, and each candidate class requires one LP feasibility check.

For Part (3), let $\varphi'(x) = A\varphi(x) + a$ with A invertible. Define

$$\Psi : \Phi_\varphi \to \Phi_{\varphi'}, \qquad \Psi(W, b) := (WA^{-1},\, b - WA^{-1}a).$$

Its inverse is

$$\Psi^{-1}(W', b') = (W'A,\ b' + W'a),$$

so Ψ is a bijection. If I write

$$\Psi(W, b) =: (W', b'),$$

then for any $x \in S$ and class $c \in C$,

$$W'_c \cdot \varphi'(x) + b'_c = (WA^{-1})_c \cdot (A\varphi(x) + a) + \big(b - WA^{-1}a\big)_c = W_c \cdot \varphi(x) + b_c.$$

Hence

$$(W, b) \in p_{x,c} \iff \Psi(W, b) \in p'_{x,c}.$$

So Ψ sends each possible fact in $\mathfrak{v}_\varphi$ bijectively to the corresponding possible fact in $\mathfrak{v}_{\varphi'}$ and preserves consistency. Therefore every pointwise extension count is unchanged, and so is PP.

Part (4) is built into the definition of the language. If $l \in L_{\mathfrak{v}_\varphi}$, then by definition

$$\bigcap_{p \in l} p \neq \emptyset.$$

Any point in that nonempty intersection is a single shared parameter vector (W, b) realising every possible fact in l at once. □

Corollary 35 (Architecture-independence on a fixed sample)**.**
Let φ_1 and φ_2 be two frozen feature maps, possibly from different architectures, that agree on every input in S. Then $\mathfrak{v}_{\varphi_1} = \mathfrak{v}_{\varphi_2}$ and $\mathrm{PP}(\varphi_1) = \mathrm{PP}(\varphi_2)$.

Proof. If $\varphi_1(x) = \varphi_2(x)$ for all $x \in S$, then the defining inequalities for every possible fact $p_{x,c}$ are identical under the two feature maps. So the vocabularies, languages, extensions, and proxies coincide. □

For a frozen feature map φ_f in Definition 57, let X_{child}, X_{anchor} and X_{probe} be disjoint. With observed child labels y_{child} and the trained network's anchor predictions $\hat{y}_f$, put

$$\pi_f = \{p^{(f)}_{x, y_{\text{child}}(x)} : x \in X_{\text{child}}\} \cup \{p^{(f)}_{a, \hat{y}_f(a)} : a \in X_{\text{anchor}}\}.$$

For an indexed list $H = (s_1, \ldots, s_m)$ of sampled demands, each demand a set of requested probe classifications, with $s_j^{(f)}$ its translation to the feature vocabulary, the score is

$$J_H(f) = \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}\{\pi_f \cup s_j^{(f)} \in L_{\mathfrak{v}_{\varphi_f}}\}.$$

Each check uses one shared hypothetical affine head for all child, anchor and requested probe labels. Different checks may use different heads. With unrestricted heads and finitely many strict inequalities, positive rescaling makes every positive score gap at least one. This rescaling is unavailable when output constraints bound score changes. J_H is the fraction of feasible demands in the list H, not the count of the full extension.

Proposition 53 (finite joint completion score)**.**
Under a uniform draw from the m indexed demands, $J_H(f)$ is the probability that the drawn joint demand is feasible with π_f. Every union $\pi_f \cup s_j^{(f)}$ that is feasible is a completion of π_f.

Proof. The quantity $J_H(f) = m^{-1} \sum_{j=1}^{m} \mathbf{1}\{\pi_f \cup s_j^{(f)} \in L_{\mathfrak{v}_{\varphi_f}}\}$ is the empirical probability of success under a uniform draw from the list H. Feasibility means $\pi_f \subseteq \pi_f \cup s_j^{(f)}$ and the union is a possible thing in $L_{\mathfrak{v}_{\varphi_f}}$, so the union lies in $\mathrm{Ext}(\pi_f)$. □

Proposition 54 (common-task compatibility)**.**
Let $\pi_1, \dots, \pi_M \in L_{\mathfrak{v}}$. If a $\mathfrak{v}$-task α has $O_\alpha \neq \emptyset$ and every $\pi_i \in \Pi_\alpha$, then $\bigcup_i \pi_i \in \Pi_\alpha$. Conversely, if $\bigcup_i \pi_i \in L_{\mathfrak{v}}$, then there exists a $\mathfrak{v}$-task with nonempty correct outputs for which every π_i is correct.

Proof. For necessity I choose $y \in O_\alpha$. By correctness, $y \in \mathrm{Ext}(\pi_i)$ for every i, so $\bigcup_i \pi_i \subseteq y$ and the union is satisfiable. Since $\mathrm{Ext}(\bigcup_i \pi_i) = \bigcap_i \mathrm{Ext}(\pi_i)$, I get $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\bigcup_i \pi_i) = \bigcap_i (\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_i)) = O_\alpha$, so $\bigcup_i \pi_i \in \Pi_\alpha$. For sufficiency let $\bar{\pi} := \bigcup_i \pi_i$ and let

$$I_\alpha := \{\bar{\pi}\}, \qquad O_\alpha := \mathrm{Ext}(\bar{\pi}).$$

Since $\pi_i \subseteq \bar{\pi}$, $\mathrm{Ext}(\bar{\pi}) \subseteq \mathrm{Ext}(\pi_i)$ and therefore

$$\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_i) = \mathrm{Ext}(\bar{\pi}) = O_\alpha.$$

□

In the common external language of total labellings $g : X \to \mathcal{C}$, child and anchor policies are jointly satisfiable iff their labels agree wherever inputs coincide; with a shared child, this requires agreement at every anchor. Agreement establishes compatibility with some common task, not correctness for a specified child task or equality of the disembodied tasks in different feature vocabularies. The archived aggregate results do not include the anchor predictions, so from those results I cannot check whether every network predicts the same class at every anchor. For $\varphi'_f = A\varphi_f + a$ with A invertible, the head bijection

$$W' = WA^{-1}, \qquad b' = b - WA^{-1}a$$

preserves all class scores, so J_H and every feasibility check with constraints on class scores are unchanged when reference inputs and sampling rules remain fixed. Numerical tolerances can still affect borderline cases.

25 Transformer vocabularies

Definition 59 (Transformer backbone, units, and selectors)**.**
Fix a transformer backbone $\mathcal{T}_\theta$. For an input sequence $x = (x_1, \dots, x_{|x|})$, let

$$h_t^\ell(x) \in \mathbb{R}^{d_\ell}$$

denote the contextual hidden state at layer ℓ and token position t.

An **evaluation unit** u is one of the following.

1. a sequence unit $u = x$
2. a token unit $u = (x, t)$
3. a prefix unit $u = x_{<t}$ for restricted next-token prediction

A **readout selector** σ maps each unit u to a contextual feature vector

$$\varphi_{\theta,\sigma}(u) \in \mathbb{R}^d.$$

When θ and σ are fixed I write $\varphi(u)$.

Let $\mathcal{U}$ be a finite set of evaluation units. I assume the support is task-homogeneous, meaning that all units in $\mathcal{U}$ belong to one task family and that label meanings are consistent across units. For each $u \in \mathcal{U}$, let C_u be a finite admissible label set. I use

$$Y := \bigcup_{u \in \mathcal{U}} C_u$$

for the finite label universe induced by the support.

Remark 5 (Default selectors)**.**
The default selector is task-dependent.

1. For sequence classification, use a pooled final-layer vector or a designated classification token.
2. For token classification, use the final-layer hidden state at the labelled token position.
3. For restricted next-token prediction, use the final hidden state at the last prefix position in a causal model.

The theory does not depend on which selector is chosen. It depends only on the resulting feature map φ on the finite support.

Remark 6 (Why the next-token case is restricted)**.**
For sequence and token classification, the label set is naturally finite. For next-token prediction, the full tokenizer vocabulary is large and makes the first operationalisation unnecessarily expensive. I therefore begin with a finite candidate set C_u for each prefix unit u. The cleanest choices are the gold token plus the top $k-1$ distractors under the frozen model, or an existing multiple-choice answer set. This keeps the embodied language finite while preserving the local commitment structure that matters for freedom.

Definition 60 (Transformer feature-classifier vocabulary)**.**
Fix a finite support $\mathcal{U}$, a feature map $\varphi : \mathcal{U} \to \mathbb{R}^d$, and the induced finite label universe Y. Let the environment be the parameter space for a shared affine readout

$$\Phi_{\varphi,Y} := \mathbb{R}^{|Y|d+|Y|},$$

whose elements are class weight vectors and biases (W, b).

For each unit $u \in \mathcal{U}$ and label $c \in C_u$, define the possible fact

$$p_{u,c} := \{(W,b) \in \Phi_{\varphi,Y} : W_c \cdot \varphi(u) + b_c > W_{c'} \cdot \varphi(u) + b_{c'} \text{ for all } c' \in C_u \setminus \{c\}\}.$$

The resulting vocabulary is

$$\mathfrak{v}_{\varphi,Y} := \{p_{u,c} : u \in \mathcal{U},\ c \in C_u\}.$$

Proposition 55 (Consistency is LP feasibility)**.**
A subset $l \subseteq \mathfrak{v}_{\varphi,Y}$ lies in $L_{\mathfrak{v}_{\varphi,Y}}$ if and only if the strict linear system with inequalities

$$W_{c_i} \cdot \varphi(u_i) + b_{c_i} > W_c \cdot \varphi(u_i) + b_c$$

for every possible fact $p_{u_i,c_i} \in l$ and every $c \in C_{u_i} \setminus \{c_i\}$ is feasible. Equivalently, $l \in L_{\mathfrak{v}_{\varphi,Y}}$ if and only if for some $\varepsilon > 0$ the LP with constraints

$$(W_{c_i} - W_c) \cdot \varphi(u_i) + (b_{c_i} - b_c) \geq \varepsilon$$

for every such pair is feasible.

Proof. By Definition 60, $\bigcap_{p \in l} p$ is the set of all (W, b) satisfying the displayed strict inequalities. So $l \in L_{\mathfrak{v}_{\varphi,Y}}$ if and only if that strict system is feasible. The standard positive-margin reformulation is equivalent because the system is finite. □

In this unconstrained affine-head construction, the finite strict system is also equivalent to the system with $\varepsilon = 1$: a strictly feasible head with minimum gap $\varepsilon_0 > 0$ rescales to $(W/\varepsilon_0, b/\varepsilon_0)$.

Corollary 36 (Architecture-independence on finite support)**.**
Let φ_1 and φ_2 be two transformer feature maps on the same finite support $\mathcal{U}$. Assume they

are paired with the same admissible label family $\{C_u\}_{u\in\mathcal{U}}$ and hence the same label universe $Y = \bigcup_{u\in\mathcal{U}} C_u$. If $\varphi_1(u) = \varphi_2(u)$ for every $u \in \mathcal{U}$, then

$$\mathfrak{v}_{\varphi_1,Y} = \mathfrak{v}_{\varphi_2,Y}.$$

Hence they induce the same satisfiable language and the same correct-policy sets for any task posed on that support. In particular, for any fixed choice of training units, probe units, and training labels on that support, they induce the same operational proxies defined below.

Proof. For each unit u and admissible label $c \in C_u$, the possible fact $p_{u,c}$ is defined by inequalities against labels in the same set $C_u \setminus \{c\}$. So $p_{u,c}$ depends only on the realised feature value $\varphi(u)$ and the admissible label family $\{C_u\}$ on the chosen support. If those coincide for φ_1 and φ_2, then every possible fact $p_{u,c}$ coincides, and therefore the two vocabularies are equal. □

Definition 61 (Transformer support concept)**.**
Fix a $\mathfrak{v}_{\varphi,Y}$-task α in the existing Stack Theory sense. Two correct policies $\pi_1, \pi_2 \in \Pi_\alpha$ are **concept-equivalent** when

$$\mathrm{Ext}(\pi_1) = \mathrm{Ext}(\pi_2).$$

A **transformer support concept** is an equivalence class

$$[\pi]_{\mathrm{Ext}} := \{\pi' \in \Pi_\alpha : \mathrm{Ext}(\pi') = \mathrm{Ext}(\pi)\}$$

of correct policies under extension equality. This relation is distinct from equal freedom in Definition 5. The support concept's freedom is

$$\mathrm{ff}([\pi]_{\mathrm{Ext}}) := |\,\mathrm{Ext}(\pi)|,$$

which is well-defined because all representatives have the same extension.

Theorem 29 (Exact transformer concept optimality)**.**
Fix a finite support $\mathcal{U}$, a feature map φ, and its vocabulary $\mathfrak{v}_{\varphi,Y}$. Let α be a task in this vocabulary with $\Pi_\alpha \neq \emptyset$, and let P_α be a distribution of future demands on

$$U := L_{\mathfrak{v}_{\varphi,Y}} \setminus \mathrm{Ext}(I_\alpha).$$

For $\pi \in \Pi_\alpha$, $B_\pi := \mathrm{Ext}(\pi) \cap U$ and $G_{P_\alpha}(\pi) = P_\alpha(S \subseteq B_\pi)$ as in Definition 20. If P_α is exchangeable, every correct policy of maximal freedom maximises G_{P_α}. If $P_\alpha(|S| = 1) > 0$, every policy maximising G_{P_α} has maximal freedom. When $I_\alpha \neq \emptyset$, policy R_α of Lemma 7 maximises G_{P_α} under every distribution of demands.

Proof. The vocabulary is finite by Definition 60. Every correct policy satisfies $\mathrm{Ext}(\pi) = O_\alpha \dot\cup B_\pi$, so freedom and buffer size induce the same ordering. Theorem 2 proves the exchangeable result and its strict case. Corollary 6 proves the final claim. □

Corollary 37 (Concept-level formulation)**.**
Under the exchangeable-demand assumptions of Theorem 29, every support concept $[\pi]_{\mathrm{Ext}}$ with maximal freedom is generalisation-optimal. If $P_\alpha(|S| = 1) > 0$, optimality holds if and only if its freedom is maximal.

Proof. All representatives share an extension, so they share a buffer, freedom and generalisation probability. Theorem 29 therefore applies to support concepts. □

Corollary 38 (Restricted next-token LLM corollary)**.**
Take the evaluation units to be prefixes $u = x_{<t}$, and let each C_u be a finite candidate next-token set. Under the assumptions of Theorem 29, any language model that selects a $<_{\mathrm{ff}}$-maximal correct policy in $\mathfrak{v}_{\varphi,Y}$ learns an optimally generalisable next-token concept on that finite support.

Proof. This is Theorem 29 specialised to prefix units. □

Remark 7 (What remains open)**.**
Full freedom requires joint feasibility across completions. The proxies below evaluate one or two new assignments at a time. Joint-completion measurements and current experiments appear in Sections 24 and 28.3.

Definition 62 (Training policy and pointwise extension counts)**.**
Fix labelled observed units

$$\mathcal{U}_{\text{train}} \subseteq \mathcal{U}, \qquad y(u) \in C_u \text{ for each } u \in \mathcal{U}_{\text{train}}.$$

Define the **training policy**

$$\pi_{\text{train}} := \{p_{u,y(u)} : u \in \mathcal{U}_{\text{train}}\}.$$

Assume $\pi_{\text{train}} \in L_{\mathfrak{v}_{\varphi,Y}}$. Let

$$\mathcal{U}_{\text{probe}} := \mathcal{U} \setminus \mathcal{U}_{\text{train}}.$$

For each probe unit $u \in \mathcal{U}_{\text{probe}}$, define the **pointwise extension count**

$$\text{ext}(u) := \left|\{c \in C_u : \pi_{\text{train}} \cup \{p_{u,c}\} \in L_{\mathfrak{v}_{\varphi,Y}}\}\right|.$$

Define the **transformer pair proxy**

$$\text{PP}(\varphi) := \sum_{u \in \mathcal{U}_{\text{probe}}} \text{ext}(u).$$

Define the **normalised transformer pair proxy**

$$\widehat{\text{PP}}(\varphi) := \frac{1}{|\mathcal{U}_{\text{probe}}|} \sum_{u \in \mathcal{U}_{\text{probe}}} \frac{\text{ext}(u)}{|C_u|}.$$

Proposition 56 (Exact single-point reduction)**.**
Fix $\varphi, \mathcal{U}, Y$, and a training policy $\pi_{\text{train}} \in L_{\mathfrak{v}_{\varphi,Y}}$.

1. For each probe unit $u \in \mathcal{U}_{\text{probe}}$, $\text{ext}(u)$ equals the number of labels $c \in C_u$ for which $p_{u,c}$ appears in some completion $l \in \text{Ext}(\pi_{\text{train}})$.
2. $\text{PP}(\varphi)$ is computable by solving $\sum_{u \in \mathcal{U}_{\text{probe}}} |C_u|$ LP feasibility problems.
3. Every $l \in L_{\mathfrak{v}_{\varphi,Y}}$ is realised by a single shared parameter vector (W, b).

Proof. Part (1) is immediate from the definition of extension and the fact that consistency is inherited by subsets. Part (2) follows from Proposition 55. Part (3) is just the meaning of $L_{\mathfrak{v}_{\varphi,Y}}$ in this vocabulary. □

Definition 63 (Two-point proxy)**.**
Fix a finite set of unordered probe pairs

$$\mathcal{P}_2 \subseteq \{\{u, v\} : u, v \in \mathcal{U}_{\text{probe}},\ u \neq v\}.$$

For each pair $\{u, v\} \in \mathcal{P}_2$, define

$$\text{ext}^{(2)}(u, v) := \left|\{(c, d) \in C_u \times C_v : \pi_{\text{train}} \cup \{p_{u,c}, p_{v,d}\} \in L_{\mathfrak{v}_{\varphi,Y}}\}\right|.$$

The **sampled two-point proxy** is

$$\text{PP}^{(2)}(\varphi) := \frac{1}{|\mathcal{P}_2|} \sum_{\{u,v\} \in \mathcal{P}_2} \frac{\text{ext}^{(2)}(u, v)}{|C_u||C_v|}.$$

Proposition 57 (Affine invariance)**.**
Let $A \in \mathbb{R}^{d\times d}$ be invertible, let $a \in \mathbb{R}^d$, and define $\varphi'(u) := A\varphi(u) + a$ for every $u \in \mathcal{U}$. Then

$$\mathrm{PP}(\varphi') = \mathrm{PP}(\varphi) \qquad \text{and} \qquad \mathrm{PP}^{(2)}(\varphi') = \mathrm{PP}^{(2)}(\varphi).$$

Proof. Define

$$\Psi(W, b) := (WA^{-1},\, b - WA^{-1}a).$$

Its inverse is $\Psi^{-1}(W', b') = (W'A,\, b' + W'a)$, so it is a bijection between readout parameter spaces. For every unit u and every class c,

$$(WA^{-1})_c \cdot (A\varphi(u) + a) + \big(b - WA^{-1}a\big)_c = W_c \cdot \varphi(u) + b_c.$$

So a class assignment is feasible for φ if and only if the corresponding class assignment is feasible for φ'. The feasible one-point and two-point assignment sets therefore agree exactly, which proves the two proxy equalities. □

Remark 8 (Relation to exact freedom)**.**
The proxies above are not the full extension cardinality $|\operatorname{Ext}(\pi_{\text{train}})|$. They are tractable summaries based on counts of feasible assignments at one or two probe units at a time. That is deliberate. The feature-classifier vocabulary gives exact single-point extension counts, but computing the full joint extension remains open Bennett [2026d]. The transformer case should therefore be framed as a staged operationalisation, not as a completed measurement theorem.

26 Embodied reinforcement learning

Definition 64 (EGRL)**.**
Fix a vocabulary $\mathfrak{v}$, a reward predicate $r : L_{\mathfrak{v}} \to \{1, 0\}$, and a finite horizon $len \in \mathbb{N}$ with $len > 0$.

The reward predicate induces a $\mathfrak{v}$-task $\mu = \langle I_\mu, O_\mu\rangle \in \Gamma_{\mathfrak{v}}$ by

$$O_\mu \;:=\; \{\, l \in L_{\mathfrak{v}} \mid r(l) = 1 \,\}, \qquad I_\mu \;:=\; \{\, i \in L_{\mathfrak{v}} \mid \exists o \in O_\mu \; (i \subseteq o) \,\}.$$

Assume $\Pi_\mu \neq \emptyset$.

I model interaction up to time t using a history task $\mathfrak{h}_{<t} \in \Gamma_{\mathfrak{v}}$. Intuitively, $I_{\mathfrak{h}_{<t}}$ are the inputs encountered so far and $O_{\mathfrak{h}_{<t}}$ are the outputs produced so far.

Given $\mathfrak{h}_{<t}$, I split it into a positive and negative subhistory by applying r to past outputs. Define

$$O_{\mathfrak{h}^1_{<t}} \;:=\; \{\, o \in O_{\mathfrak{h}_{<t}} \mid r(o) = 1 \,\}, \qquad O_{\mathfrak{h}^0_{<t}} \;:=\; O_{\mathfrak{h}_{<t}} \setminus O_{\mathfrak{h}^1_{<t}}.$$

For $b \in \{0, 1\}$, let

$$I_{\mathfrak{h}^b_{<t}} \;:=\; \{\, i \in I_{\mathfrak{h}_{<t}} \mid \exists o \in O_{\mathfrak{h}^b_{<t}} (i \subseteq o) \,\}, \qquad \mathfrak{h}^b_{<t} \;:=\; \langle I_{\mathfrak{h}^b_{<t}}, O_{\mathfrak{h}^b_{<t}}\rangle.$$

A **pair proxy** $<^{\mathfrak{pp}}$ is a binary relation on the set of admissible policy pairs

$$\mathfrak{pp} \;:=\; \{(m, n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \operatorname{Ext}(m) \cap \operatorname{Ext}(n) = \emptyset\}.$$

Define the unseen region of the current history by

$$U_{\mathfrak{h}_{<t}} \;:=\; L_{\mathfrak{v}} \setminus \operatorname{Ext}(I_{\mathfrak{h}_{<t}}).$$

For such pairs, **learning** means choosing $(\pi, n) \in \mathfrak{pp}$ that is maximal under $<^{\mathfrak{pp}}$. Given a current input $i_t \in I_\mu$, **inference** means choosing an output $o_t \in \operatorname{Ext}(i_t) \cap \operatorname{Ext}(\pi)$.

Given $(\pi, n), (j, k) \in \mathfrak{pp}$, the **freedom pair proxy** is

$$(j, k) <^{\mathfrak{pp}}_{\mathrm{ff}} (\pi, n) \quad \text{iff} \quad |(\operatorname{Ext}(j) \cup \operatorname{Ext}(k)) \cap U_{\mathfrak{h}_{<t}}| < |(\operatorname{Ext}(\pi) \cup \operatorname{Ext}(n)) \cap U_{\mathfrak{h}_{<t}}|\,.$$

Proposition 58 (Weakness pair proxy optimality)**.**
Let $\mathfrak{h}_{<t}$ be a history-task and let $\mathfrak{h}^1_{<t}$ and $\mathfrak{h}^0_{<t}$ denote the induced positive and negative history-tasks (as in Definition 64). Consider the admissible set of policy pairs

$$\mathfrak{pp} \;:=\; \{(\pi, n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(\pi) \cap \mathrm{Ext}(n) = \varnothing\}.$$

Define the *effective completion set* of a pair as $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$, and write

$$U_{\mathfrak{h}_{<t}} \;:=\; L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_{\mathfrak{h}_{<t}})$$

for the unseen region of the current history. Under the same maximally uninformative extension model as in Proposition 10, apply the model to the effective completion set $E_{\pi,n}$. Then the probability that a future demand set lies in $E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}$ is maximised by choosing $(\pi, n) \in \mathfrak{pp}$ with maximal $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$. Equivalently, the freedom pair proxy is optimal under that model. If, in addition, $(E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}}))$ is the same set for every admissible pair, then maximising $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi,n}|$.

Proof. The proof is the same counting argument as in Proposition 10, with $\mathrm{Ext}(\pi)$ replaced by the effective completion set $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$.

Under the maximally uninformative model, the parent selects an additional set $S \subseteq U_{\mathfrak{h}_{<t}}$ uniformly at random. An admissible pair (π, n) is consistent with that parent extension precisely when all selected unseen outputs lie inside the effective completion set, i.e., when $S \subseteq E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}$. Therefore,

$$\mathbb{P}\big((\pi, n) \text{ generalises} \mid \mathfrak{h}_{<t}\big) \;=\; \frac{2^{|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|}}{2^{|U_{\mathfrak{h}_{<t}}|}},$$

which is strictly increasing in $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$. That proves optimality of the freedom pair proxy.

For the final equivalence, suppose the seen component $E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}})$ is the same set for every admissible pair. Then

$$|E_{\pi,n}| = |E_{\pi,n} \cap \mathrm{Ext}(I_{\mathfrak{h}_{<t}})| + |E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$$

differs across admissible pairs only through the unseen term. So maximising $|E_{\pi,n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi,n}|$. □

The algorithms below assume nonempty sets wherever a choice is taken. A failed search returns $\emptyset$ for unconstrained exploration.

27 Physical quantities

In this section I use the notation of Section 15. Continuation sets $\mathcal{C}_{\tau,t}$ and their probability measure $P_{\mathcal{C}}$, test intervals J_t, tenures[28] ρ_{c,J_t}, buffers B_π, novelty, damage and repair are defined there.

A timeline is a sequence of distinctions, which are changes without durations, energies or other measured quantities. They are atomic. A physical model determines which changes can follow which, and assigns measured quantities to the permitted changes. In Definition 29, the continuation set $\mathcal{C}_{\tau,t}$ is restricted to the continuations that the chosen physical model permits. In this section I state that restriction in terms of standard physical measurements.

A tick of objective time is a position in the order of changes, and thus has no duration. A physical model is a layer with the clock of Theorem 10 and a unit for that clock (Definition 48), so each span of that clock has a duration in seconds. Many ticks of objective time can pass

[28]Tenure is the period of objective time for which a thing persists.

Algorithm 1 EGRL system (schematic)

```
If S is a set, then S.Choose() returns an element of S.
If S is a set, then S.Add(x) := S ∪ {x}.
function EGRL_Loop(μ, <^pp, r, len)
    t ← 0
    𝔯_{ℌ<t} ← 0
    ℌ_{<t} ← ⟨∅, ∅⟩
    i_t ← I_μ.Choose()
    while t < len do
        π ← learning(ℌ_{<t}, <^pp, i_t, r)          ▷ see learning methods
        o_t ← inference(π, i_t)
        𝔯_{ℌ<t} ← 𝔯_{ℌ<t} + r(o_t)                  ▷ feedback
        I_{ℌ<t} ← I_{ℌ<t}.Add(i_t)
        O_{ℌ<t} ← O_{ℌ<t}.Add(o_t)
        t ← t + 1
        i_t ← I_μ.Choose()
    end while
    return 𝔯_{ℌ<t}
end function

function Inference(π, i_t)
    E ← Ext(π) ∩ Ext(i_t)
    return E.Choose()
end function
```

between two consecutive ticks of that clock's subjective time, and the number can differ from one span to the next. Objective time remains the order of the changes.

Fix a bounded physical region $\mathcal{R}$ as a control volume, meaning a region with a boundary that energy and matter can cross. Let a and b be physical-clock readings in seconds with $b > a$. Let $E_{\mathcal{R}}(a)$ and $E_{\mathcal{R}}(b)$ be the internal, kinetic, and potential energy stored in the region under the standard engineering convention. Let Q_{in} and Q_{out} be the heat flowing into the region and out of the region over $[a, b]$. Let W_{in} and W_{out} be non-flow work done on the region and by the region. Let $E_{\text{mat,in}}$ and $E_{\text{mat,out}}$ be the enthalpy, kinetic energy, and potential energy transported by the matter flowing into the region and by the matter flowing out of the region. Enthalpy is internal energy plus flow work, which is work needed to push matter across a boundary. This convention includes flow work in the stream enthalpy and excludes it from W_{in} and W_{out}. The convention does not include rest energy. All these quantities are measured in joules. The first-law balance is

$$E_{\mathcal{R}}(b) - E_{\mathcal{R}}(a) = Q_{\text{in}} - Q_{\text{out}} + W_{\text{in}} - W_{\text{out}} + E_{\text{mat,in}} - E_{\text{mat,out}}.$$

The gross incoming energy is

$$E_{\text{draw}} := Q_{\text{in}} + W_{\text{in}} + E_{\text{mat,in}}.$$

Dividing an energy transfer by $b - a$, I get its average power in watts. Time-dependent transfer rates are instantaneous power.

Material flow is recorded on the same interval. For any conserved material constituent i, let $m_i(t)$ be the amount inside $\mathcal{R}$ and let $m_{i,\text{in}}$ and $m_{i,\text{out}}$ be the amounts crossing its boundary. Then

$$m_i(b) - m_i(a) = m_{i,\text{in}} - m_{i,\text{out}}.$$

The amounts may be measured in kilograms or moles. Chemical species may be transformed

Algorithm 2 LEARNING METHODS

For a $\mathfrak{v}$-task α, $\alpha.expand() := \{\omega \in \Gamma_{\mathfrak{v}} : \omega \sqsubset \alpha, \neg\exists\zeta \in \Gamma_{\mathfrak{v}}(\omega \sqsubset \zeta \sqsubset \alpha)\}$ gets the set of all α's immediate children (i.e. children which are only one generation away from α, meaning I exclude grandchildren etc)

$\mathfrak{q}$ is a prioritised queue. If (n, obj) is in $\mathfrak{q}$, then n is inverse priority (smaller n is higher priority). $\mathfrak{q}.put(n, obj)$ places (n, obj) in the queue, and $\mathfrak{q}.get()$ returns the highest priority item. $\mathfrak{q}.notempty()$ returns true iff $\mathfrak{q}$ is not empty.

If $\mathcal{V}$ is a set then $\mathcal{V}.add(\mathfrak{c}) := \mathcal{V} \cup \{\mathfrak{c}\}$.

function CLASSIFICATION($\mathfrak{h}_{<t}$, r) ▷ used by LEARNING()

  $O_{\mathfrak{h}^1_{<t}} \leftarrow \{o \in O_{\mathfrak{h}_{<t}} : r(o) = 1\}$ ▷ relatively "attractive" memories

  $O_{\mathfrak{h}^0_{<t}} \leftarrow O_{\mathfrak{h}_{<t}} - O_{\mathfrak{h}^1_{<t}}$

  $I_{\mathfrak{h}^1_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^1_{<t}}(i \subseteq o)\}$

  $I_{\mathfrak{h}^0_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^0_{<t}}(i \subseteq o)\}$

  $\mathfrak{h}^0_{<t} \leftarrow \langle I_{\mathfrak{h}^0_{<t}}, O_{\mathfrak{h}^0_{<t}}\rangle$

  $\mathfrak{h}^1_{<t} \leftarrow \langle I_{\mathfrak{h}^1_{<t}}, O_{\mathfrak{h}^1_{<t}}\rangle$

  return $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t}$

end function

Example 1 : A simple example showing how learning works without noise.

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)

  $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t}$ = CLASSIFICATION($\mathfrak{h}_{<t}, r$)

  $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$ ▷ discard policies that don't apply

  $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j,k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$ ▷ remove ambiguity

  return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$

end function

Algorithm 3 LEARNING METHODS CONTINUED...

Example 2 : Learning is possible even with noise (the only drawback being that it is harder to understand) by selectively "forgetting" outputs when there is no valid policy. Given only noise, there would be a policy for each output (rote learning means $|\,\mathrm{Ext}(\pi)\,| \to 1$*).*

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)
  $\mathfrak{q} \leftarrow \mathfrak{q}.put(0, \mathfrak{h}_{<t})$
  $\mathcal{V} \leftarrow \{\mathfrak{h}_{<t}\}$ ▷ set of visited nodes
  while $\mathfrak{q}.notempty()$ **do**
    $n, \mathfrak{h} \leftarrow \mathfrak{q}.get()$
    $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t} =$ CLASSIFICATION$(\mathfrak{h}, r)$
    $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$
    $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j,k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$
    if $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \neq \emptyset$ **then**
      return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$
    end if
    for all $\mathfrak{c} \in \mathfrak{h}.expand()$ **do**
      if $\mathfrak{c} \notin \mathcal{V}$ **then**
        $\mathfrak{q}.put(n+1, \mathfrak{c})$
        $\mathcal{V} \leftarrow \mathcal{V}.add(\mathfrak{c})$
      end if
    end for
  end while
  return $\emptyset$ ▷ try uninformed exploration when there is no valid policy
end function

inside the boundary, so this balance applies directly to conserved constituents. A reaction model may add its stoichiometric species balances.

Total energy and energetic availability differ. Total energy is transferred and transformed under the first-law balance. The capacity of that energy to perform work can decrease. Fix a reference environment with specified temperature $\Theta_0 > 0$, pressure, and chemical composition. Let $\mathrm{Ex}_{\mathcal{R}}(t)$ be the exergy of the region, meaning the maximum work obtainable as it equilibrates with that reference environment, excluding work required to displace the environment. Let $\mathrm{Ex}_{\mathrm{in}}$ and $\mathrm{Ex}_{\mathrm{out}}$ be the exergy transferred across the boundary over $[a, b]$ by heat, work, and matter. Let $S_{\mathrm{gen}} \geq 0$ be the entropy generated inside the region over $[a, b]$. These transfers follow from the boundary conditions under the physical model. The aggregate exergy balance is

$$\mathrm{Ex}_{\mathcal{R}}(b) - \mathrm{Ex}_{\mathcal{R}}(a) = \mathrm{Ex}_{\mathrm{in}} - \mathrm{Ex}_{\mathrm{out}} - \Theta_0 S_{\mathrm{gen}}.$$

Let $S_{\mathcal{R}}(t)$ be the thermodynamic entropy inside the region. Let S_{in} and S_{out} be the entropy transported into and out of it by heat and matter. Thermodynamic entropy obeys

$$S_{\mathcal{R}}(b) - S_{\mathcal{R}}(a) = S_{\mathrm{in}} - S_{\mathrm{out}} + S_{\mathrm{gen}}.$$

The stored energetic availability is $\mathrm{Ex}_{\mathcal{R}}$. The gross availability drawn from the surroundings is $\mathrm{Ex}_{\mathrm{in}}$. The availability leaving through the outputs is $\mathrm{Ex}_{\mathrm{out}}$. The net exergy transfer into the region is $\mathrm{Ex}_{\mathrm{in}} - \mathrm{Ex}_{\mathrm{out}}$. It equals the increase in stored exergy plus the exergy $\Theta_0 S_{\mathrm{gen}}$ destroyed by irreversibility. These exergy quantities and $\Theta_0 S_{\mathrm{gen}}$ are measured in joules. The entropy quantities are measured in joules per kelvin. Accordingly, E_{draw} measures gross boundary input, while $\Theta_0 S_{\mathrm{gen}}$ measures irreversible loss of energetic availability. Total energy remains conserved.

These balances are conditions on the existing continuation set. For each $c \in \mathcal{C}_{\tau,t}$ and test interval J_t, the chosen physical model assigns measured times and a record of the flows across the region's boundary. These assignments determine quantities such as $E_{\mathrm{draw}}(c; J_t)$, $\mathrm{Ex}_{\mathrm{in}}(c; J_t)$, $S_{\mathrm{gen}}(c; J_t)$, physical duration, and material input. A continuation lies in $\mathcal{C}_{\tau,t}$ only when it satisfies the physical balances[29] and any stated bounds on stored exergy, input power, heat transfer, or material flow. For any measured physical quantity $Y(c; J_t)$, its expected value on continuations through which policy π persists is

$$\mathbb{E}_{c \sim P_{\mathcal{C}}}[Y(c; J_t) \mid \rho_{c,J_t}(\pi) = |J_t|],$$

when the conditioning event has positive probability. For $Y = E_{\mathrm{draw}}$, this is expected energy drawn in joules. For $Y = \mathrm{Ex}_{\mathrm{in}}$, it is expected energetic availability drawn. For $Y = S_{\mathrm{gen}}$, it is expected entropy production. For $Y = m_{i,\mathrm{in}} - m_{i,\mathrm{out}}$, it is expected net material draw. For Y equal to E_{draw} divided by the measured duration of J_t on c, it is expected average input power.

Freedom and thermodynamic entropy differ. Freedom $|\operatorname{Ext}(\pi)|$ is the number of completions of π in $L_{\mathfrak{v}}$. Those possible things may have overlapping truth sets and need not be physical microstates. A coarse-graining is a fixed grouping of microstates into macrostates. For a macrostate whose accessible physical microstates are equiprobable under a fixed coarse-graining, let N_{micro} be their number. Let k_B be the Boltzmann constant and let S_{phys} be the macrostate's thermodynamic entropy. Boltzmann's expression is

$$S_{\mathrm{phys}} = k_B \ln N_{\mathrm{micro}}.$$

For nonuniform microstate probabilities p_i, Gibbs entropy $S_{\mathrm{phys}} = -k_B \sum_i p_i \ln p_i$ replaces $S_{\mathrm{phys}} = k_B \ln N_{\mathrm{micro}}$. Formula

$$S_{\mathrm{phys}} = k_B \ln |\operatorname{Ext}(\pi)|$$

[29] Each physical balance is an equation for a quantity like energy or some sort of matter. Physical balance means that the quantity flowing into a region must equal that which flows out, plus the difference in quantity the region contains. These equations hold because energy and matter are never created or destroyed. They go somewhere.

has a thermodynamic interpretation only when the chosen physical model identifies freedom $|\operatorname{Ext}(\pi)|$ with the number of equiprobable microstates under a fixed coarse-graining. Stack Theory alone contains no such identification. In Section 12 I weight each possible thing by the equilibrium probability of the microstates that it describes. Weighted freedom is then the probability that a possible thing is true, and I derive the least free energy that a commitment requires. A system preserves its identity when the possible thing that defines that identity stays true along the physical continuation, or becomes true again through repair. For viability, that possible thing is μ. A system may export entropy while preserving its identity.

Change and energy have different mathematical roles. Each distinction is a change when ordered. A timeline is a sequence of changes. Energy is a measured state quantity in a chosen physical model. A transition can change many facts while conserving total energy. Energy, exergy, entropy, duration and material flow connect physics to Stack Theory by restricting which continuations occur and which policies can remain true on them. When a demand set S satisfies $S \subseteq B_\pi$, the current policy accommodates those demands without changing. Novelty then entails no Stack-Theoretic requirement for additional energy expenditure. Any ongoing maintenance cost or cost of action, adaptation, or repair is part of the physical continuation.

Resources can be formalised by what they can be converted into. In the book I claim that energy, information and material are stored freedom of function. Living things exchange present freedom of function for future freedom of function, converting one thing into another. Here is the formal counterpart. A resource theory specifies which states of a system can be converted into which, under the operations that the theory includes. States are then compared by that relation [Gonda and Spekkens, 2023]. I use only the relation. Let $R \subseteq \Phi$ be a finite set of distinctions, and let $\to$ be a relation on R such that $r \to r$ for every $r \in R$, and such that $r \to s$ and $s \to t$ together imply $r \to t$. Here $r \to s$ means that r can be converted into s. The set of distinctions obtainable from r is

$$\operatorname{from}(r) := \{t \in R \mid r \to t\}.$$

Each $\operatorname{from}(r)$ is a possible fact, and $\mathfrak{v} := \{\operatorname{from}(r) \mid r \in R\}$ is a vocabulary.

Proposition 59 (Conversion adds commitments)**.**
Let R, $\to$ and $\mathfrak{v}$ be as above. For each $r \in R$, let $\pi_r := \operatorname{Enc}_{\mathfrak{v}}(r)$ be the encoding of r in the vocabulary $\mathfrak{v}$. Then π_r is a possible thing, and its truth set is $T(\pi_r) = \operatorname{from}(r)$. For all $r, s \in R$,

$$r \to s \iff \pi_r \subseteq \pi_s \iff \operatorname{Ext}(\pi_s) \subseteq \operatorname{Ext}(\pi_r).$$

Hence $r \to s$ implies $\operatorname{ff}(\pi_r) \geq \operatorname{ff}(\pi_s)$, with equality if and only if $s \to r$. For a finite measure ν on the subsets of $L_{\mathfrak{v}}$, let $\operatorname{ff}_\nu(l) := \nu(\operatorname{Ext}(l))$ be the ν-freedom of Definition 22. Then $r \to s$ if and only if $\operatorname{ff}_\nu(\pi_r) \geq \operatorname{ff}_\nu(\pi_s)$ for every such ν. If every pair of distinctions in R is convertible in at least one direction, then $r \to s$ if and only if $\operatorname{ff}(\pi_r) \geq \operatorname{ff}(\pi_s)$. If some pair of distinctions in R is convertible in neither direction, then no function f from R to the real numbers satisfies the equivalence $r \to s \iff f(r) \geq f(s)$ for all $r, s \in R$. In particular, no single ν determines $\to$.

Proof. By Definition 8, $\pi_r = \{\operatorname{from}(a) \mid a \in R,\ r \in \operatorname{from}(a)\} = \{\operatorname{from}(a) \mid a \to r\}$. Since $r \to r$, $\operatorname{from}(r) \in \pi_r$, so $T(\pi_r) \subseteq \operatorname{from}(r)$. If $r \to t$ and $a \to r$, then $a \to t$, so $t \in \operatorname{from}(a)$ for every $\operatorname{from}(a) \in \pi_r$, which gives $\operatorname{from}(r) \subseteq T(\pi_r)$. Thus $T(\pi_r) = \operatorname{from}(r)$, which contains r, so $\pi_r \in L_{\mathfrak{v}}$.

If $r \to s$ and $\operatorname{from}(a) \in \pi_r$, then $a \to r$ and so $a \to s$, giving $\operatorname{from}(a) \in \pi_s$. Hence $\pi_r \subseteq \pi_s$. Conversely, if $\pi_r \subseteq \pi_s$ then $\operatorname{from}(r) \in \pi_s$, so $r \to s$. If $\pi_r \subseteq \pi_s$, every completion of π_s contains π_r, so $\operatorname{Ext}(\pi_s) \subseteq \operatorname{Ext}(\pi_r)$. Conversely, $\pi_s \in \operatorname{Ext}(\pi_s)$, so $\operatorname{Ext}(\pi_s) \subseteq \operatorname{Ext}(\pi_r)$ gives $\pi_r \subseteq \pi_s$.

The extensions are finite sets, so the inclusion $\operatorname{Ext}(\pi_s) \subseteq \operatorname{Ext}(\pi_r)$ gives $\operatorname{ff}(\pi_r) \geq \operatorname{ff}(\pi_s)$, with equality if and only if the two extensions are equal. By the displayed equivalences, the two extensions are equal if and only if $r \to s$ and $s \to r$. For a finite measure ν, the same inclusion gives $\operatorname{ff}_\nu(\pi_r) \geq \operatorname{ff}_\nu(\pi_s)$ whenever $r \to s$. For the converse, let ν be the measure giving unit mass to π_s and no mass to the other possible things. Then $\operatorname{ff}_\nu(\pi_s) = 1$, while $\operatorname{ff}_\nu(\pi_r) = 1$ if

$\pi_s \in \mathrm{Ext}(\pi_r)$, meaning $\pi_r \subseteq \pi_s$, and $\mathrm{ff}_\nu(\pi_r) = 0$ otherwise. So $\mathrm{ff}_\nu(\pi_r) \geq \mathrm{ff}_\nu(\pi_s)$ for this ν implies $r \to s$.

Suppose every pair of distinctions in R is convertible in at least one direction, and let $\mathrm{ff}(\pi_r) \geq \mathrm{ff}(\pi_s)$. If $r \not\to s$ then $s \to r$, and this conversion cannot be reversed, so $\mathrm{ff}(\pi_s) > \mathrm{ff}(\pi_r)$ by the equality case above, a contradiction. Hence ff alone determines $\to$. Finally, suppose some function f from R to the real numbers satisfies the equivalence $r \to s \iff f(r) \geq f(s)$ for all $r, s \in R$. For any $r, s \in R$, either $f(r) \geq f(s)$ or $f(s) \geq f(r)$, so $r \to s$ or $s \to r$. Hence no such f exists when some pair of distinctions in R is convertible in neither direction. If a single ν determined $\to$, then $f(r) := \mathrm{ff}_\nu(\pi_r)$ would be such a function, so no single ν determines $\to$. □

Adaptation is an exchange of freedom of function in the present for freedom of function in the future, done by gathering resources. When I speak of a **resource** I mean stored freedom of function, and energy, information and material all serve this one purpose. I prove that each of the three is exchanged for freedom of function. Committing to a policy requires free energy, meaning energy and material available for work, and the least amount is $-k_B\Theta_0$ times the logarithm of the fraction of viable weighted freedom that remains after the commitment (Proposition 17). Irreducible regret is weighted freedom lost, and a memory that stores more information has no greater irreducible regret (Theorem 11, Proposition 24). What is gained is a policy that meets unseen demands with a probability that rises with its freedom and falls with the irreducible regret of its memory (Theorem 4, Corollary 18). Any resource is stored freedom in the vocabulary of what it can be converted into, and a conversion that cannot be undone lowers that freedom (Proposition 59).

28 Experimental results

28.1 Neural freedom

Measurement, selection, and training are all addressed in separate studies. The papers reporting these studies are under review as I write [Bennett, 2026h,d,g], so the numbers in this appendix serve only to illustrate the theory, and may change. Each frozen feature map induces another embodied language, so I do not compare networks within a cohort by Bagi. Under a proved common disembodied task, a cohort's list of future demands gives the lower bound $\widehat{\beta}_{\mathcal{T}}$ on the semantic Unagi score (Section 14). Without proof that all networks share one disembodied task, that number is only how many of those demands each network survives. The correlations in Table 6 are low because of how the measurement was implemented, which is a problem I fixed in later experiments. In the later vocabulary-selection study I calibrated measurements using separate networks trained on portions of its final child samples, fixed a rule, and applied that rule to select fresh networks trained on complete child samples. I say which samples were used for which purpose in Section 28.3. In the XM study I applied freedom inside generative training at a fixed number of candidates inspected per update.

I trained matched 100-network cohorts on MNIST, Fashion-MNIST, and fixed random labels. A frozen feature map and one replacement affine head determined whether each joint demand was feasible with the empirical policy. The joint completion score was compared with raw Hessian trace, final-layer relative flatness, spectral-product margin, and weight norm.[30] The joint completion score correlated with test accuracy on both the MNIST and the Fashion-MNIST cohort. Margin had larger point estimates, and random-label joint completion score did not predict random-label accuracy. A separate experiment compared MNIST training regimes with

[30]Raw Hessian trace sums curvature of training loss along each parameter coordinate. Relative flatness adjusts final-layer curvature for weight and feature scale [Petzka et al., 2021]. Spectral-product margin divides class-score separation by a product of layers' spectral norms, where a spectral norm is a weight matrix's largest singular value [Bartlett et al., 2017].

Table 5: Additional reported finite-world and controlled-policy results. These are empirical illustrations, not further theorem hypotheses.

Study	Reported outcome and scope
Controlled classifiers	Ten benchmark families, 48 constructed child-correct policies each: Spearman correlations 0.541–0.883, pooled within-benchmark $\rho = 0.725$. Freest selection exceeded the 48-policy mean in every family. Responses beyond the child were deliberately constructed [Bennett, 2026h].
Prediction with memory	Freedom-guided selection reduced normalised regret by 58% for delayed cues and 69% for longer delays relative to selection by trained-task regret [Bennett, 2026c].
Causal selves	Finite-world setting: freedom regret 0, simplicity 13.042 bits. Under Beta$(0.7, 2)$: 0.225 versus 7.274 bits, with optimal choices in 92.3% versus 69.7% of cases [Bennett, 2026a].
Audience decoders	With two probes allowed, stationary-decoder success was 0.185 without probing, 0.926 with probing and 0.961 for an oracle. With episode-varying decoders: 0.282, 0.928 and 0.959 respectively [Bennett, 2026a].
Trust and commitment	At horizon one, silence/nonbinding trust was 0 versus binding 0.442. Binding reached 0.950 at horizon five. Informative promises gave 0.633 at horizon one and binding 0.750 at horizon two [Bennett, 2026a].
Alignment ablations	In the matched constructions, removing the relevant self caps success at 1/2; the intact construction attains 1 [Bennett, 2026a,i].
Collective restriction	Three-part synthetic collectives: cancer rates 94.0% under maximal restriction, 80.8% under random selection and 59.9% under freest selection. Surveillance plus termination reduced persistent cancer from 79.1% to 7.4% [Bennett, 2026j].
Collective lineages	Twelve generations, controls succeeding with probability 0.8: unseen survival rose from 0.3% to 95.4%, with 2.19 inherited changes on average. Survival required every part viable and a common policy, and transient cancer still occurred in almost every controlled lineage [Bennett, 2026j].
Finite novelty	Freedom selection gained 1.674 bits in mean base-two log survival over random viable choice under sampled demand sets [Bennett, 2026b].

different batch sizes and learning rates on training sets from 500 to 60,000 examples. Function-preserving rescaling then tested which statistics changed while the learned function stayed fixed. I also obtained PAC-Bayes survival certificates for future demands from sampled paths, which I describe at the end of this section.

The frozen-feature cohorts used 100 trained networks per dataset, 250 labelled child examples, 32 anchors and 128 probes. The common demand list had 128 independent paths, each comprising 16 distinct probe inputs with uniformly sampled hypothetical labels. Prefixes of sizes 2, 4, 8 and 16 gave 512 indexed demands. Prefixes within a path were dependent, and paths were the independent units. Settings were fixed without calibration against accuracy. Unresolved feasibility checks gave lower and upper fractions of feasible demands, and these coincided in the reported cohorts.

I also compared training regimes across training-set sizes. Table 7 gives the large-batch test-accuracy advantage on MNIST at each reported training-set size. The advantage peaked at 2,000 examples and shrank as the training set grew. Large-batch networks also had lower Hessian trace at every scale. I varied batch size and learning rate together in this comparison [Bennett, 2026d].

Raw Hessian trace suffers from another malady. Function-preserving rescaling between adjacent ReLU layers leaves every output and prediction unchanged while changing raw Hessian trace by up to 99× in the reported experiment [Bennett, 2026d]. This proves that raw trace is not identifiable from the learned input-output function. It does not show that curvature in fixed coordinates cannot record an optimisation path, and it does not refute reparameterisation-adjusted curvature measures.

In the two cohorts below, no setting of the joint completion score was calibrated against training set accuracy or anything like that (as later experiments had). One hundred matched networks were trained on MNIST and another hundred on Fashion-MNIST, and only the seed changed within each cohort. Table 6 gives each cohort's correlation ρ between the joint completion score and test accuracy, with its interval and its Holm-corrected p-value q, where Holm's correction bounds a test family's chance of any false rejection [Holm, 1979]. Both natural-label correlations were positive with intervals that excluded zero. Table 6 also lists split-half Spearman–Brown reliability, which compares scores from two halves of the list H and adjusts their agreement to the full length of the list. The correlations were the same under lower and upper linear-programming feasibility bounds, and no solver outcome was unresolved [Bennett, 2026d].

Table 6: Joint completion score in matched 100-network cohorts. Correlation ρ is with test accuracy, with its 95% interval and Holm-corrected p-value q. Reliability is split-half Spearman–Brown reliability. Margin ρ is spectral-product margin's correlation with test accuracy. Mean score is the mean joint completion score. Bound is the PAC-Bayes lower bound on expected path score, and the two bounds are valid jointly at confidence at least 0.95.

Cohort	ρ	95% interval	q	Reliability	Margin ρ	Mean score	Bound
MNIST	0.290	$[0.094, 0.468]$	0.004	0.940	0.504	0.287	0.241
Fashion-MNIST	0.468	$[0.301, 0.607]$	4.00×10^{-5}	0.956	0.546		0.269
Random labels	0.058		0.567			0.420	

As Table 6 shows, spectral-product margin had larger point estimates in both cohorts. However I must emphasise that joint completion score here was uncalibrated, and the calibrated version used in later experiments performed many times better. Smaller weight norm also had a positive association with test accuracy that survived Holm correction. Paired intervals established a stronger correlation for joint completion score than for raw Hessian trace and final-layer relative flatness on Fashion-MNIST, while no other paired difference was resolved. Margin, which is

Table 7: The large-batch test-accuracy advantage on MNIST, in percentage points, at the reported training-set sizes.

Training examples	Large-batch advantage
2,000	1.6
24,000	0.08
60,000	0.02

quicker to compute, may be a surrogate for part of what the joint completion score measures.

The random-label control illustrates a limit of the estimator. As Table 6 shows, the random-label joint completion score had a small association with test accuracy, and the data did not determine its sign. Random-label networks had a greater mean joint completion score than natural-label MNIST networks, so I concluded that a higher joint completion score need not track random-label test accuracy. The task, learned policy, representation, and relevant distribution of demands changed together. So the control did not test the freest correct ordering inside one fixed task and language, and it showed that a uniform distribution over arbitrary labels gives no single cross-task scale of network quality.

I also certified expected path scores. A path is a sequence of four demands, each containing the one before. I used $n = 128$ independent paths, and each path was scored by the fraction of its four demands that were feasible. Across all paths, these prefixes formed the 512 demands. I chose a uniform prior over the 100 frozen networks of a cohort independently of path sampling, then a posterior concentrated on whichever network ranked first by joint completion score. So the posterior had KL divergence $\log 100$ from the prior. Applying Theorem 8 with $\delta = 0.025$ per cohort, I obtained the expected path-score lower bounds in Table 6. Both bounds are valid jointly at confidence at least 0.95. These bounds concern new paths drawn under this sampling rule, while image-classification error is a separate target.

28.2 Freedom-guided XM

For finite nonempty contexts C and finite output set Y, let Φ_{gen} contain all permission profiles $F : C \to 2^Y \setminus \{\emptyset\}$. The exclusion vocabulary and its policy are

$$e_{c,y} := \{G \in \Phi_{\mathrm{gen}} : y \notin G(c)\}, \qquad \mathfrak{v}_{\mathrm{perm}} := \{e_{c,y} : c \in C, y \in Y\}, \qquad \pi_F := \{e_{c,y} : y \notin F(c)\}.$$

Writing $a_c = |F(c)|$, each completion retains a nonempty subset of $F(c)$ independently at each context, whence

$$\mathrm{ff}(\pi_F) = \prod_{c \in C} (2^{a_c} - 1).$$

The measured log freedom uses estimated output support. K instead counts sampled training candidates [Bennett, 2026g].

Table 8 gives the efficiency gains reported for Explorative Modeling, the XM of Chapter VIII of the book, on control tasks [Gladstone et al., 2026].

The XM study, reported in a paper under review as I write, had three experiments [Bennett, 2026g]. Experiment 1 used sample-based Forward XM to train a generator with twelve output cells, which are regions of output space, one around each possible output. At each update and for each target, Forward XM draws K latent candidates, which are random inputs to the generator. It generates one output from each candidate. It then updates the network only through the candidate with the least reconstruction loss, whose output lies closest to the target. Experiment 1 ran $K \in \{1, 2, 4, 8, 16, 32\}$ under two target functions. A target function assigns a target output to each context. One target function was uniform over the outputs, and the other depended on

Table 8: Efficiency gains reported for XM on control tasks.

Quantity	Reported gain
Data efficiency	6.2×
FLOP efficiency	4.1×
Parameter efficiency	47%
Inference steps	up to 256× fewer

the context. Each combination of K and target function had twelve paired repetitions. The full run contained 144 jobs. After training, I estimated the permission profile from fresh draws at each audit context, meaning a context reserved for measurement and absent from training. I then recorded three quantities. Best-of-eight hit is the chance that at least one of eight draws lands in the target's cell. Active output support is the number of cells the draws occupy. Log freedom at audit context c is $\log(2^{a_c} - 1)$, where $a_c = |F(c)|$ is the number of outputs the estimated permission profile F permits at c. I set log freedom to zero at an audit context where no draw landed in an output cell. As Table 9 shows, mean log freedom under the context-dependent target function rose with K throughout. Best-of-eight hit instead peaked at $K = 8$ and then declined. I concluded that XM is a means of increasing freedom, which is why it increases accuracy, and that exploration is a means only, since beyond $K = 8$ each further candidate raised freedom while accuracy fell.

Table 9: Mean log freedom of the induced permission profile in XM Experiment 1, under a context-dependent target function, at the reported values of K.

K	Mean log freedom
1	approximately 0
8	5.03
32	8.21

Experiment 2 tested selection among candidates trained by the same procedure when the demanded output frequencies changed, across thirty final worlds, each a randomly generated training problem. Each world had twelve output cells and a child sample whose labels followed a long-tailed distribution, in which a few outputs were frequent and the rest were rare. Each world also had a balanced parent, which contained the child sample and new contexts on which the outputs were weighted uniformly. Each world contained 48 trained candidates from $K \in \{4, 8, 16, 32\}$, four model families, and three checkpoints. Under child validation I selected the candidate with the greatest best-of-eight hit on the child sample. Under freedom selection I admitted candidates whose child hit lay within a tolerance of the best, then ranked them by a rule fitted over a set of candidate formulas. I fixed that rule by calibration on separate worlds, with the first 240 formulas of the set. The fixed rule's child-fit tolerance was 0.970, so a candidate was eligible when its child hit was at least that fraction of the best. A rule's separation rate is the fraction of calibration worlds containing a pair of eligible candidates at least one unit of measured log freedom apart. The fixed rule's separation rate was 0.917. The predeclared requirement was 0.80. The freedom selector depended on features of the outputs on 256 unlabelled parent contexts, and never on the parent hit, which was the outcome measure. The selector features and the reported parent hit came from the same 240 fresh latent samples drawn at each context after training. The primary selected configurations differed in training computation because, as Table 10 shows, child validation and freedom selection led to different K in almost every world. In the fixed-setting post hoc check I averaged, inside each world, the

paired gains of six combinations of $K \in \{16, 32\}$ and updates in $\{1250, 1875, 2500\}$.

Balanced-parent hit rate is the best-of-eight hit on the added parent contexts. It rose under freedom selection. Table 10 gives the worlds won by freedom selection. The fixed-setting post hoc check was positive in every world.

Table 10: Outcomes of XM Experiments 2 and 3. The baseline is child-validation selection in Experiment 2 and standard XM in Experiment 3. Image metrics are from the final evaluation at classifier-free guidance 1.5, meaning sampling with a strengthened class condition.

Experiment	Quantity	Baseline	Freedom
2	Balanced-parent hit rate	0.3169	0.3892
2	K chosen, and in how many of the 30 worlds	8, in all 30	16, in 29
2	Worlds won by freedom		29 of 30
2	Paired sign-flip p-value		1.86×10^{-9}
2	Post hoc worlds with positive gain		30 of 30
2	Post hoc mean gain		0.00646
3	FID	46.9803	43.7753
3	Inception Score	18.17	19.02
3	Recall	0.214	0.239
3	Precision	0.586	0.562
3	Mean eligible-set size		3.76
3	Added mean reconstruction loss		0.00559

Experiment 3 used XJumpy Small, the public XM implementation, on ImageNet-1K at 256×256 resolution. The standard run and the freedom-guided run both used $K = 25$. They shared the same architecture, data order, optimiser schedule, seed, GPU model, sampler and 200,000-update stopping point. In the standard run I chose the minimum-reconstruction-loss candidate among the draws. In the freedom-guided run I first identified an eligible set, meaning the candidates whose reconstruction loss lay within a tolerance of the minimum. I then chose within it by a fixed rule based on class-conditioned log freedom. Hence I inspected no extra candidates. Table 10 gives the mean size of the eligible set and the reconstruction loss added by choosing within it. I evaluated FID, the Fréchet Inception Distance of Chapter VIII of the book, at 150,000 updates and at the stopping point. At the final evaluation I also recorded classifier-free-guidance FID, Inception Score, and precision and recall. Inception Score is higher when generated images are classified confidently and cover many classes. Precision is the share of generated images that fall within the reference distribution, and recall is the share of the reference distribution covered by the generated images. Table 10 gives these values. FID, Inception Score and recall improved under freedom guidance while precision fell.

ImageNet candidate choice used fixed regions. A projection of cached image-compressor outputs to 64 dimensions defined the prototype regions. Each class had sixteen. I recorded recent winning candidates in a decayed class-by-region table. Its activity threshold determined which regions were included in the current log freedom. Among candidates inside the reconstruction-loss tolerance, the rule's score had a primary term that rose with any increase in log freedom. Bounded secondary terms depended on underuse, prototype margin, prototype fit, candidate dispersion and reconstruction-loss gap. For samples whose class condition was dropped, I applied standard XM selection and left this class-conditioned table unchanged. I calibrated the rule in the two stages of Table 11, continuing the lowest-FID formulas of the first stage into the second, and then fixed the selected formula for paired training. Final evaluation used 50,000 generated images [Bennett, 2026g].

Table 11: Calibration stages of the ImageNet candidate-choice rule in XM Experiment 3.

Stage	Formulas	Updates	Seed
1	12	5,000	33
2	4	10,000	44

Table 12: Official test accuracy in percent, as means over twelve matched trials per benchmark, with sample standard deviations in parentheses. Random is mean accuracy of all 48 networks in freedom's generation zero. Other columns give accuracy of each criterion's final pick.

Benchmark	Random	Freedom	Marginal likelihood	GdScore
KMNIST	44.28 (2.49)	47.02 (3.60)	46.86 (3.20)	32.24 (2.95)
Fashion-MNIST binary	78.23 (1.64)	78.80 (1.57)	78.13 (2.37)	78.14 (1.91)
Rotten Tomatoes	51.85 (0.50)	55.99 (1.54)	53.82 (2.05)	55.46 (1.88)
AG News binary	50.24 (0.19)	59.84 (2.98)	50.00 (0.01)	58.23 (3.34)
20NG binary	60.65 (1.07)	67.22 (2.39)	66.56 (3.83)	66.72 (2.85)

28.3 Learned-vocabulary selection

In this section I give methods and results per trial for the evolutionary experiment of Chapter VIII of the book [Bennett, 2026h]. The experiment tested selection by freedom, and the paper reporting it is under review as I write, so its numbers illustrate the theory and may change. Results come first, followed by data and measurement, calibration, evolution and competitors, and reproducibility details.

Proposition 10 concerns policies correct for one task. It shows that, under uniform unseen demands, generalisation probability rises strictly with freedom among such policies. Here each requested classification has one external meaning, namely which input receives which class, but each network interprets it in its own feature-classifier vocabulary. At each network's anchors, I also imposed that network's own predictions or score constraints. Applying Proposition 10 across networks therefore additionally requires compatible anchor commitments (Proposition 54) and a task construction establishing correctness. A passing fraction estimates feasibility under a declared sampling design. The selection experiments tested whether that estimate guided evolution toward accuracy.

28.3.1 Results

Paired comparisons share training samples. Mean initial accuracy is the average over the 48 initial networks of every trial. Each criterion's evolution within a trial is an arm. Initial-population means were recorded per arm, because matching data, specifications and training seeds does not imply numerically identical trained networks across hardware and library versions. Table 13 gives the two benchmarks whose arm means differed, while the other benchmarks had identical initial means across criteria. Table 14 gives paired differences and outcomes, and differences at each competitor's stopping generation were positive against both competitors on all five benchmarks.

Freedom's selection score rose on all five benchmarks. Table 15 shows how freedom's selected networks changed from generation one to twenty. Mean accuracy rose on KMNIST, binary AG News and binary 20NG, and fell slightly on Rotten Tomatoes and binary Fashion-MNIST. I concluded that higher measured freedom and higher accuracy need not move together between every pair of generations. Each trial's training sample was reused across that trial's arms, one

Table 13: Initial-population mean accuracy in percent, per arm, on the two benchmarks whose arms differ.

Benchmark	Freedom arm	Competitor arm
KMNIST	44.2798	44.2107
20NG binary	60.6526	60.6629

Table 14: Freedom minus each competitor, in mean percentage points over twelve paired trials, with wins, ties and losses at final picks. First pick is selected from generation zero, before offspring. Matched is measured at each competitor's stopping generation.

Benchmark	Competitor	First pick	Final	Matched	Wins, ties, losses
KMNIST	Marginal likelihood	+1.23	+0.16	+0.16	5, 0, 7
KMNIST	GdScore	+9.60	+14.78	+14.78	12, 0, 0
Fashion-MNIST binary	Marginal likelihood	+0.80	+0.67	+0.67	7, 0, 5
Fashion-MNIST binary	GdScore	+1.05	+0.65	+0.65	8, 1, 3
Rotten Tomatoes	Marginal likelihood	+2.46	+2.17	+2.17	11, 0, 1
Rotten Tomatoes	GdScore	+0.77	+0.53	+0.53	9, 0, 3
AG News binary	Marginal likelihood	+6.33	+9.84	+9.84	12, 0, 0
AG News binary	GdScore	+3.33	+1.61	+1.61	7, 0, 5
20NG binary	Marginal likelihood	+0.45	+0.66	+0.41	7, 0, 5
20NG binary	GdScore	+0.86	+0.51	+0.48	8, 0, 4

calibration was shared across freedom's trials, and official test sets were reused. So I report the results in this subsection as descriptive paired results, without asserting independence of trajectories or converting standard deviations into inferential claims.

Table 15: Freedom-selected trajectories as means over twelve trials. V is freedom's selection score. Subscripts give selection generations. Text scores are passing fractions, and image scores are passing fractions raised or lowered according to other passing fractions. Accuracy changes are in percentage points. Validation is the official test accuracy of the network selected once from freedom's generation zero using 512 separate labelled examples.

Benchmark	V_1	V_{20}	Accuracy$_1$	Accuracy$_{20}$	Change	Validation
KMNIST	0.8883	0.9000	45.56	47.02	+1.47	48.69
Fashion-MNIST binary	0.7837	0.8417	78.90	78.80	−0.10	79.25
Rotten Tomatoes	0.6386	0.6979	56.32	55.99	−0.34	55.42
AG News binary	0.3138	0.4836	56.33	59.84	+3.51	56.13
20NG binary	0.8446	0.8634	66.13	67.22	+1.10	67.77

Validation-selection served as an additional control or reference. I selected once from freedom's generation zero by accuracy on 512 separate labelled examples from the training split, then stopped, and Table 15 gives the official test accuracy of the selected networks beside freedom's trajectory. Freedom's final pick exceeded the validation-selected network on Rotten Tomatoes and binary AG News, and trailed it on the other benchmarks. This reference depended on 512 labels that the freedom criterion never used, while freedom's calibration cross-validated on trial training samples instead, scoring each fold of a sample in turn with networks trained on the other folds. I stopped validation selection after one choice because repeated offspring training

and selection against one fixed labelled set can favour its peculiarities.

28.3.2 Data and measurement

Each benchmark had twelve trials. Table 16 gives the training and test inputs of each benchmark. For KMNIST I used torchvision's partitions, with images resized to 32×32 [Clanuwat et al., 2018]. In binary Fashion-MNIST, original classes zero (T-shirt/top) and six (Shirt) became zero and one, with images at native 28×28 [Xiao et al., 2017]. Its pixel values were divided by 255. Rotten Tomatoes came from `cornell-movie-review-data/rotten_tomatoes` [Pang and Lee, 2005]. Its training texts were deduplicated after normalising case and whitespace, empty texts were excluded, and each trial's labels were split equally between classes. Official test evaluation covered every row of the official test split, and official validation data were unused. In binary 20NG, `sci.crypt`, `sci.electronics`, `sci.med` and `sci.space` formed class zero, and `rec.autos`, `rec.motorcycles`, `rec.sport.baseball` and `rec.sport.hockey` formed class one [Lang, 1995]. For binary 20NG I loaded scikit-learn's training and test splits, removed headers, footers and quotes, and excluded empty documents. Binary AG News was restricted to original classes zero (World) and one (Sports) [Zhang et al., 2015]. Its loader discarded test labels before passing inputs to any criterion.

Table 16: Training and test inputs of each benchmark. The Rotten Tomatoes training split is the official one after deduplication.

Benchmark	Training inputs	Test inputs
KMNIST	60,000	10,000
Fashion-MNIST binary	12,000	2,000
Rotten Tomatoes	deduplicated	1,066
AG News binary	60,000	3,800
20NG binary	4,586	3,032

Text inputs used fixed hashing. I lowercased each text, hashed its words and adjacent word pairs into nonnegative coordinates, and normalised to Euclidean length one. Rotten Tomatoes and binary 20NG used 20,000 input coordinates. Binary AG News had 12,000. A multilayer perceptron mapped hashed inputs to learned features through two hidden layers of equal width. Each hidden layer was followed by a rectified linear unit. The learned features were followed by the head, a linear classifier. Table 17 gives the number of learned features on every benchmark. Feature extraction beyond hashing was learned.

Each trial reserved a pool of unlabelled inputs from its dataset's training split. Pool members were chosen by hashing each input's index, without reading labels, and excluded that trial's training sample and the 512 separate validation examples. Anchors and probes came from this pool, and no input served as both within one check. Probe labels were sampled hypothetical classes, and no pool label was read. Table 17 gives sizes. Anchor numbers were 1/8, 1/4 and 1/2 of one more than feature dimension, rounded upward. Joint demand sizes were 1/64, 1/32 and 1/16 of one more than feature dimension, rounded upward and capped at eight. Reference set R contained one more input than the feature dimension, all drawn from training and pool inputs.

Scores were normalised before measurement. For each network I subtracted each input's mean class score, then divided every centred score by one common standard deviation of centred scores across training and pool inputs. One output-constraint bound of 0.35 then permitted a comparable change for every network. All inequalities below use these normalised units. With features fixed, every requirement constrains only a replacement head's weights W and biases b. I use $z_c(x) = W_c\varphi_f(x) + b_c$ for a replacement head's score for class c, $z_{f,c}(x)$ for its trained

Table 17: Data and measurement sizes per trial. Pool inputs are unlabelled. Reference inputs number feature dimension plus one.

Benchmark	Classes	Labels	Pool	Features	Anchor numbers	Joint demand sizes
KMNIST	10	40	160	32	5, 9, 17	1, 2, 3
Fashion-MNIST binary	2	32	128	32	5, 9, 17	1, 2, 3
Rotten Tomatoes	2	128	512	128	17, 33, 65	3, 5, 8
AG News binary	2	32	128	64	9, 17, 33	2, 3, 5
20NG binary	2	256	1,024	128	17, 33, 65	3, 5, 8

network's score, and $\hat{y}_f(x)$ for its trained network's predicted class. Training labels and requested probe labels require

$$z_y(x) - z_k(x) \geq 10^{-5} \qquad \text{for every class } k \neq y.$$

At label anchors, I imposed this same margin using each anchor's predicted class $\hat{y}_f(a)$. At score anchors, used on text benchmarks only, a head must instead satisfy, for every class $k \neq \hat{y}_f(a)$,

$$\left|(z_{\hat{y}_f(a)}(a) - z_k(a)) - g_0\right| \leq 10^{-6} + 0.025\,|g_0|,$$

where $g_0 := z_{f,\hat{y}_f(a)}(a) - z_{f,k}(a)$ is its trained network's current gap. When g_0 is close to zero, this tolerance can include a tie or reversal. An output constraint on reference input $x \in R$ requires, for every class $k \neq \hat{y}_f(x)$,

$$\left|(z_{\hat{y}_f(x)}(x) - z_k(x)) - (z_{f,\hat{y}_f(x)}(x) - z_{f,k}(x))\right| \leq 0.35.$$

I imposed no separate bound on weights or biases. One head must satisfy every condition within a check.

Freedom here is unchanged by affine reparameterisation of features. In Section 24, I matched an invertible feature change $\varphi'_f(x) = A\varphi_f(x) + a$ with head map $W' = WA^{-1}$, $b' = b - WA^{-1}a$, which preserves every class score. Training labels, anchors, probes, margins and output constraints are all inequalities on class scores, and normalisation depends only on class scores. So every check's outcome is unchanged under such a reparameterisation while reference inputs and sampling rules stay fixed.

Interpretation. Mixing or shifting feature coordinates changes how a head is written, while every behaviour a head can produce stays available. Freedom measured this way is unchanged by such a reparameterisation, with no prior imposed and no volume of parameters measured.

Unresolved checks stay separate. A setting is one combination of anchor kind, anchor number and joint demand size. I ran $N = 256$ checks for each setting. I use n_+, n_- and $n_?$ for verified passes, verified failures and unresolved checks, where a check is unresolved when a solver returns neither a feasible head nor a proof of infeasibility. When $n_+ + n_- > 0$, a setting's passing fraction and its bounds from unresolved checks are

$$q = \frac{n_+}{n_+ + n_-}, \qquad q_{\text{low}} = \frac{n_+}{N}, \qquad q_{\text{high}} = \frac{n_+ + n_?}{N}.$$

These bounds concern unresolved outcomes in sampled checks, rather than sampling confidence. Earlier feasible heads and infeasibility proofs were reused where supported.

28.3.3 Calibration

A selection rule may adjust its primary passing fraction. Let q be a primary setting's passing fraction. Let e_j be further measurements, m_j and $\sigma_j > 0$ their calibration centres and scales, a_j fixed weights and c a cap. Selection uses

$$V = q + c \min\{q, 1-q\} \tanh\left(\sum_j a_j \frac{e_j - m_j}{\sigma_j}\right).$$

With $c = 0$, selection uses q directly. Rules used at most two adjustment measurements and caps $c \in \{0, 0.25, 0.5, 0.75\}$. In stage one of calibration, I fitted adjustment weights using accuracies on the fold not used in training, and fixed them before stage two.

Interpretation. A rule may raise or lower its primary passing fraction q according to other passing fractions, by at most $c \min\{q, 1-q\}$. Factor $\min\{q, 1-q\}$ shrinks each nudge near zero and one, so a fraction below one stays below one.

Calibration cross-validated on trial training samples. Cross-validation trains networks on three of a sample's four folds and scores them on its fourth, rotating until every fold has been the scored fold once. Stage one used trials zero through seven, whose four folds gave 32 populations, one per fold not used in training. In stage one I evaluated 24 selection rules and advanced a shortlist to stage two. Stage two used trials eight through eleven, giving 16 populations. For each fold, all four shortlisted rules started from one population trained on the other three folds. Stage two ran initial parent selection followed by two rounds of offspring training and selection, ending at generation three. Final networks then trained on complete trial training samples. Calibration added no separate labelled examples, but it reused labels across final trials. Per-trial sample sizes therefore describe final-network training, rather than the total number of labels used by a calibrated procedure.

Calibration ranks rules by a fixed score, whose terms Table 18 lists. Best-network inclusion means selecting at least one network with maximal accuracy, on the fold not used in training, within its population. During stage one, stability measures the overlap in parent choices across the evaluations on the four folds. It defaults to zero when that overlap is unavailable. During stage two, stability measures the mean fraction of parents surviving from one generation to the next. Full measurement saturation means that all four parents reach primary score one and all their tie-breaking measurements also reach maximum. Hence measurement can no longer separate the four parents. I assess ties after measurement-based tie-breaks. Search limits and scoring weights are shared across benchmarks.

Two of the benchmarks received longer fitting during calibration. For binary AG News and binary Fashion-MNIST only, calibration fitting allows two further 140-epoch training phases at learning-rate multipliers 0.1 and 0.01. A network becomes eligible for calibration selection only after classifying every fitted training example correctly with minimum margin above 10^{-8}. This eligibility rule applies only to calibration, leaving final training and both competitors unchanged.

Ties are resolved in fixed orders. I resolve ties in V by the number of passing checks, comparing larger joint demand sizes first, then following fixed anchor and anchor-kind orderings. Current parents win remaining ties. Tables 19 and 20 give each fixed rule.

28.3.4 Evolution and competitors

Networks trained from mutable specifications. Image specifications ranged over a convolutional network, an image multilayer perceptron and a shallow convolutional network. Text specifications covered one multilayer perceptron family. Width, learning rate, weight decay, batch size and dropout varied within Table 21. Matrix parameters used Muon, which orthogonalises each momentum update [Jordan et al., 2024], and remaining parameters used momentum stochastic

Table 18: Calibration ranking score. A rule's score sums each term multiplied by its weight.

Term	Weight
Mean accuracy, on the fold not used in training, of a rule's first-ranked network	1
Mean accuracy of its four selected parents on that fold	0.25
Tenth percentile of parent accuracy across splits	0.10
Best-network inclusion rate	0.05
Selection stability	0.05
Rate at which all four parents reach primary score one	−0.15
Rate at which all their tie-breaking measurements also reach maximum	−0.35
Rate at which rankings contain ties within selected parents or across a selection cutoff	−0.10

Table 19: Fixed primary settings. Both image benchmarks raise or lower the passing fraction, with cap c.

Benchmark	Anchor kind	Anchors	Joint demand size	Cap c
KMNIST	Label	9	3	0.75
Fashion-MNIST binary	Label	9	2	0.5
Rotten Tomatoes	Label	65	8	0
AG News binary	Score	9	5	0
20NG binary	Label	33	5	0

Table 20: Fixed adjustment parameters, rounded. KMNIST's adjustment uses a label-anchor setting with nine anchors and joint demand size one (`label_r9o1`), and an aggregate of label-anchor settings at nine anchors (`label_r9`). Binary Fashion-MNIST's adjustment uses an aggregate lower bound from unresolved checks (`label_bound`) and an aggregate of label-anchor settings at joint demand size three (`label_o3`).

Benchmark	Measurement	a_j	m_j	σ_j
KMNIST	`label_r9o1`	0.61598866	0.89843750	0.02316562
KMNIST	`label_r9`	−0.78775502	0.82421875	0.02371077
Fashion-MNIST binary	`label_bound`	0.73652309	0.47020449	0.12973043
Fashion-MNIST binary	`label_o3`	−0.67641240	0.47716503	0.13272961

gradient descent, identically for every criterion. Admission required full training accuracy and a positive margin.

Table 21: Training-specification grids. Image settings apply to both image benchmarks. RT denotes Rotten Tomatoes. Epoch and attempt limits apply to ordinary training.

Setting	Images	RT and binary 20NG	Binary AG News
Width	192, 256, 384, 512	256, 384, 576	192, 256, 288
Learning rate	0.0003, 0.001, 0.002	0.001, 0.002, 0.003	0.0005, 0.001, 0.0015
Weight decay	0, 10^{-5}, 10^{-4}, 10^{-3}	0, 10^{-5}, 10^{-4}	0, 10^{-5}, 10^{-4}
Batch size	32, 64, 128	16, 32, 64	32, 64, 128
Dropout	0, 0.05, 0.10	0	0
Epoch limit	280	320	240
Attempt limit	8	10	10

Each parent had twelve offspring. Offspring inherited specifications rather than weights, and trained from fresh random starting values. One left its specification unchanged with a new training seed. Ten raised or lowered width, learning rate, weight decay, batch size or dropout by one grid step, reflecting at grid boundaries. One changed architecture. Where architecture had only one choice, as on text, that offspring raised width and learning rate. Hyperparameters with one value, such as zero dropout on text, stayed unchanged. Initial specifications, training seeds and offspring schedules matched across criteria. Parent choices determined subsequent specifications, so matching seeds did not imply identical later networks.

Numbers of network trainings are upper limits. Excluding calibration, twenty selection generations corresponded to at most 960 network trainings per trial. Across a benchmark's trials, the maximum was 11,520. Calibration trained at most 8,448 networks per benchmark. Stopping could reduce executed work. A criterion's patience is how many generated generations it allows without strict improvement in its selection score before stopping. On binary 20NG both competitors had patience three, and Table 22 gives their stopping generations by trial. On the other benchmarks competitors had patience twenty and reached generation twenty. Freedom's rule permitted early stopping only at full measurement saturation, and freedom reached generation twenty in every trial on every benchmark. Generation-matched comparisons used freedom's selected network at each competitor's stopping generation. This matched the number of generations without altering calibration, criterion inputs or training costs.

Table 22: Stopping generations of the two competitors on binary 20NG, by trial, under patience three.

Trial	0	1	2	3	4	5	6	7	8	9	10	11
Marginal likelihood	20	14	12	8	11	16	14	6	16	16	12	15
GdScore	14	7	10	20	10	7	12	8	8	11	7	5

Marginal likelihood uses a last-layer Kronecker-factored Laplace calculation [Immer et al., 2021]. For a network with d learned features and K classes, let $\mathbf{X} \in \mathbb{R}^{n \times (d+1)}$ contain its n training examples' features with a bias coordinate appended, $\mathbf{P} \in \mathbb{R}^{n \times K}$ their softmax probabilities, and $\theta \in \mathbb{R}^{K(d+1)}$ a network's flattened trained head parameters. I define

$$\mathbf{A} = \frac{\mathbf{X}^\top \mathbf{X}}{n}, \qquad \mathbf{B} = \frac{\operatorname{diag}\left(\sum_i \mathbf{P}_{i,:}\right) - \mathbf{P}^\top \mathbf{P}}{n}, \qquad \mathbf{H} = n\,(\mathbf{B} \otimes \mathbf{A}).$$

Here $\otimes$ is a Kronecker product, and $\mathbf{H}$ approximates head curvature. Let $h_1, \ldots, h_r$ be eigenvalues of $\mathbf{H}$, with $r = K(d+1)$, and let $\ell(\theta)$ be training negative log likelihood. For prior precision

$\xi > 0$, scoring uses

$$E(\xi) = -\ell(\theta) - \frac{\xi}{2}\|\theta\|^2 + \frac{r}{2}\log\xi - \frac{1}{2}\sum_{j=1}^{r}\log(h_j + \xi).$$

I ranked networks by maximum $E(\xi)$ over 49 logarithmically spaced precisions from 10^{-6} to 10^{6}. Each score used a fitted representation and fitted head, and head parameters were not refitted for each precision. Each score approximated the evidence at the fitted parameters. Marginal likelihood integrates likelihood against a prior, and a change of coordinates preserves that integral when its prior transforms consistently. Here ξ specifies an isotropic Gaussian prior, with equal variance in every direction of the head's own parameter coordinates. A general affine reparameterisation of features turns an isotropic prior into an anisotropic one outside this family.

Interpretation. Marginal likelihood is higher for heads that fit training labels while sitting in a broad region of prior probability. Its prior is written in whatever coordinates a network happens to use, so reparameterising features can change its score while every prediction is unchanged.

GdScore adapts gradient-based unsupervised accuracy estimation to a fitted final layer [Xie et al., 2025]. On unlabelled official test inputs, I computed softmax probabilities $\mathbf{P}$ and a pseudo-label matrix $\mathbf{Y}$. A prediction received its predicted class when its maximum probability was at least 0.5, and a lower-confidence prediction received a seeded random class. On binary tasks maximum probability was always at least 0.5. For a batch of m feature vectors, I formed matrix $\mathbf{X}$ without a bias coordinate and computed a classifier-weight gradient and its entrywise quasi-norm,

$$\mathbf{G} = \frac{(\mathbf{P} - \mathbf{Y})^\top \mathbf{X}}{m}, \qquad \|\mathbf{G}\|_{0.3} = \Big(\sum_{j,k} |\mathbf{G}_{jk}|^{0.3}\Big)^{1/0.3}.$$

Smaller average $\|\mathbf{G}\|_{0.3}$ was preferred. Batches had size 128, with equal averaging over batches including a final partial batch. Random seed one was reset for each network, bias gradients were excluded, and scoring updated no weights. For any $c > 0$, replacing features $\mathbf{X}$ by $c\mathbf{X}$ and classifier weights W_f by W_f/c preserves every score and pseudo-label, yet multiplies $\mathbf{G}$ and $\|\mathbf{G}\|_{0.3}$ by c. A function-preserving rescaling of features therefore changes GdScore while leaving freedom's checks unchanged.

28.3.5 Reproducibility

My Chapter VIII folder on GitHub contains source code, twelve-trial result files for each benchmark and criterion, and scripts that recompute every table here from recorded results [Bennett, 2025c]. Recomputing tables requires only Python's standard library. Retraining requires a GPU and each benchmark's dataset. Trial $t \in \{0, \ldots, 11\}$ has task seed $91001 + 1000003t$.

Each stage has its own code. Measurement code is in `wam/surf.py`, `wam/fspace.py`, `wam/fgeom.py`, `wam127_partc_weakm/full.py` and `wam127_partc_weakm/lp.py`. Calibration code is in `wam/partf_core.py`, and each result file contains full-precision coefficients, centres, scales and tie settings under `calibration.f_b`. Source code in `wam127_partc_weakm/net.py` specifies every layer, and restart and stopping behaviour for unsuccessful training attempts. Code for both competitors is in `wam/partf_comparator_scores.py`.

Affine invariance was checked numerically. I subtracted each input's mean class score, normalised the score scale, and constructed affine feature coordinates using a singular-value decomposition and pivoted factorisation. I rounded these arrays to twelve decimal places and checked that they reconstructed scores within 5×10^{-9}. Finite rank thresholds and solver tolerances can affect borderline cases, so this invariance concerns mathematical feasibility rather than bitwise equality of numerical implementations.

Hardware and library versions differ between some arms. Freedom's binary 20 Newsgroups arm ran on Apple's MPS backend while its competitors ran on CUDA. KMNIST's freedom arm also has different recorded hardware and library versions from its GdScore arm, and KMNIST's marginal-likelihood arm did not record library versions. These comparisons share data and specifications, and their initial networks need not be bitwise identical. Each result file contains a record of its environment.

28.4 Persistence and repair

In the book I simulated the loop of repair and renewal. A body has twelve possible facts, each intact or disrupted. A demand set is three possible facts drawn at random, all of which must be intact. So with n intact possible facts, the chance of completing a task is $\binom{n}{3}/\binom{12}{3}$, which is the chance under exchangeable demands that a demand set is inside a buffer, as in the proof of Theorem 2, with the intact possible facts as the buffer. The simulation advances in cycles, and I call one cycle a simulation tick. Unless I state otherwise, a tick in this section and its tables is a simulation tick. In each simulation tick, background damage disrupts intact possible facts. A task is then attempted, and the attempt disrupts intact possible facts too. A completed task adds a fixed gain to the reserve. Repair then restores disrupted possible facts at one unit of capacity each, for as long as capacity lasts. Viability requires a minimum number of intact possible facts after repair, and Table 23 lists every setting. I selected three values of the gain per completed task to set the loop's renewal against the expected damage at full health, which is about 0.8 disruptions per tick, and Table 24 shows the outcome of each. In the burst test, I disrupted a chosen number of possible facts at once in timelines viable at the burst tick and recorded survival over the following 200 ticks. A young system is one hit at tick 5, before the system has stored much. A mature system is one hit at tick 300, after a system in the surplus regime has stored a large reserve. Table 25 gives two burst sizes, and a figure in the book gives the rest.

The surplus setting also admits a resource calculation. For this calculation, I take independent damage draws across possible facts, steps and cycles, with the possible facts of each demand set chosen independently of those draws. One simulation tick is one acquisition and repair cycle. Its disruption and restoration steps can be represented by successive ticks of the embodied timeline, so one simulation tick spans several ticks of that timeline. Viability is checked after repair. From a fully restored body, the disruption probability and expected cost of restoring every tracked possible fact are

$$d = 1 - (1 - 0.02)(1 - 0.05) = 0.069, \qquad \mathbb{E}[C_n] = 12d = 0.828.$$

A task is completed when the three possible facts of its demand set survive both damage steps, whether completion is assessed before or after the damage from the task attempt. The surplus gain in Table 23 therefore gives

$$\gamma \geq 1.6(1 - d)^3 - 12d \approx 0.463 > 0.$$

Assessing success before task damage only increases this mean. The operating requirement $a = 12$ suffices to restore every possible fact in each cycle. For an intact initial body, a finite initial phase reaches a reserve above this requirement: seven damage-free cycles give

$$A_7 = 2 + 7(1.6) = 13.2 = a + 1.2, \qquad p_{\mathrm{start}} = (1 - d)^{84} > 0.$$

Proposition 27 then gives positive probability of perpetual operation from that initial condition, and Corollary 20 shows that reserve per cycle converges almost surely on that event. These calculations concern the fixed body and the demands of the illustration. Open-ended policy generation depends on the further demand assumption of Theorem 17.

Table 23: Settings of the loop simulation.

Setting	Value
Possible facts	12
Possible facts in a demand set	3
Disruption chance per intact possible fact per tick, background	0.02
Disruption chance per intact possible fact per task attempt	0.05
Capacity spent per possible fact repaired	1
Intact possible facts required for viability	6
Initial capacity	2
Gain per completed task, deficit	0.5
Gain per completed task, break-even	0.9
Gain per completed task, surplus	1.6
Timelines per setting	300
Horizon in ticks	2,000
Burst tick, young	5
Burst tick, mature	300
Ticks observed after a burst	200

Table 24: Outcome of the loop simulation at each rate of renewal.

Regime	Gain per task	Fraction viable at 2,000 ticks	Mean stored capacity at the end
Deficit	0.5	0.00	0.3
Break-even	0.9	0.17	13.3
Surplus	1.6	0.99	1,347

Table 25: Chance of surviving the 200 ticks after one burst, among timelines viable when the burst occurs. At break-even, 83 timelines were viable at the mature burst tick.

Possible facts disrupted	Break-even, young	Break-even, mature	Surplus, young	Surplus, mature
6	0.02	0.70	0.61	1.00
9	0.00	0.69	0.22	1.00

Table 26: Binary addition.

	c_{ff}				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.11	.039	.75	.008	.10	.037	.48	.012
10	.27	.064	.91	.006	.13	.048	.69	.009
14	.68	.106	.98	.005	.24	.097	.91	.006

Table 27: Binary multiplication.

	c_{ff}				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.05	.026	.74	.009	.01	.011	.58	.011
10	.16	.045	.86	.006	.08	.034	.78	.008
14	.46	.061	.96	.003	.21	.050	.93	.003

28.5 Binary arithmetic

I ran some toy binary arithmetic experiments for my 2023 paper on weak hypotheses [Bennett, 2023a]. For each finite arithmetic task, four bits encoded an input and four an addition or multiplication output. A child had six, ten or fourteen sampled correct outputs. I implemented a variant of A^* to search the same child for a freest correct and a minimum-description-length correct policy. Success meant reconstructing the full sixteen-input parent's output set. There were 75–256 trials per setting. The reported freedom-to-description-length generalisation-rate ratios range from 1.10 to 5.00 [Bennett, 2023a, 2025a]. For each trial and each selector $c \in \{c_{\mathrm{ff}}, c_{mdl}\}$, I define

$$O_{\mathrm{recon}}(c) := \mathrm{Ext}(c) \cap \mathrm{Ext}(I_n).$$

The table's Rate is the fraction of trials with $O_{\mathrm{recon}}(c) = O_n$; its ±95% column gives the Wald confidence-interval half-width. AvgExt is the trial mean of $|O_{\mathrm{recon}}(c) \cap O_n|/|O_n|$, and StdErr is the standard error of that mean.

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